Z + b-tagged jet in Summer22: three taggers at the Loose working point

All kinematic distributions for ParticleNet, DeepJet and RobustParT, each with its own BTV scale factors. All corrections, including the muon changes of 8 November.

8 November 2026

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Analysis setup 1/2: data, simulation, event selection

Data

- 2022 runs C–D, EGamma and Muon datasets, 22Sep2023 reprocessing, NanoAODv12.
- Certified lumisections only (golden JSON Cert_Collisions2022_355100_362760).
- Test sample: 20 files per channel = 389 pb^{-1} (ee) and 742 pb^{-1} (μμ), measured with brilcalc (normtag_PHYSICS) on the lumisections actually read.

Simulation (Run3Summer22, NanoAODv12)

- Drell–Yan: DYto2L-2Jets, MadGraph5_aMC@NLO with FxFx merging (0, 1, 2 partons at NLO), m($\ell\ell$) > 50 GeV. Inclusive sample used for generator pT($\ell\ell$) < 100 GeV, pT($\ell\ell$)-binned samples above (100–200, 200–400, 400–600, > 600).
- Backgrounds: $t\bar{t} \to 2\ell 2\nu$ (POWHEG), WW, WZ, ZZ (Pythia8).
- Each sample scaled to cross section × luminosity / sum of generator weights. 20 files per sample.

Jet flavour in MC

- Leading jet's hadronFlavour: b (a B hadron in the jet), c (a C hadron, no B), light (neither).

Triggers

- Z→ee: HLT_Ele30_WPTight_Gsf. Z→μμ: HLT_IsoMu24.

Leptons

- Electrons: pT > 25 GeV, |η| < 2.4, cut-based Tight ID, barrel–endcap gap 1.442 < |η_SC| < 1.566 removed, impact parameter |dxy| < 0.05 (0.1) cm and |dz| < 0.1 (0.2) cm in the barrel (endcap).
- Muons: pT > 26 GeV, |η| < 2.4, Tight ID, PF isolation (ΔR = 0.4) < 0.15 (Tight).

Z + jet selection (applied in this order)

- Trigger fired; two leptons of the channel, leading lepton pT > 35 GeV; opposite charge; 71 < m($\ell\ell$) < 111 GeV.
- Missing transverse momentum (type-1 PUPPI MET) < 50 GeV, to remove $t\bar{t}$.
- At least one AK4 jet (PUPPI): pT > 30 GeV, |η| < 2.5, Tight-lepton-veto ID (with the NanoAODv12 fix), ΔR > 0.4 from the leptons. The leading jet is the one we study and tag.

Analysis setup 2/2: corrections, scale factors, b/c tagging

All corrections from the CMS jsonPOG files (/cvmfs/cms-griddata.cern.ch/cat/metadata), Summer22 versions.

Jets and MET

- Jet energy corrections Summer22_22Sep2023_V4 (data per run era and MC); jet energy resolution smearing Summer22_22Sep2023_JRV2 (MC).
- Type-1 PUPPI MET recomputed with the corrected jets.
- Jet veto map Summer22_23Sep2023_RunCD_V1: event dropped if a jet (pT > 15 GeV, tight ID, EM fraction < 0.9, no muon nearby) is in a vetoed region.

Leptons

- Electron energy scale (data) and smearing (MC), EGM.
- Muon momentum scale (data) and smearing (MC), MUO. In Run 3 this replaces the Rochester correction.

Event cleaning

- MET filters in data and MC: goodVertices, globalSuperTightHalo2016, EcalDeadCellTriggerPrimitive, BadPFMuon, BadPFMuonDz, hfNoisyHits, eeBadSc.

Event weights in MC

- Pileup: LUM puWeights.json.gz (2022 C–D golden profile, 69.2 mb).
- Electrons: reconstruction and Tight ID SFs (EGM 2022 Re-reco BCD).
- Muons: Tight ID SF (NUM_TightID_DEN_TrackerMuons) and Tight isolation SF (NUM_TightPFIso_DEN_TightID), MUO.
- Trigger: either lepton can fire, so SF = $[1 - (1-\varepsilon^{1}\text{data})(1-\varepsilon^{2}\text{data})] / [1 - (1-\varepsilon^{1}\text{MC})(1-\varepsilon^{2}\text{MC})]$ with the POG data and MC efficiencies of both leptons (EGM Ele30 Tight ID; MUO IsoMu24 for Tight ID + Tight isolation).
- Not applied: L1 prefiring weight (not needed in Run 3), DY Z pT reweighting (switched off).

b and c tagging (leading jet only)

- Taggers: ParticleNet, DeepJet, RobustParT. Official BTV Summer22 Loose/Medium/Tight working points.
- b-tag: B score > cut. c-tag: CvL > cut and CvB > cut. b-tag and c-tag are separate selections.
- MC: the tagged jet is weighted by the BTV SF for its true flavour (b-tag: _comb for b and c, _light for light; c-tag: _wc for c, _tnp for b, _light for light). Outside the SF range the SF at the edge is used.

Uncertainties and software

- Statistical only (data and MC). All SFs are applied with their central values; their uncertainties and the other systematic uncertainties are not included yet.
- coffea 0.7, BTVNanoCommissioning (Zb_Zc_ratio workflow), correctionlib.

How to read the numbers in these slides

Z selection	Trigger, two leptons, opposite charge, 71 < m($\ell\ell$) < 111 GeV. No jet or MET requirement.
Full selection	Z selection + MET < 50 GeV + at least one jet. All plots use this unless stated.
data/MC	Data events divided by the MC prediction (all corrections and weights). 1 = perfect agreement.
Shape	data/MC in one range divided by the overall data/MC: the normalisation is taken out, only the shape is left.
Slope	Straight-line fit of data/MC versus pT, quoted in % per 100 GeV.
Loose b-tag	The leading jet's b score is above the BTV Loose working point (about 10% of light jets pass).
Tagged	Fraction of Z + jet events whose leading jet is tagged.
Tag ratio	Tagged fraction in data divided by tagged fraction in MC. Tests the tagging and its SFs, independent of the overall normalisation.
Flavour share	Share of all MC events whose leading jet is a true b, c or light jet (DY), or $t\bar{t}$.
Errors	Statistical only (data and MC). Only 20 files per sample, so small differences are not significant.

What this deck shows

The same Z + jet selection as before, with one extra requirement: the leading jet passes the Loose b-tag of one of three taggers. Each tagger is a separate set of plots.

tagger (b score cut)	Loose	Medium	Tight	BTV SFs used in MC
ParticleNet	0.047	0.245	0.6734	particleNet_comb / _light
DeepJet	0.0583	0.3086	0.7183	deepJet_comb / _light
RobustParT (Robust Particle Transformer)	0.0849	0.4319	0.8482	robustParticleTransformer_comb / _light

Only Loose is used here; Medium and Tight are listed for reference. The SF is applied to the tagged jet according to its true flavour (_comb for b and c jets, _light for light jets).

How to read the plots

- Left panel Z→ee, right panel Z→μμ. Black points: data. Coloured stack: MC, split by the true flavour of the leading jet (DY+b, DY+c, DY+light) plus $t\bar{t}$ and dibosons (VV).
- Lower panel: data/MC. The hatched band is the MC statistical uncertainty only; the BTV SFs are applied, but their uncertainties and other systematic uncertainties are not in the band yet.
- The overall data/MC is about 0.93–0.96 in every region, so points sit slightly below 1 everywhere; look at the shape.
- The order of the plots is the same for each tagger, so the same page of each section can be compared directly.

The three taggers side by side

Flavour shares are of all MC. Tagged = fraction of Z + jet events whose leading jet passes the Loose b-tag. Tag ratio = tagged fraction in data / in MC (1 = data and MC agree).

	data/MC	DY+b	DY+c	DY+light	tagged (data)	tag ratio
no tag, Z→ee	0.949	4%	8%	87%	100%	–
no tag, Z→μμ	0.943	4%	8%	87%	100%	–
PNet, Z→ee	0.947 ± 0.031	21%	23%	54%	16.9%	0.998 ± 0.031
PNet, Z→μμ	0.925 ± 0.021	22%	21%	55%	17.6%	0.980 ± 0.022
DeepJet, Z→ee	0.926 ± 0.029	18%	22%	58%	18.4%	0.976 ± 0.029
DeepJet, Z→μμ	0.924 ± 0.021	20%	21%	57%	18.9%	0.979 ± 0.022
RobustParT, Z→ee	0.962 ± 0.031	20%	22%	55%	17.4%	1.013 ± 0.031
RobustParT, Z→μμ	0.981 ± 0.023	22%	21%	54%	17.9%	1.040 ± 0.024

mean BTV SF of the tagged jet (DY MC)	b jets	c jets	light jets	also c-tagged, MC	data μμ
PNet	0.95	0.94	1.21	77%	83%
DeepJet	0.95	0.95	1.19	80%	83%
RobustParT	0.96	0.96	1.18	78%	84%

All three select a similar sample at Loose: about 1 in 6 Z + jet events, of which only ~20% are real b jets.

Light jets are mistagged more often in data than in MC (light SF ≈ 1.2); the SFs correct for it. Most Loose b-tagged jets also pass the Loose c-tag of the same tagger (last two columns).

ParticleNet, Loose b-tag

Leading jet PNet B score > 0.047; MC weighted with the particleNet BTV scale factors.

	data/MC	DY+b	DY+c	DY+light	tt̄	tagged MC	tagged data
Z→ee	0.947 ± 0.031	21%	23%	54%	2%	16.9%	16.9%
Z→μμ	0.925 ± 0.021	22%	21%	55%	2%	17.9%	17.6%

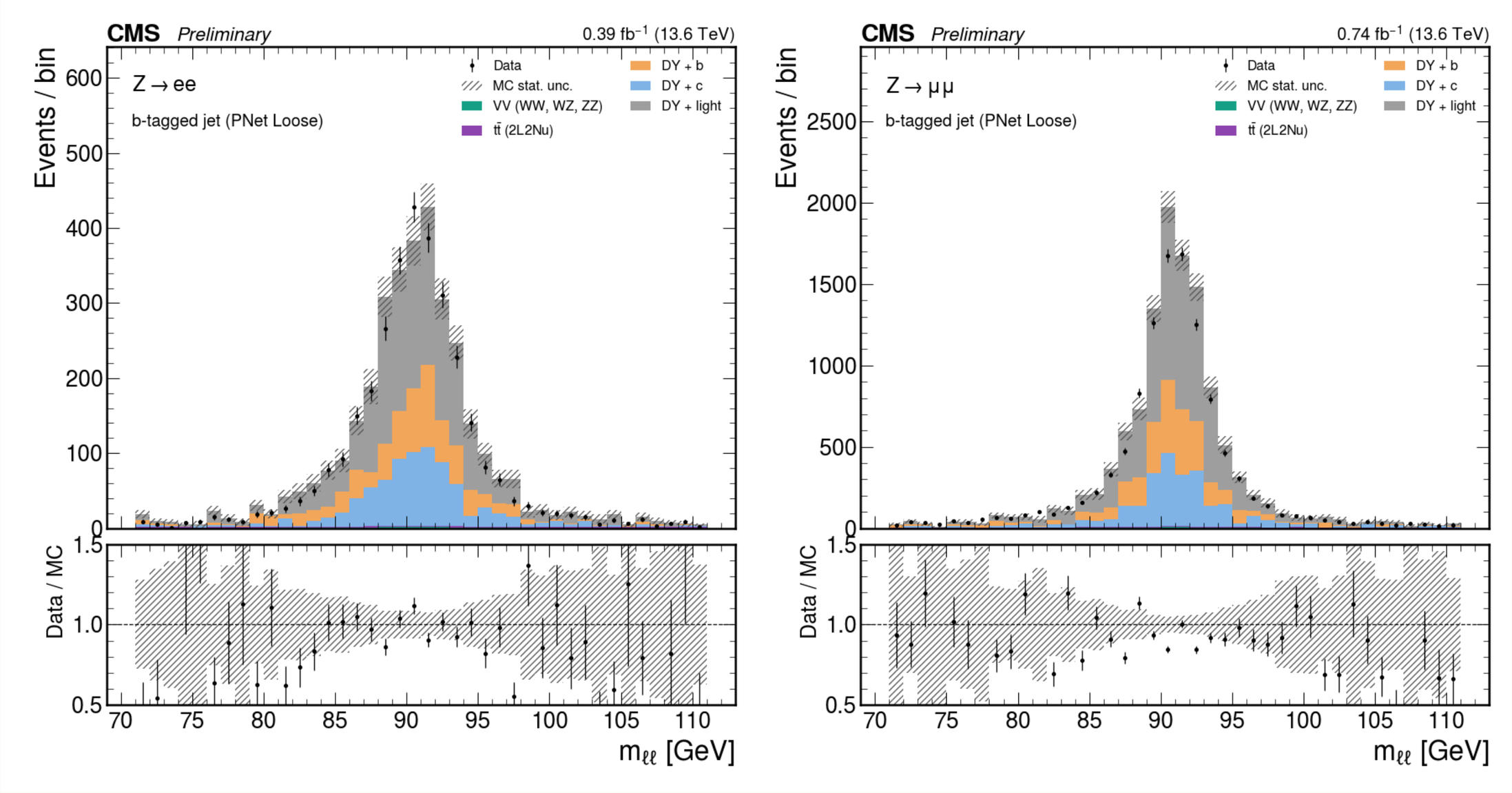
Flavour shares are of all MC. Tagged = fraction of Z + jet events whose leading jet passes the tag. Statistical errors only.

42 plots follow, in this order: Z and leptons, jets, Z–jet, MET and vertices, generator Z pT, the PNet scores, the lepton η–φ map. Left Z→ee, right Z→μμ.

ParticleNet · Loose b-tag (B > 0.047)

Mass of the lepton pair, m(ℓℓ)

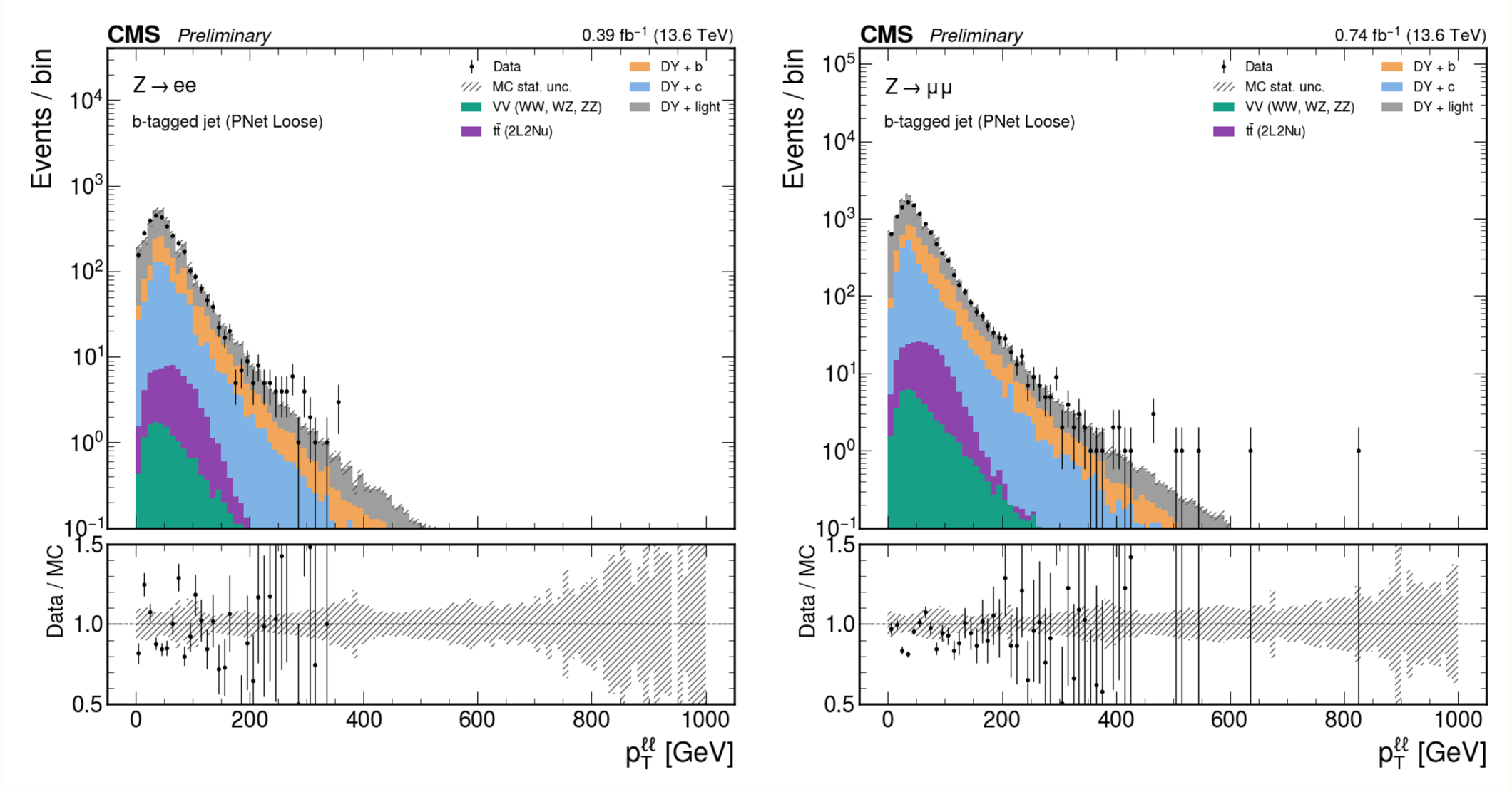
The Z peak, linear scale. Selection: 71–111 GeV. Tests the lepton energy scale and resolution.



ParticleNet · Loose b-tag (B > 0.047)

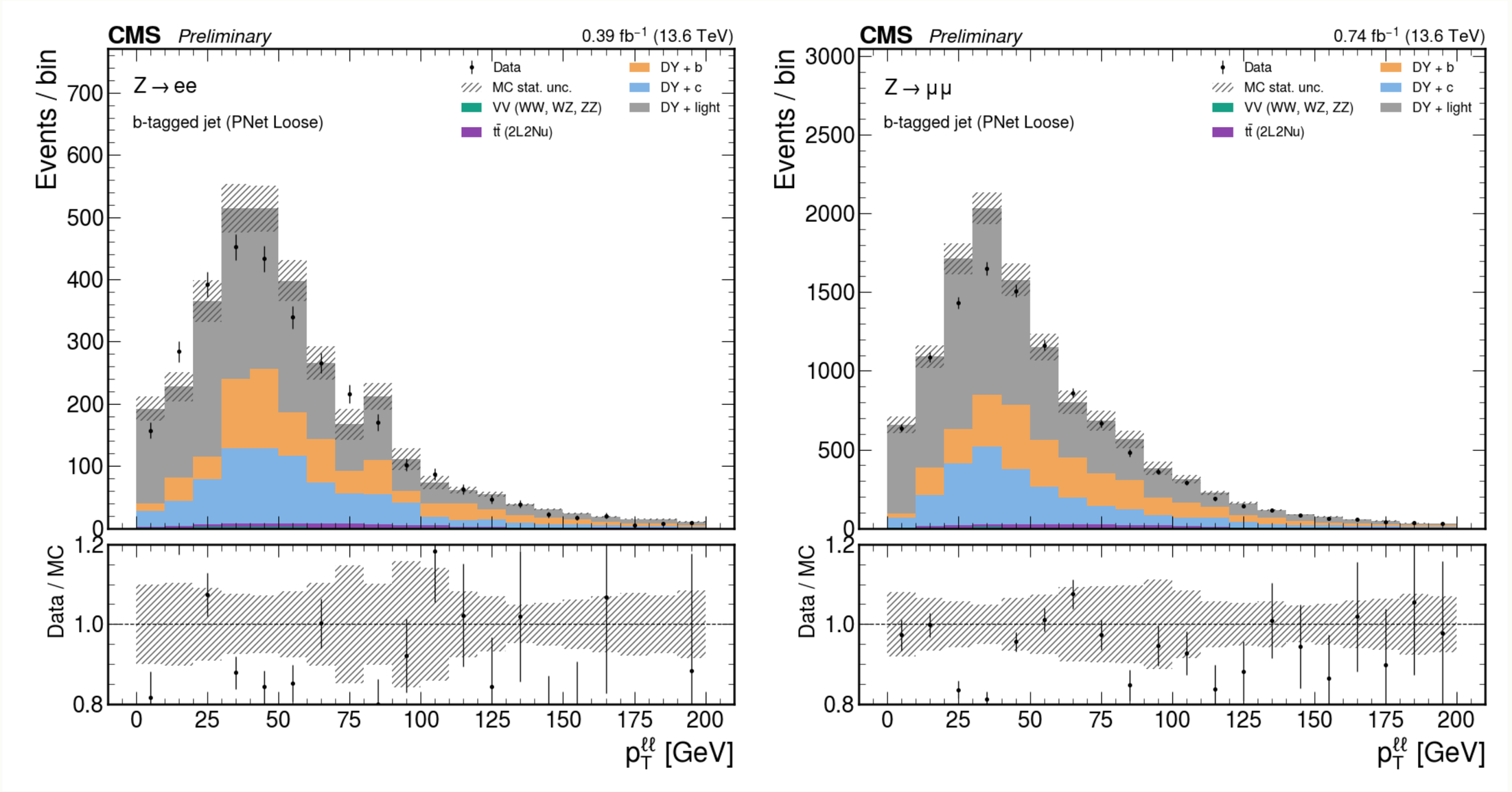
Transverse momentum of the lepton pair, pT(ℓℓ) = Z pT

Log scale up to 1 TeV. This is the distribution with the pT trend discussed in the follow-up slides (low at 30–60 GeV, high at 100–150 GeV).



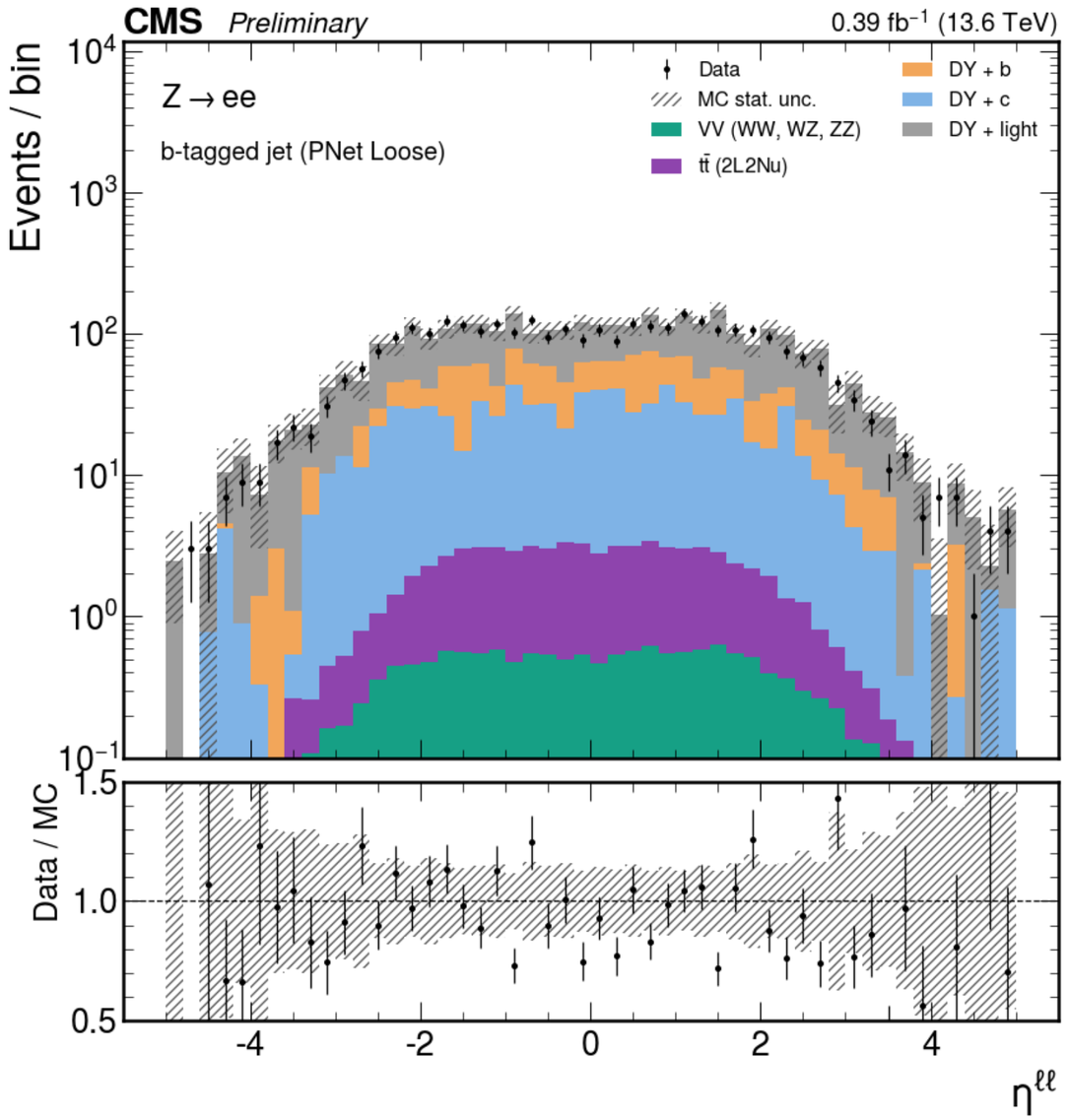
Z pT, zoom on 0–200 GeV

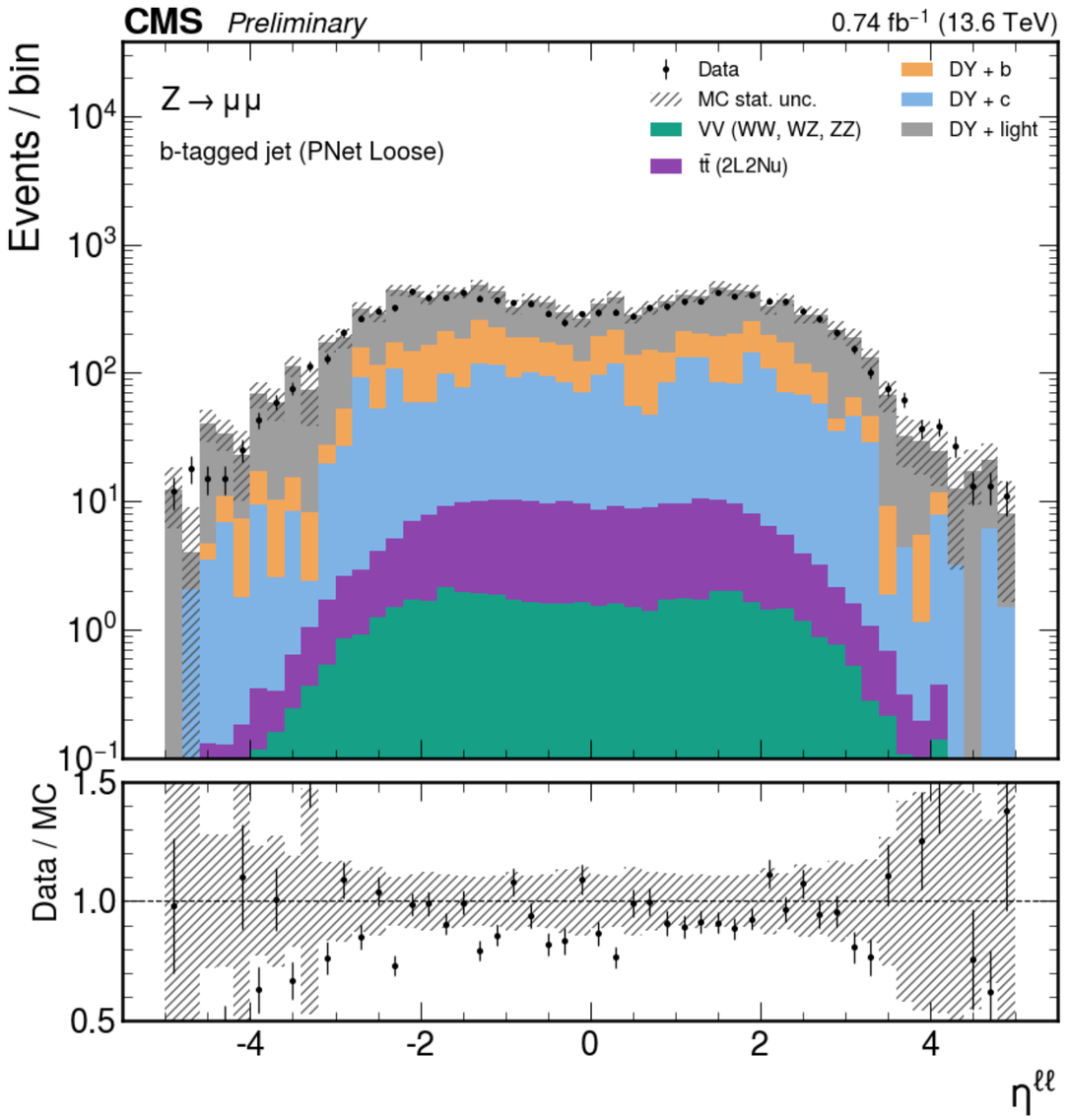
Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.



ParticleNet · Loose b-tag (B > 0.047)

Pseudorapidity of the lepton pair, η(ℓℓ)

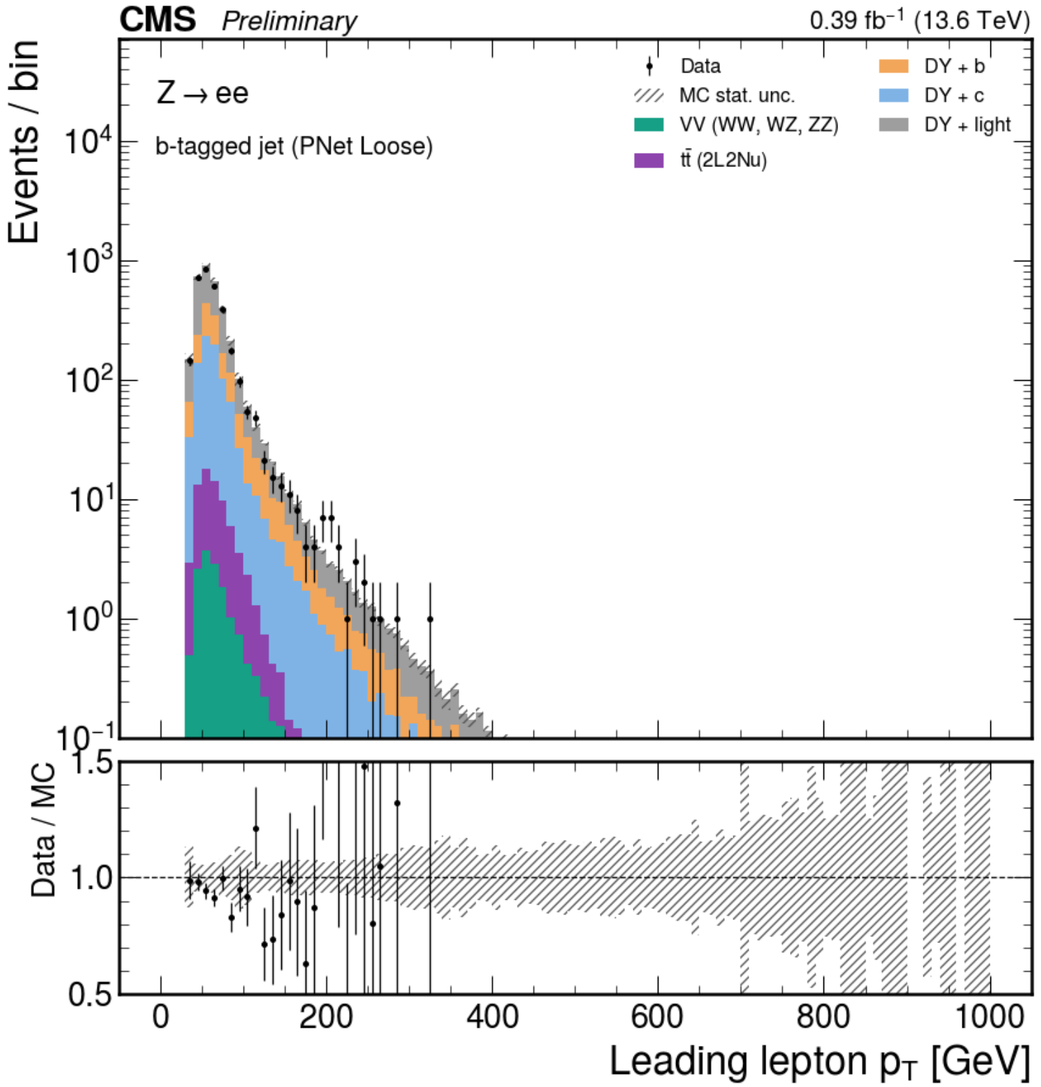


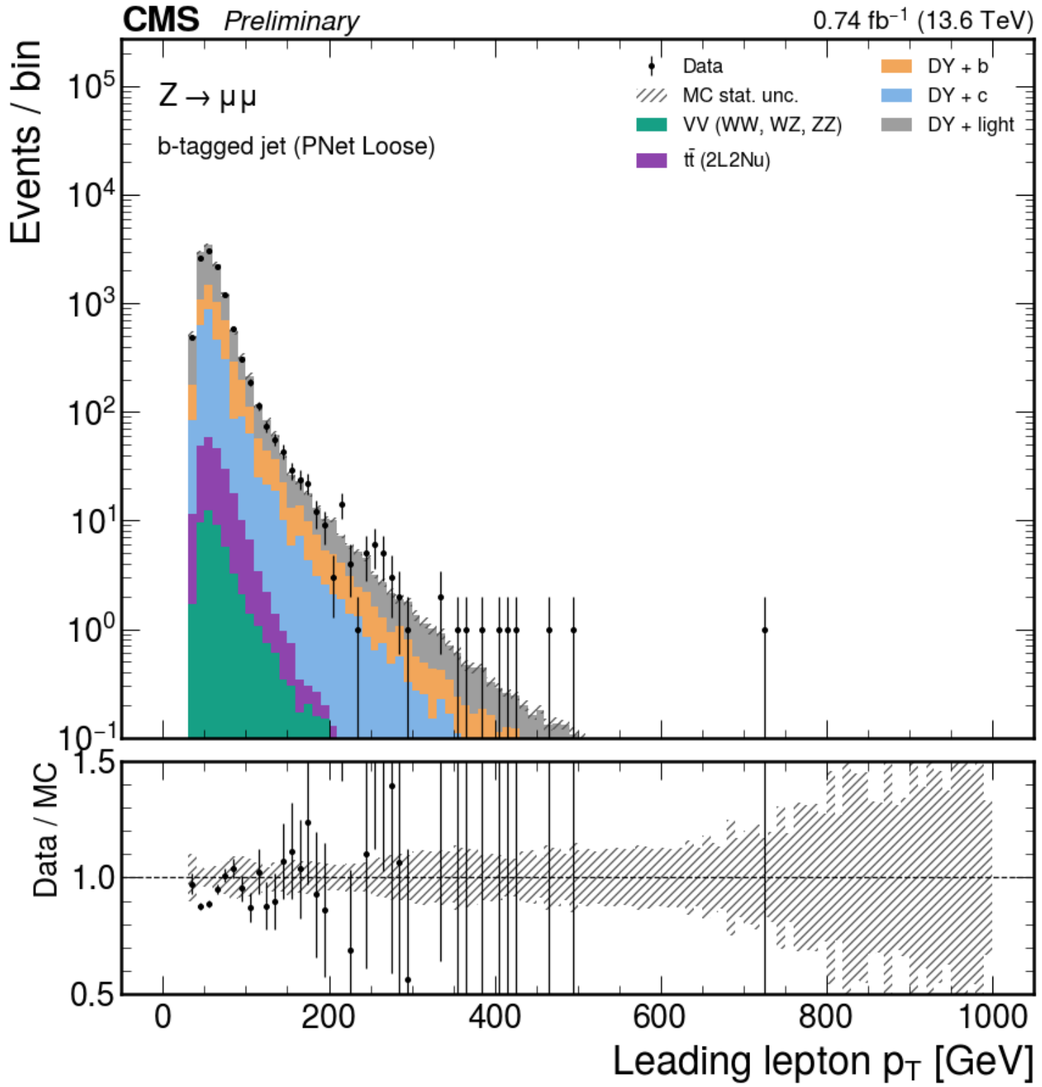


ParticleNet · Loose b-tag (B > 0.047)

Leading-lepton pT

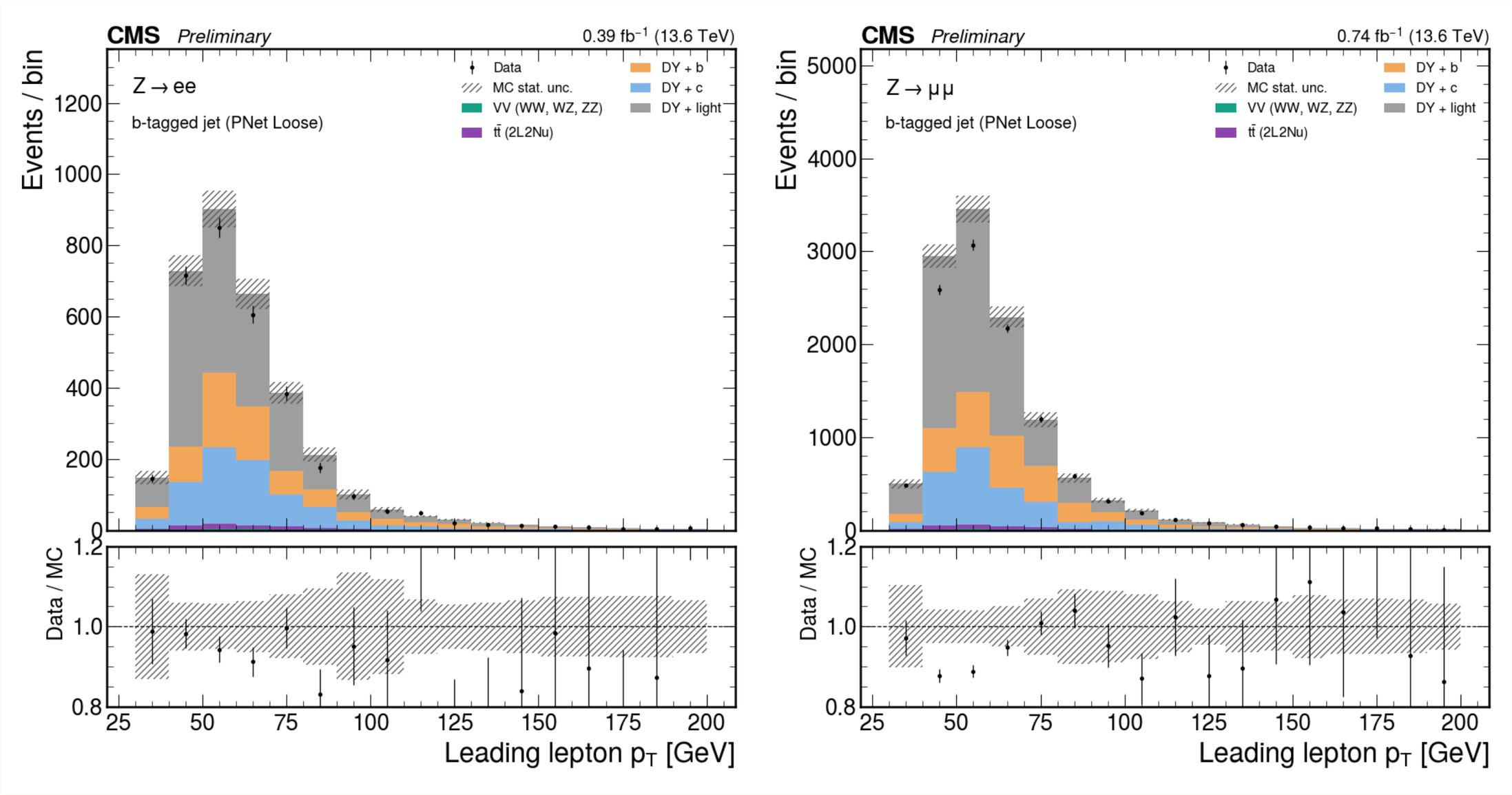
The selection requires pT > 35 GeV for the leading lepton (on the trigger plateau).





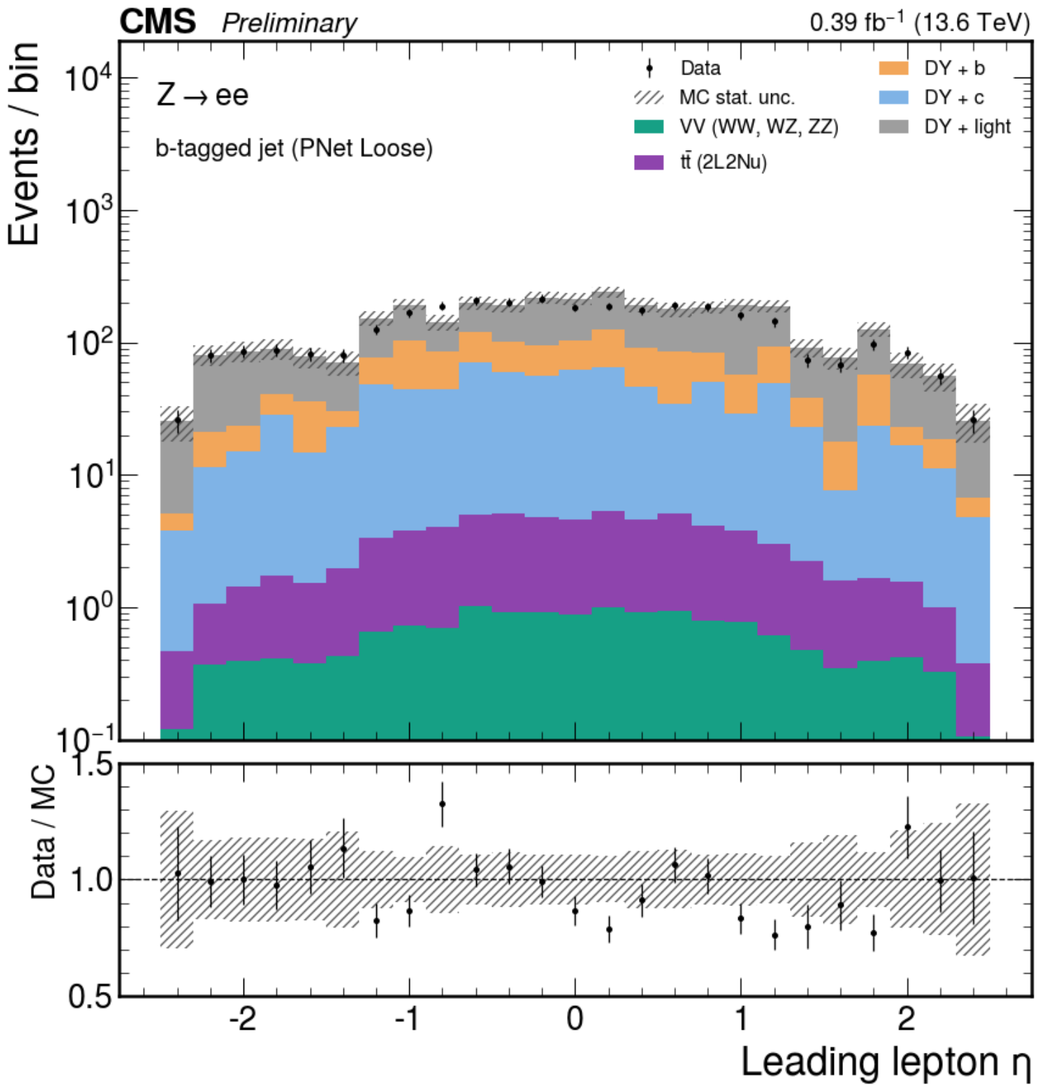
Leading-lepton pT, zoom on 30–200 GeV

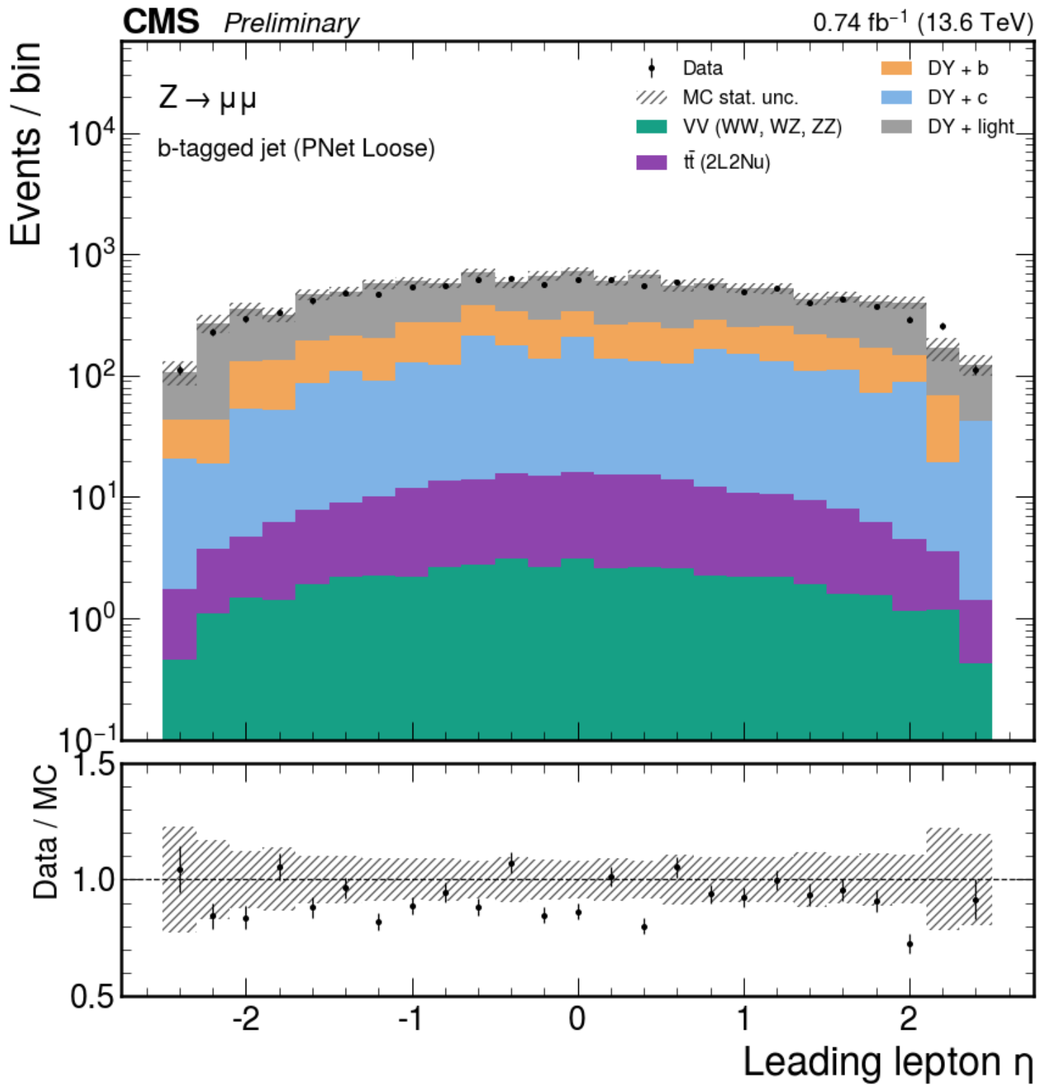
Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.



Leading-lepton η

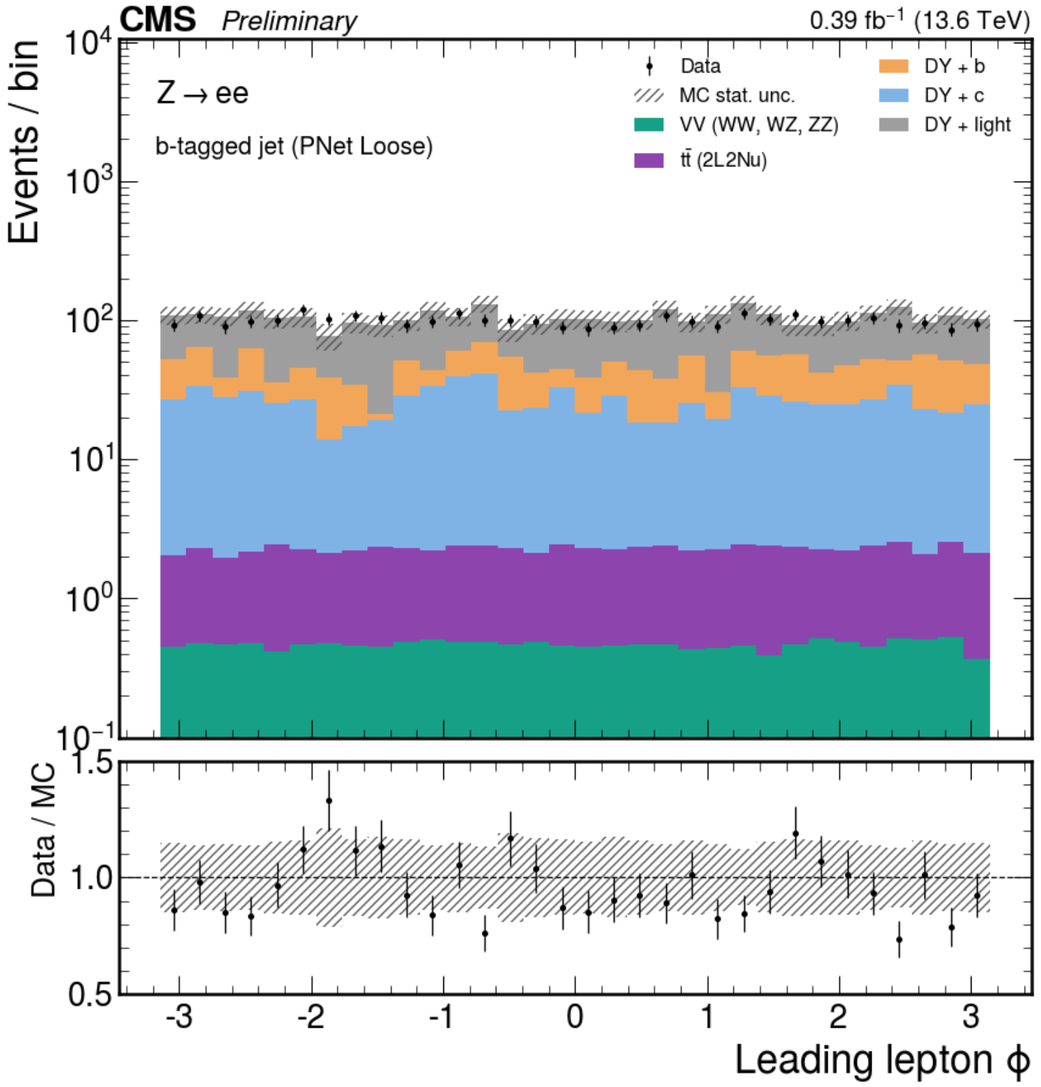
|η| < 2.4. For electrons the barrel–endcap gap (1.44 < |η| < 1.57) is removed, so that bin is low.

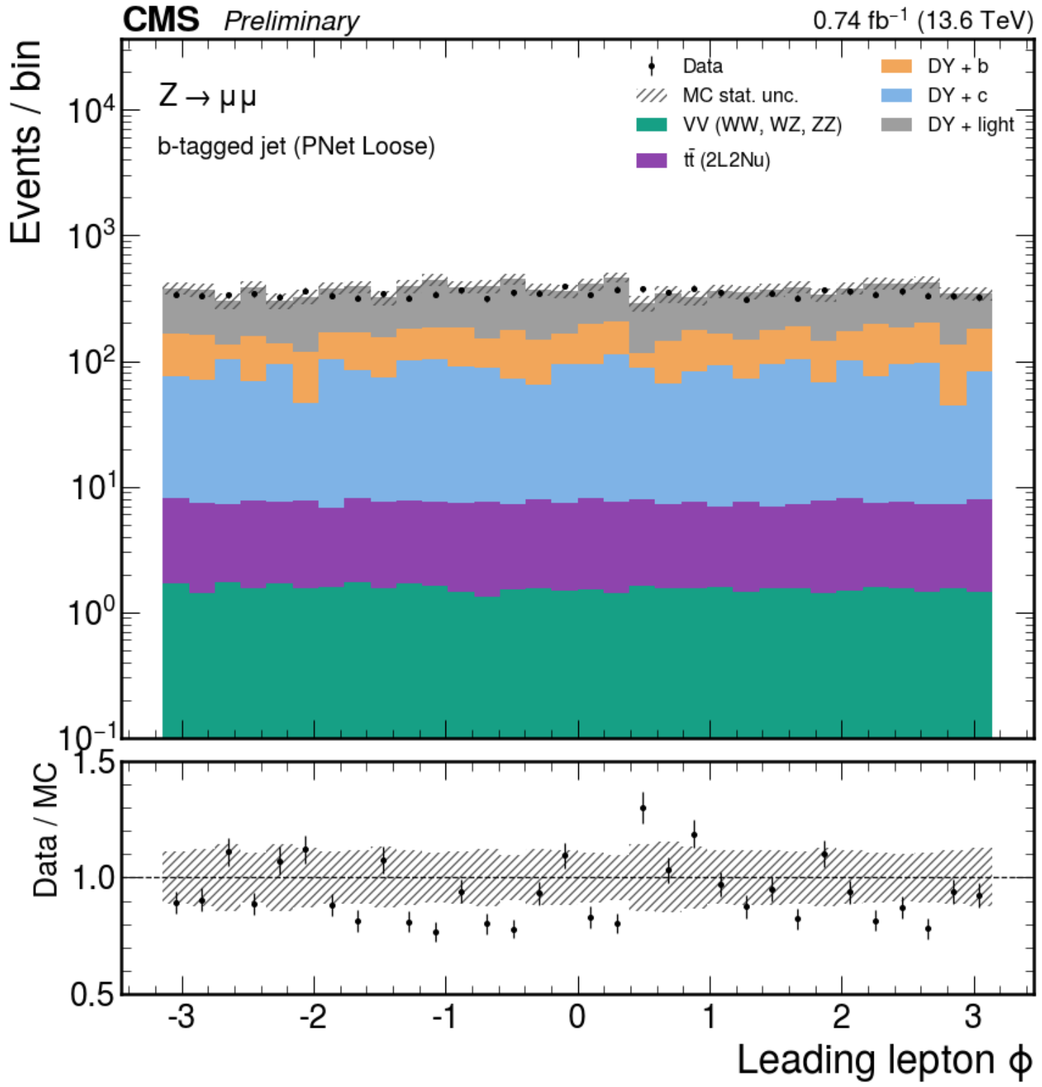




Leading-lepton φ

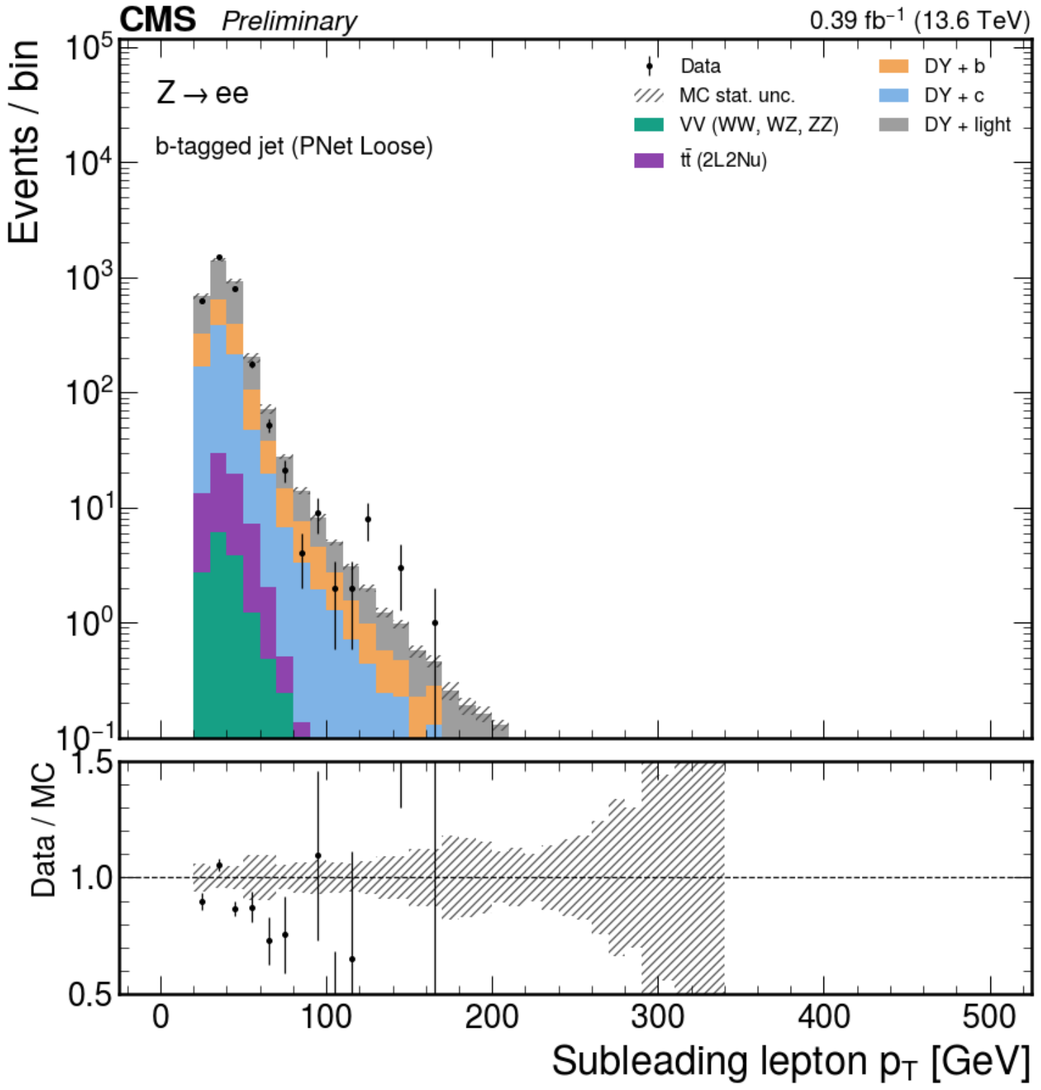
Expected to be flat; dips would point to detector problems.

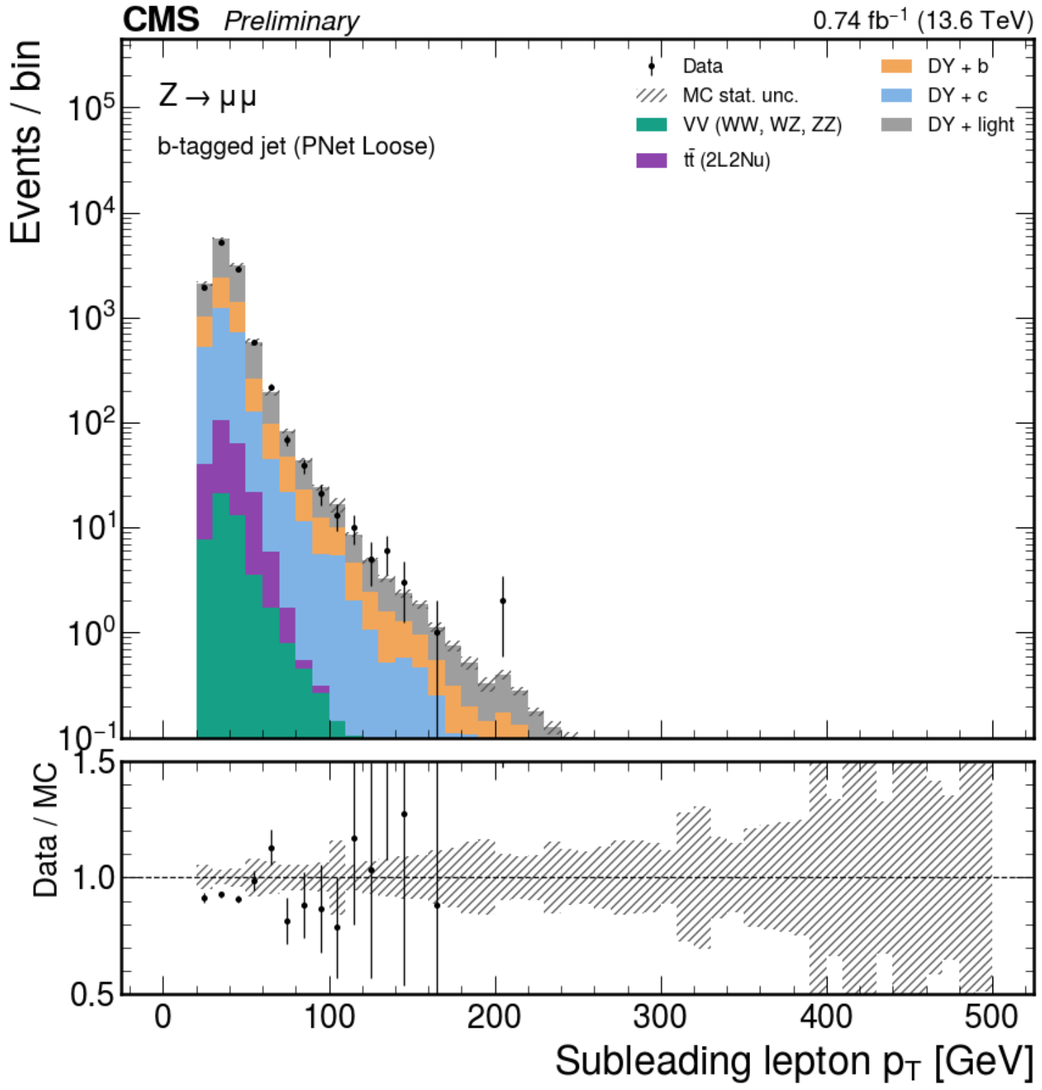




Second-lepton pT

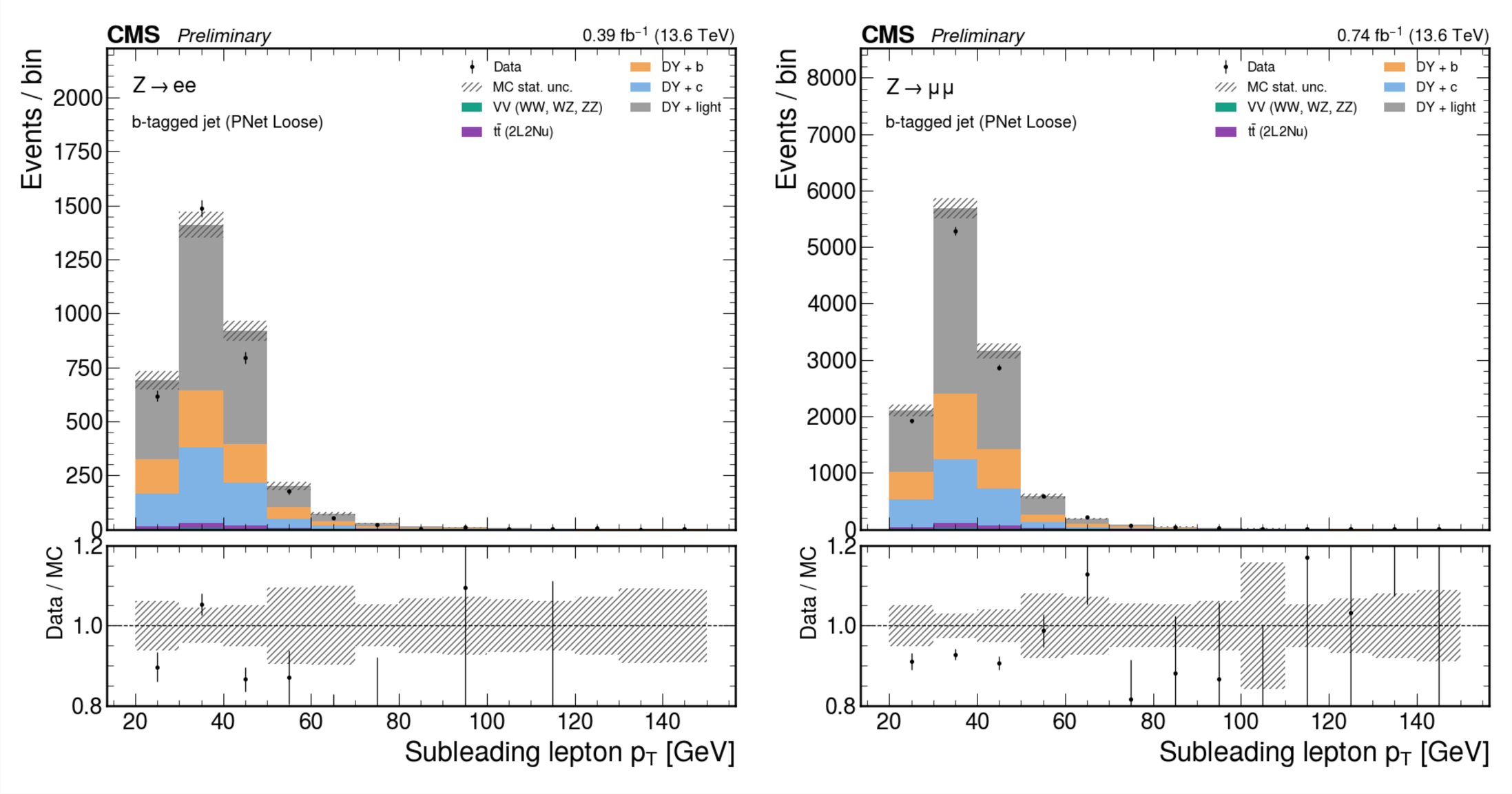
The selection requires pT > 25 GeV for electrons and > 26 GeV for muons.



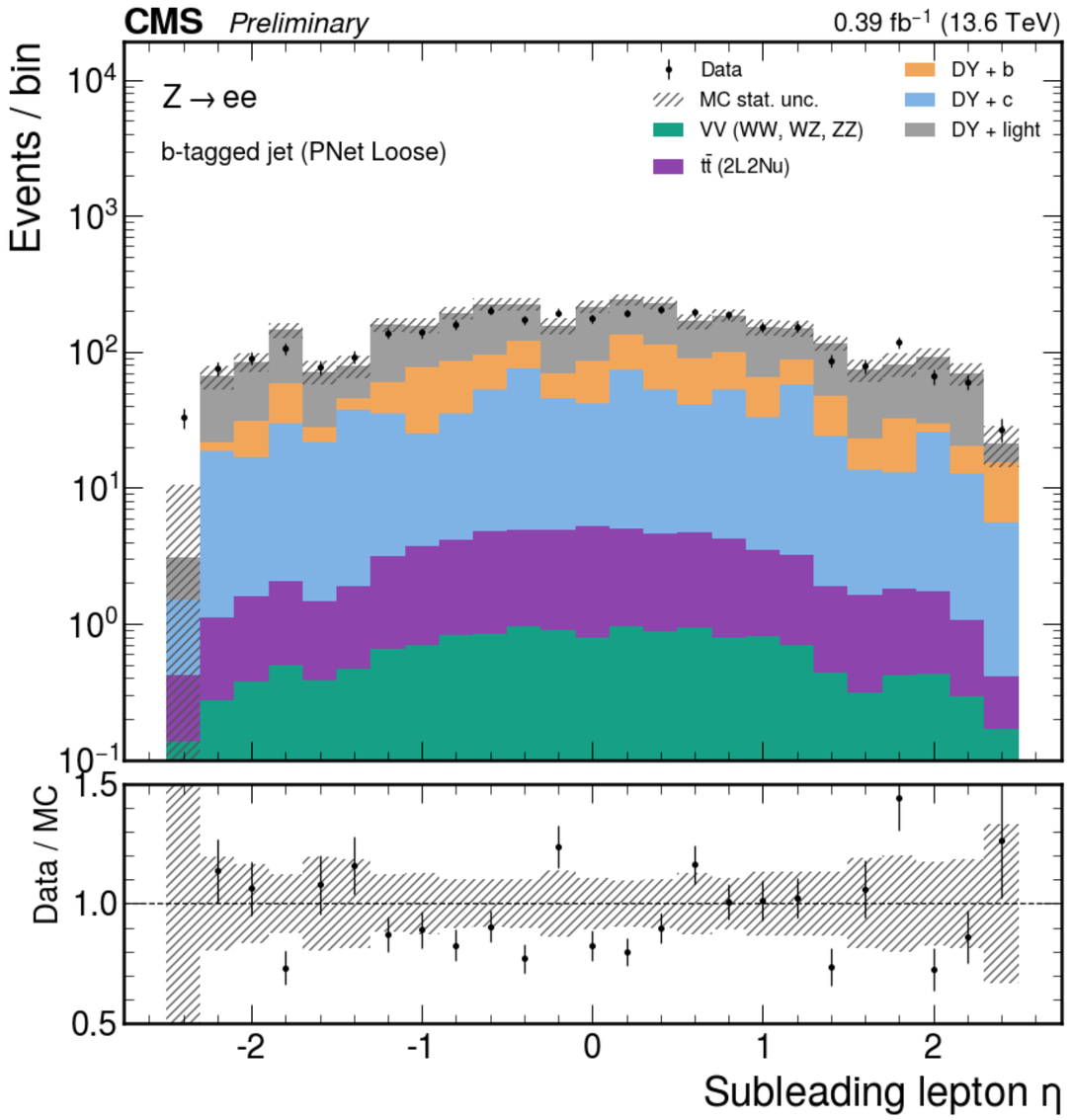


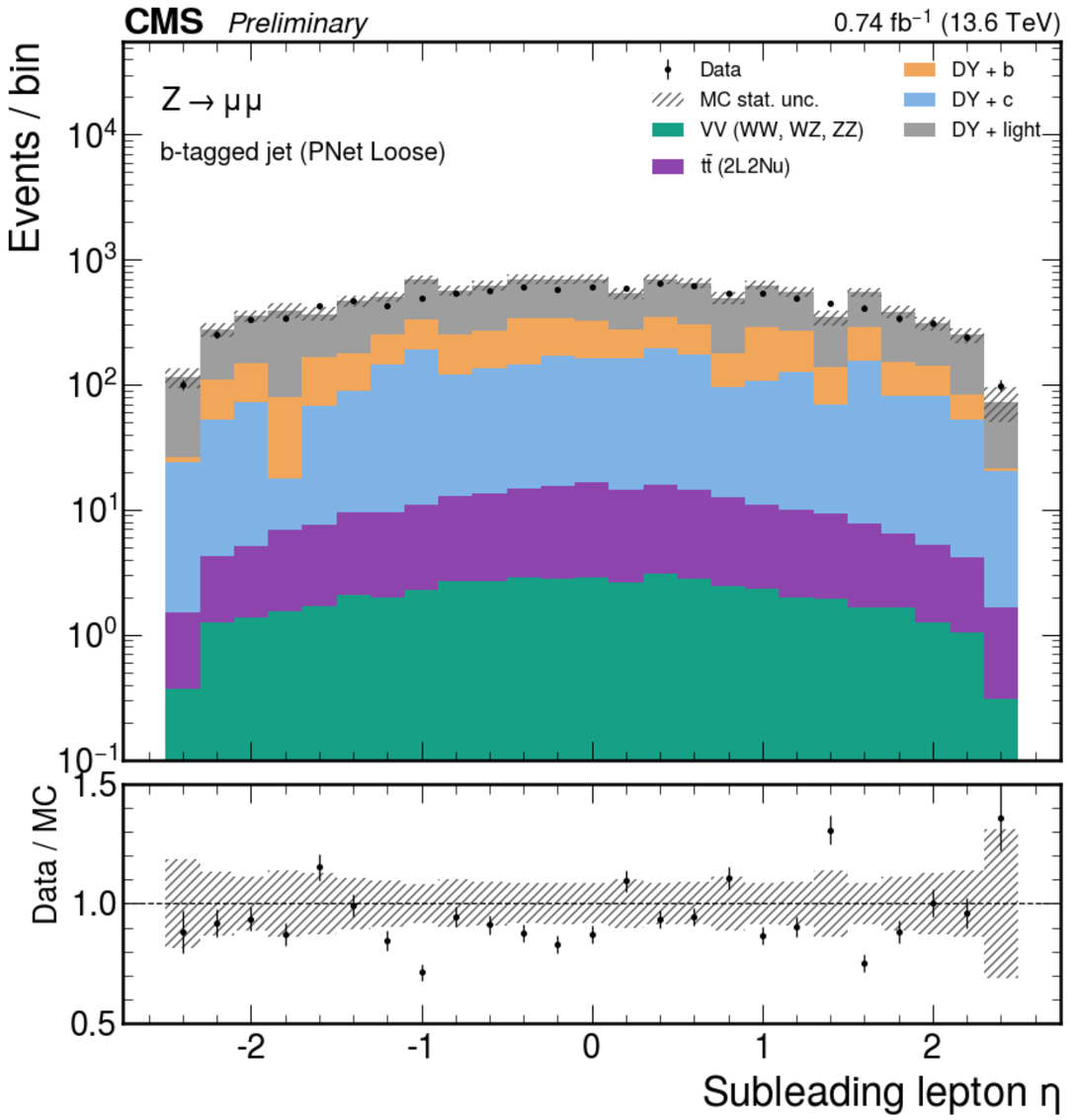
Second-lepton pT, zoom on 20–150 GeV

Zoom: 20–150, linear scale, data/MC from 0.8 to 1.2.

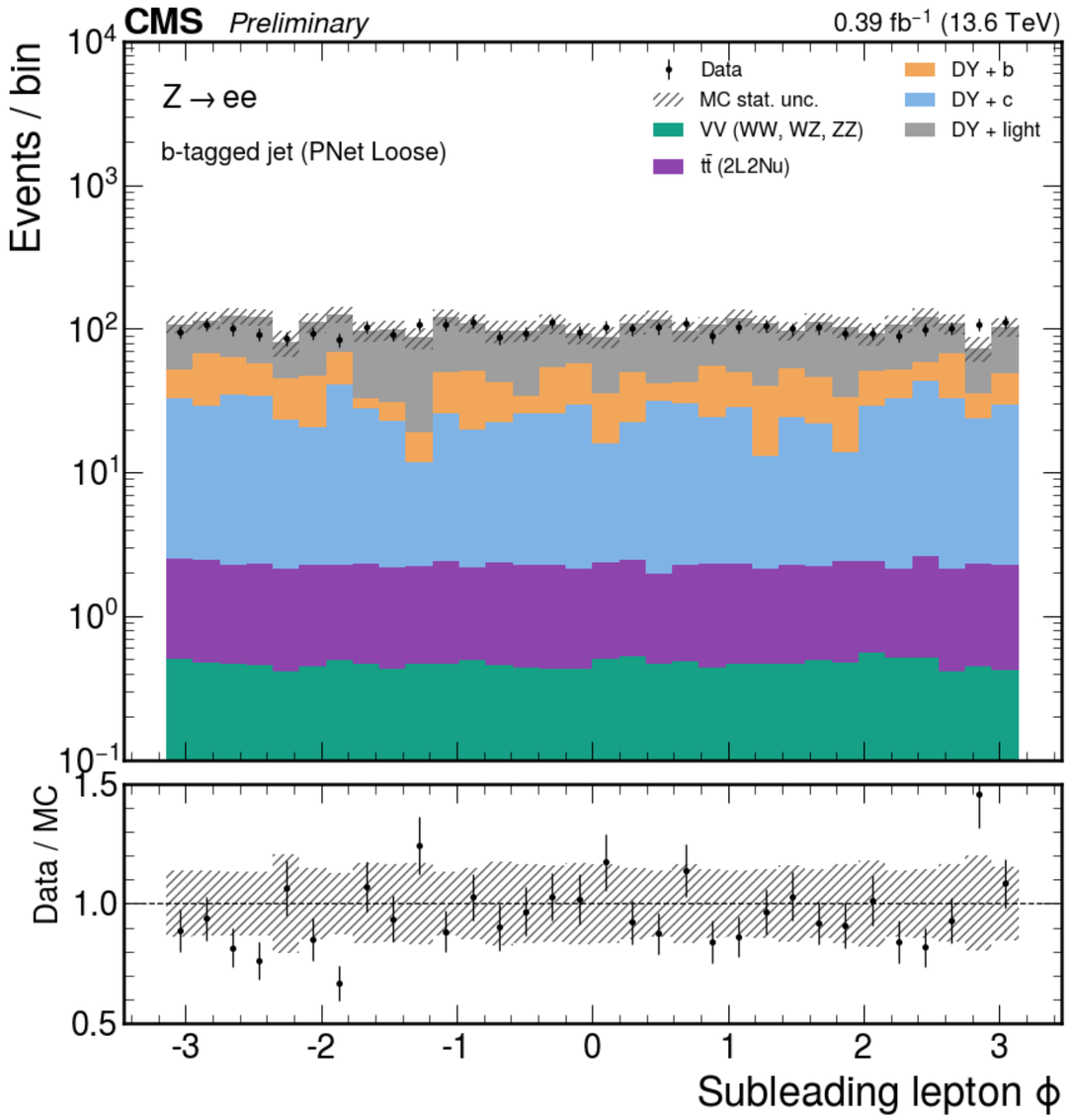


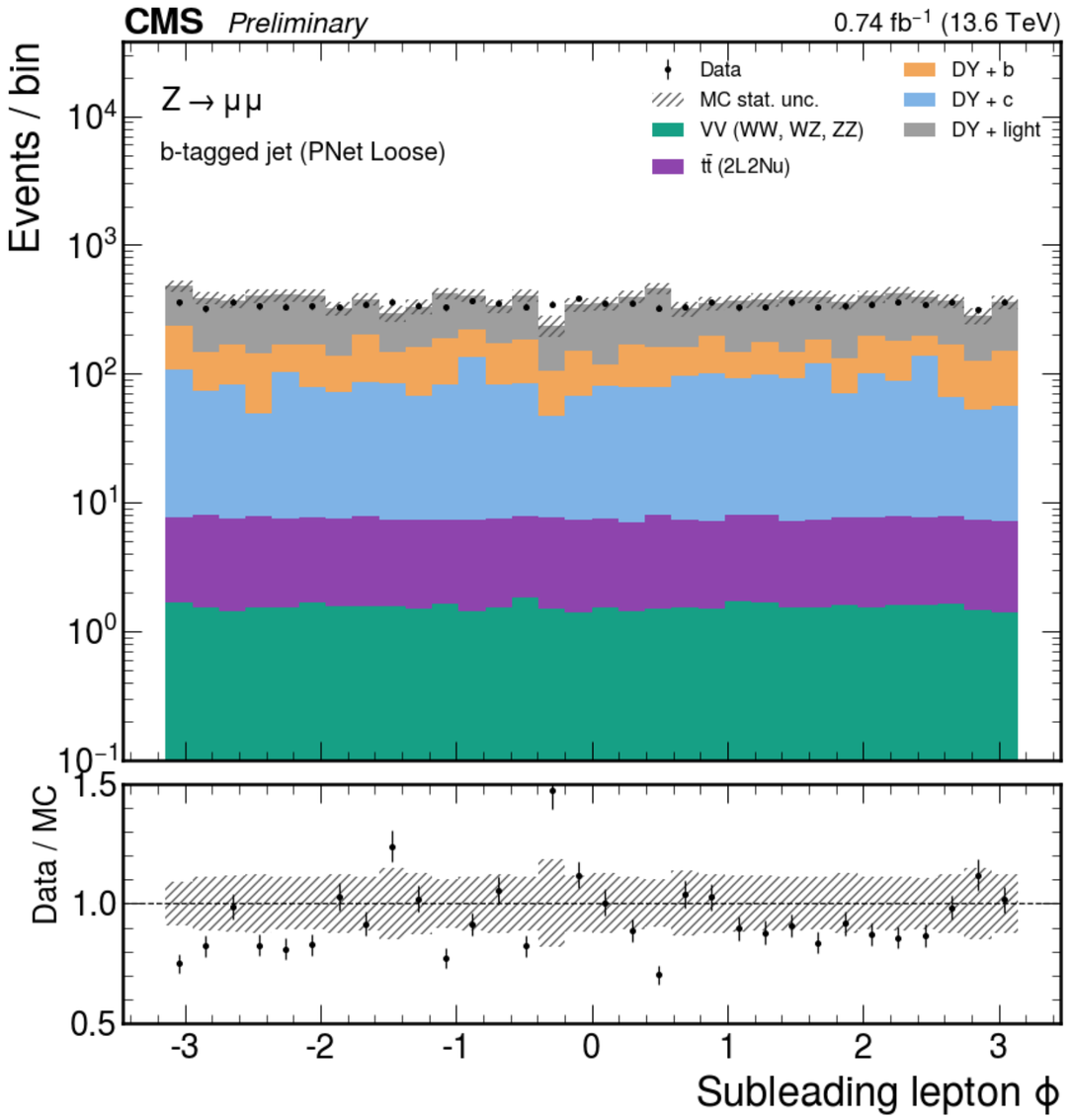
Second-lepton η





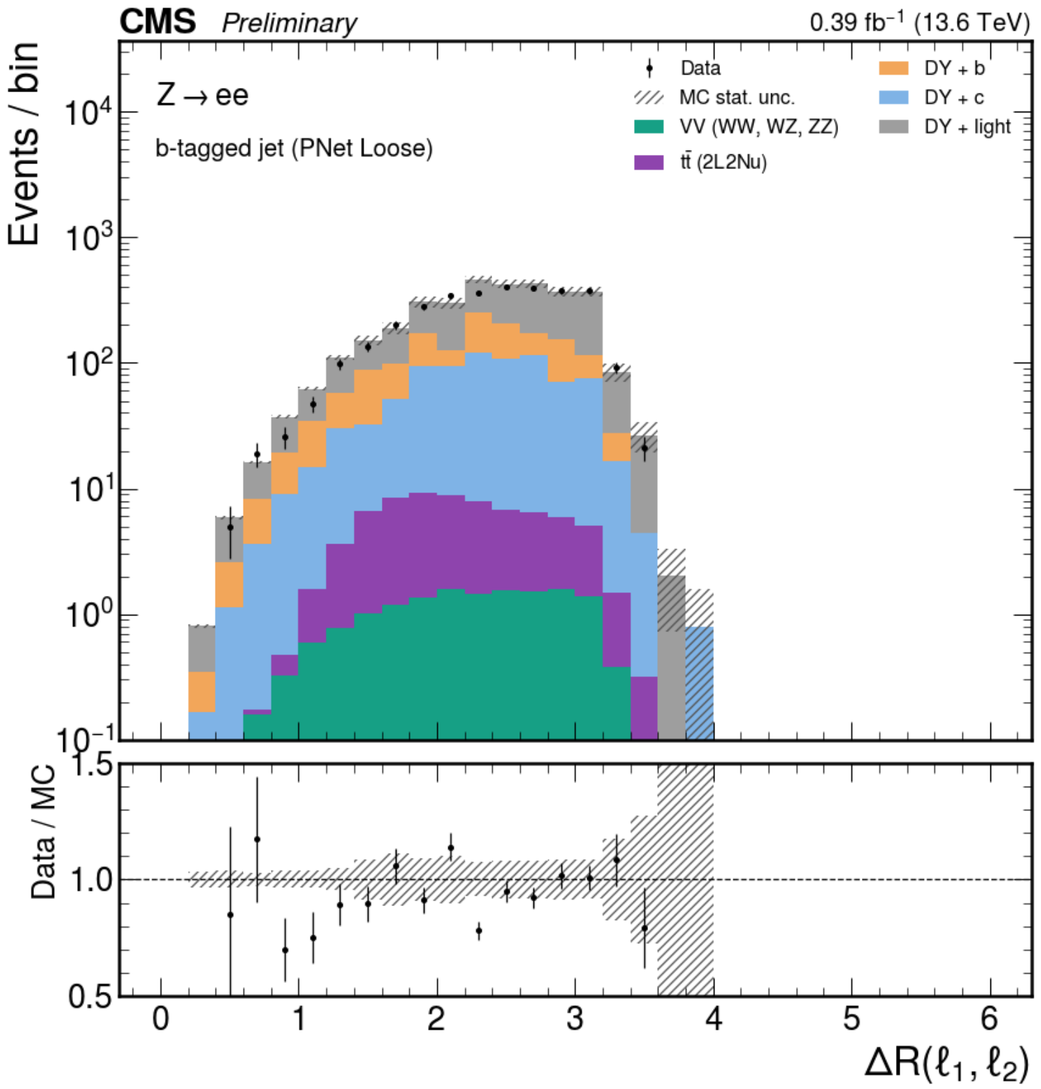
Second-lepton φ

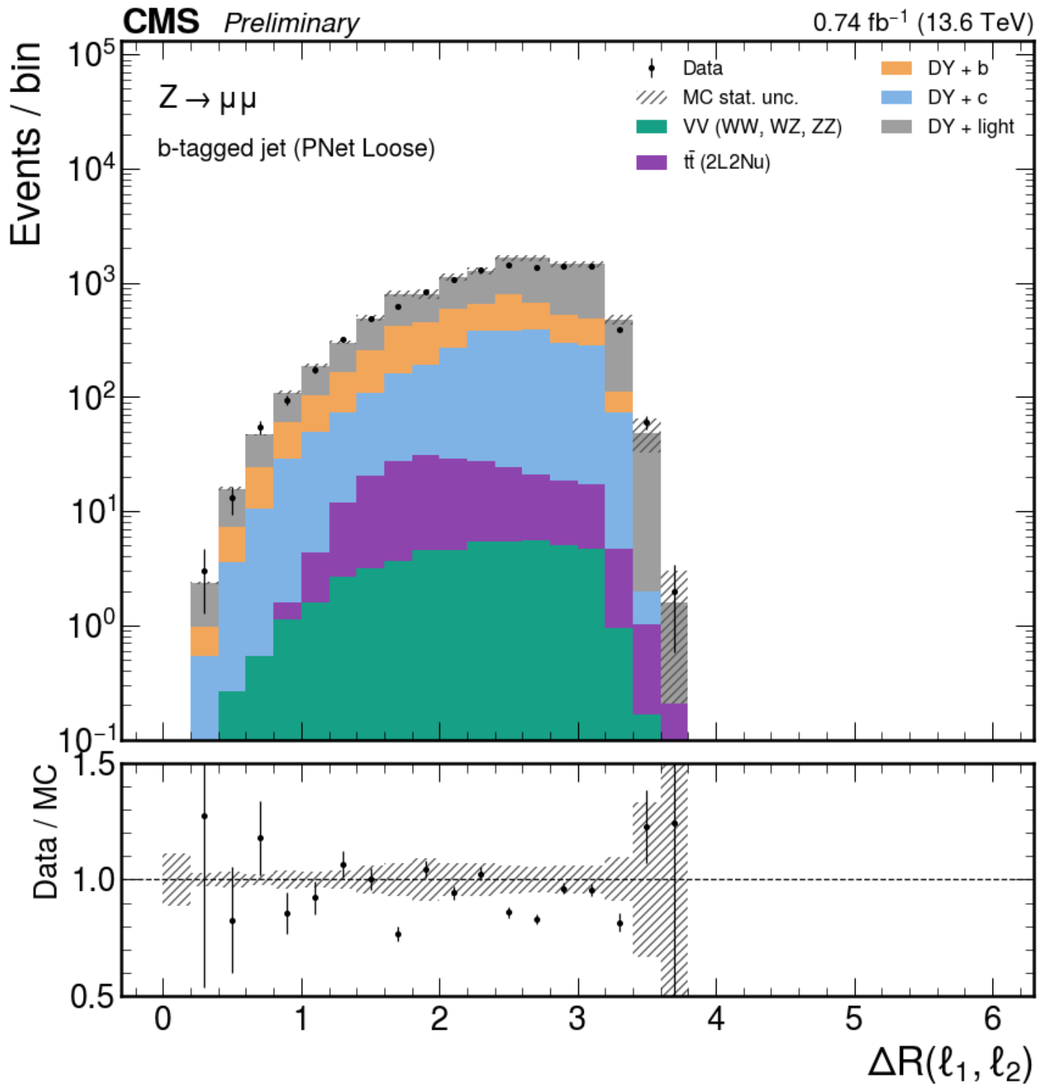




Angular distance between the two leptons, ΔR(ℓℓ)

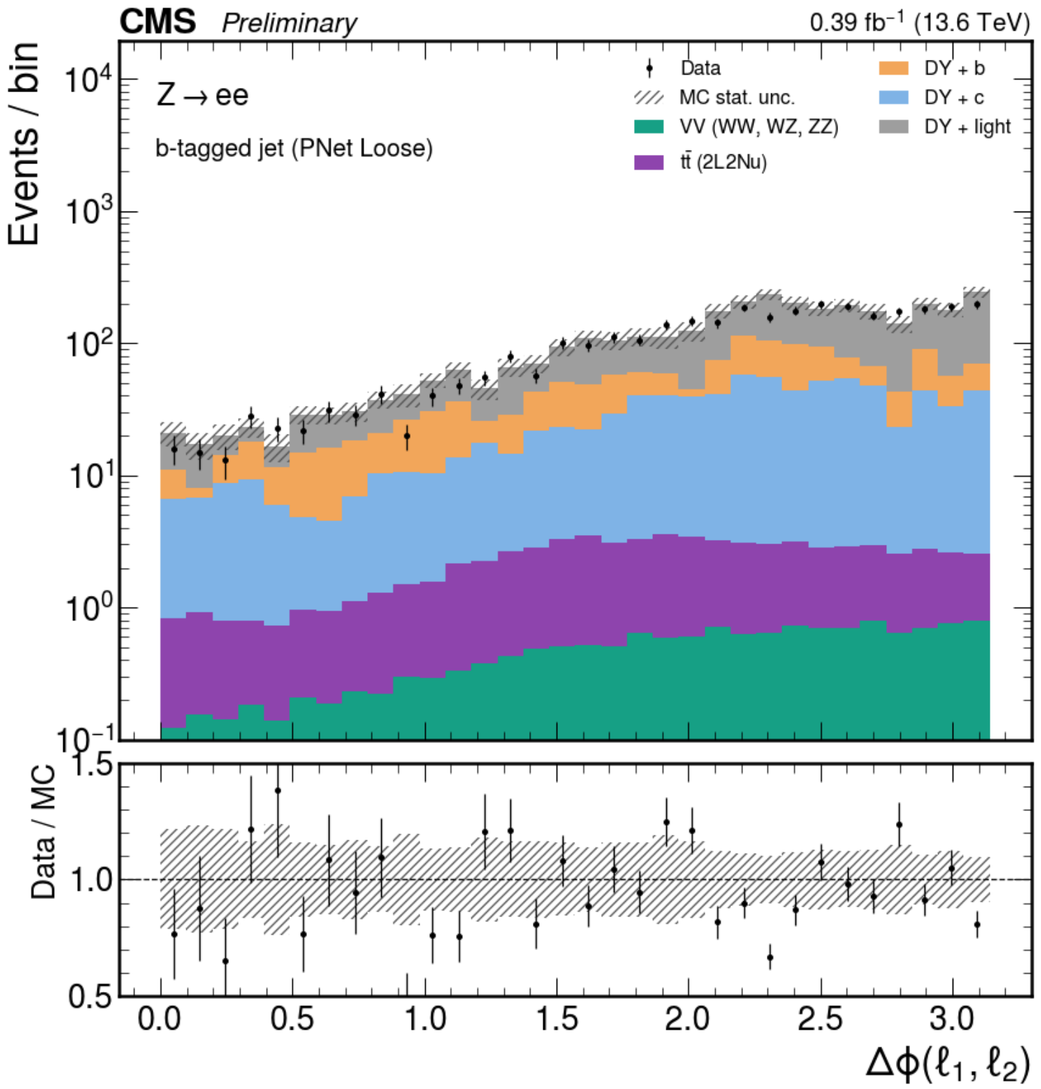
ΔR = √(Δη² + Δφ²). Small ΔR = boosted Z.

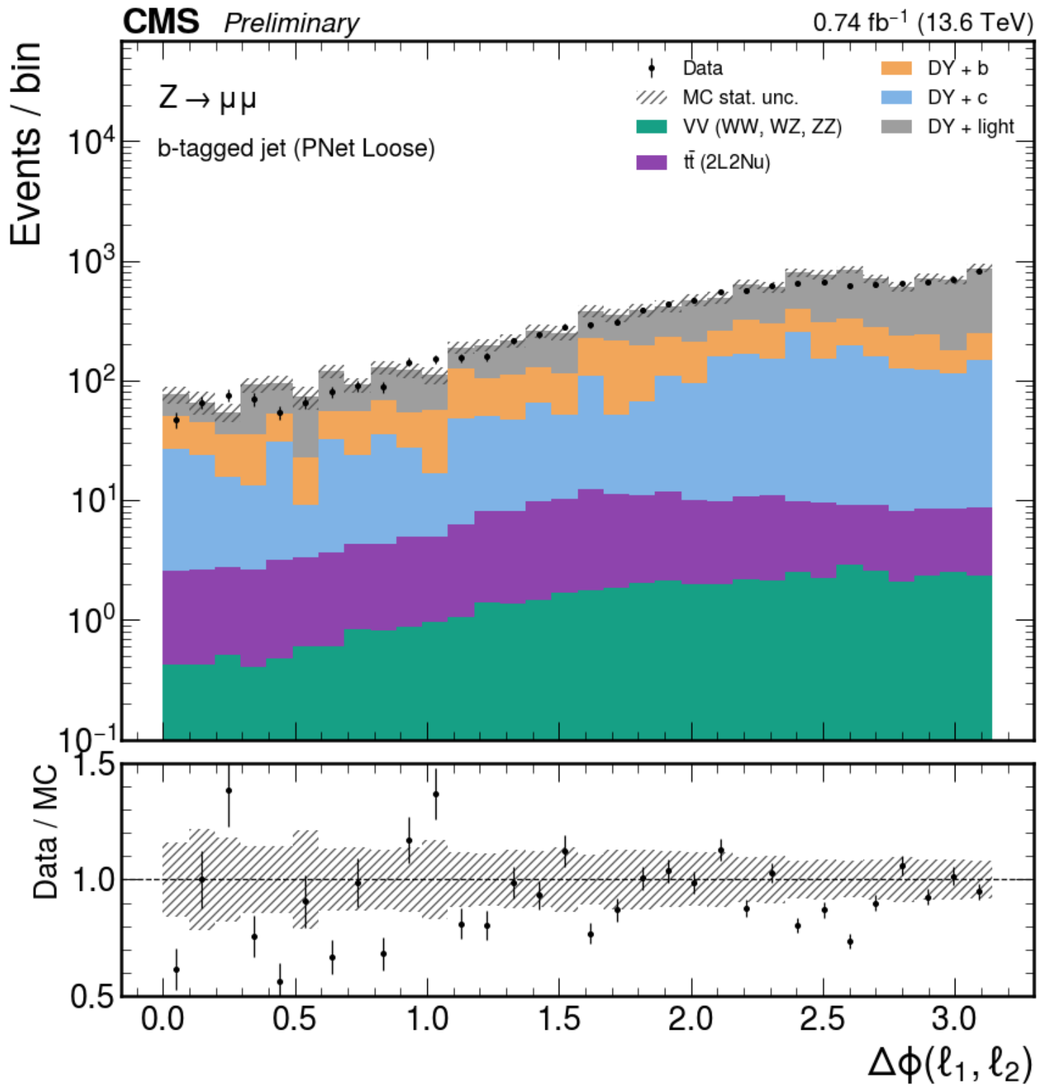




Azimuthal angle between the two leptons, Δφ(ℓℓ)

Close to π (back to back) when the Z has low pT.

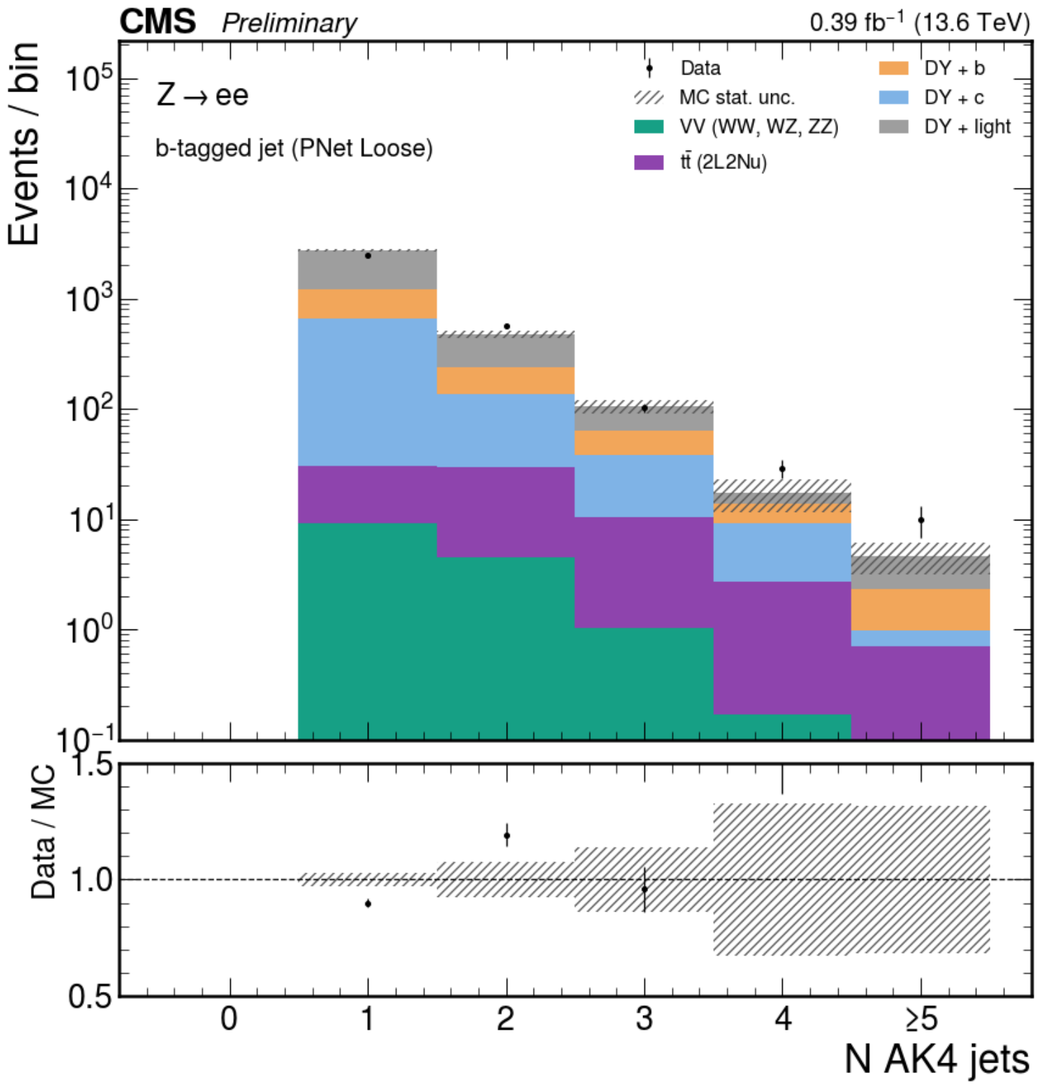


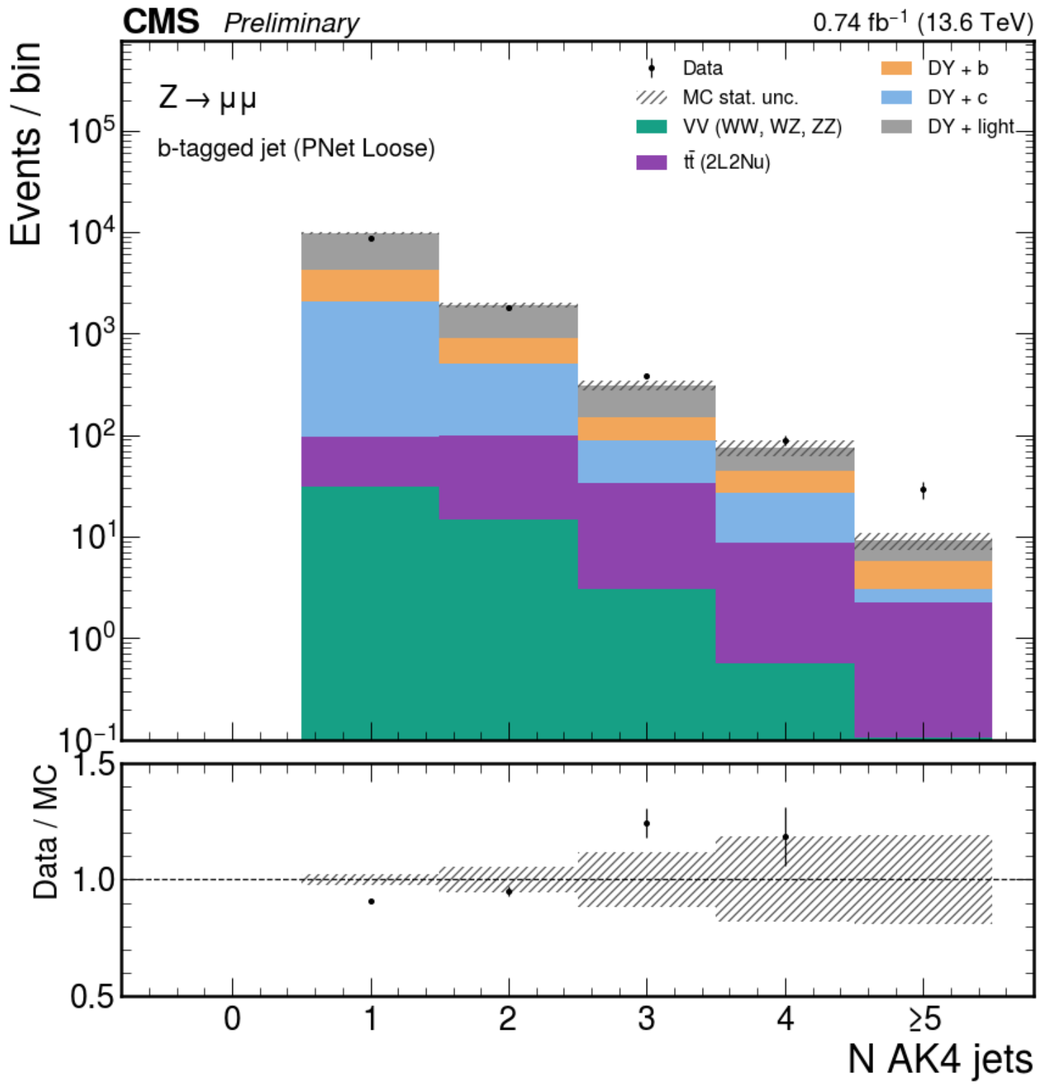


ParticleNet · Loose b-tag (B > 0.047)

Number of selected jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. The leading jet is b-tagged here, so every event has at least one jet.

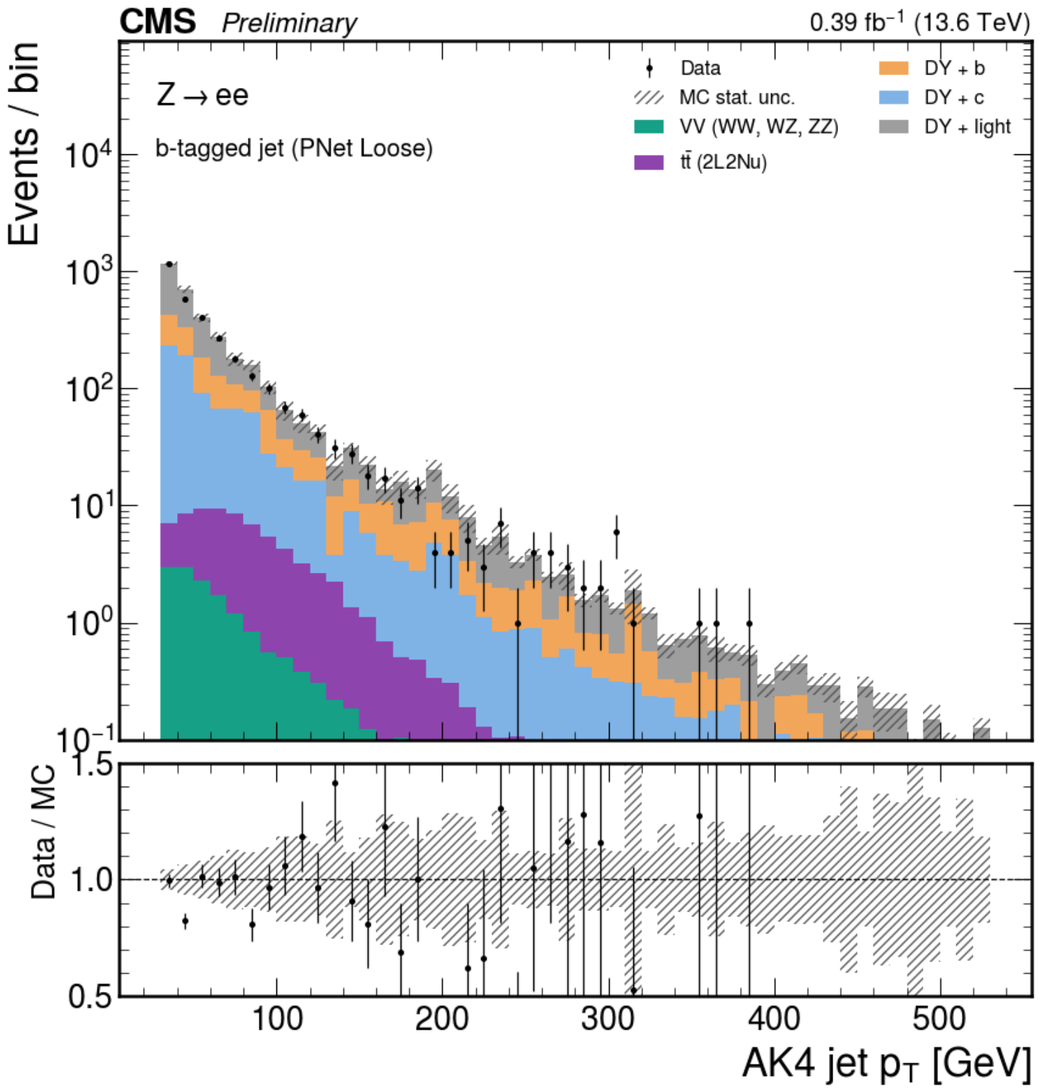


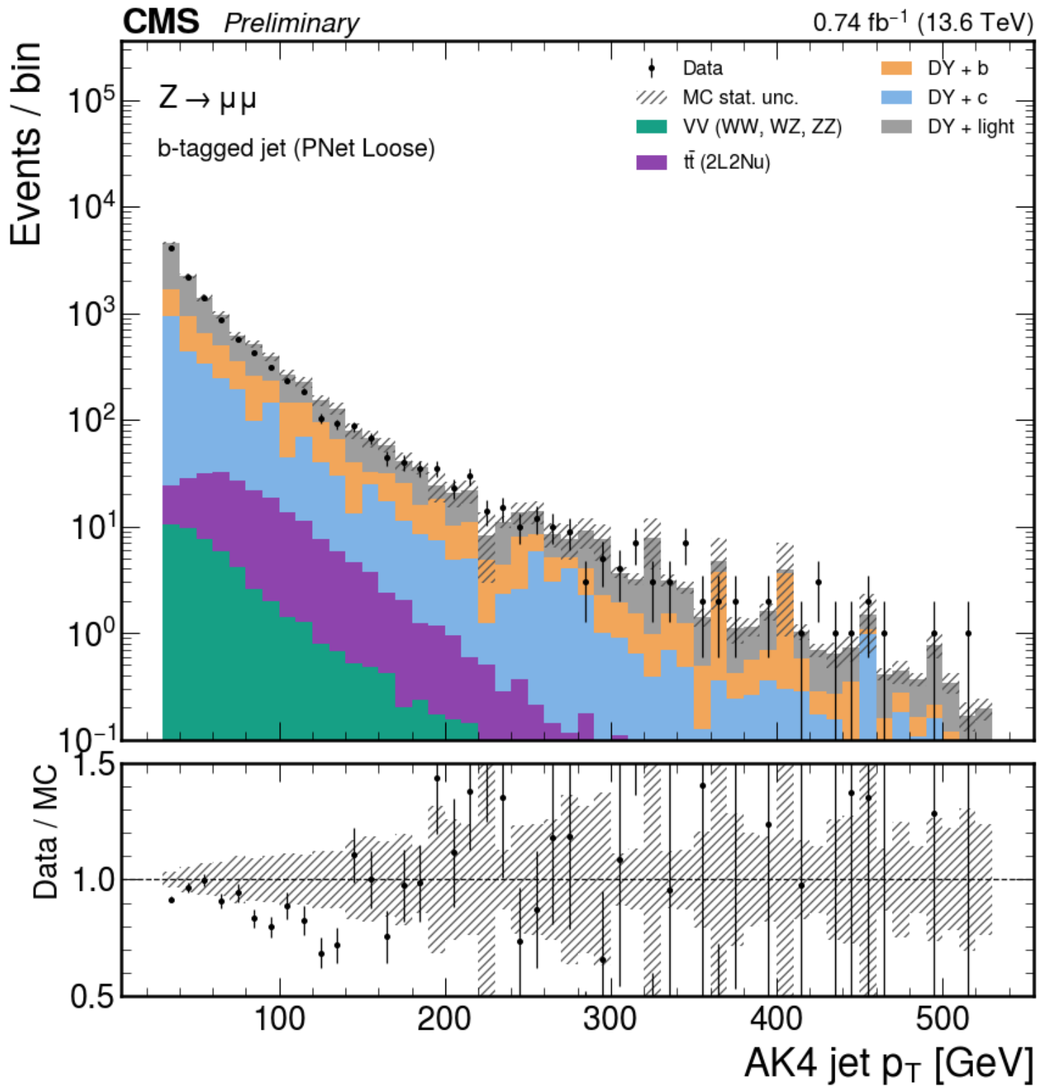


ParticleNet · Loose b-tag (B > 0.047)

Leading-jet pT (this is the b-tagged jet)

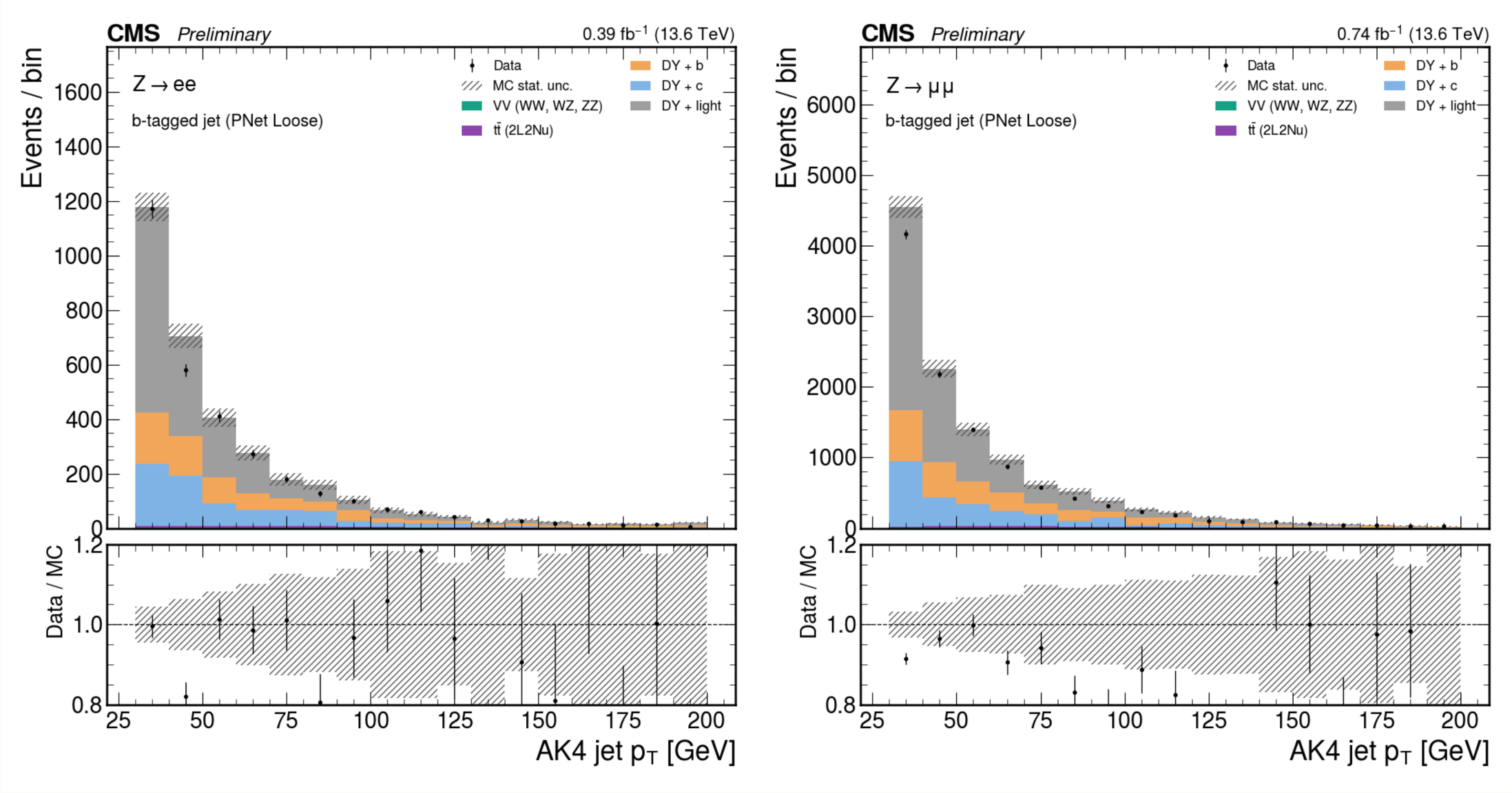
pT > 30 GeV.





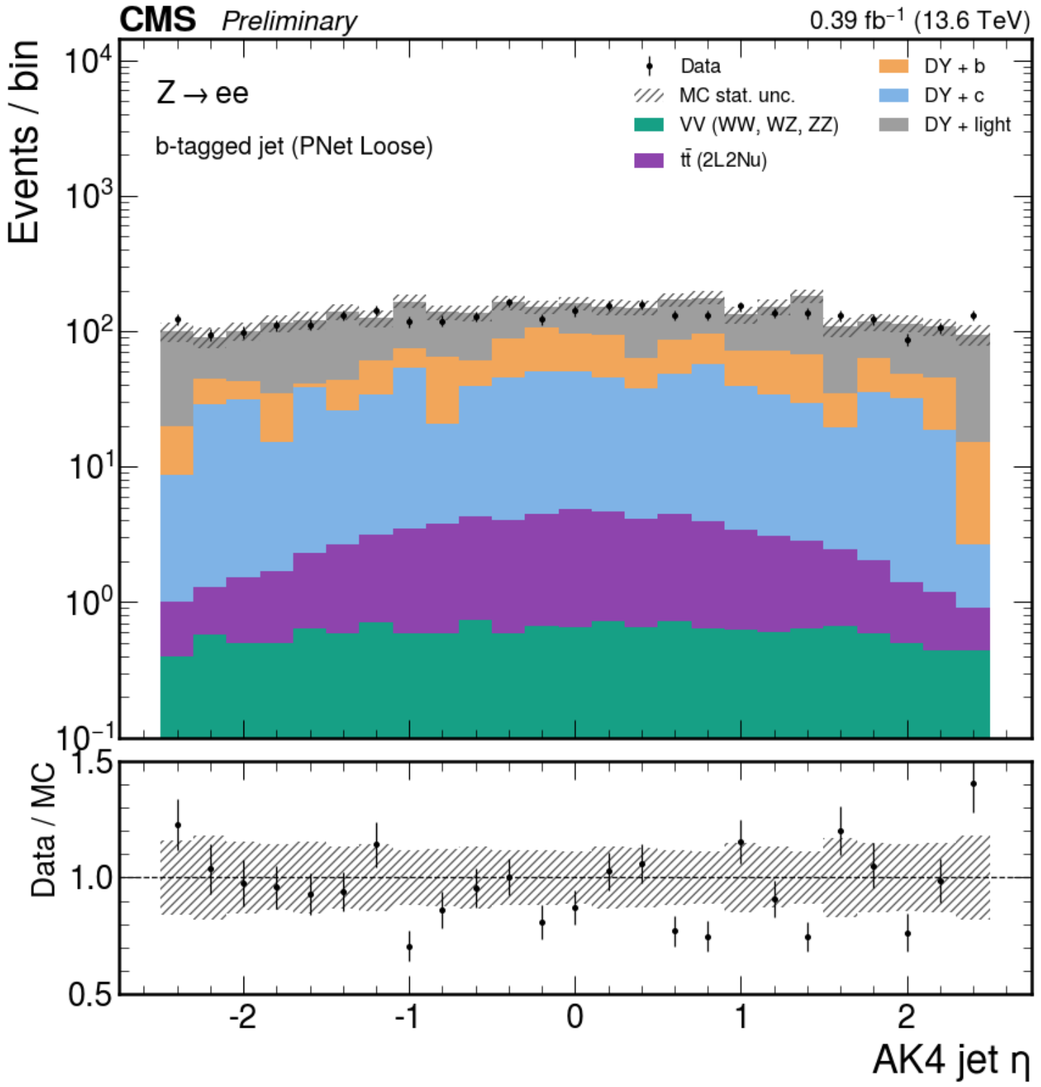
Leading-jet pT, zoom on 30–200 GeV

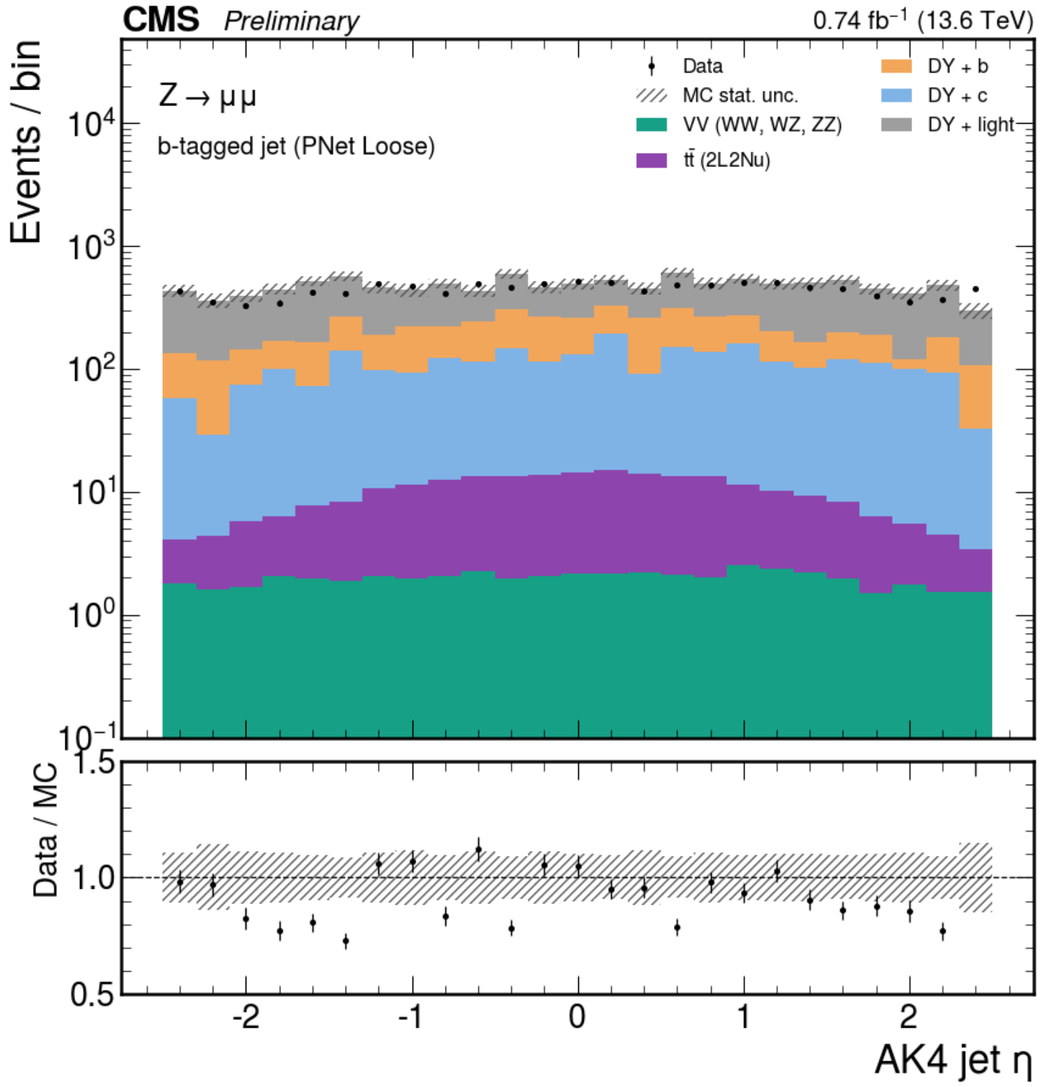
Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.



Leading-jet η

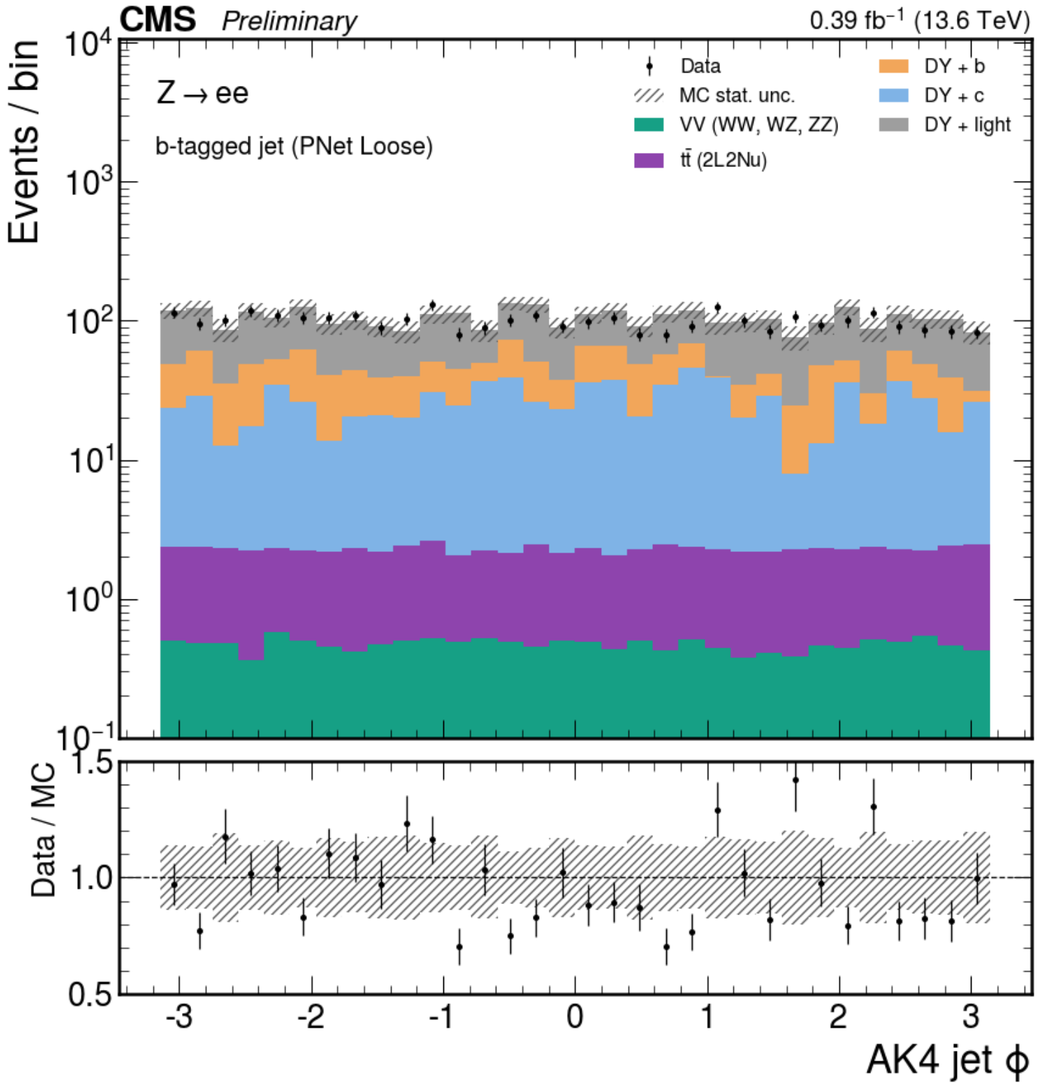
|η| < 2.5 (tracker coverage, needed for b-tagging).

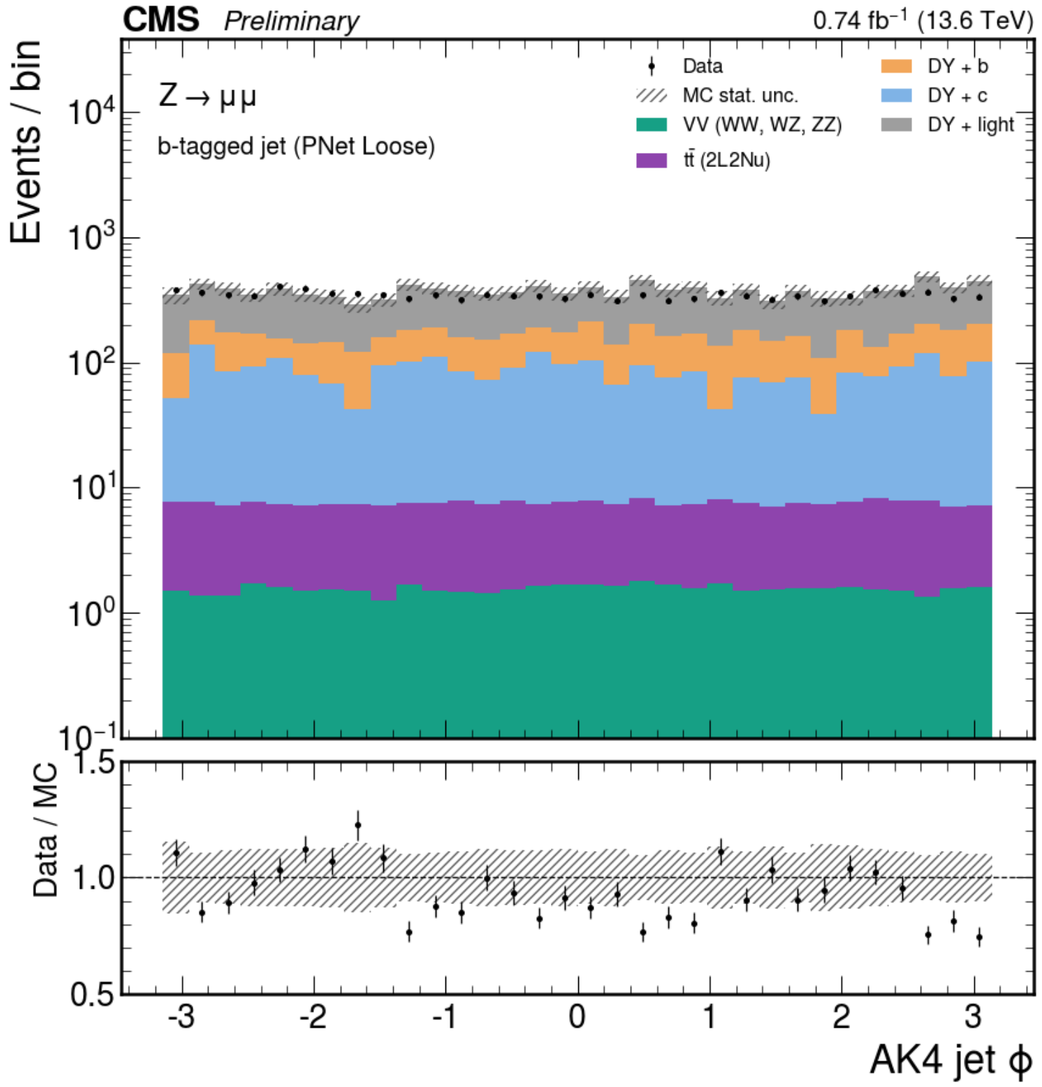




Leading-jet φ

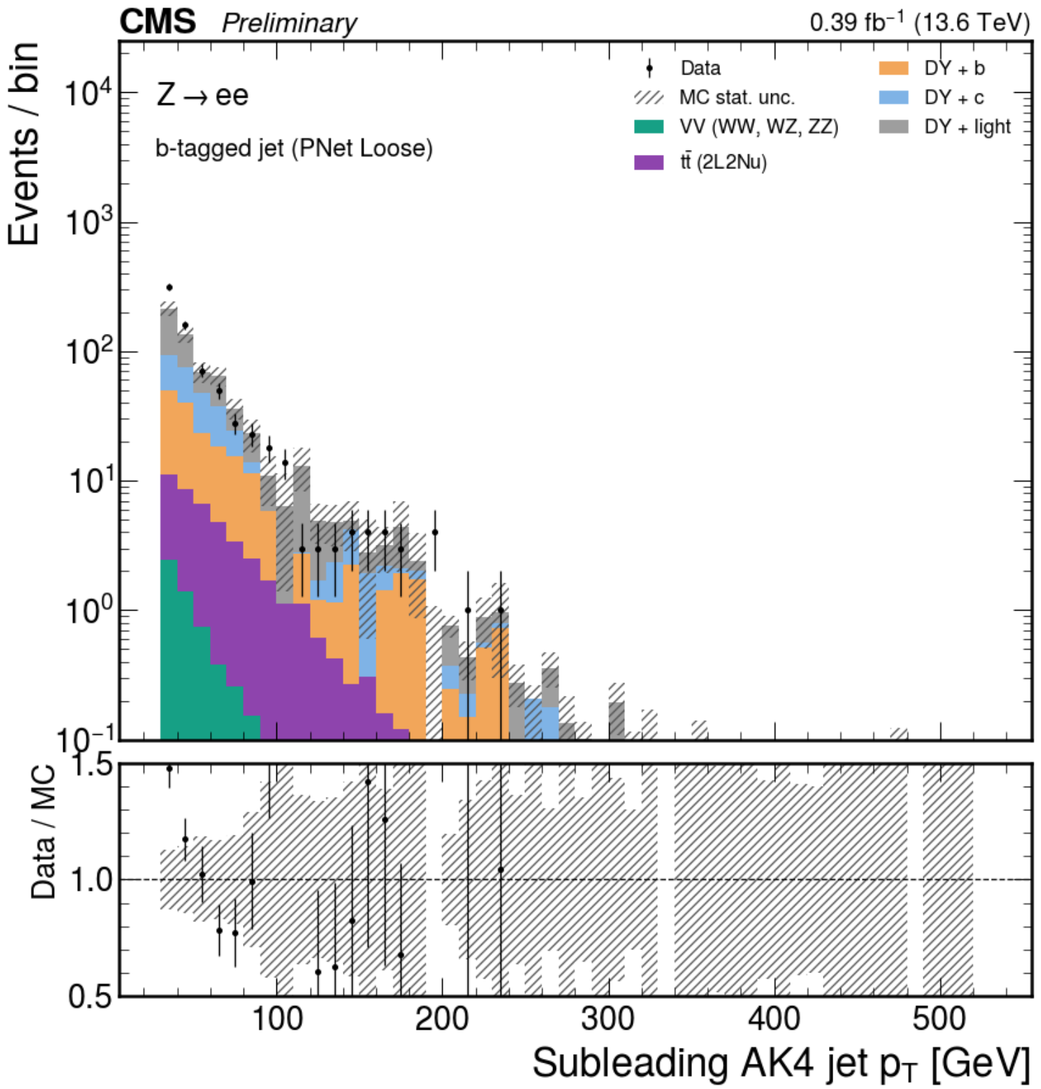
Expected to be flat.

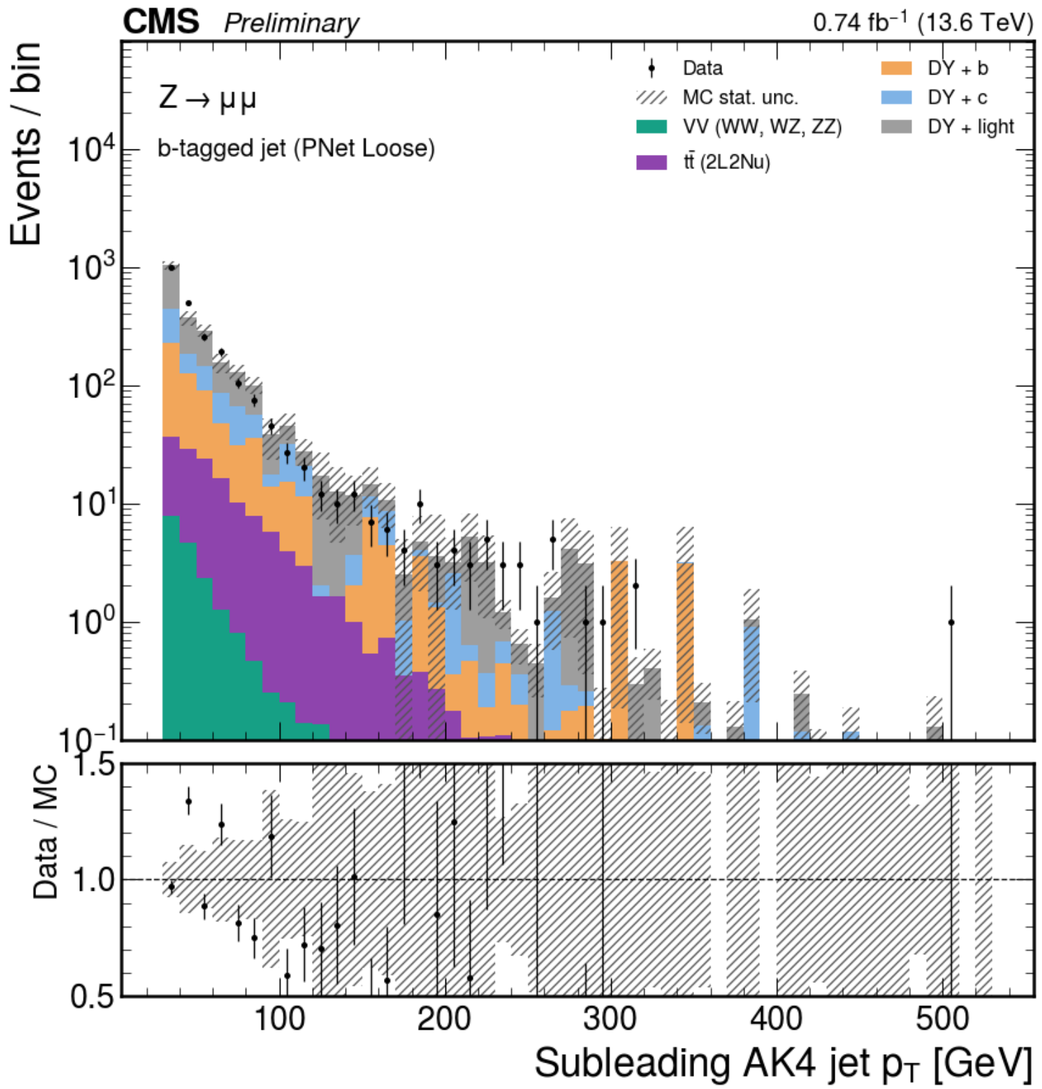




Second-jet pT

Only events with at least two jets. The second jet is not tagged.

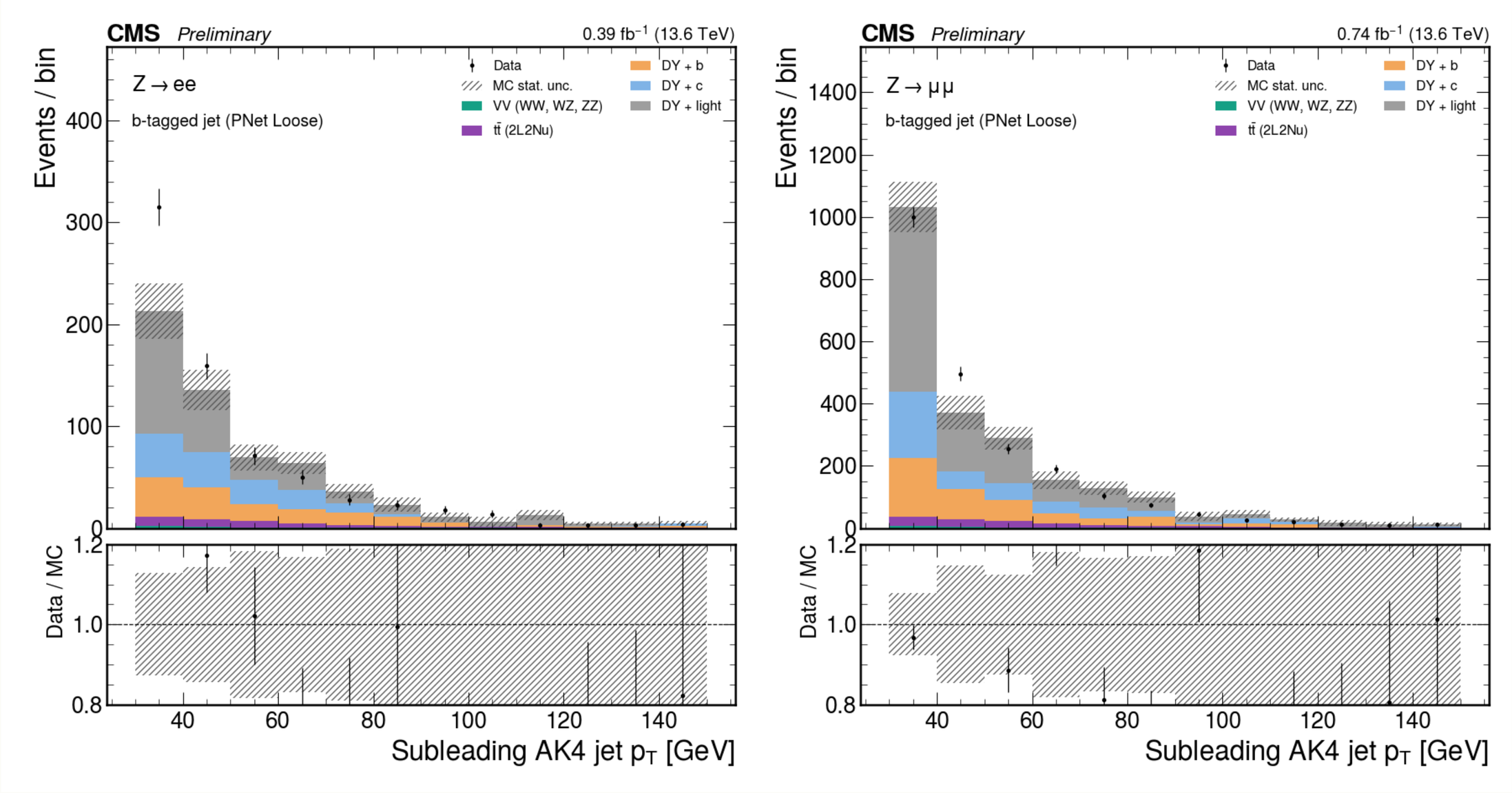




ParticleNet · Loose b-tag (B > 0.047)

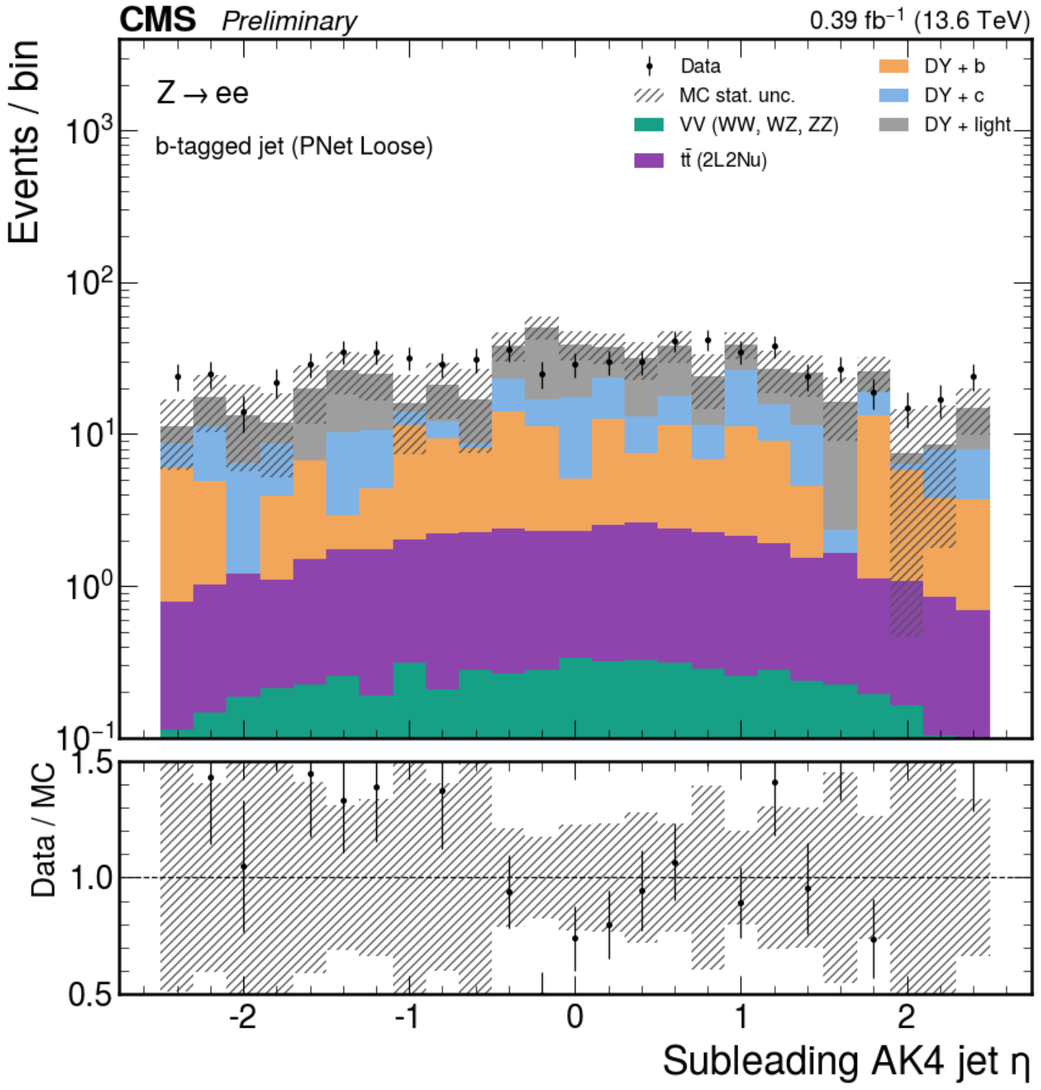
Second-jet pT, zoom on 30–150 GeV

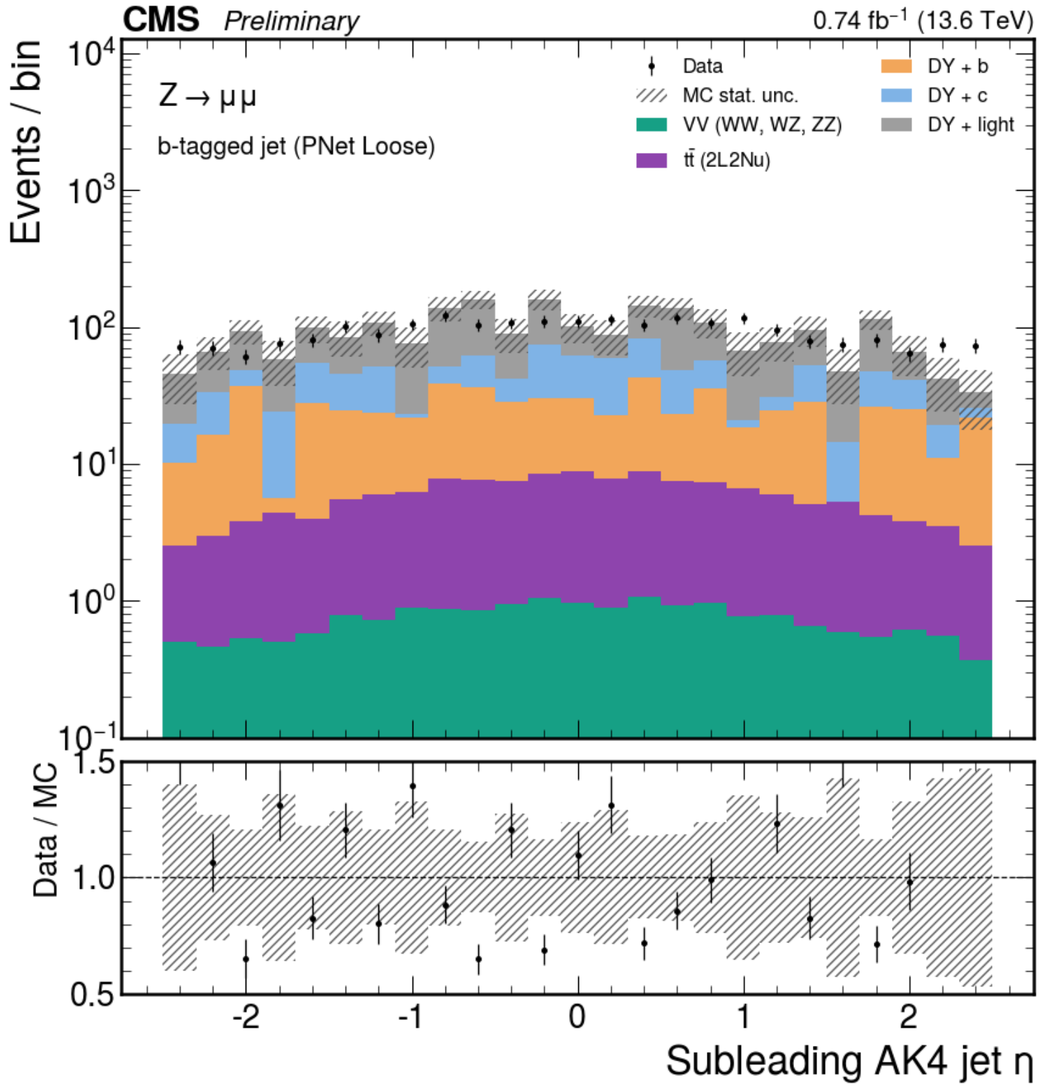
Zoom: 30–150, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.



Second-jet η

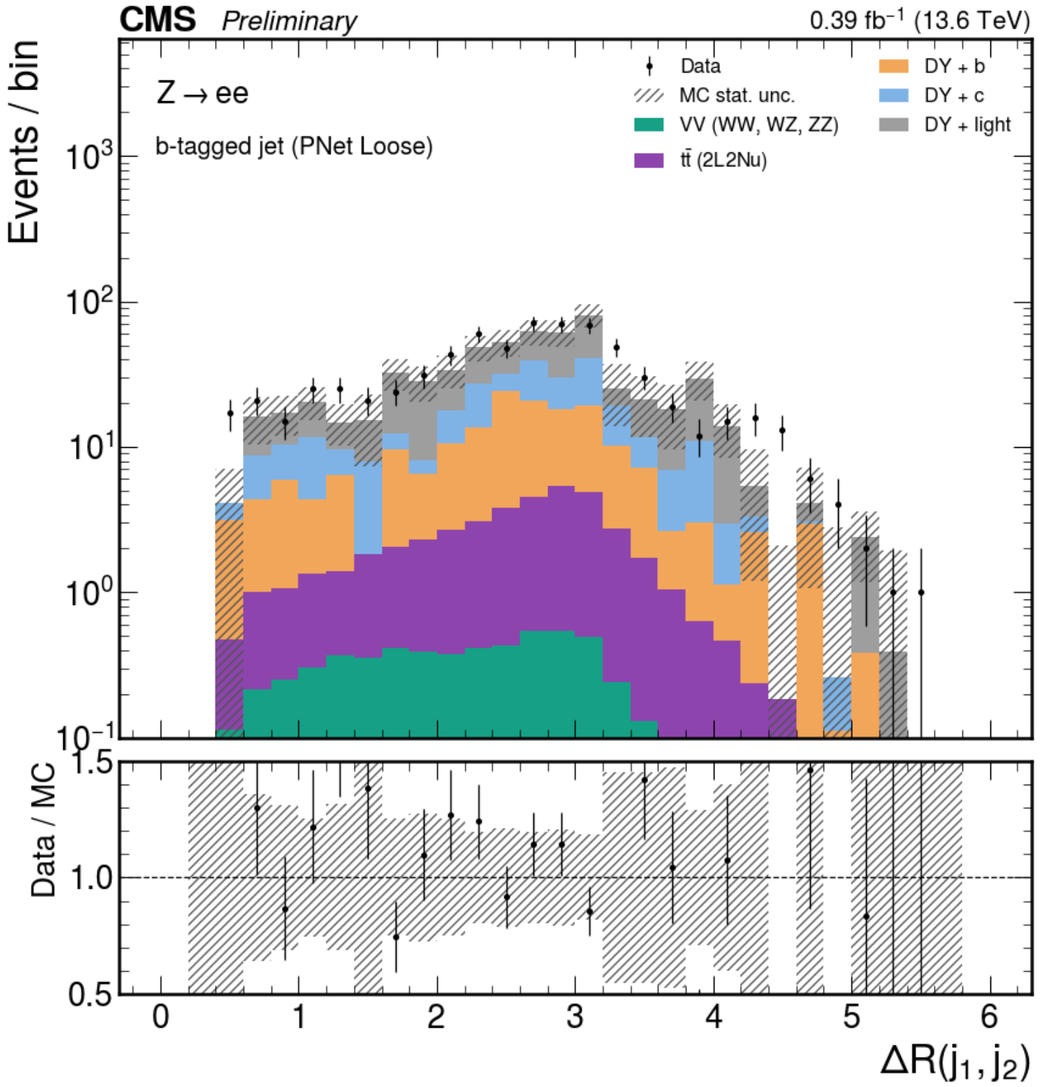
Only events with at least two jets.

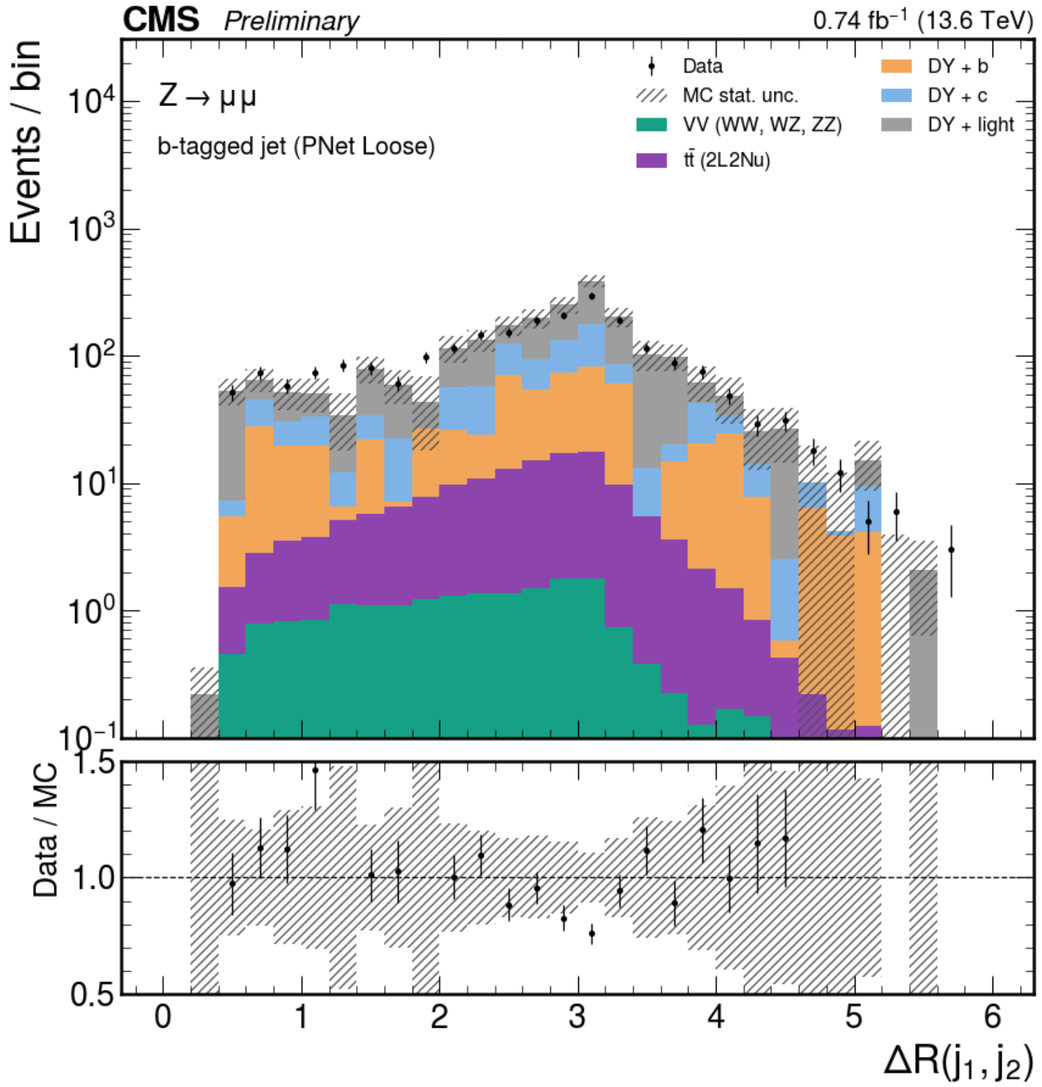




Angular distance between the two leading jets, ΔR(jj)

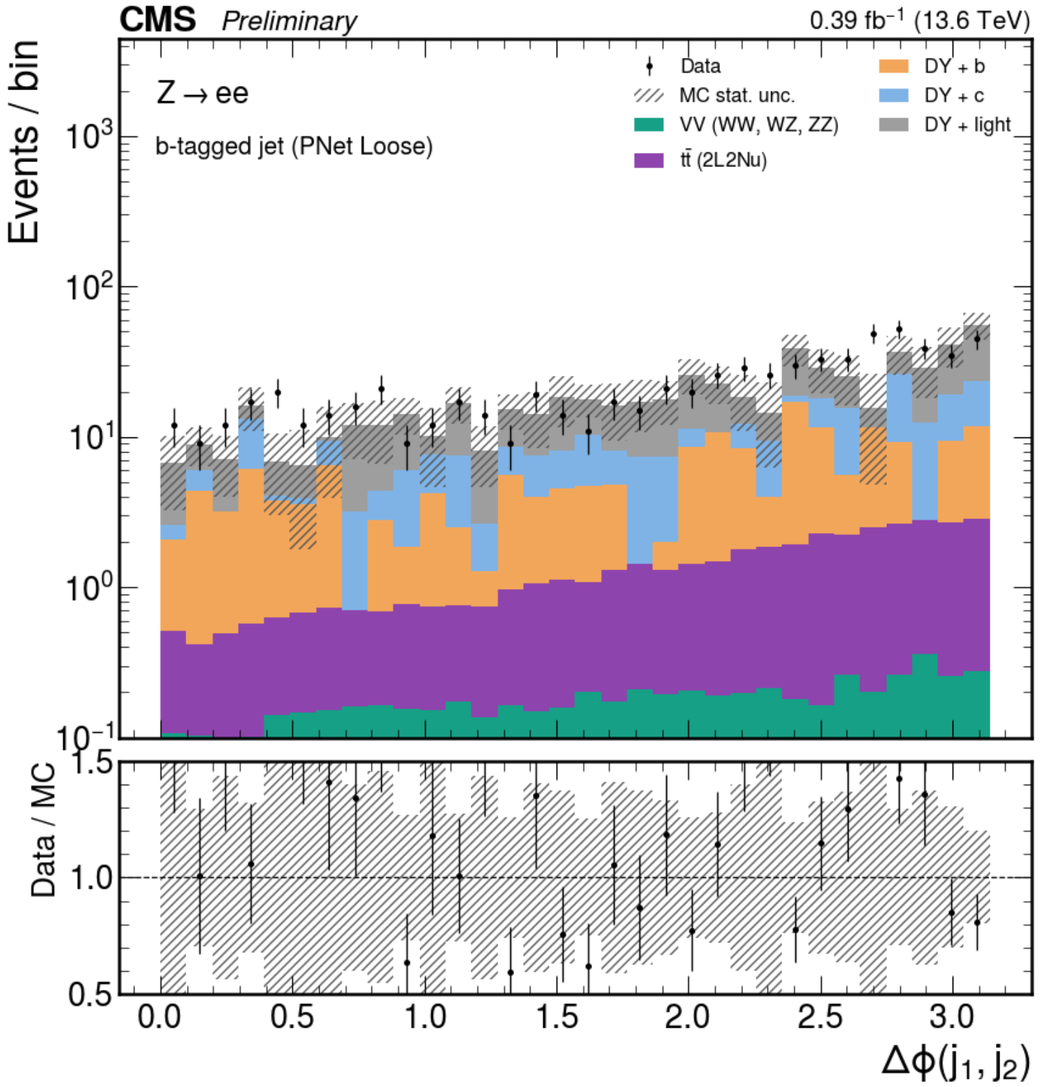
Only events with at least two jets.

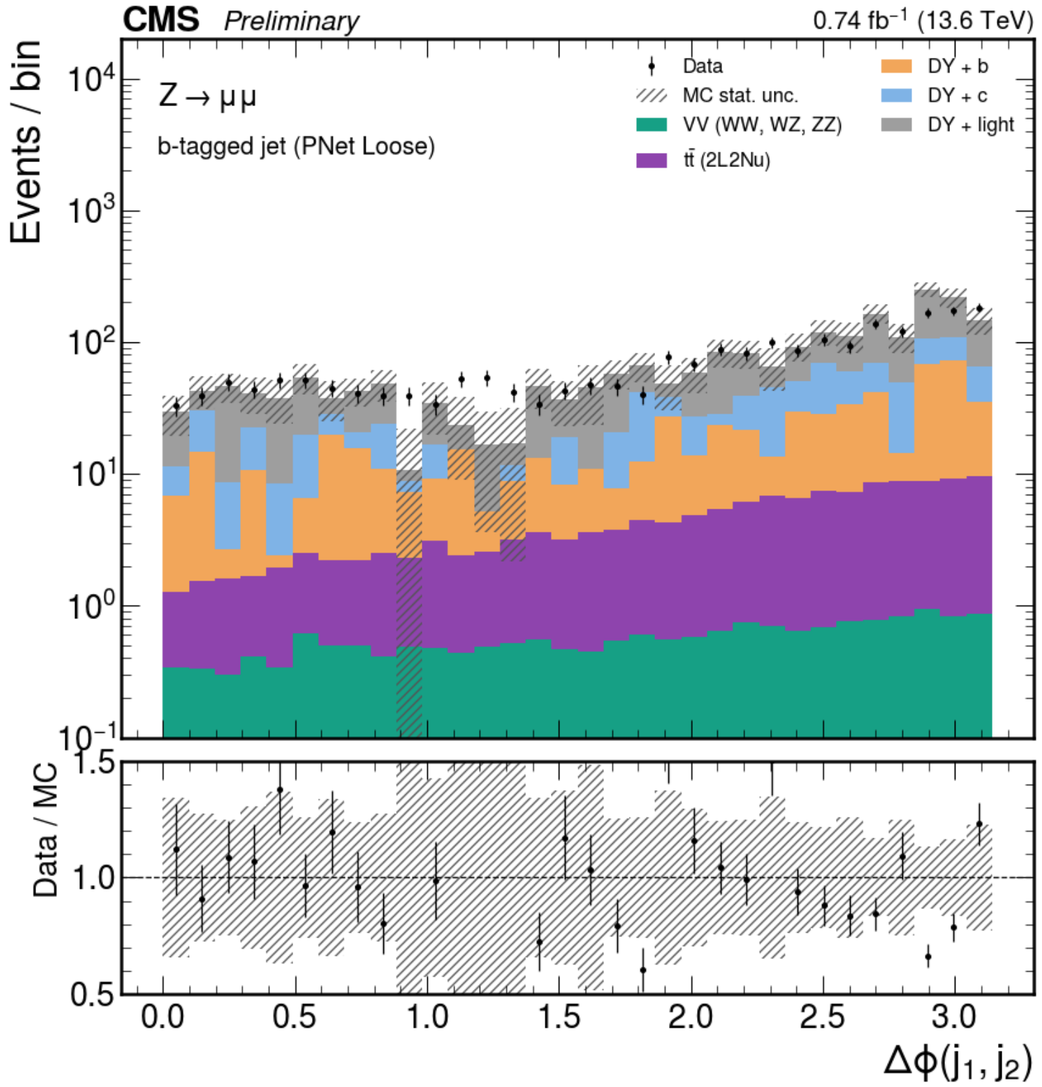




Azimuthal angle between the two leading jets, Δφ(jj)

Only events with at least two jets.

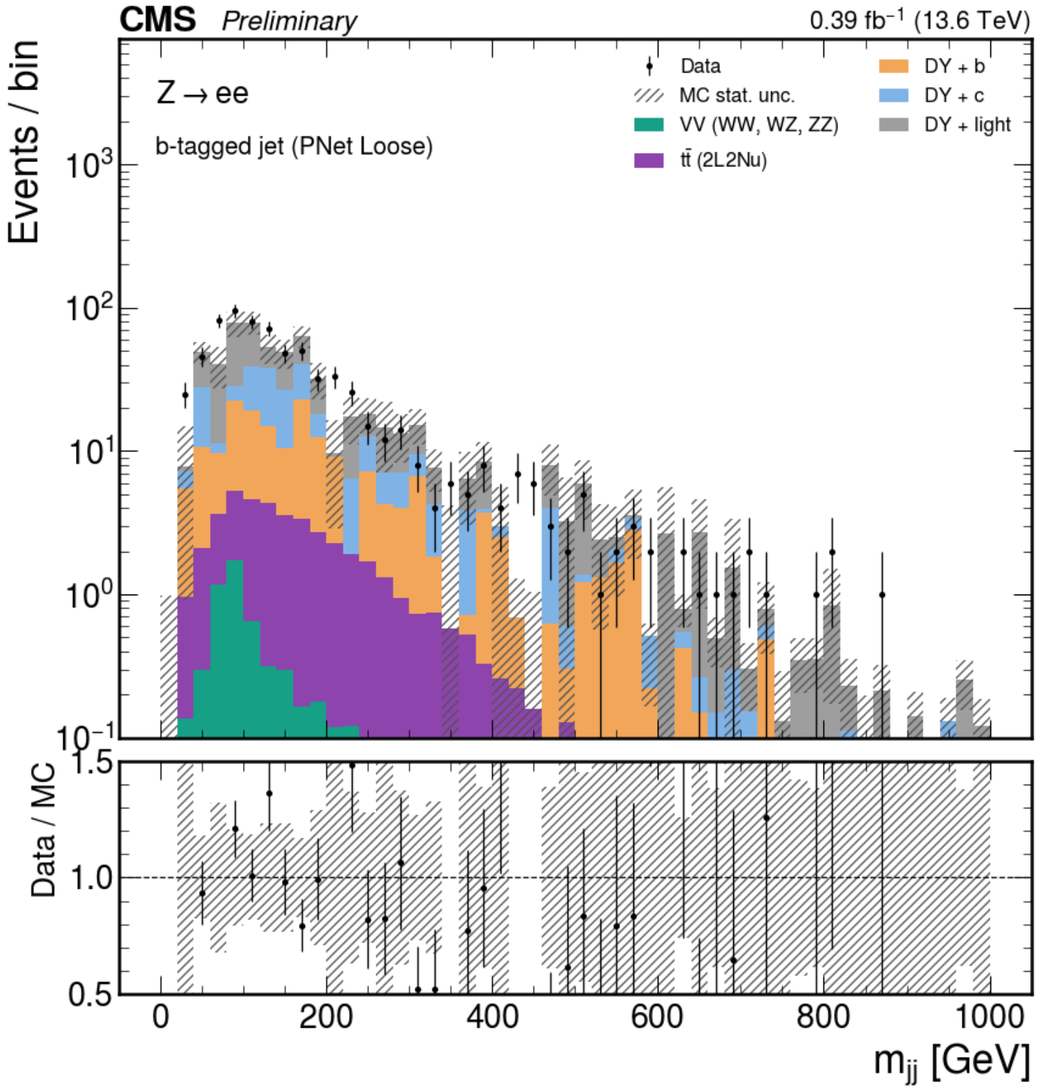


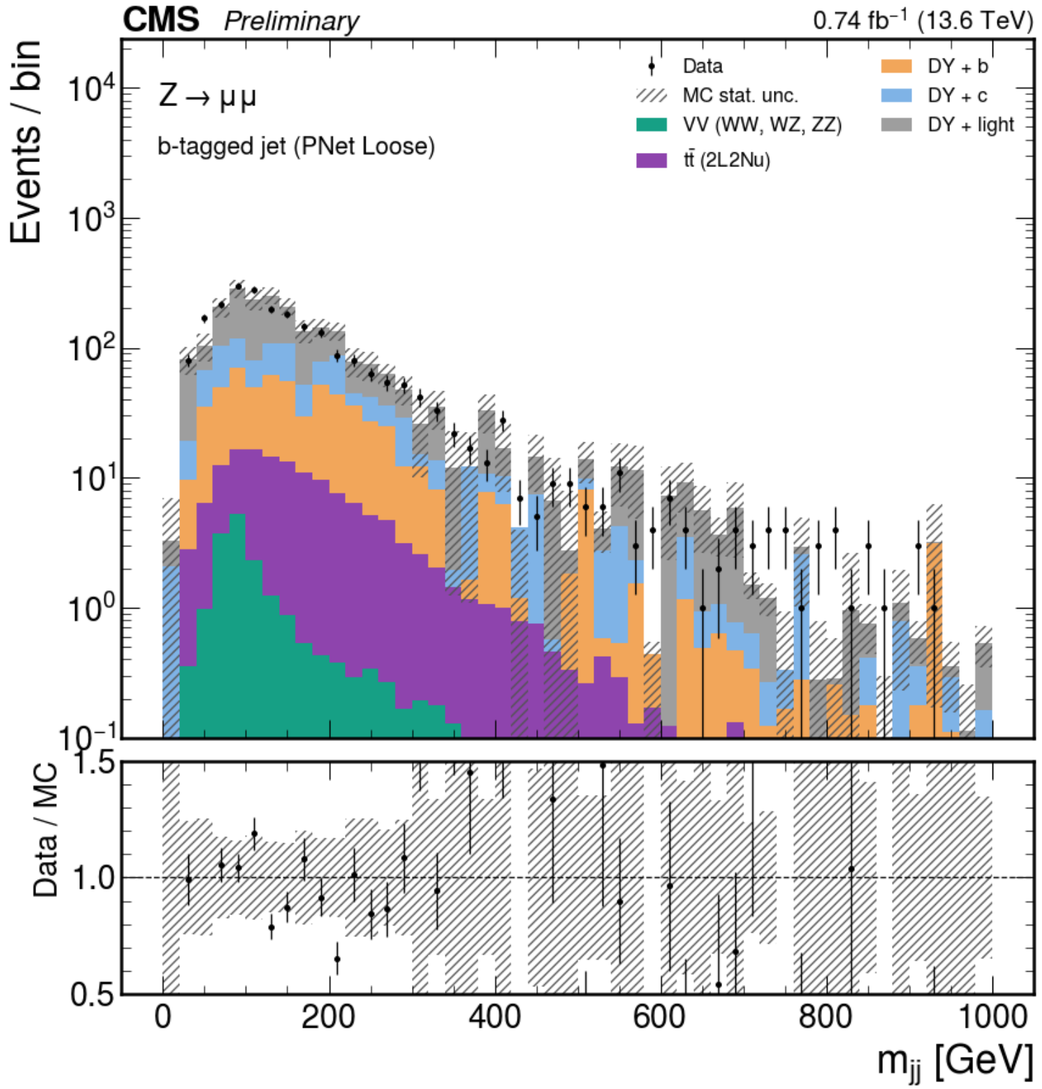


ParticleNet · Loose b-tag (B > 0.047)

Mass of the two leading jets, m(jj)

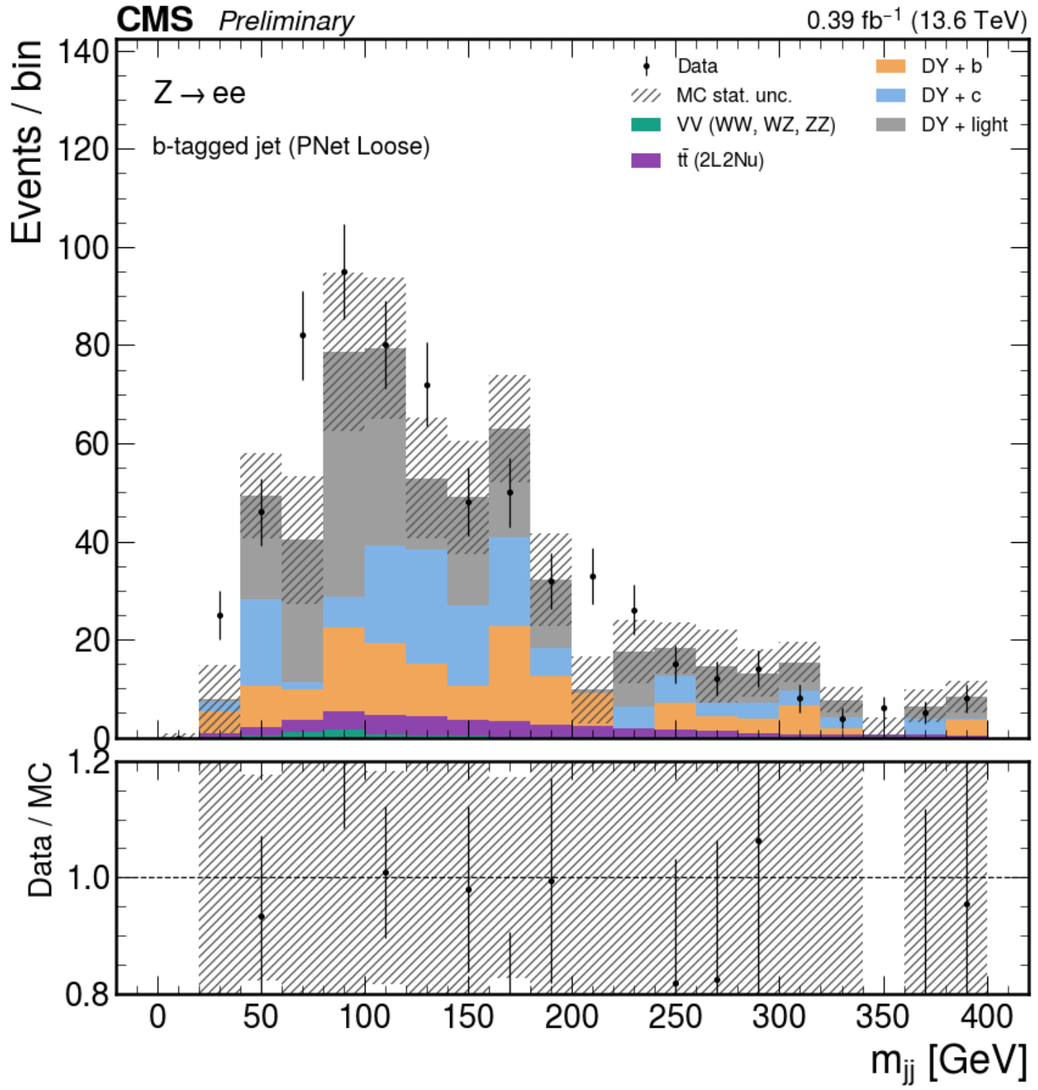
Only events with at least two jets.

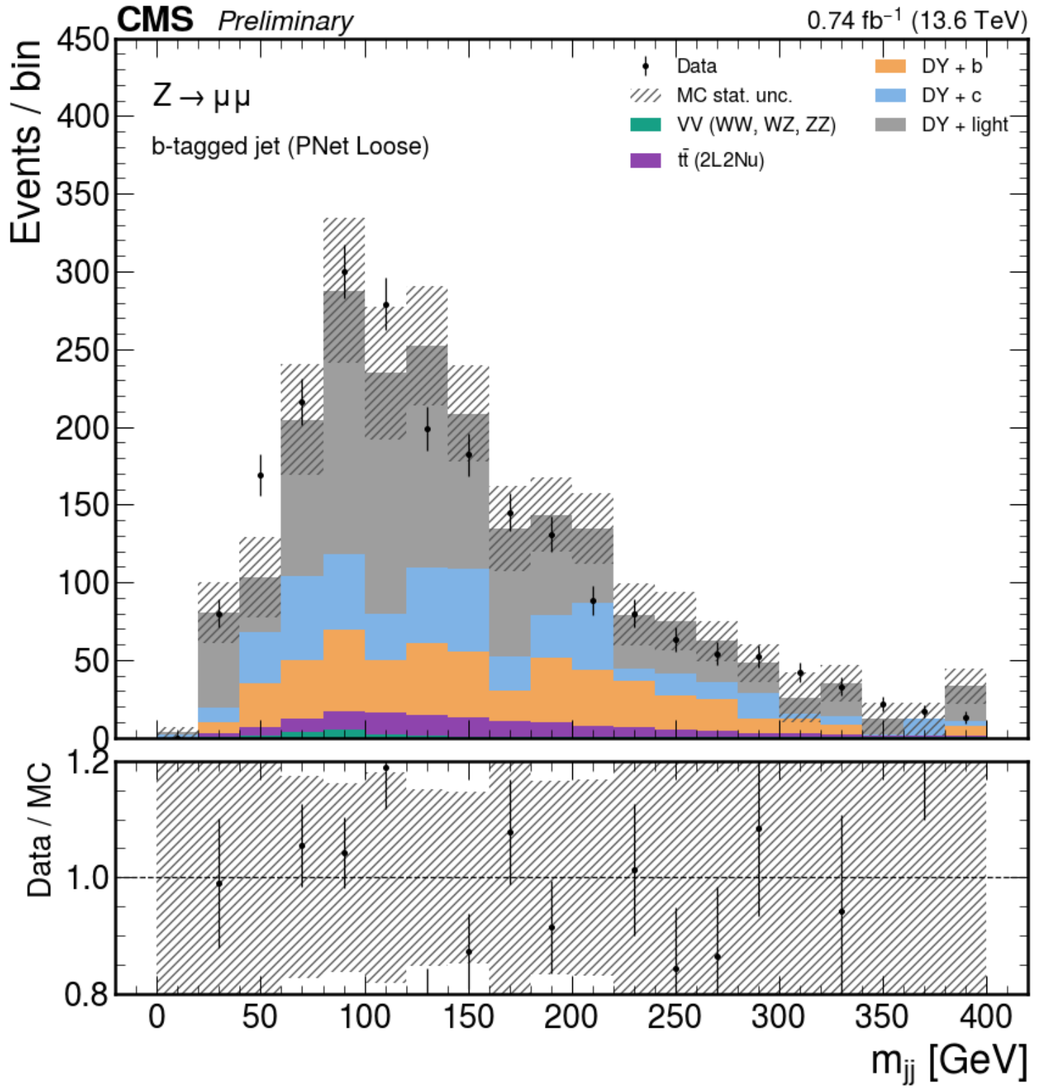




m(jj), zoom on 0–400 GeV

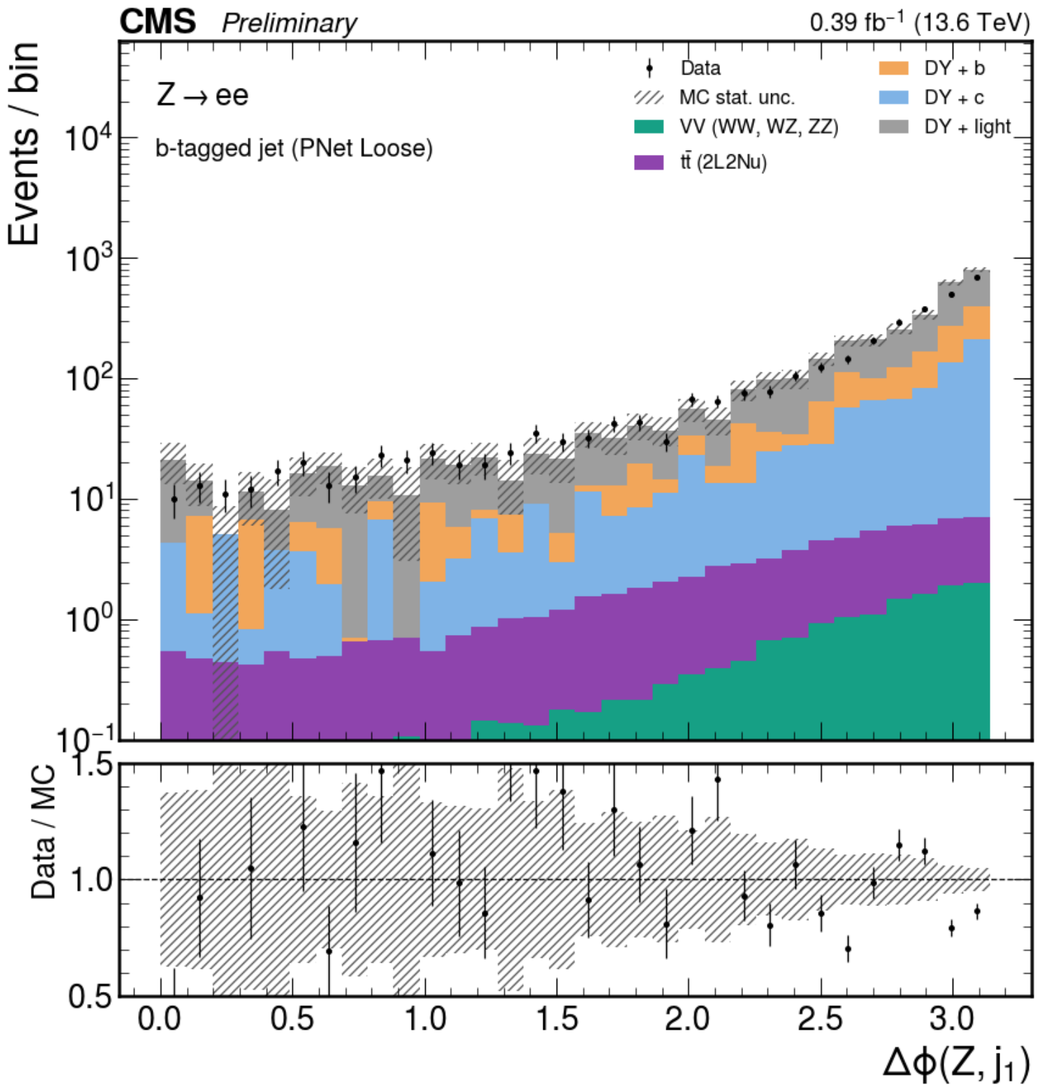
Zoom: 0–400, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.

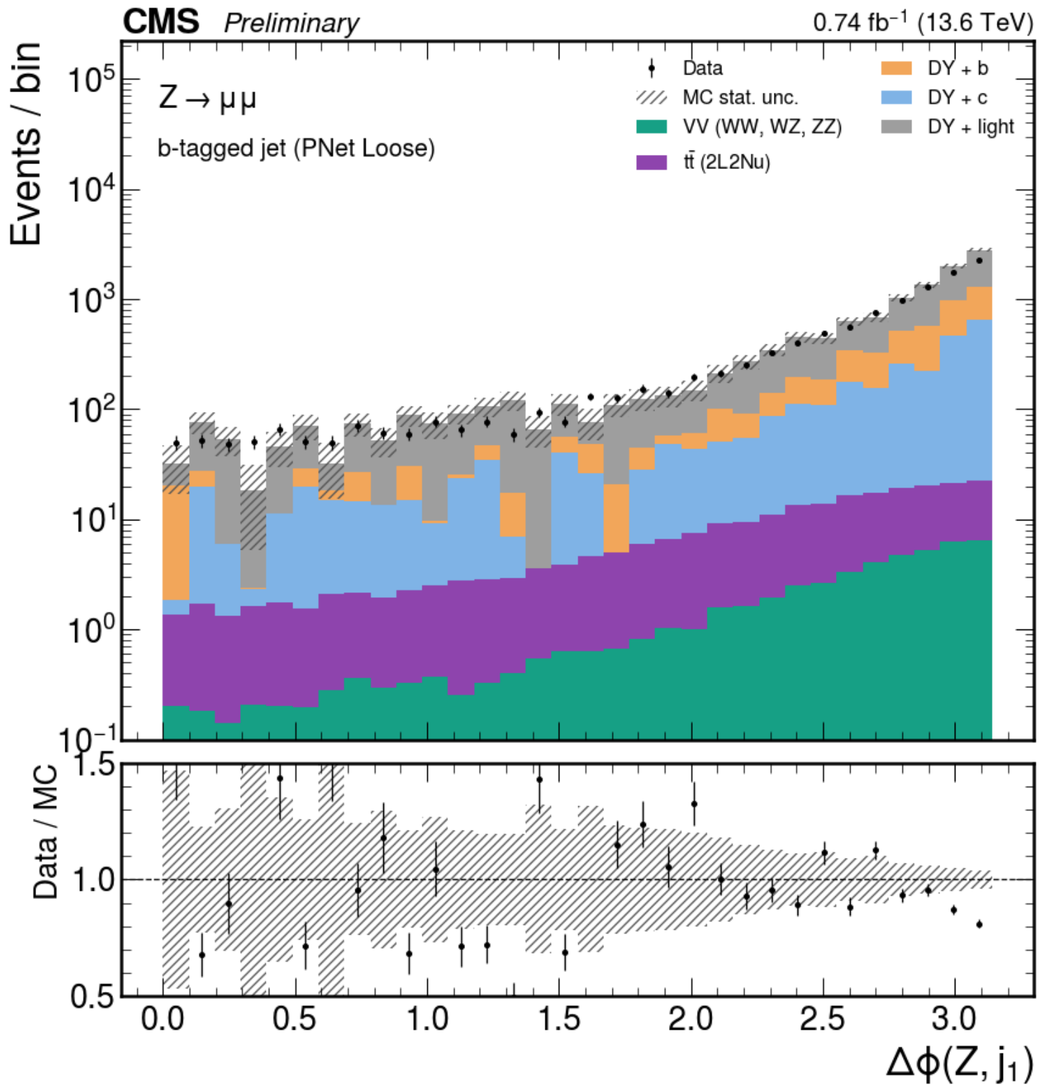




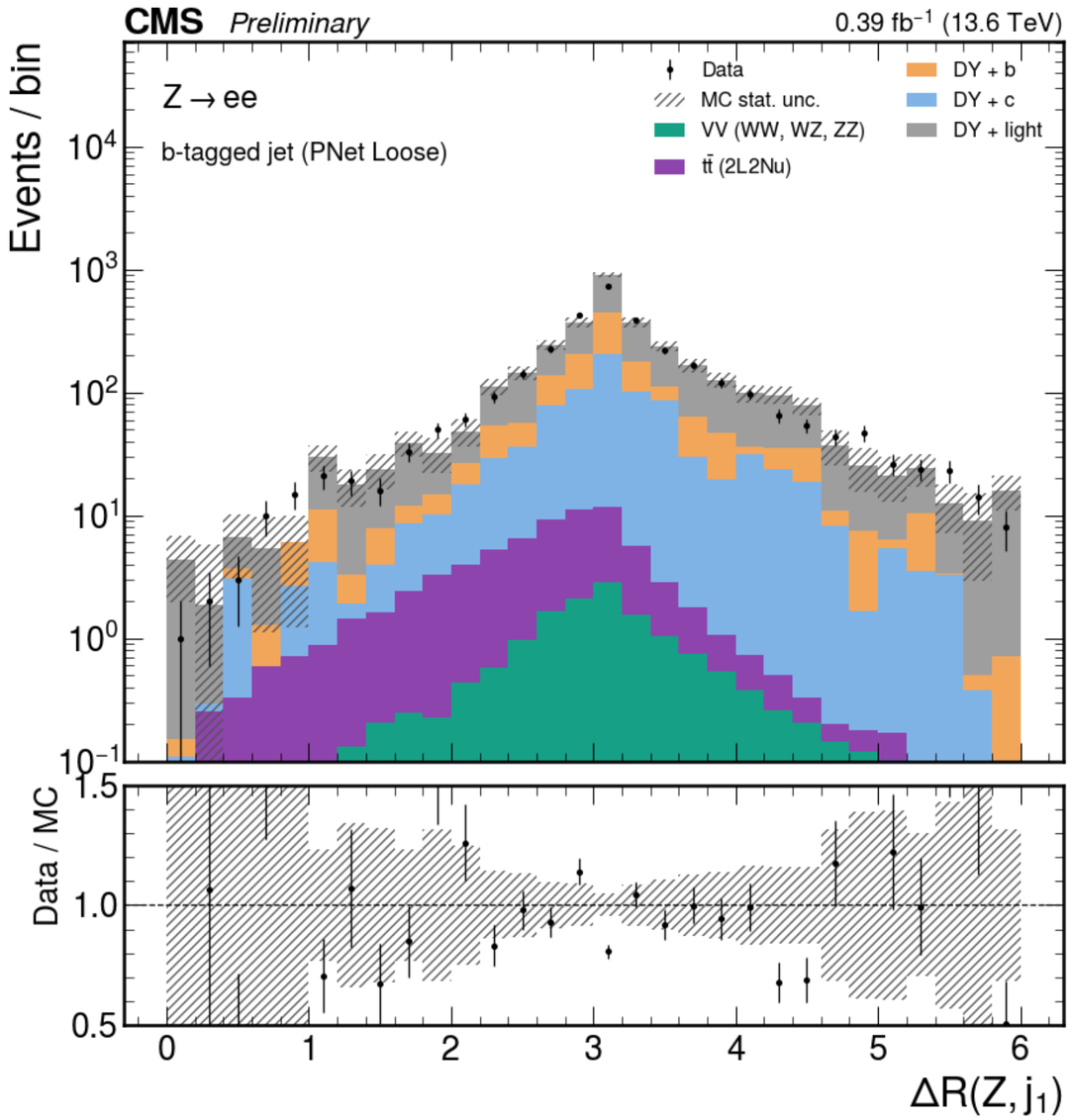
Azimuthal angle between the Z and the leading jet, Δφ(Z, j)

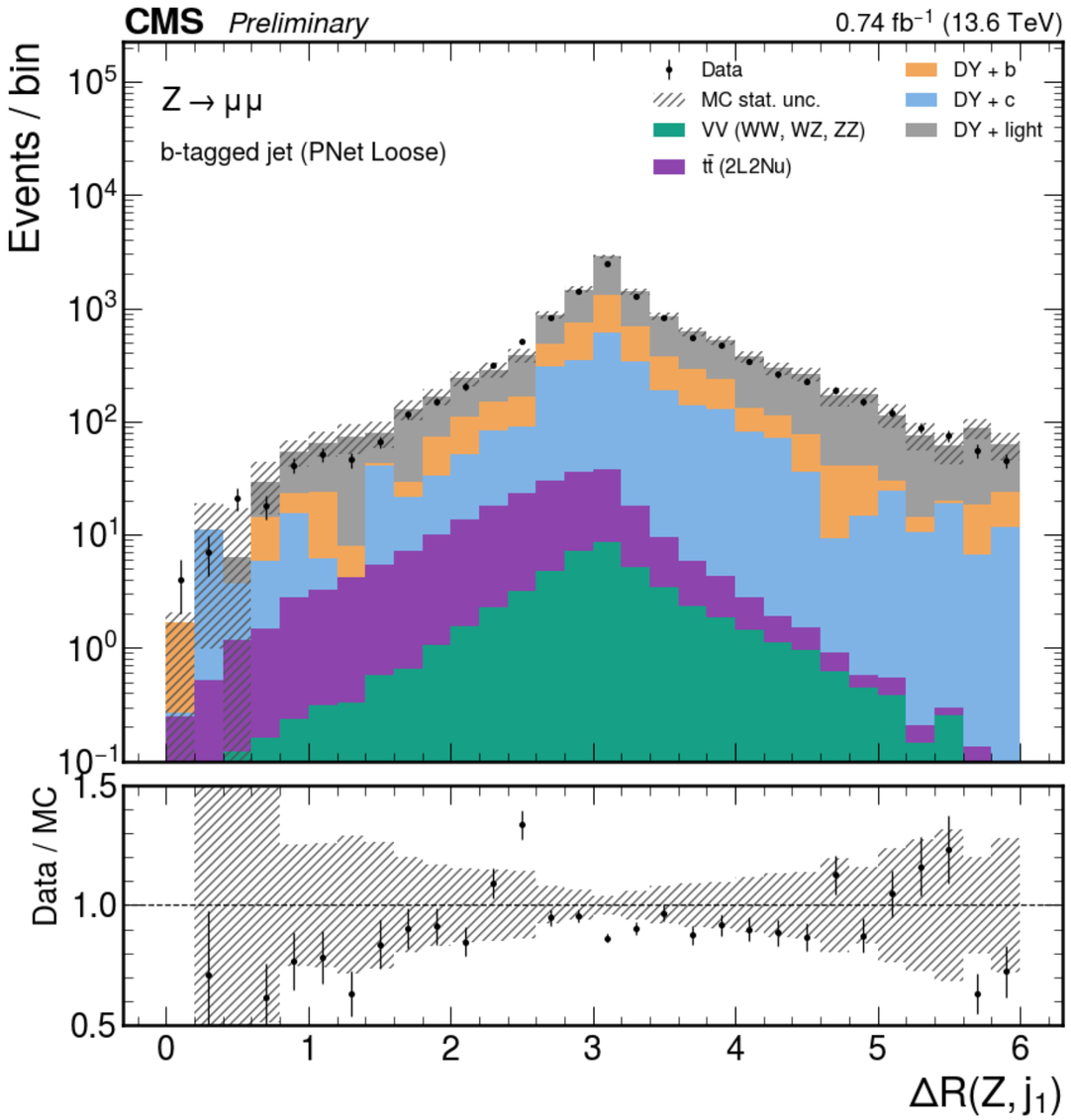
Close to π when the jet recoils against the Z.





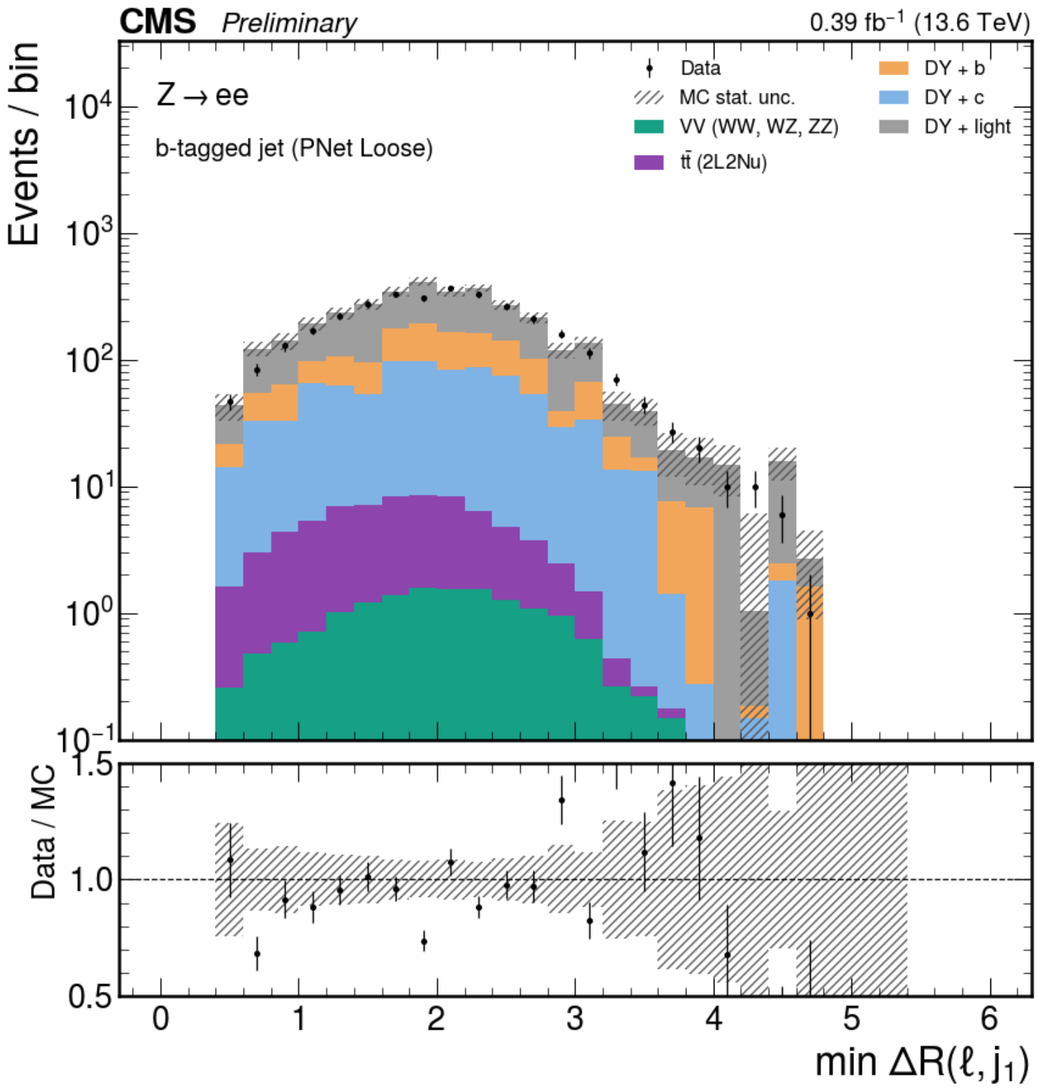
Angular distance between the Z and the leading jet, ΔR(Z, j)

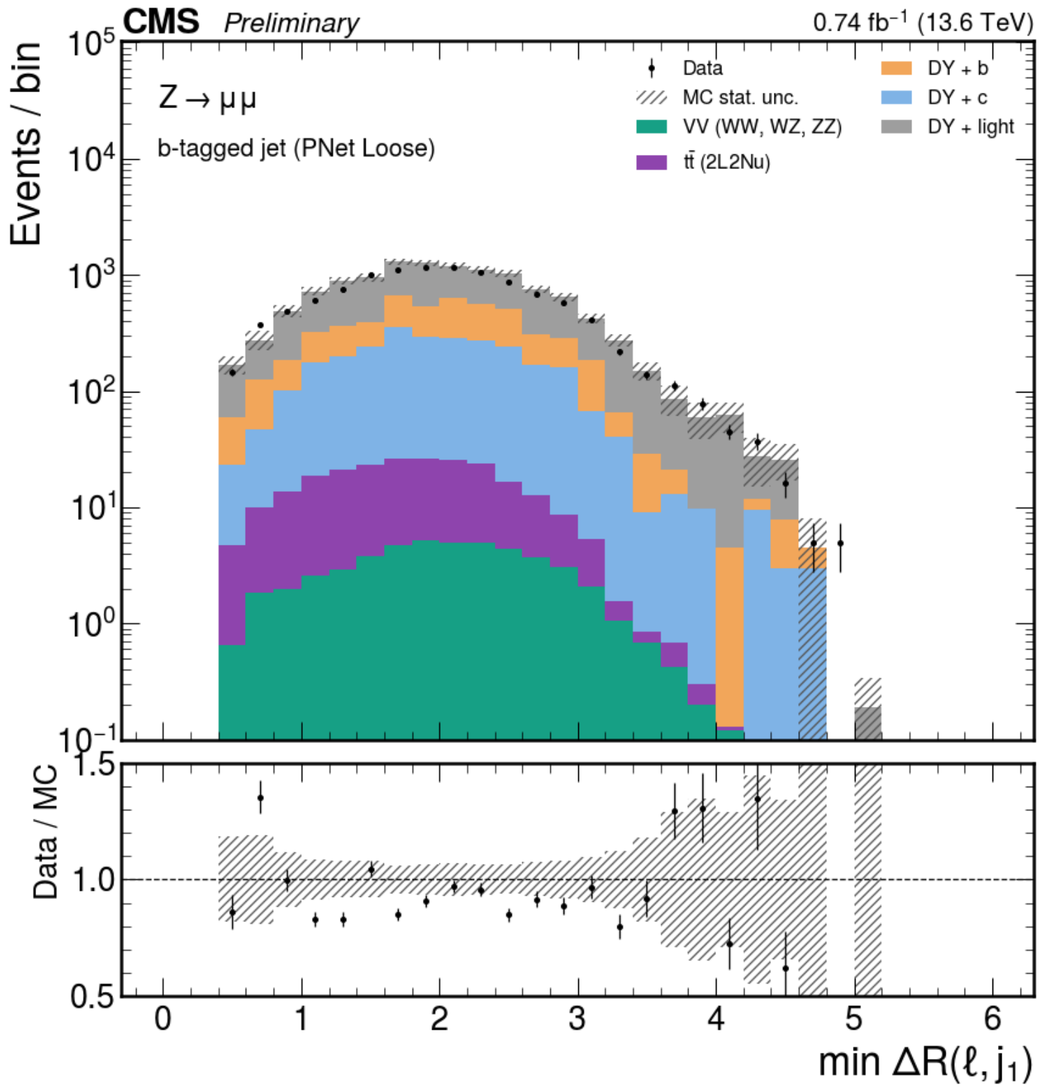




Smallest ΔR between a lepton and the leading jet

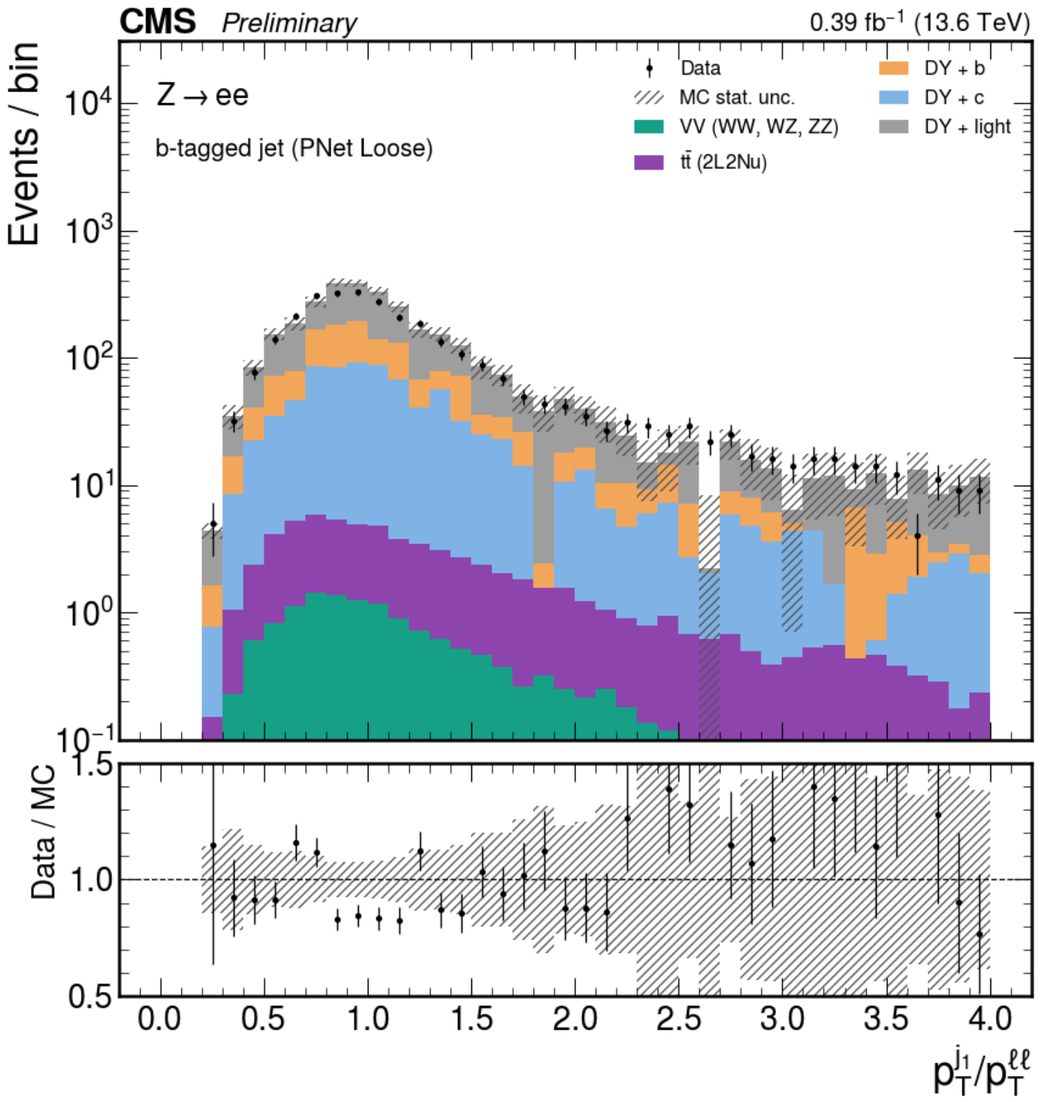
Must be > 0.4 by selection (jets overlapping a lepton are removed).

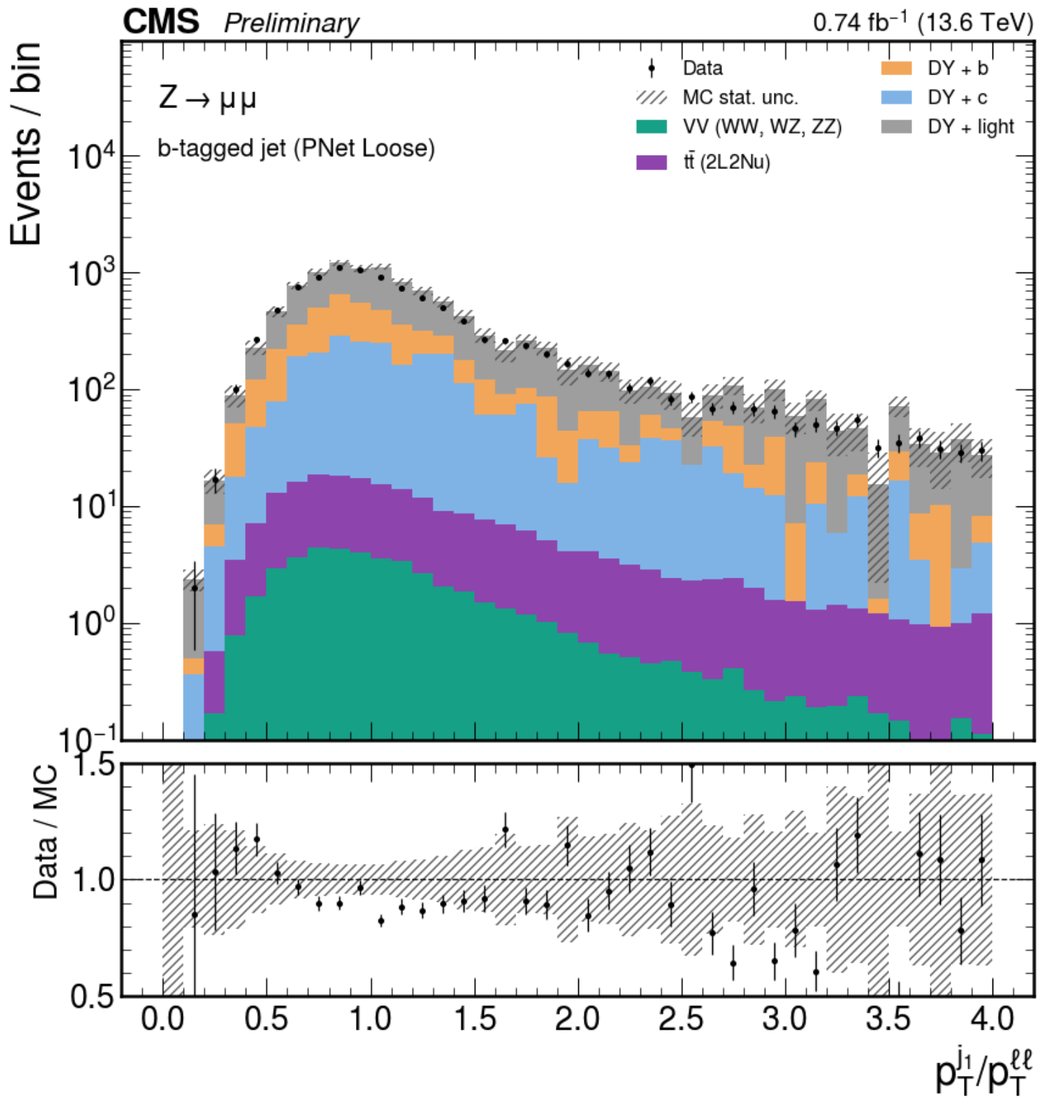




pT balance: leading-jet pT / Z pT

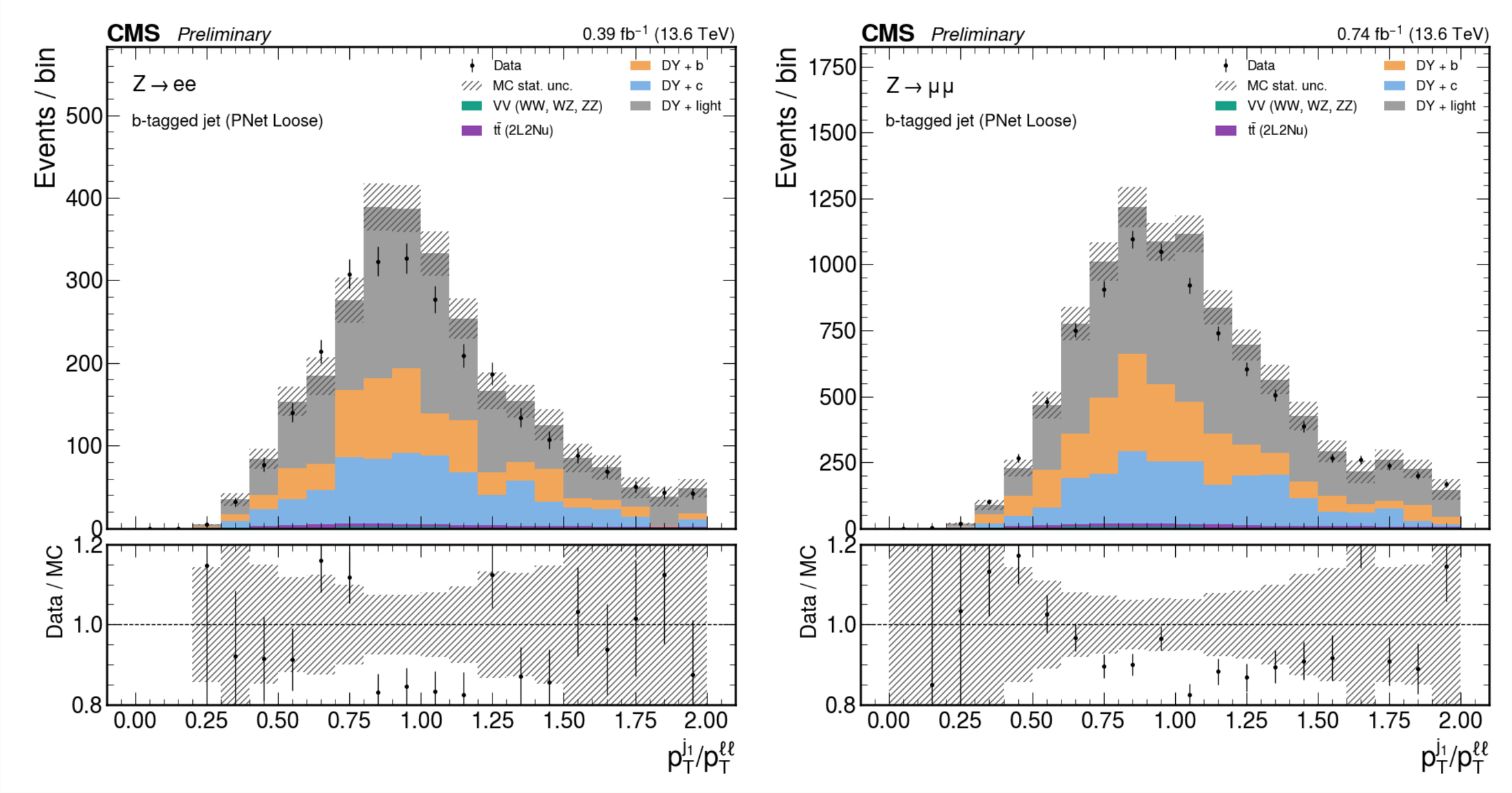
Close to 1 when the jet alone balances the Z. Sensitive to the jet energy scale and to extra jets.





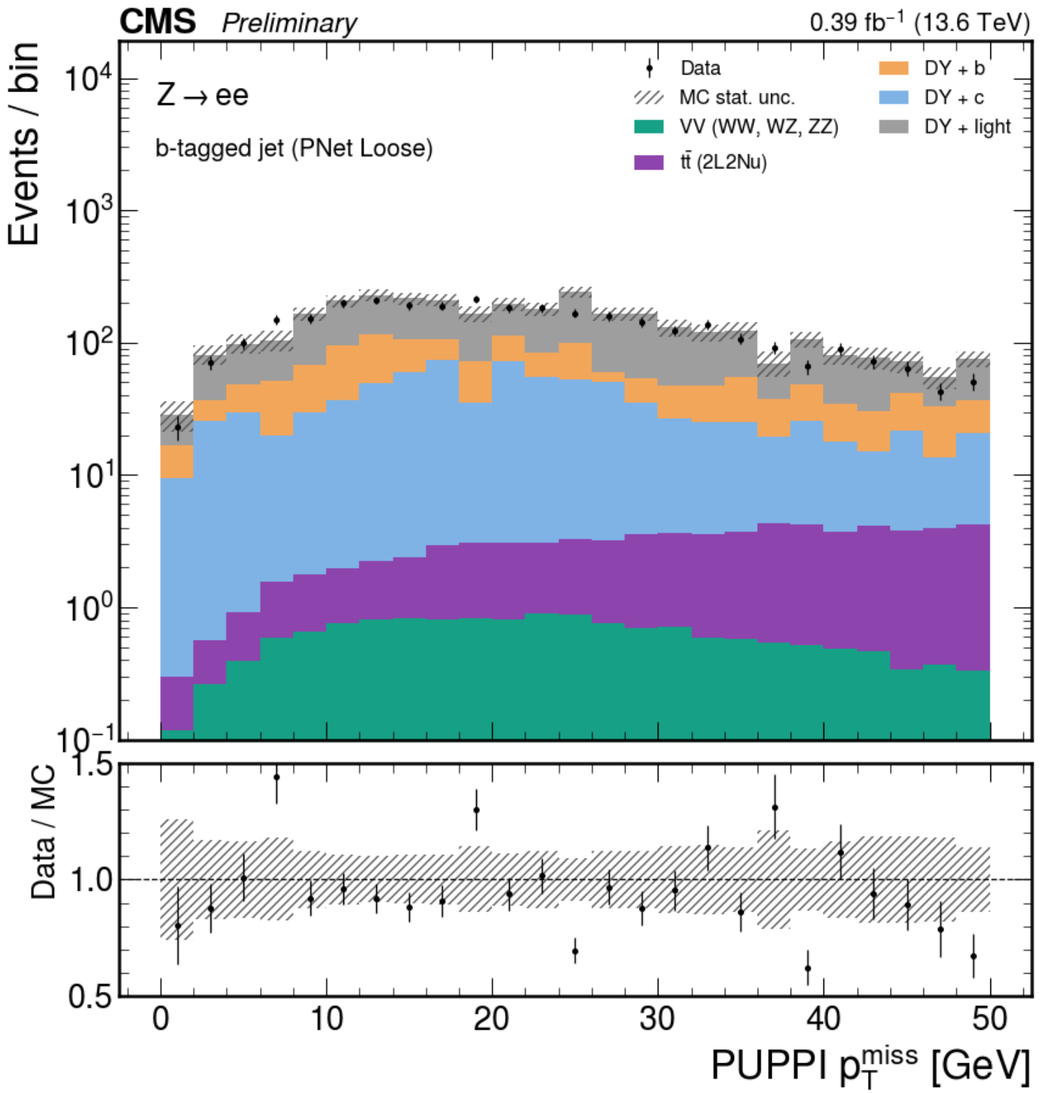
pT balance, zoom on 0–2

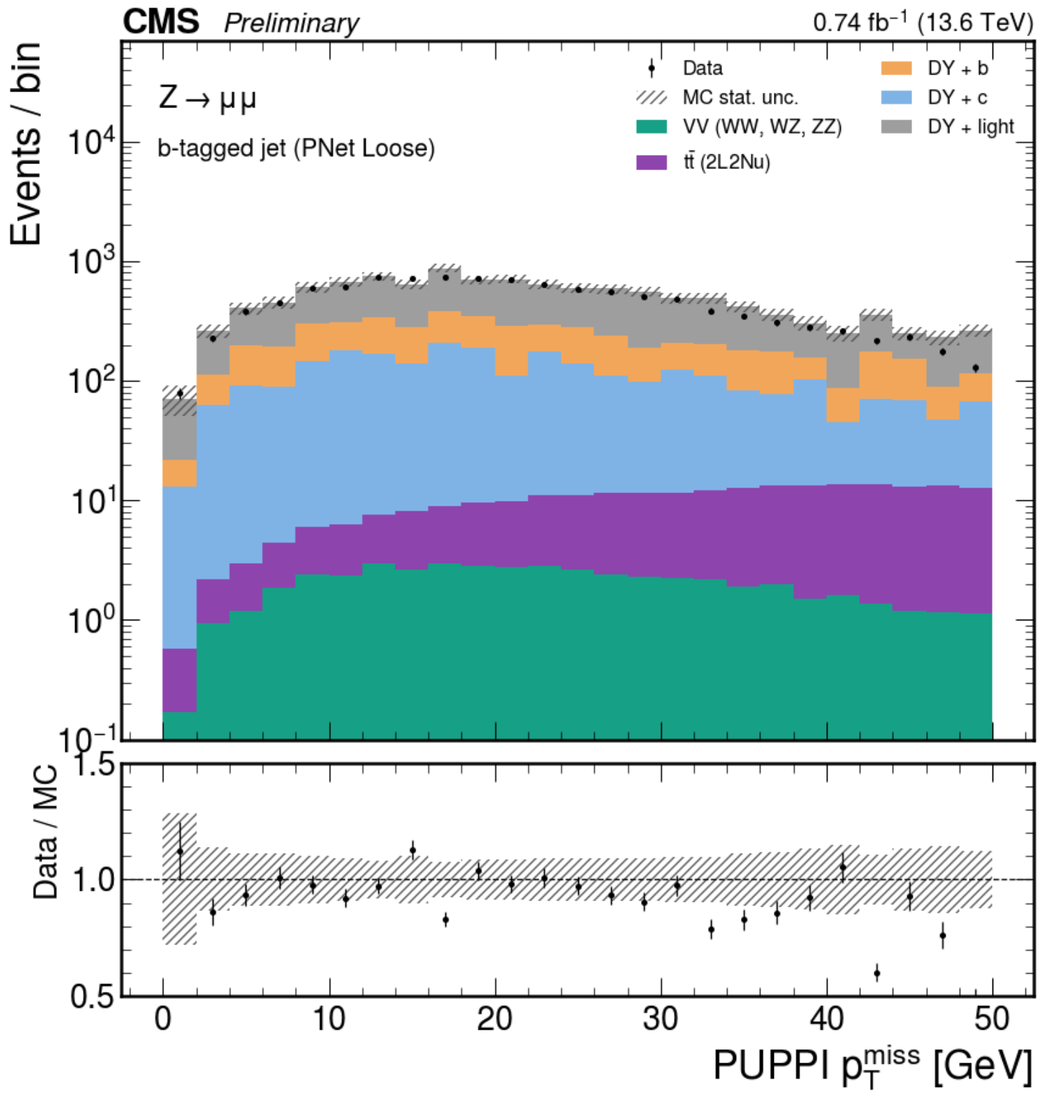
Zoom: 0–2, linear scale, data/MC from 0.8 to 1.2.



Missing transverse momentum (type-1 PUPPI MET), after the MET cut

The selection requires MET < 50 GeV, so the plot stops there.

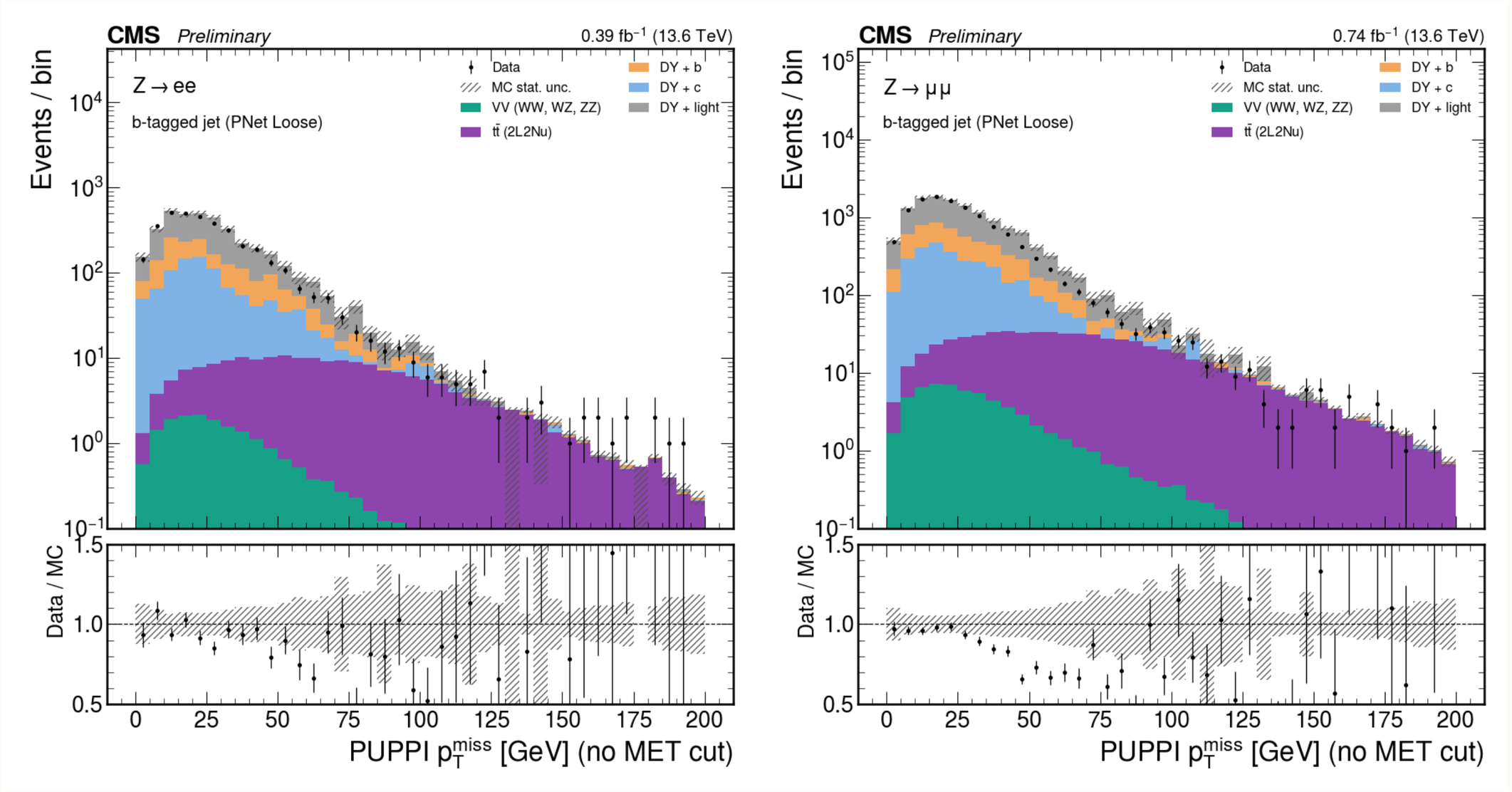




ParticleNet · Loose b-tag (B > 0.047)

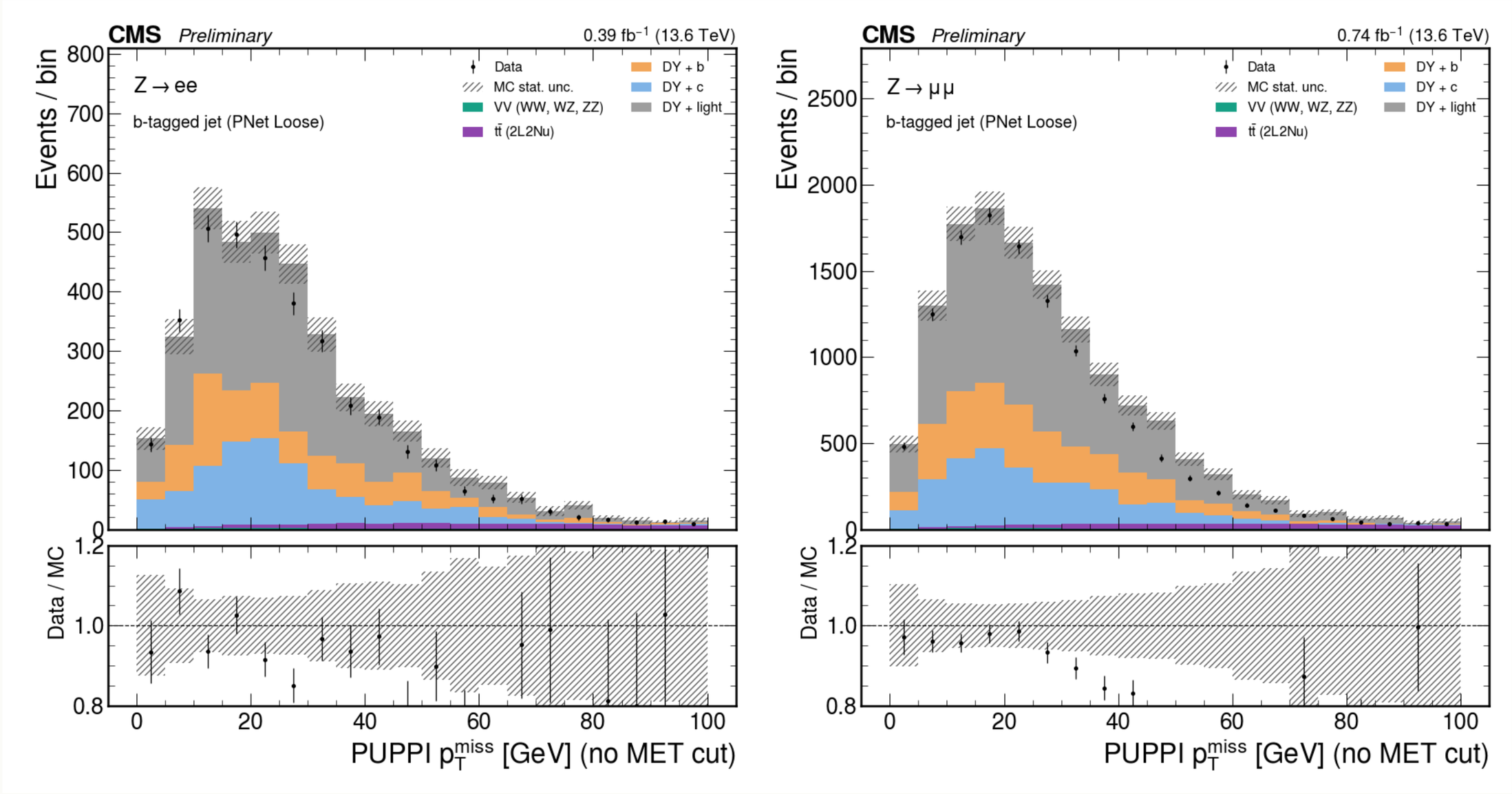
MET without the MET cut, up to 200 GeV

All other cuts applied, so the region removed by MET < 50 GeV is visible. $t\bar{t}$ dominates at high MET. MC is above data at 40–120 GeV, as without tagging (known 2022 jet/MET issue).



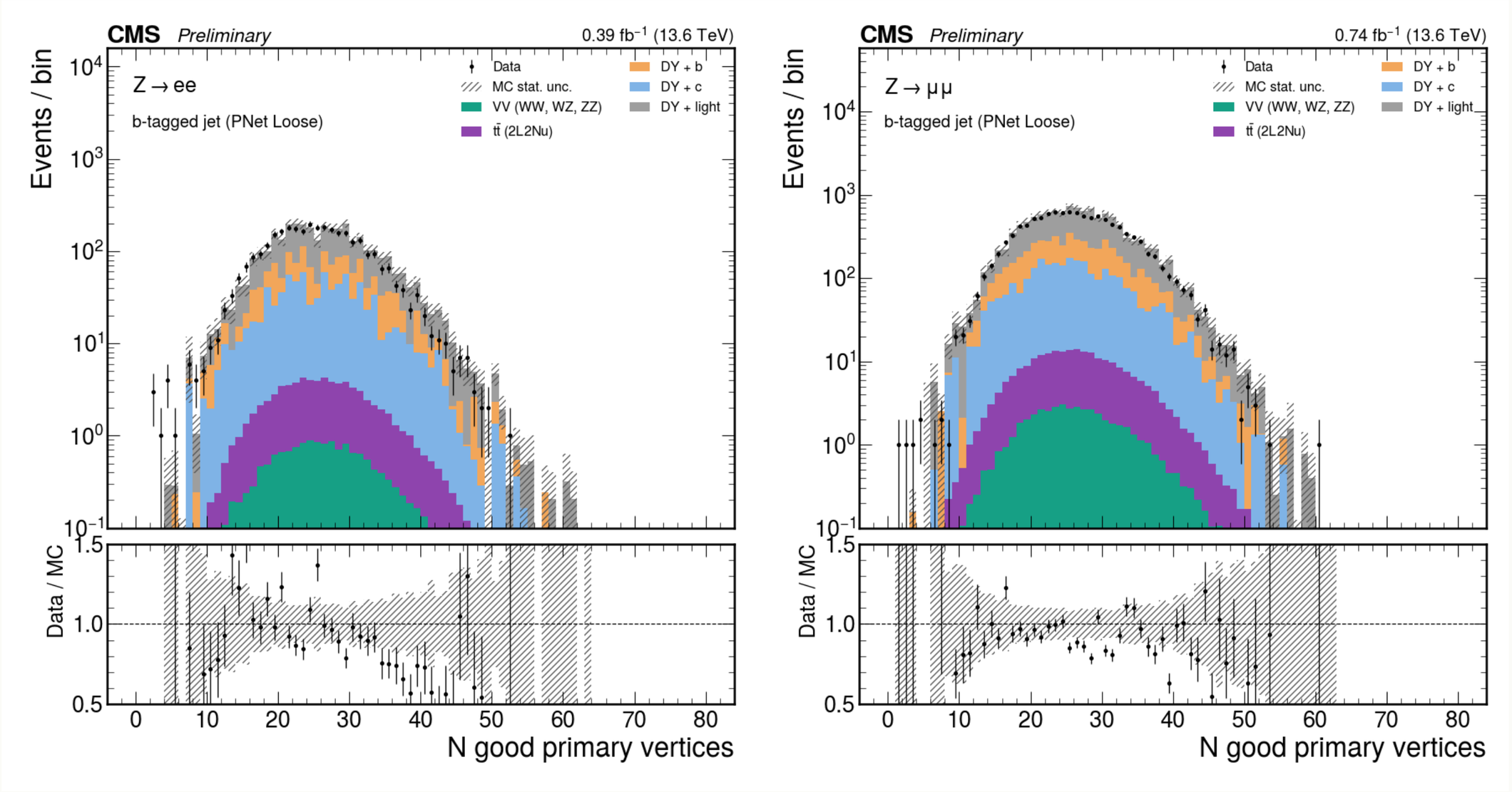
MET without the MET cut, zoom on 0–100 GeV

Zoom: 0–100, linear scale, data/MC from 0.8 to 1.2.



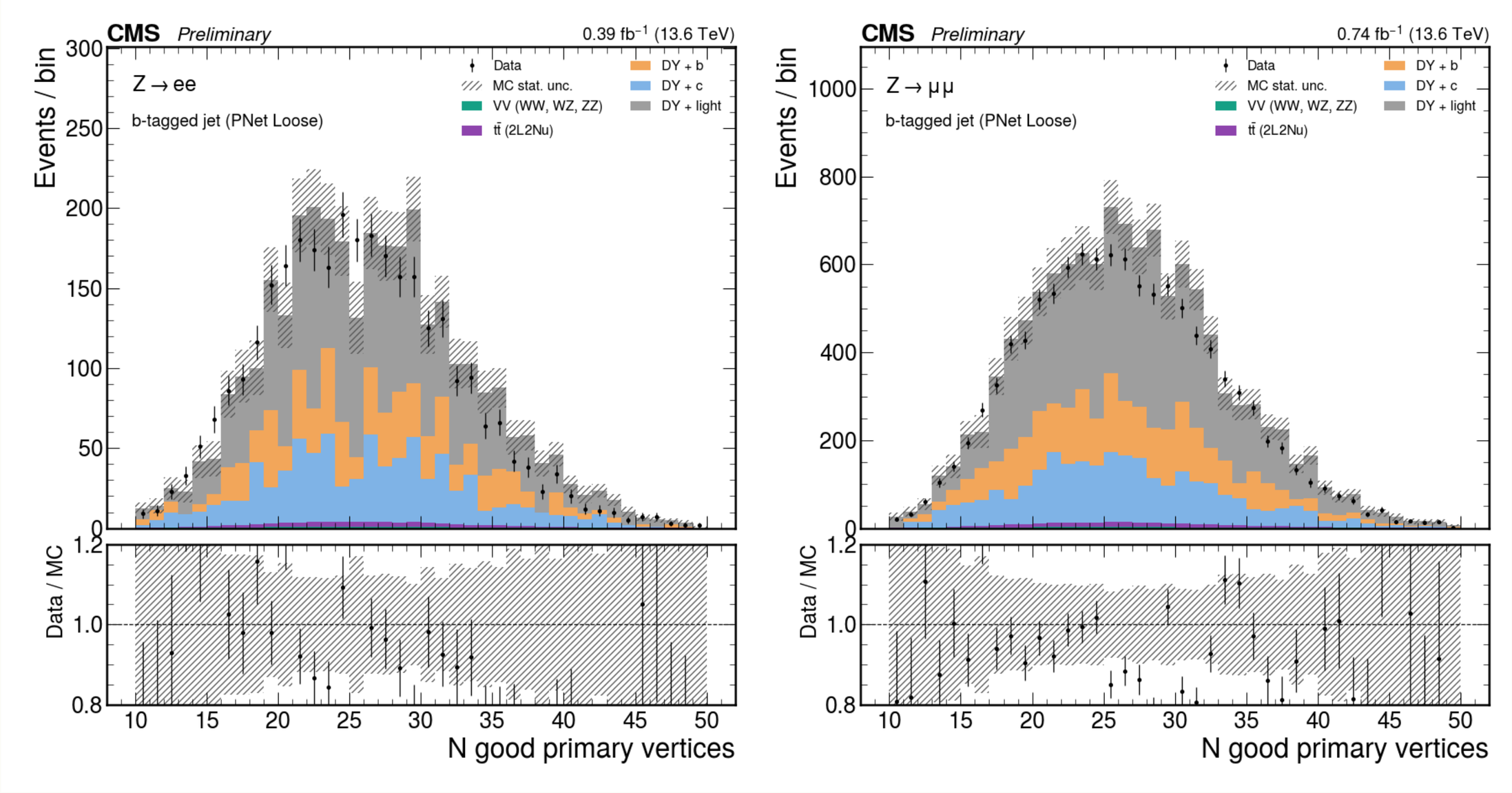
Number of good primary vertices

After the pileup weights. Data and MC differ slightly because our 20 test data files cover runs with less (ee) or about the same (μμ) pileup as all of 2022 C–D, which the MC is weighted to.



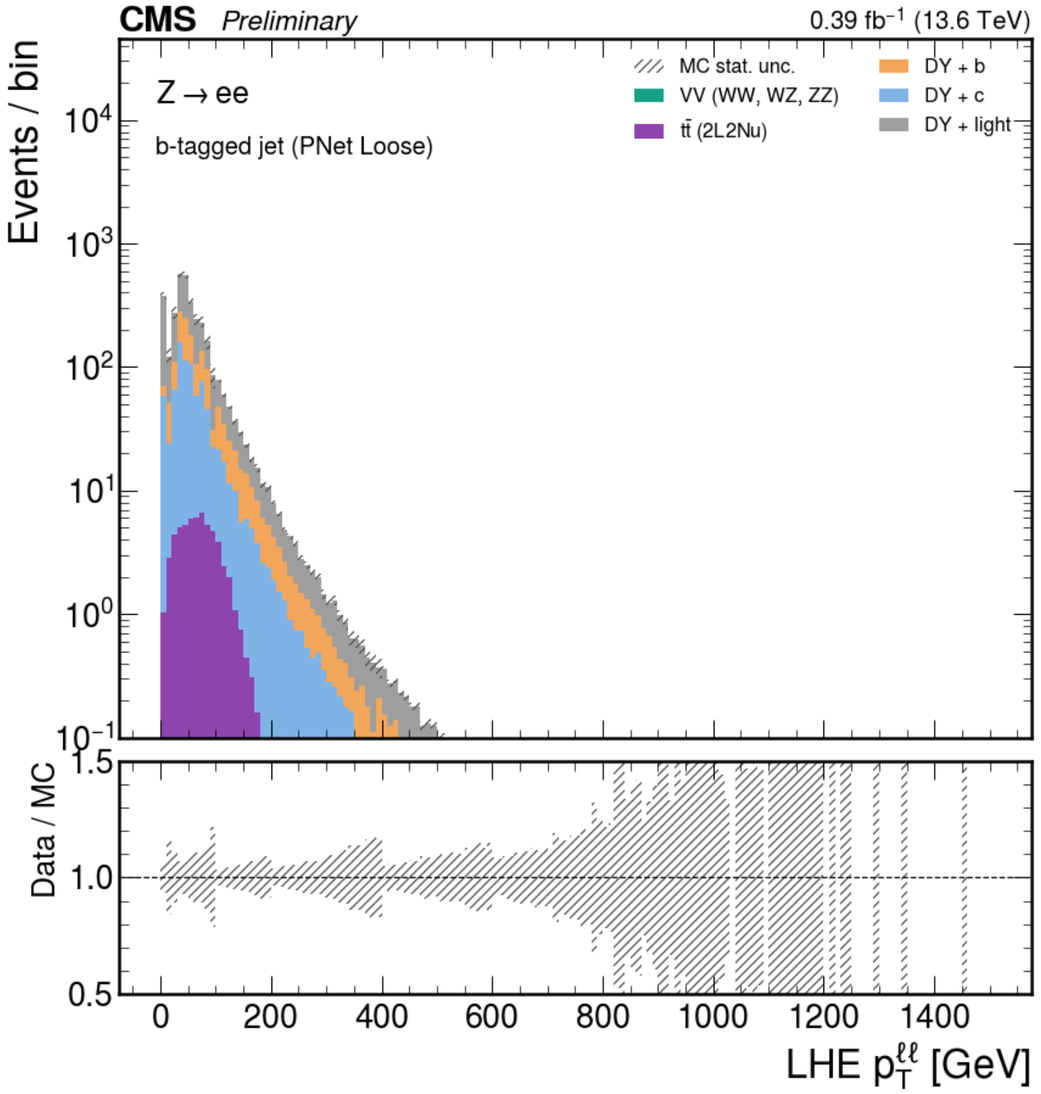
Number of good primary vertices, zoom on 10–50

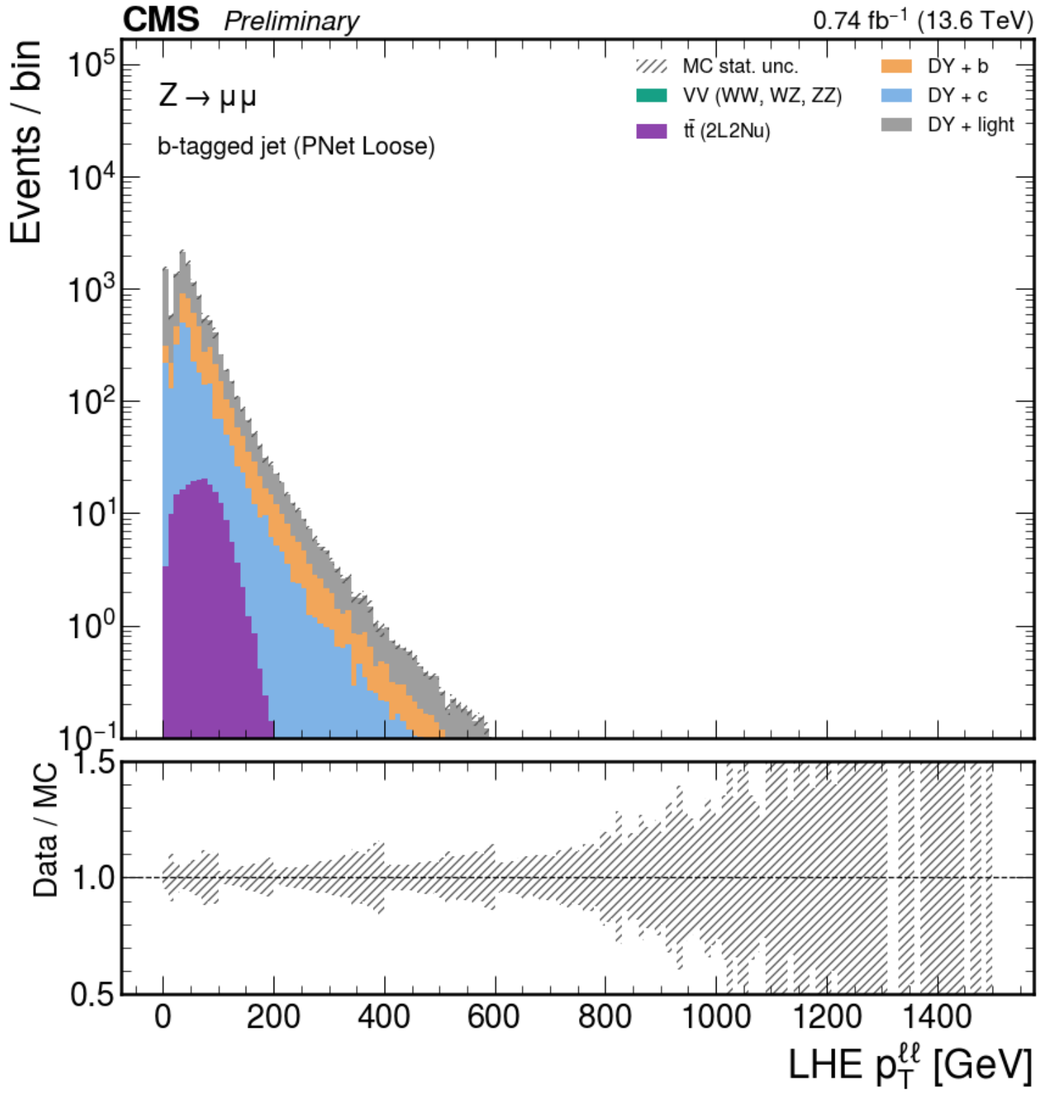
Zoom: 10–50, linear scale, data/MC from 0.8 to 1.2.



Generator-level Z pT (simulation only)

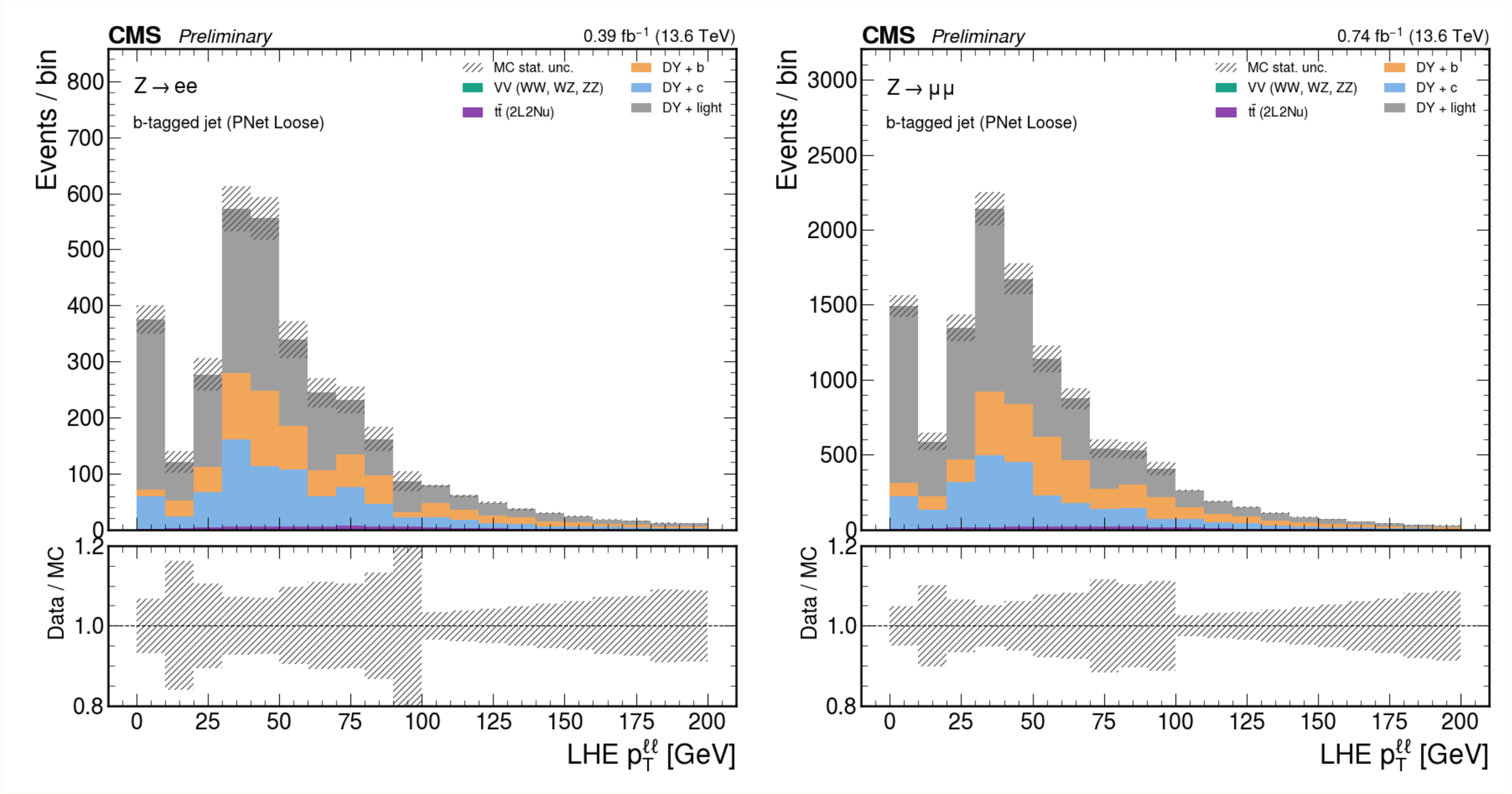
Z pT from the event generator, before any detector effect; no data. Shows which pT(ℓℓ)-binned DY sample contributes where.





Generator-level Z pT, zoom on 0–200 GeV

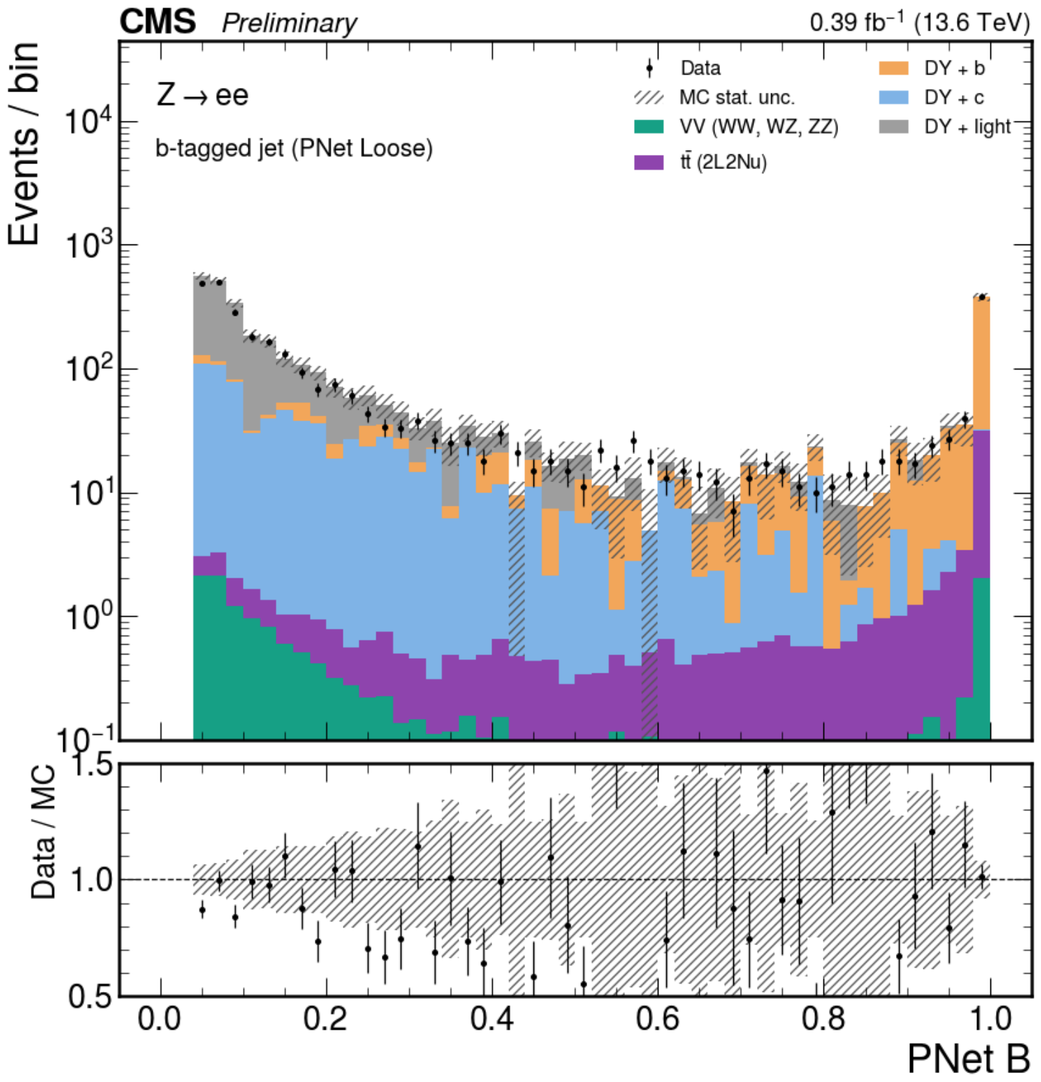
Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.

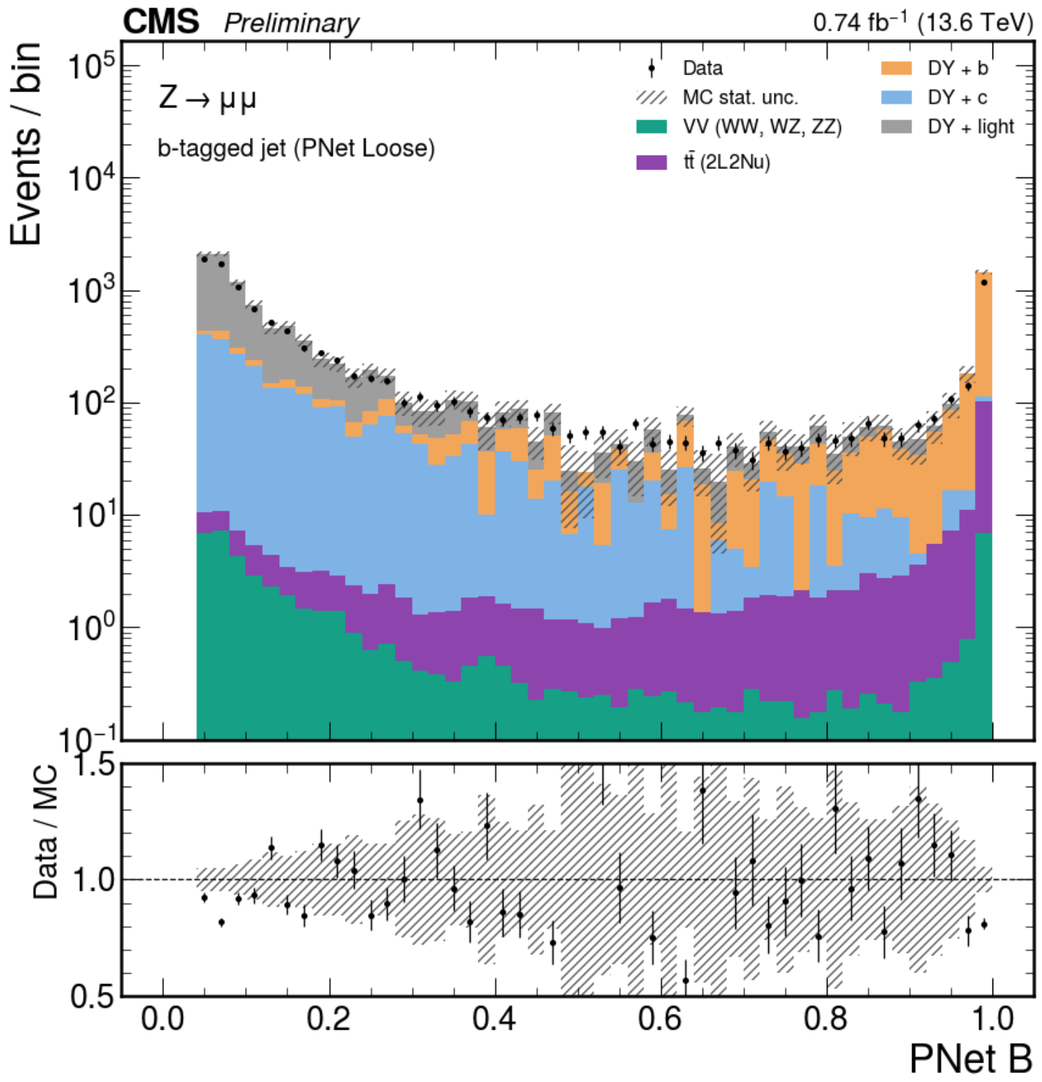


ParticleNet · Loose b-tag (B > 0.047)

Leading-jet PNet B score (b vs everything)

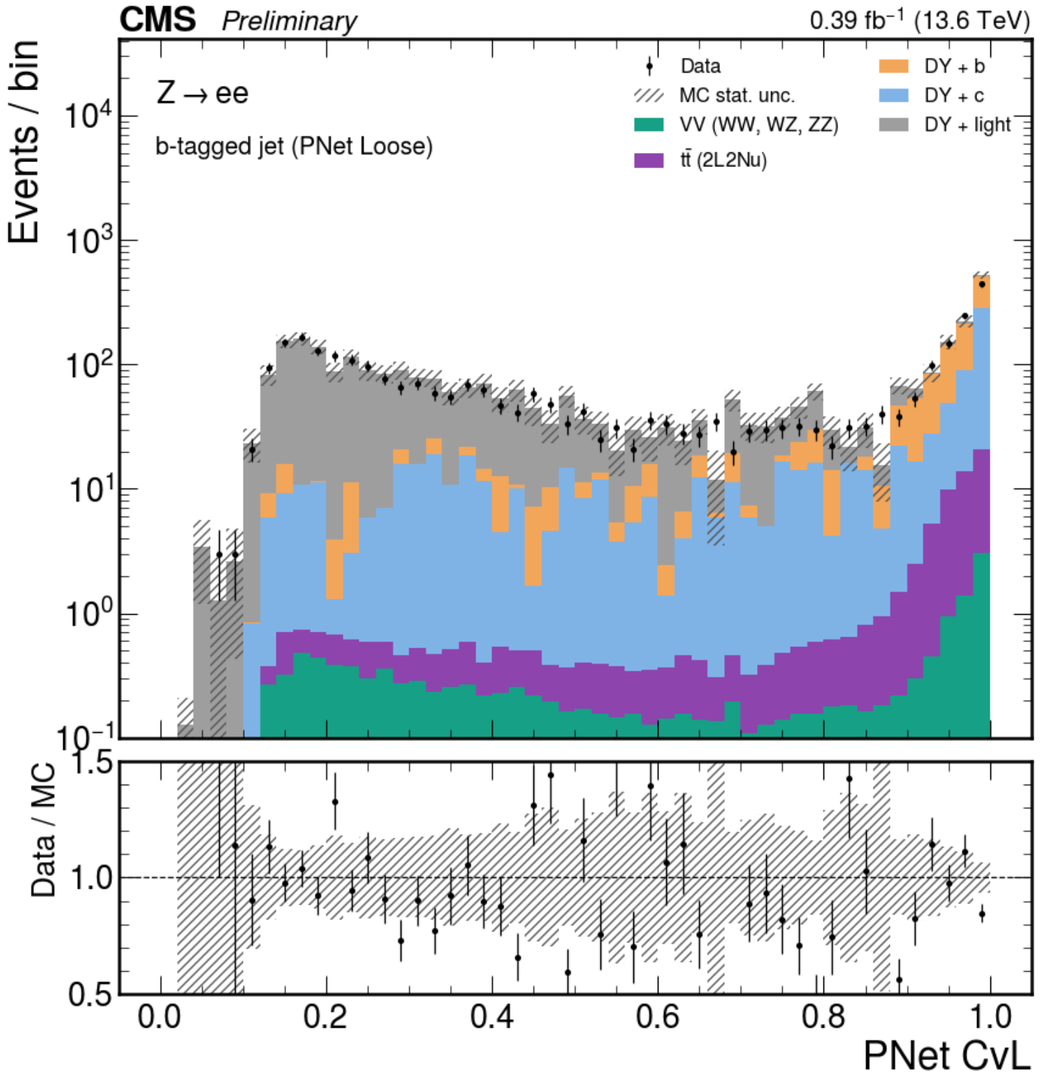
The score used for the selection: only jets above 0.047 enter this region.

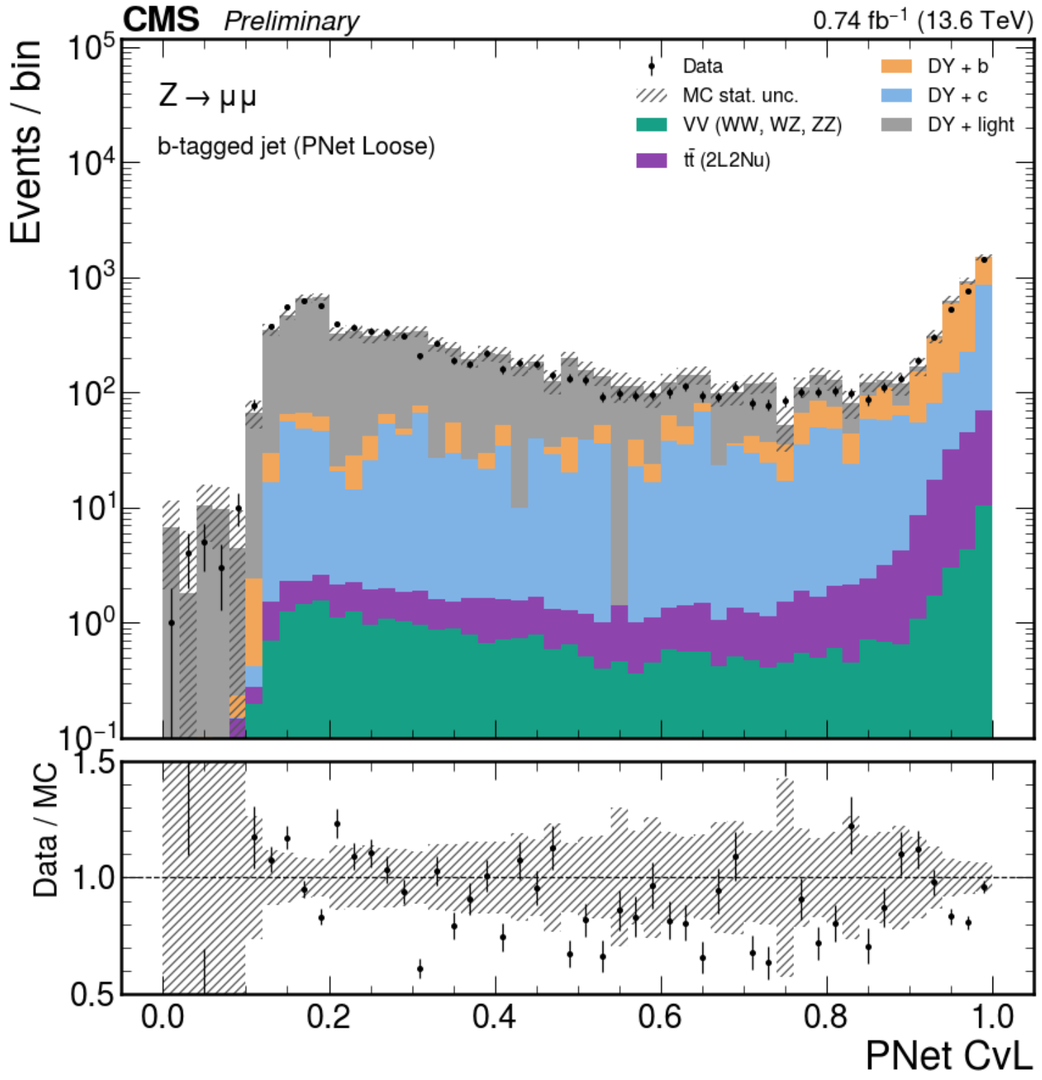




Leading-jet PNet CvL score (c vs light)

Not used in this b-tag selection; shown for comparison.

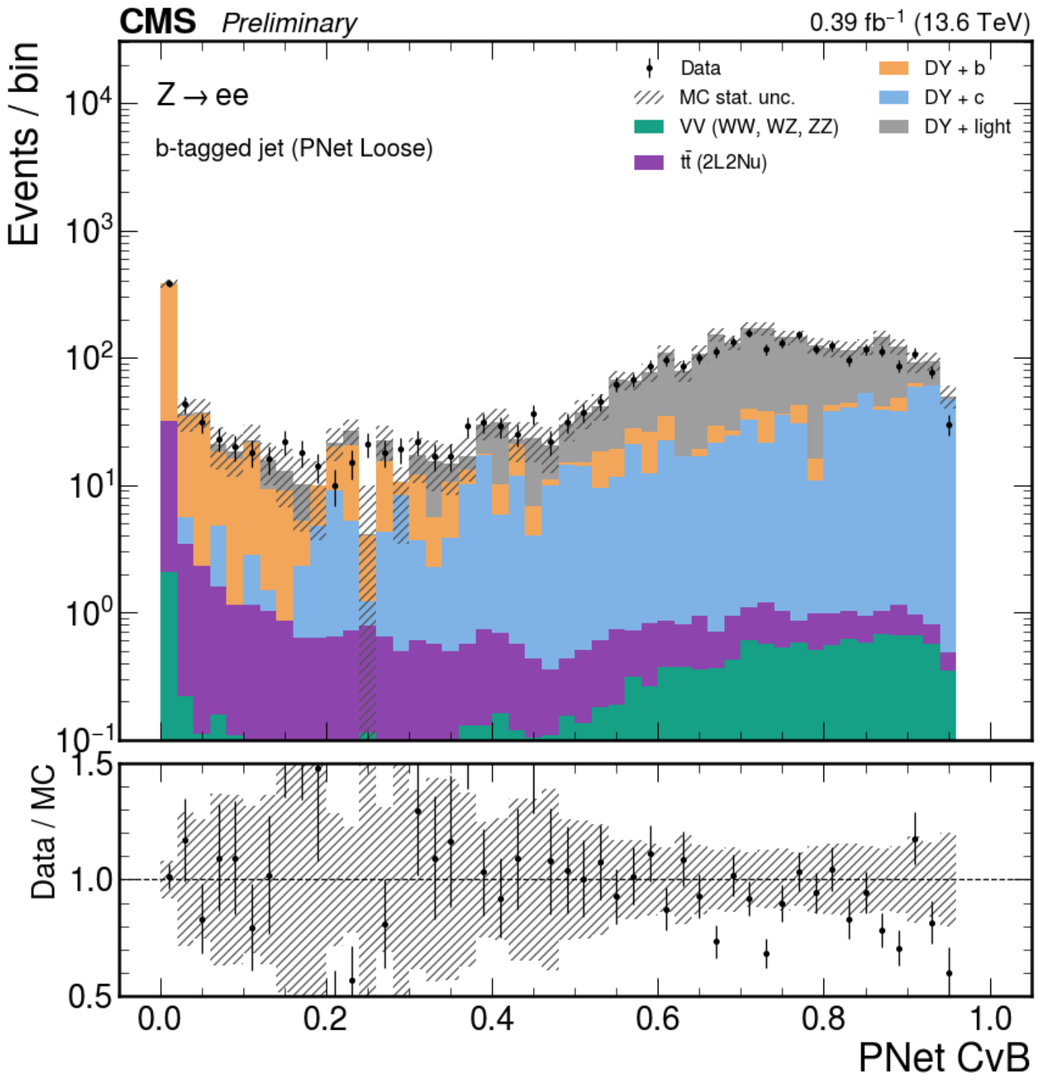


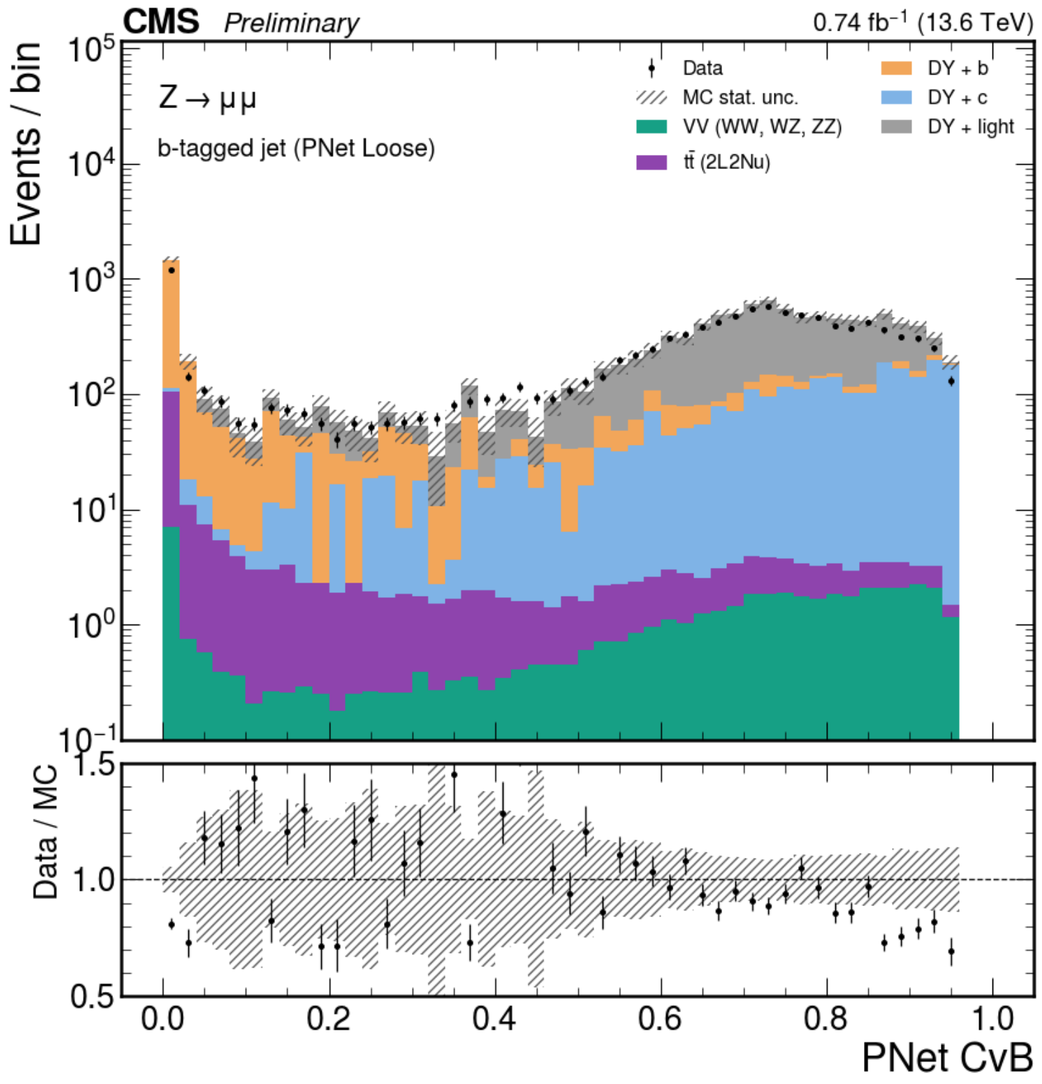


ParticleNet · Loose b-tag (B > 0.047)

Leading-jet PNet CvB score (c vs b)

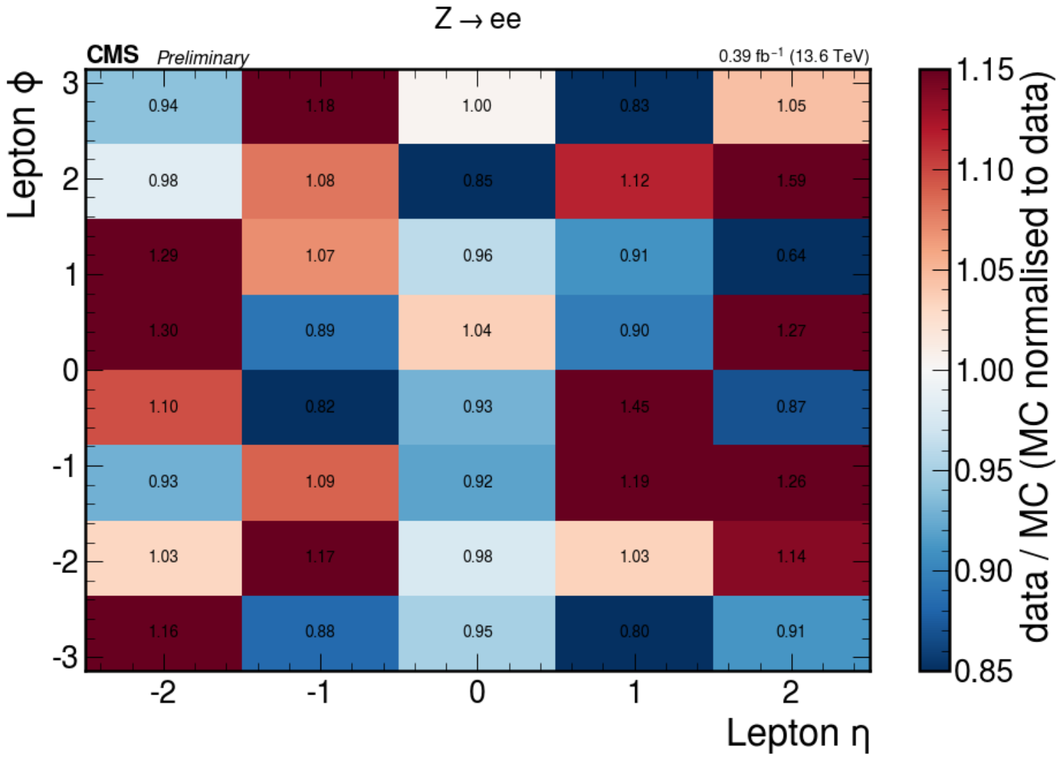
Not used in this b-tag selection. Low values = b-like.

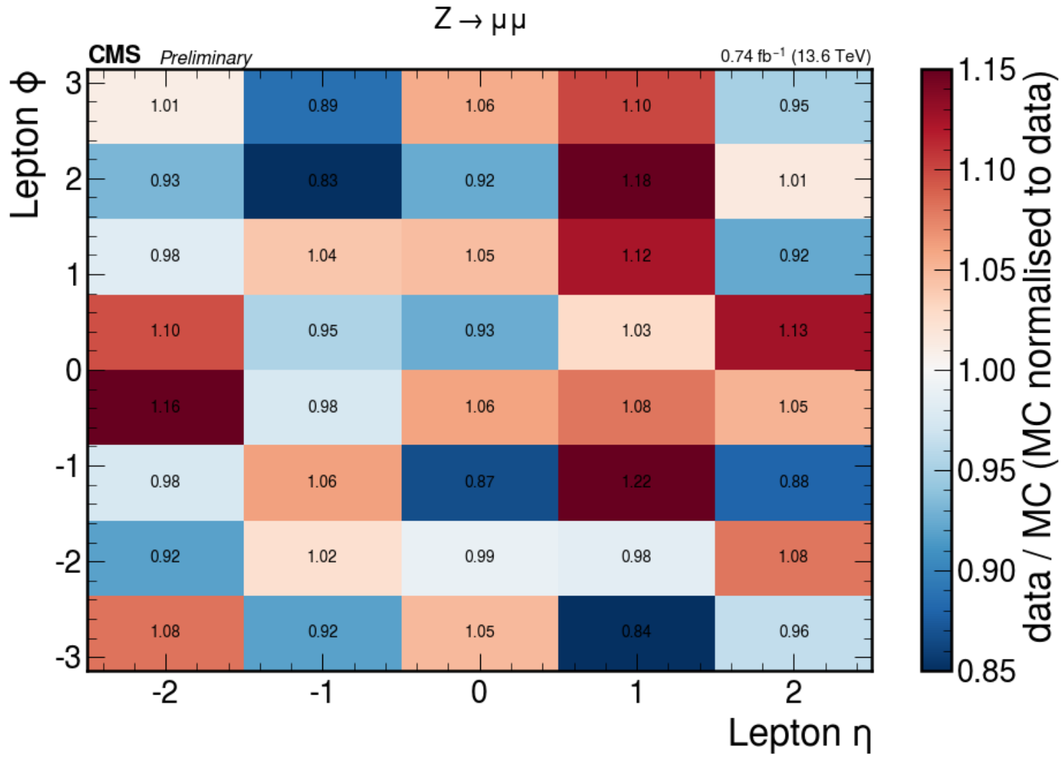




Lepton η–φ map: data / MC

Both Z leptons, MC normalised to data in each map, cells of 1.0 in η × π/4 in φ. Colour scale from 0.85 to 1.15; the number in each cell is data/MC. Few events per cell in a tagged region.





DeepJet, Loose b-tag

Leading jet DeepJet B score > 0.0583; MC weighted with the deepJet BTV scale factors.

	data/MC	DY+b	DY+c	DY+light	tt̄	tagged MC	tagged data
Z→ee	0.926 ± 0.029	18%	22%	58%	2%	18.9%	18.4%
Z→μμ	0.924 ± 0.021	20%	21%	57%	1%	19.3%	18.9%

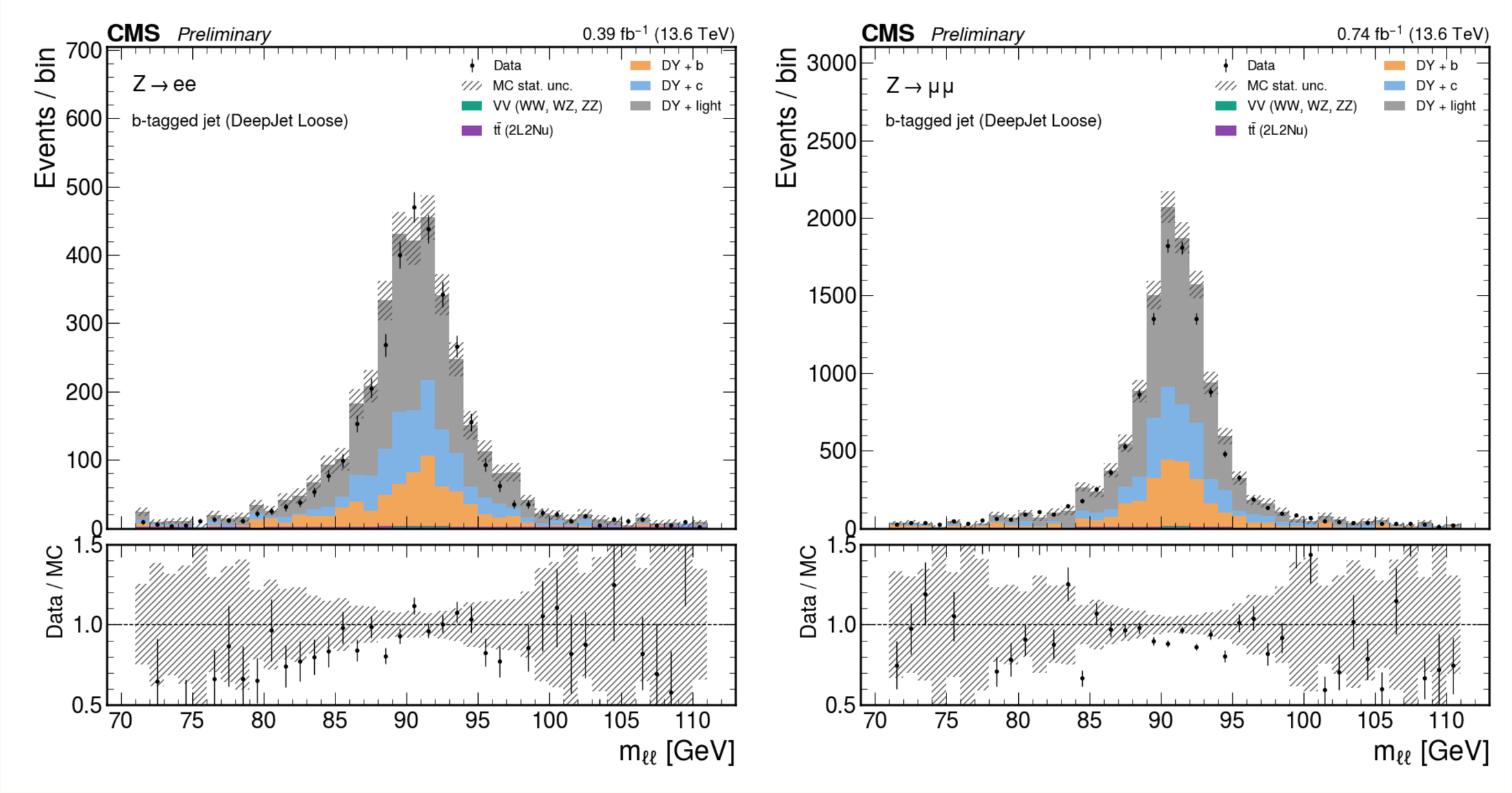
Flavour shares are of all MC. Tagged = fraction of Z + jet events whose leading jet passes the tag. Statistical errors only.

42 plots follow, in this order: Z and leptons, jets, Z–jet, MET and vertices, generator Z pT, the DeepJet scores, the lepton η–φ map. Left Z→ee, right Z→μμ.

DeepJet · Loose b-tag (B > 0.0583)

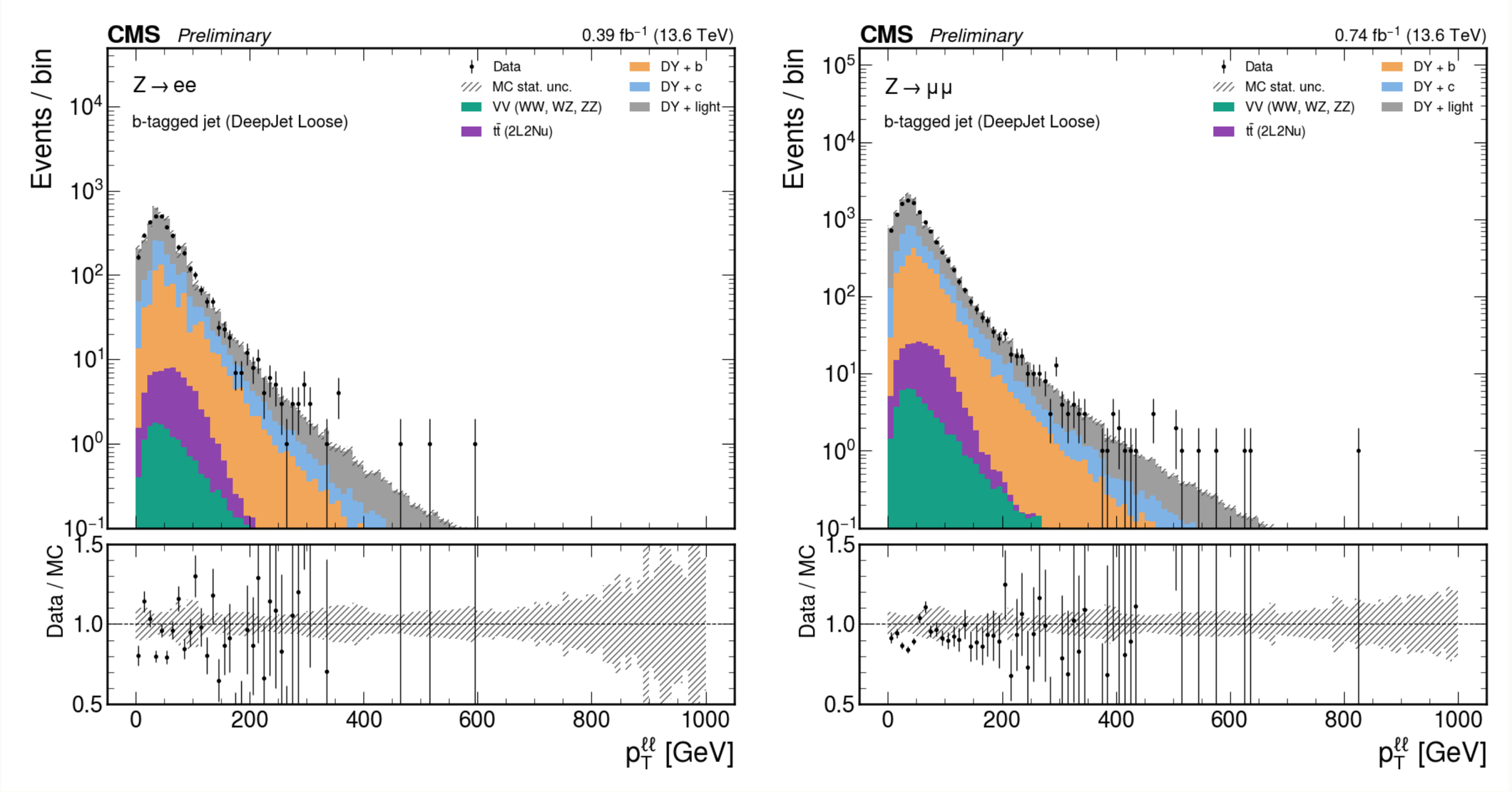
Mass of the lepton pair, m(ℓℓ)

The Z peak, linear scale. Selection: 71–111 GeV. Tests the lepton energy scale and resolution.



Transverse momentum of the lepton pair, pT(ℓℓ) = Z pT

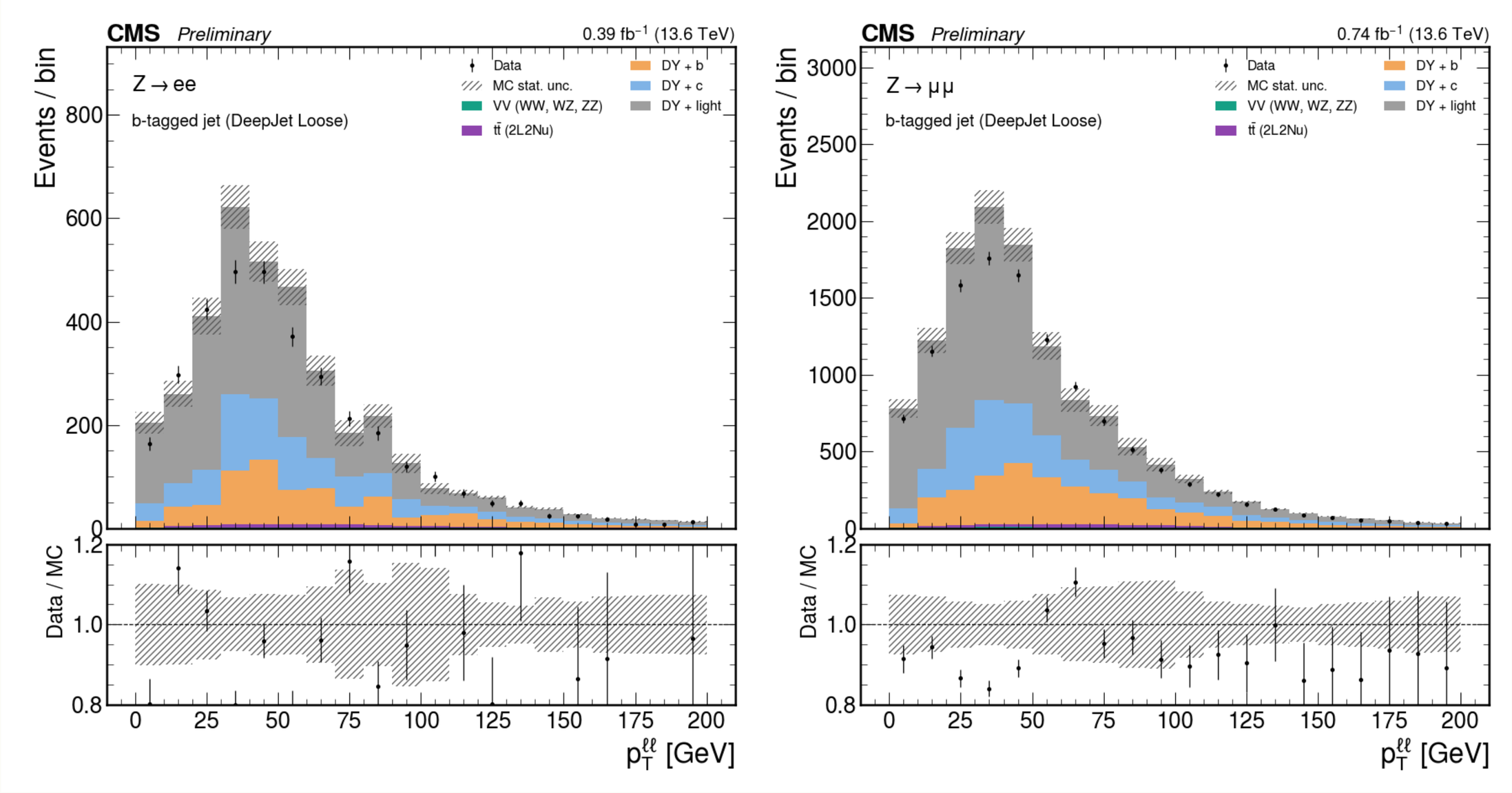
Log scale up to 1 TeV. This is the distribution with the pT trend discussed in the follow-up slides (low at 30–60 GeV, high at 100–150 GeV).



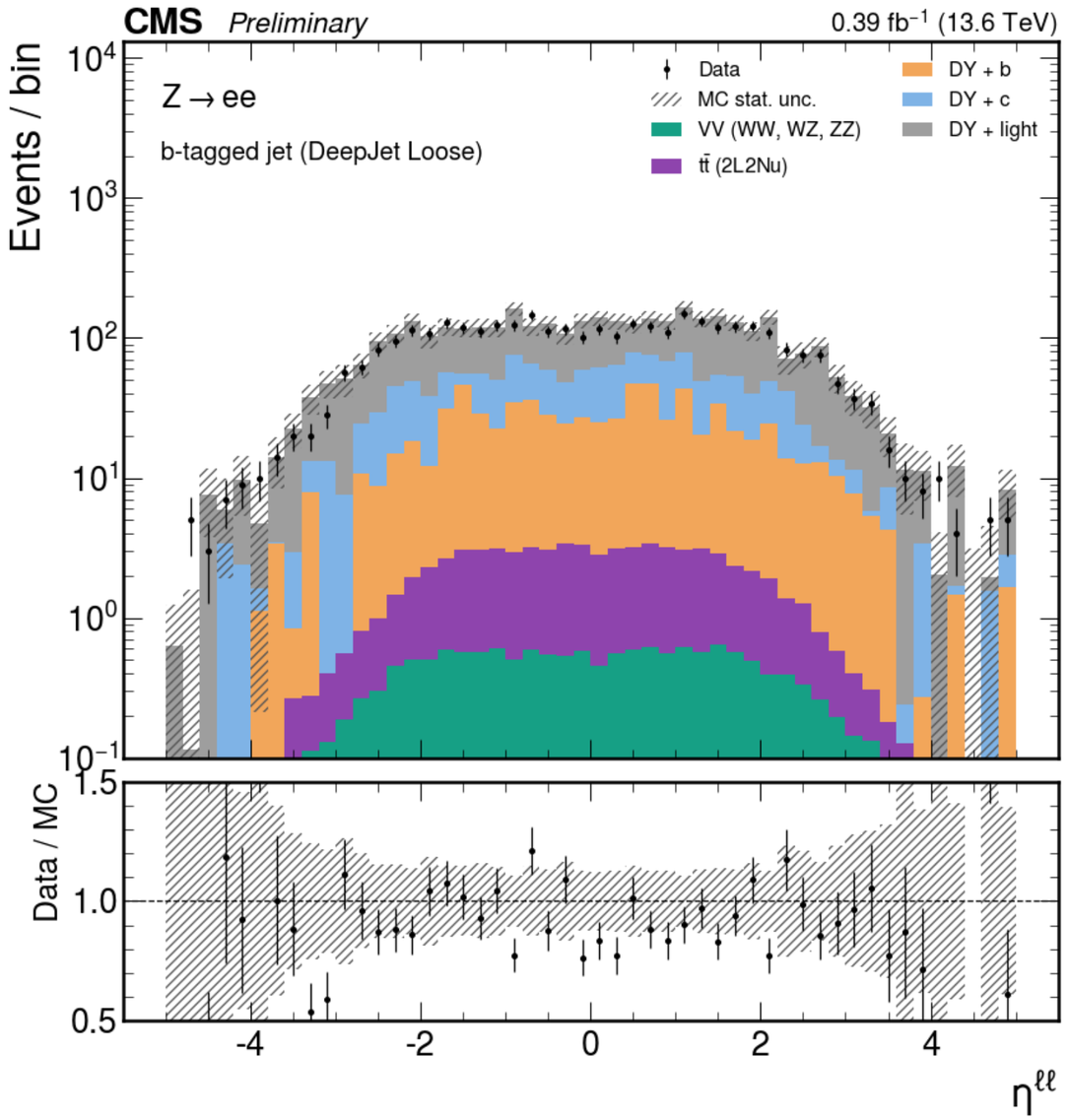
DeepJet · Loose b-tag (B > 0.0583)

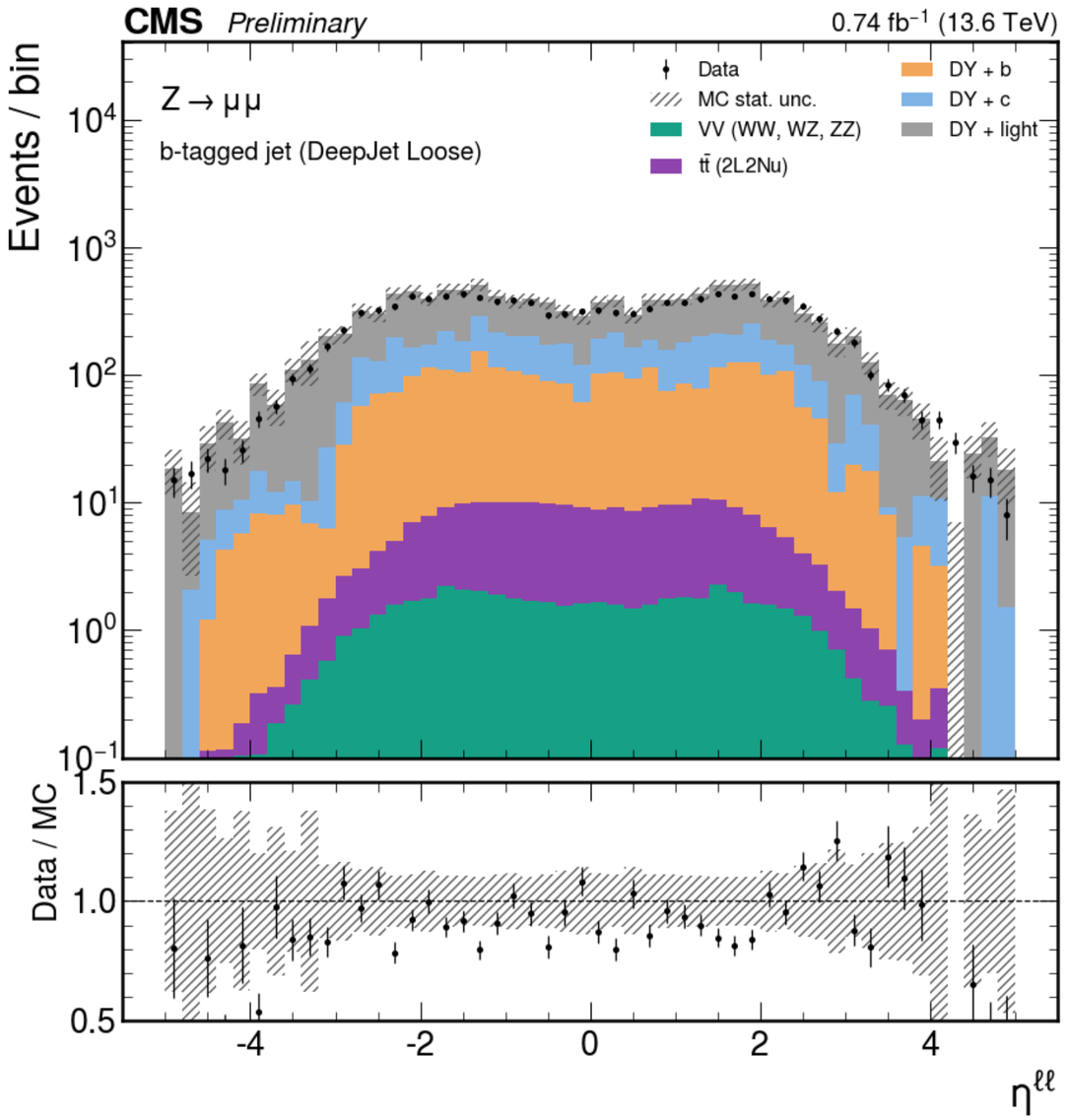
Z pT, zoom on 0–200 GeV

Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.



Pseudorapidity of the lepton pair, η(ℓℓ)

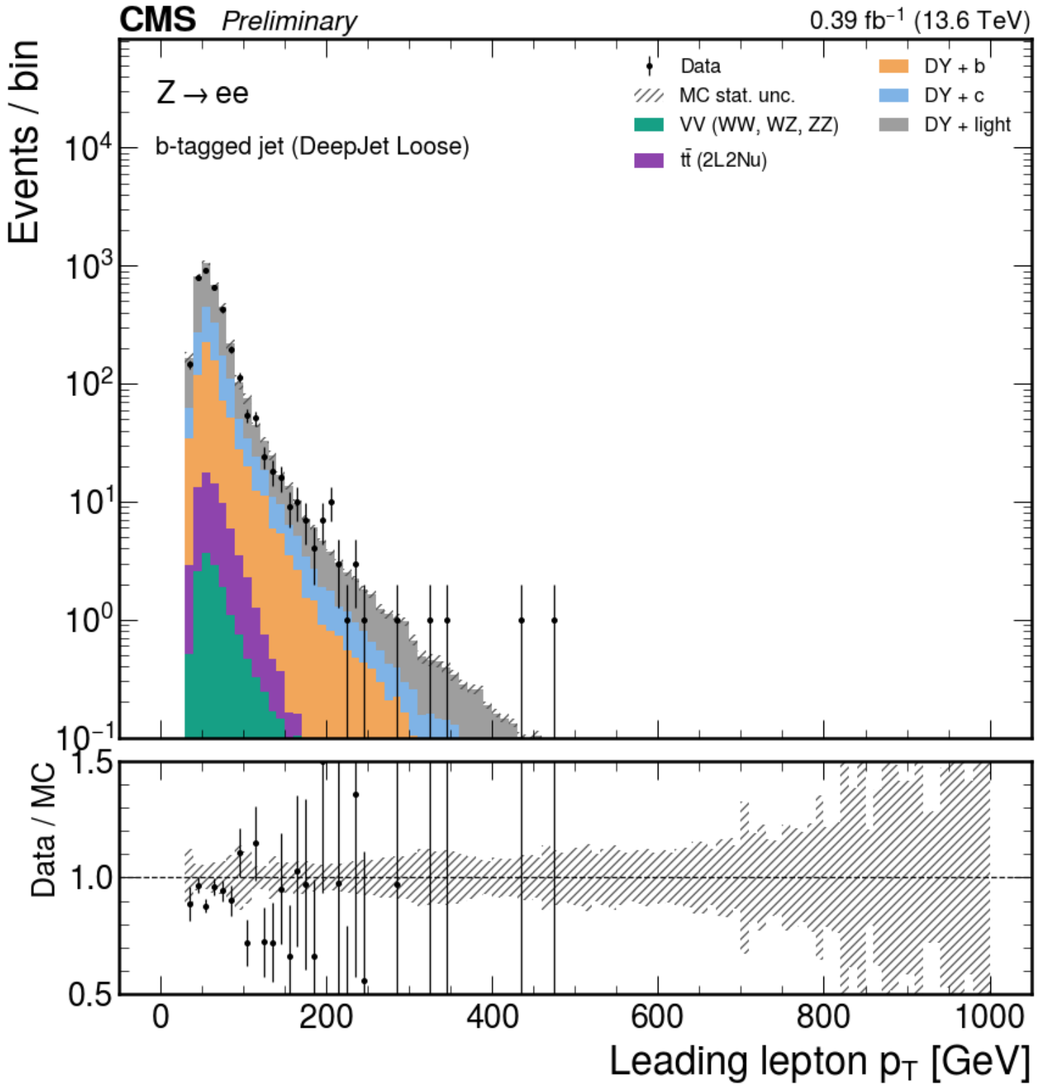


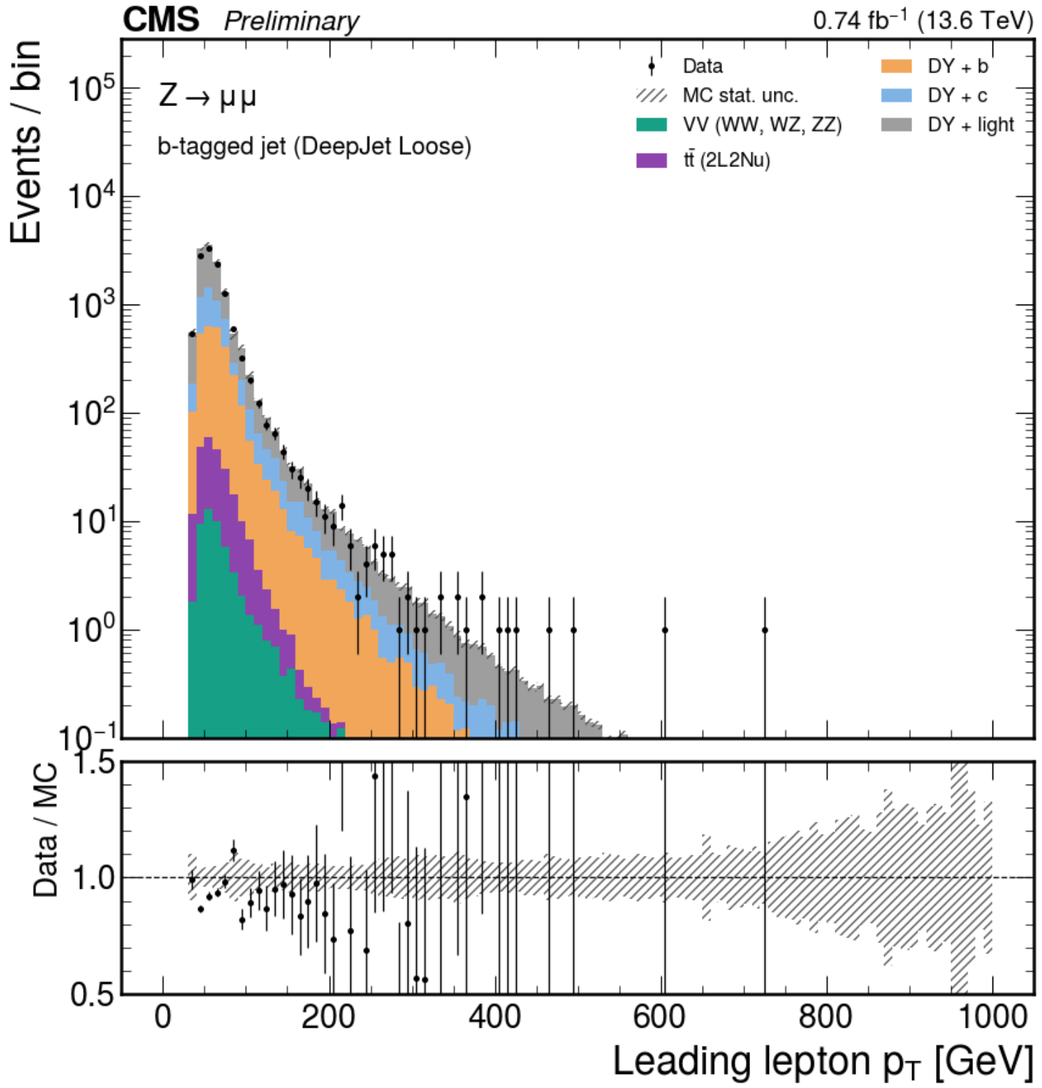


DeepJet · Loose b-tag (B > 0.0583)

Leading-lepton pT

The selection requires pT > 35 GeV for the leading lepton (on the trigger plateau).

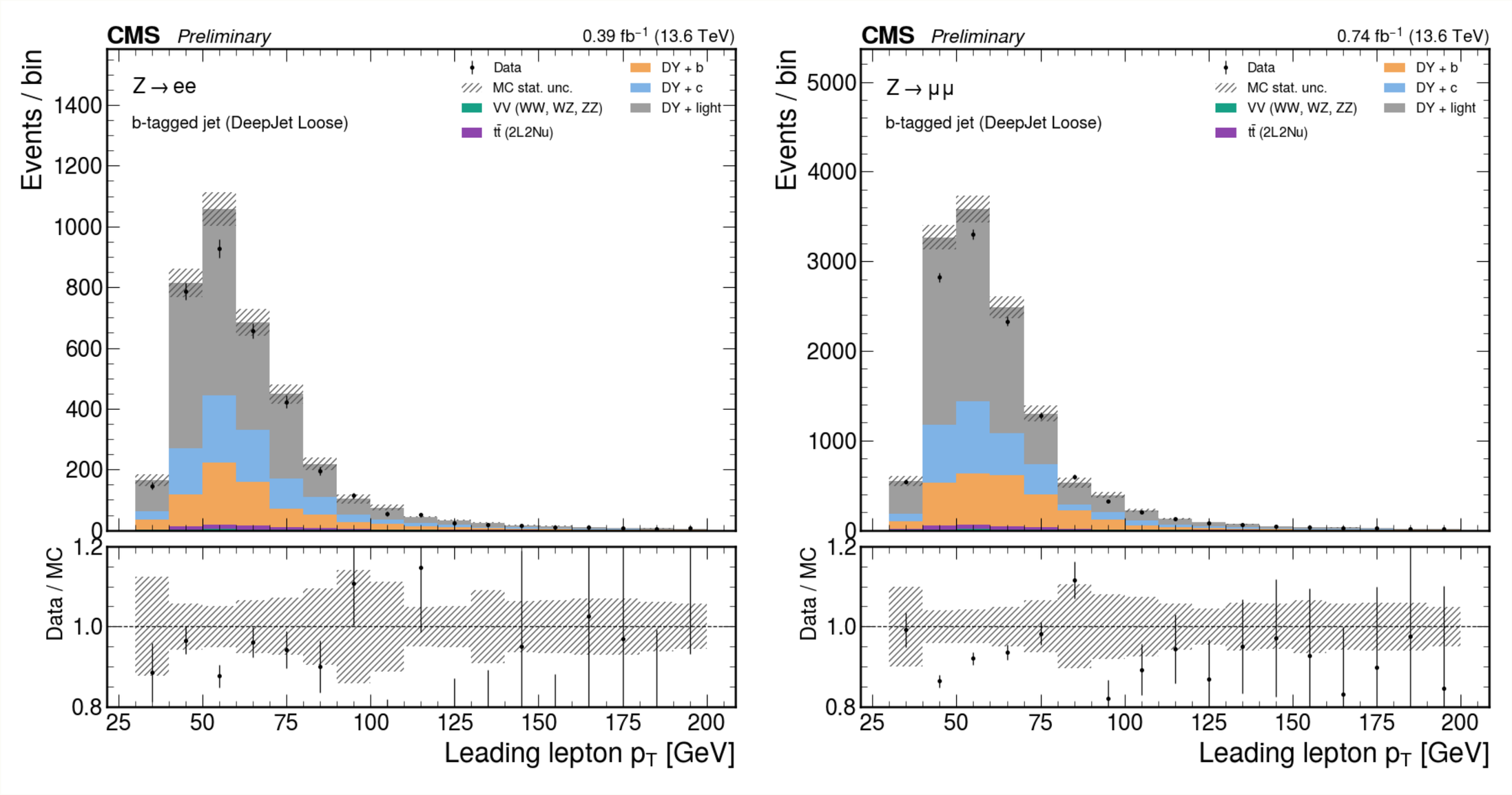




DeepJet · Loose b-tag (B > 0.0583)

Leading-lepton pT, zoom on 30–200 GeV

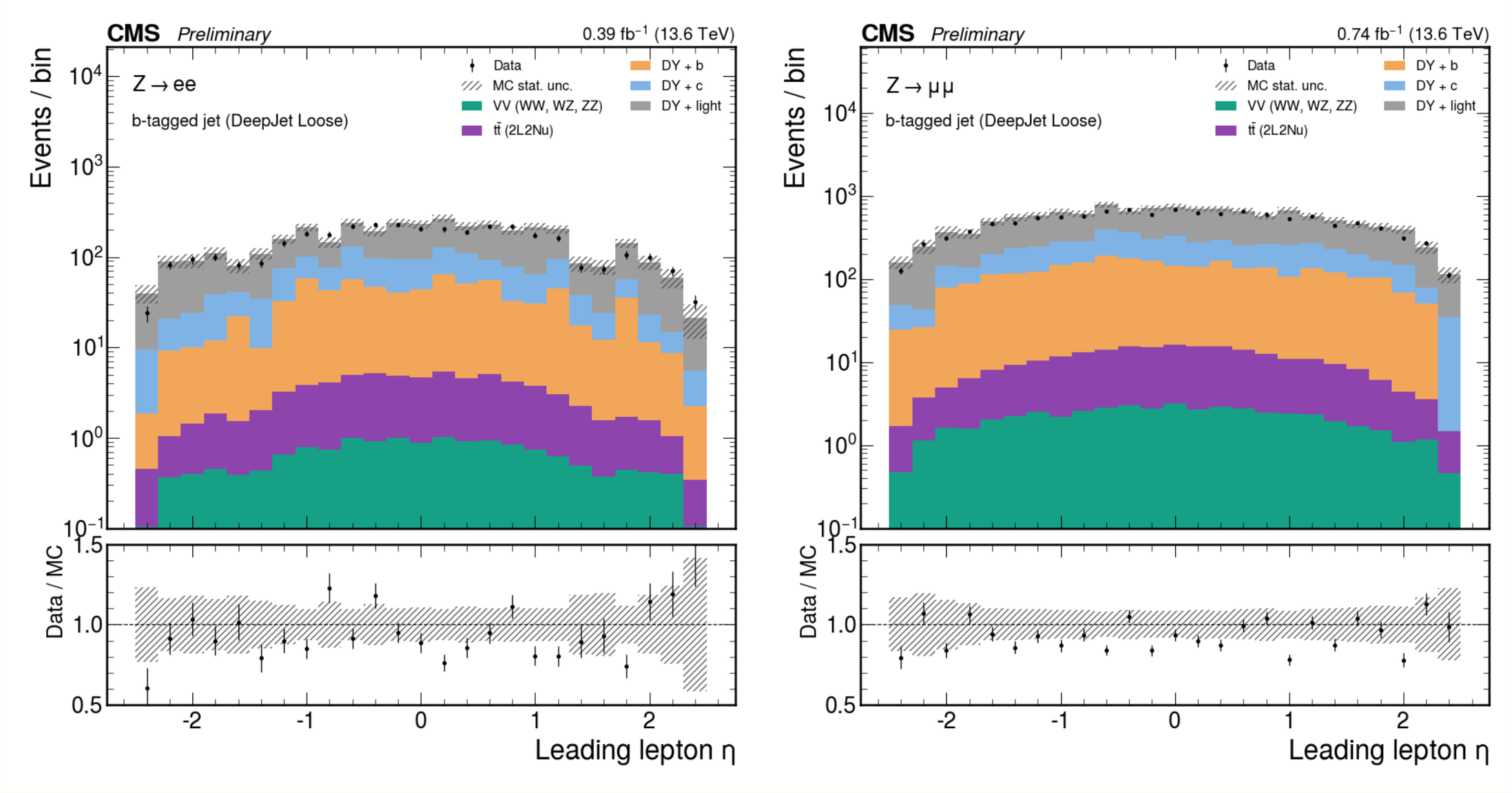
Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.



DeepJet · Loose b-tag (B > 0.0583)

Leading-lepton η

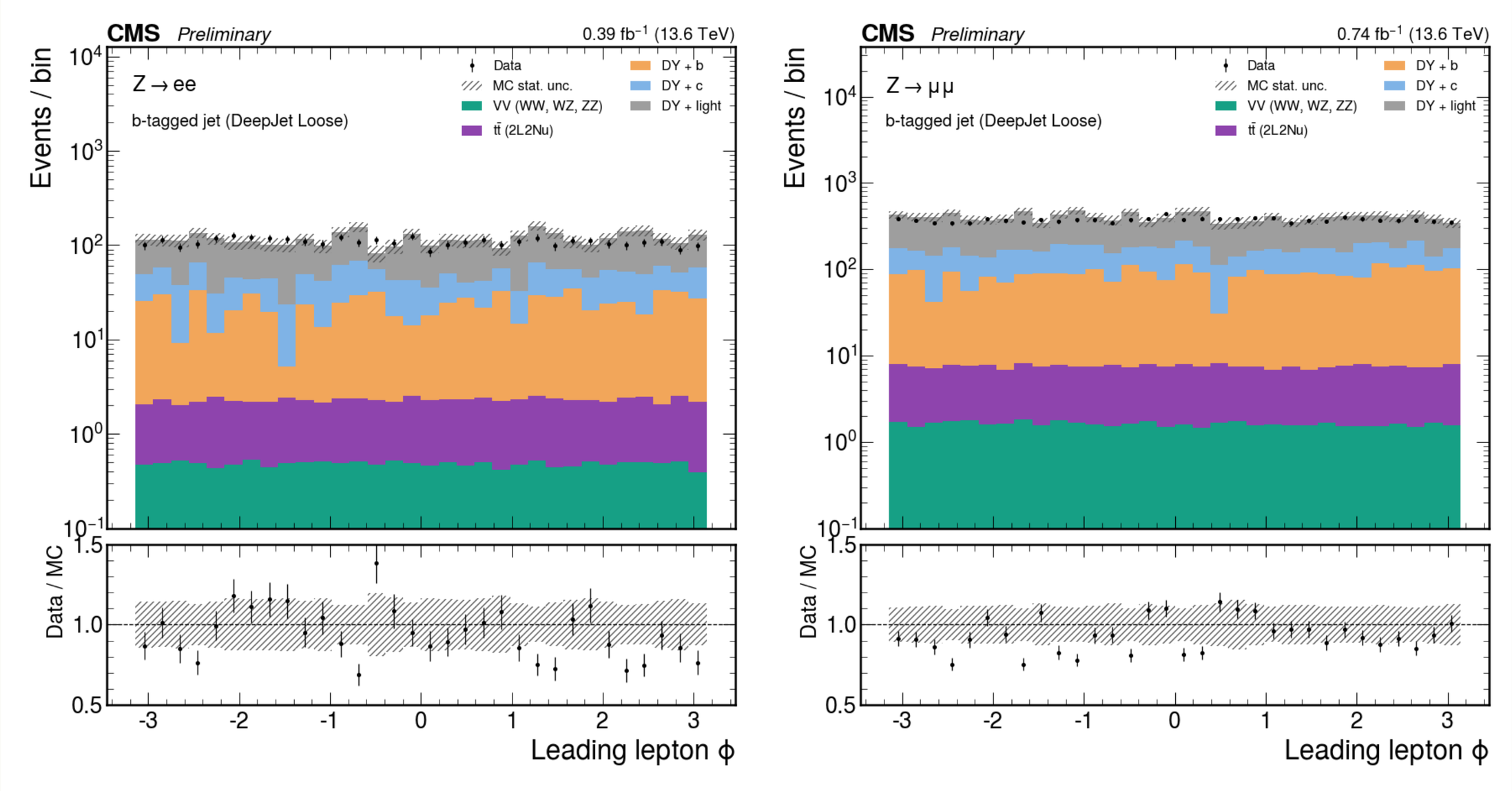
|η| < 2.4. For electrons the barrel–endcap gap (1.44 < |η| < 1.57) is removed, so that bin is low.



DeepJet · Loose b-tag (B > 0.0583)

Leading-lepton φ

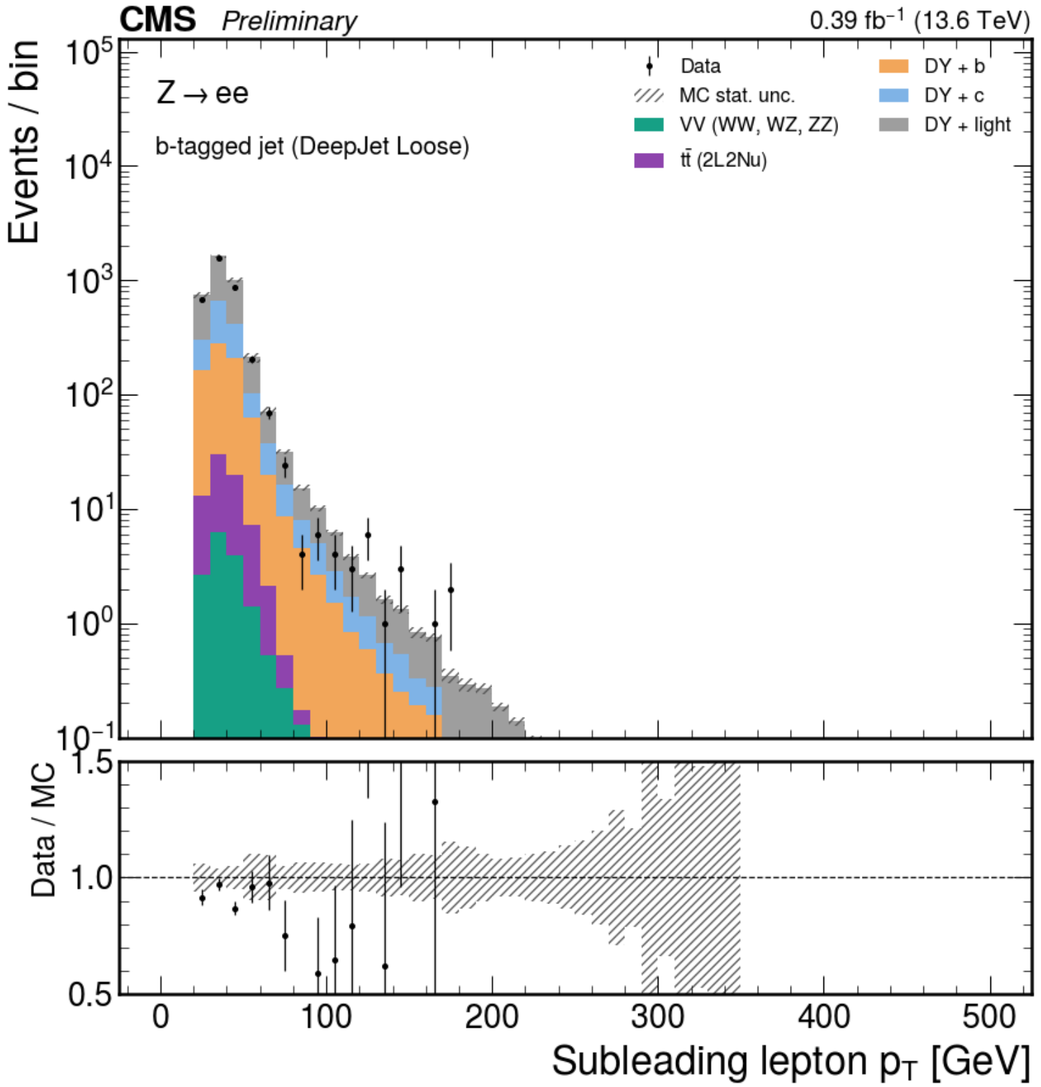
Expected to be flat; dips would point to detector problems.

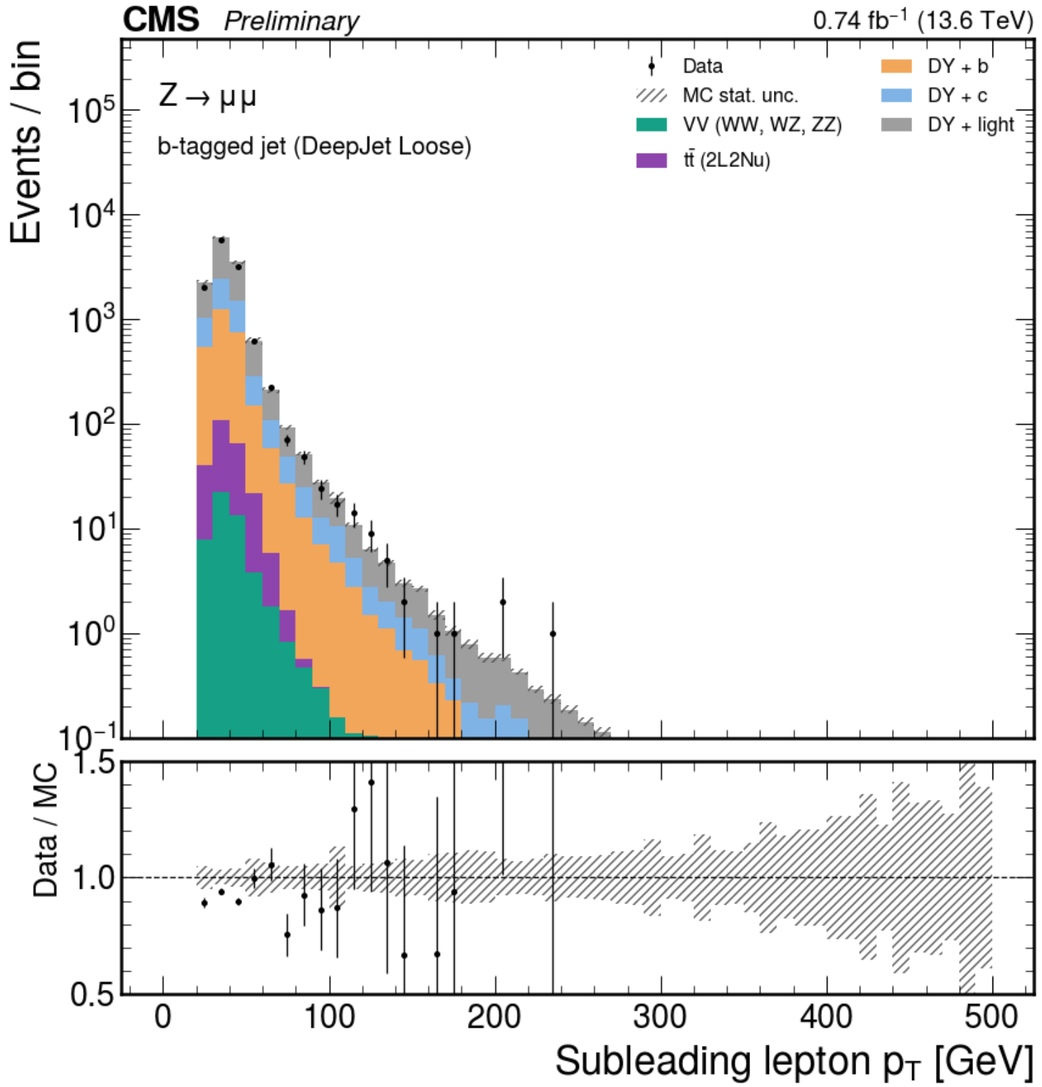


DeepJet · Loose b-tag (B > 0.0583)

Second-lepton pT

The selection requires pT > 25 GeV for electrons and > 26 GeV for muons.

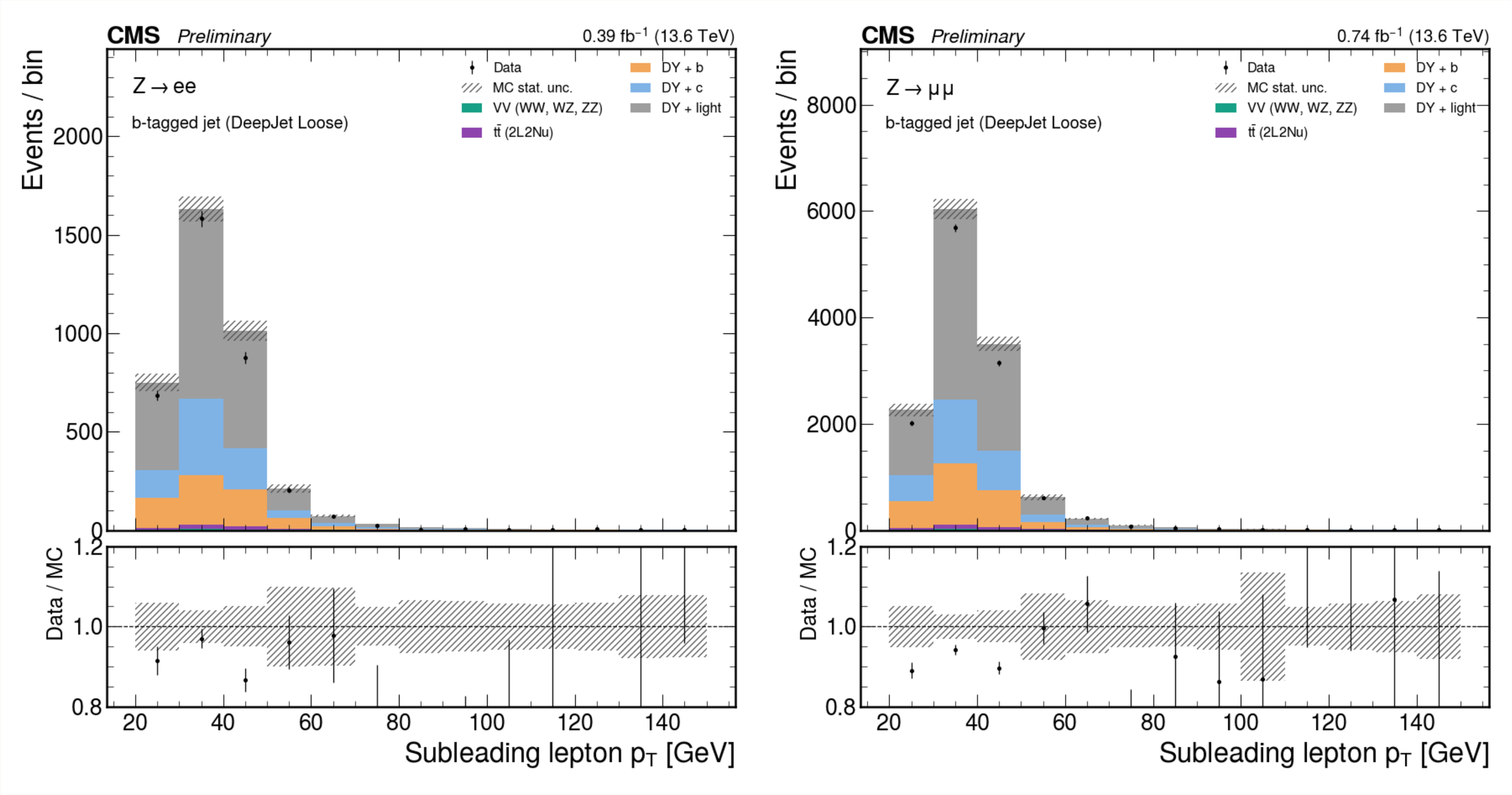




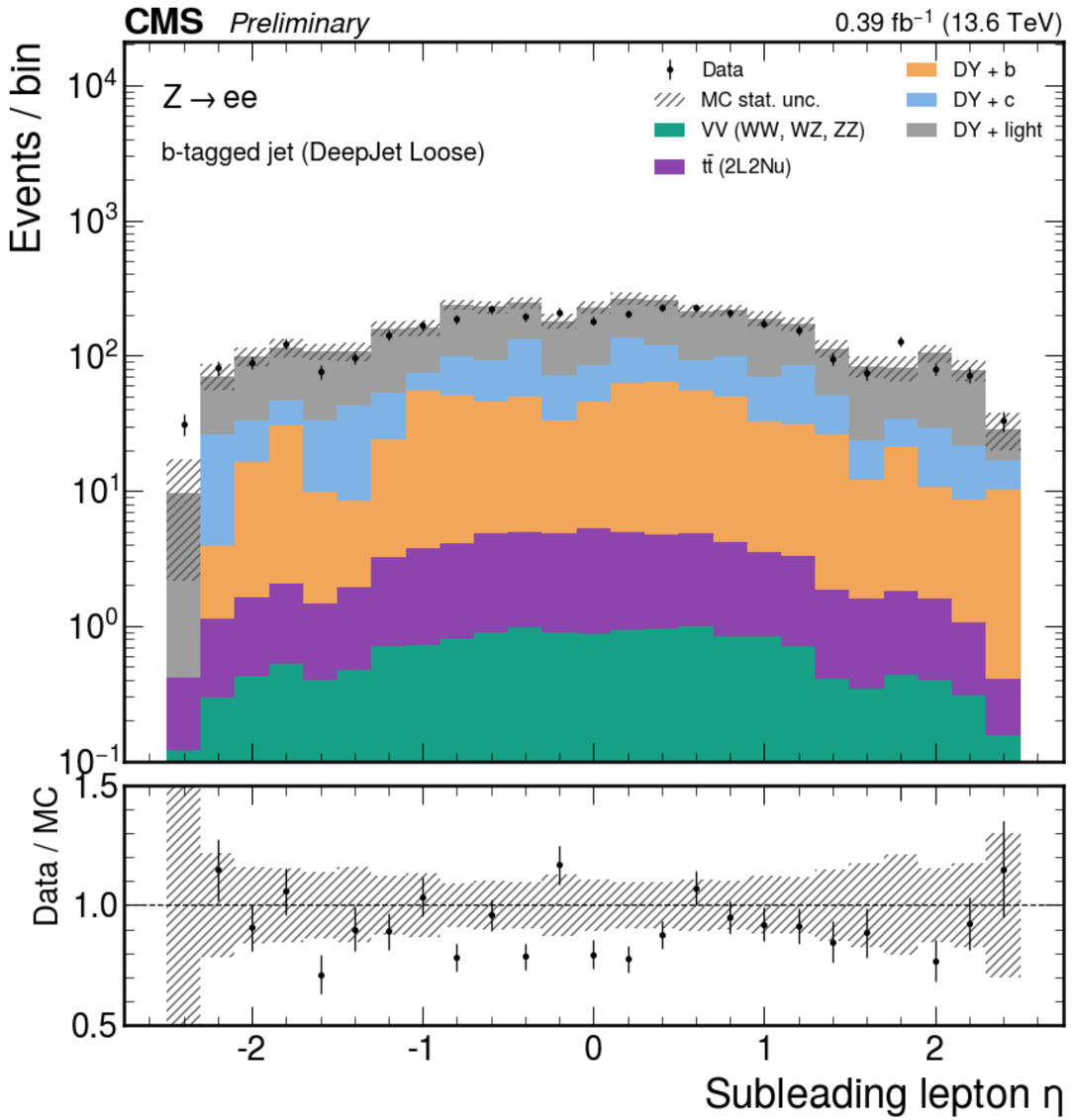
DeepJet · Loose b-tag (B > 0.0583)

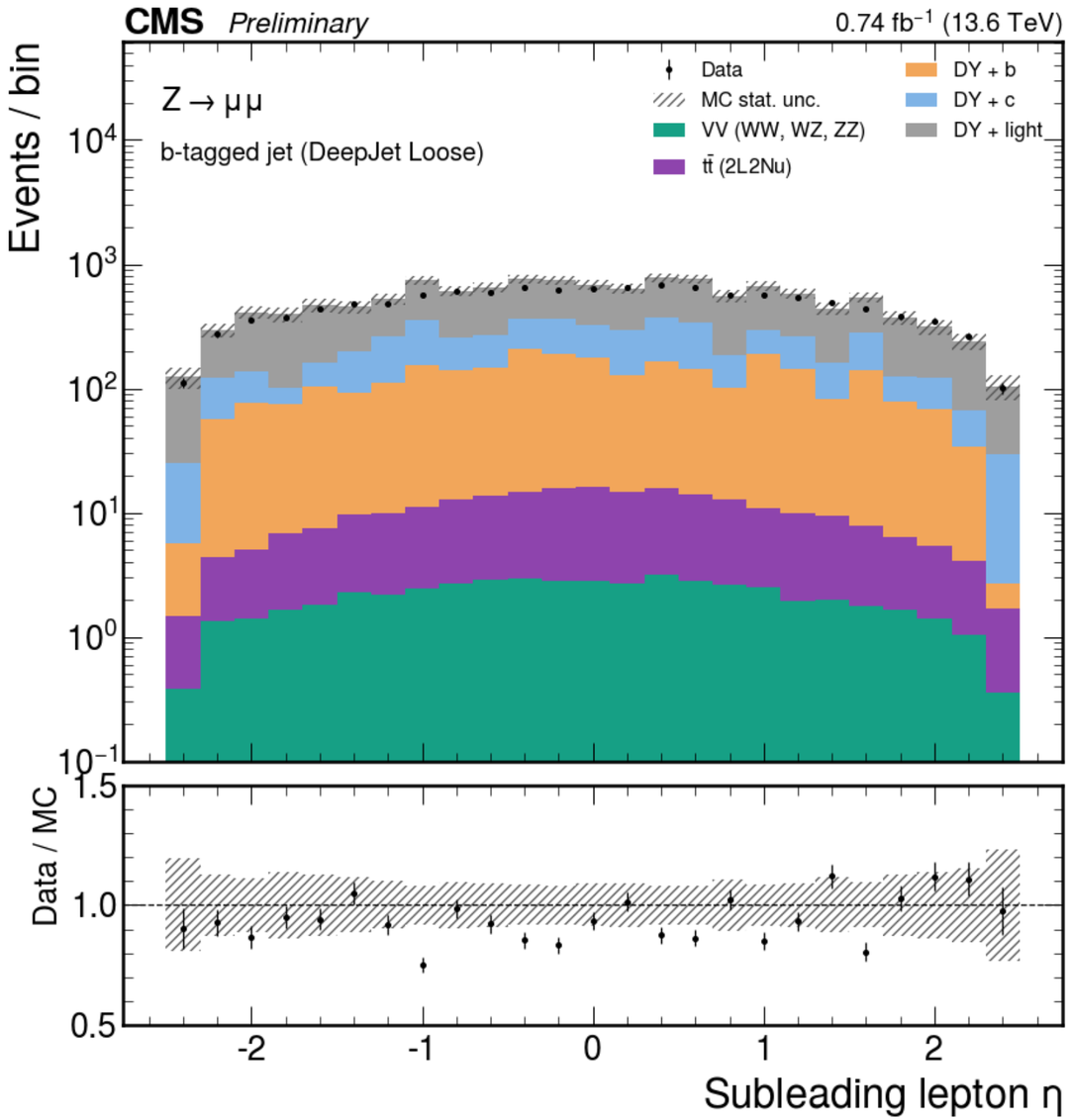
Second-lepton pT, zoom on 20–150 GeV

Zoom: 20–150, linear scale, data/MC from 0.8 to 1.2.

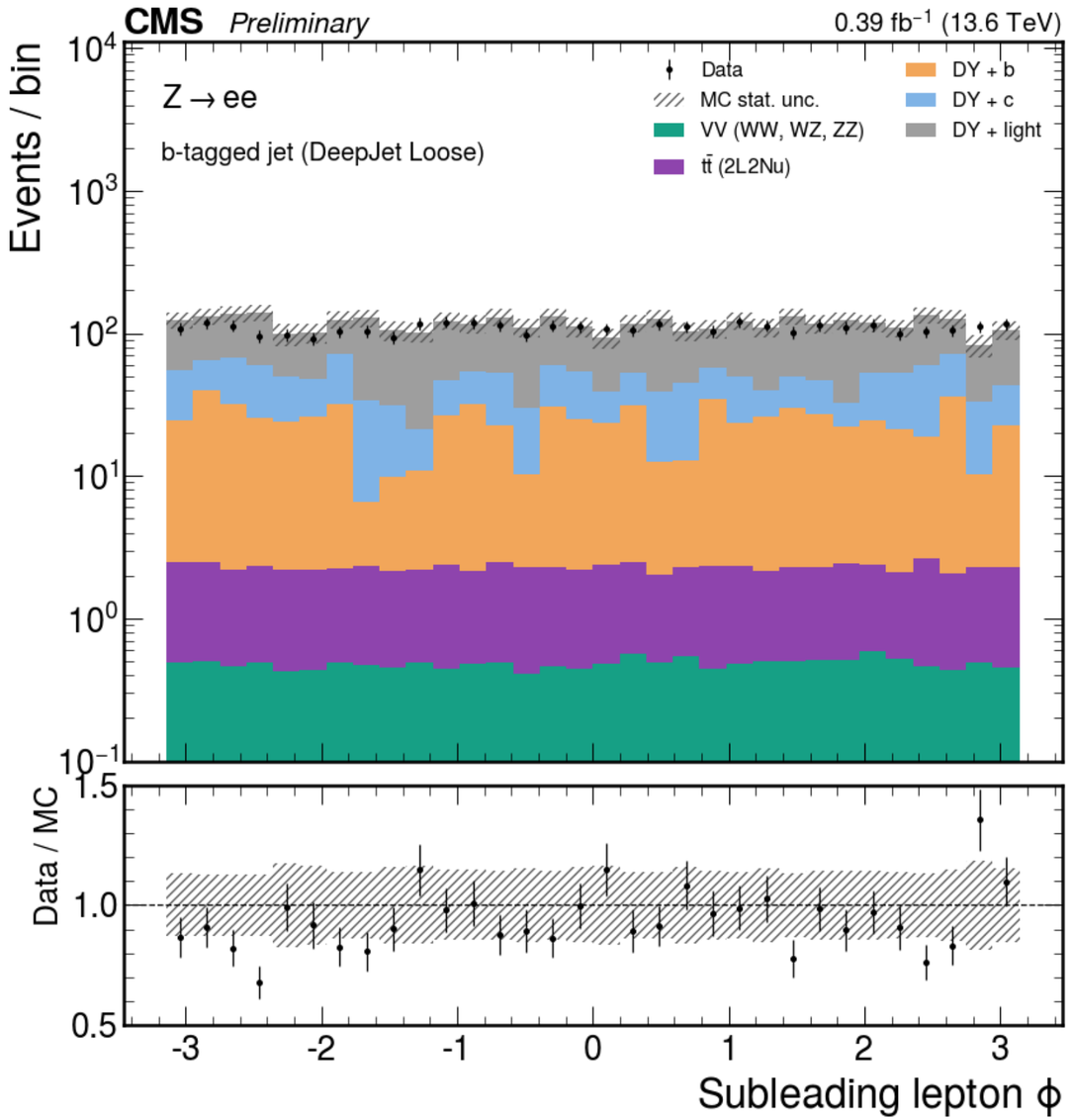


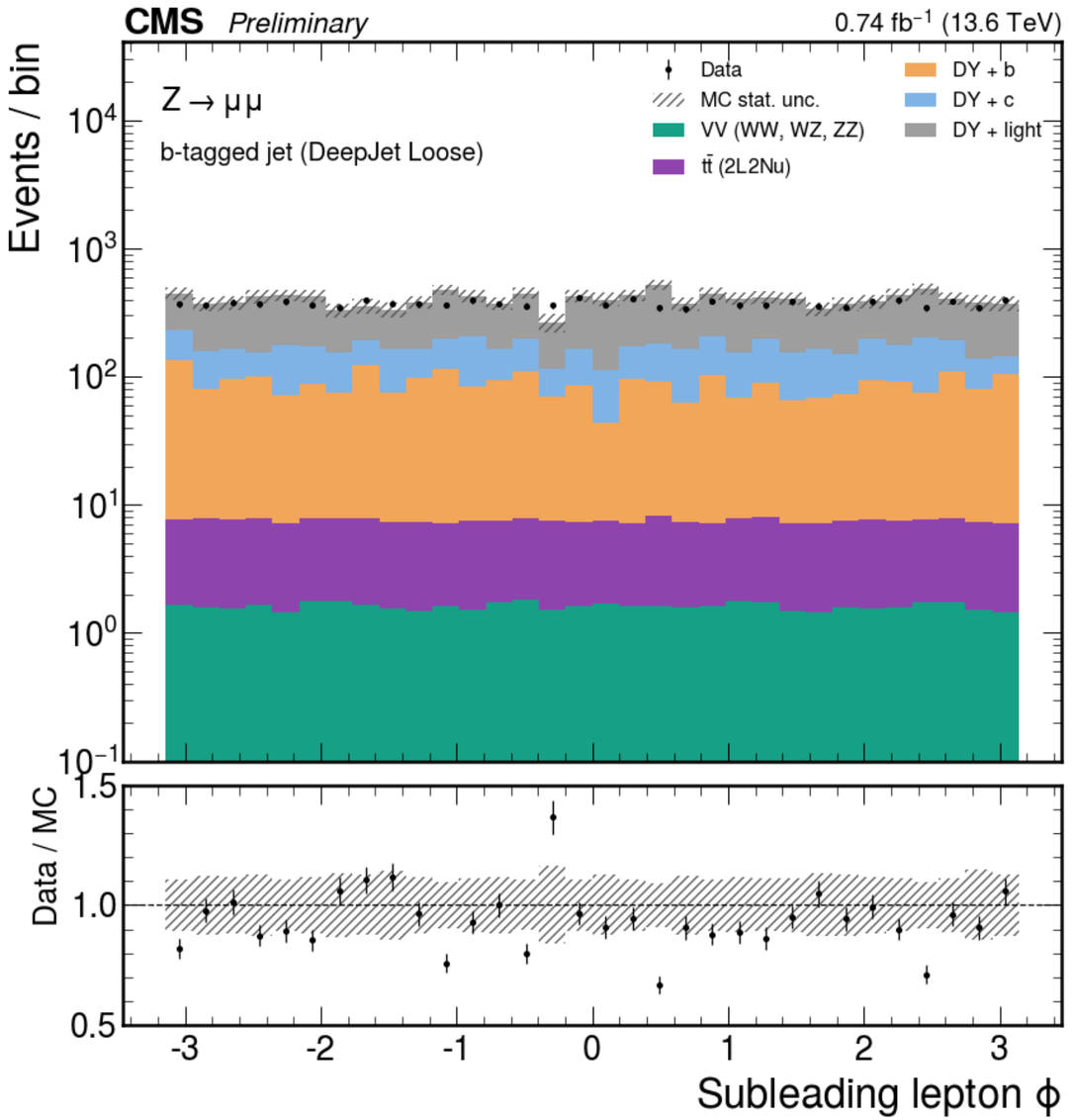
Second-lepton η





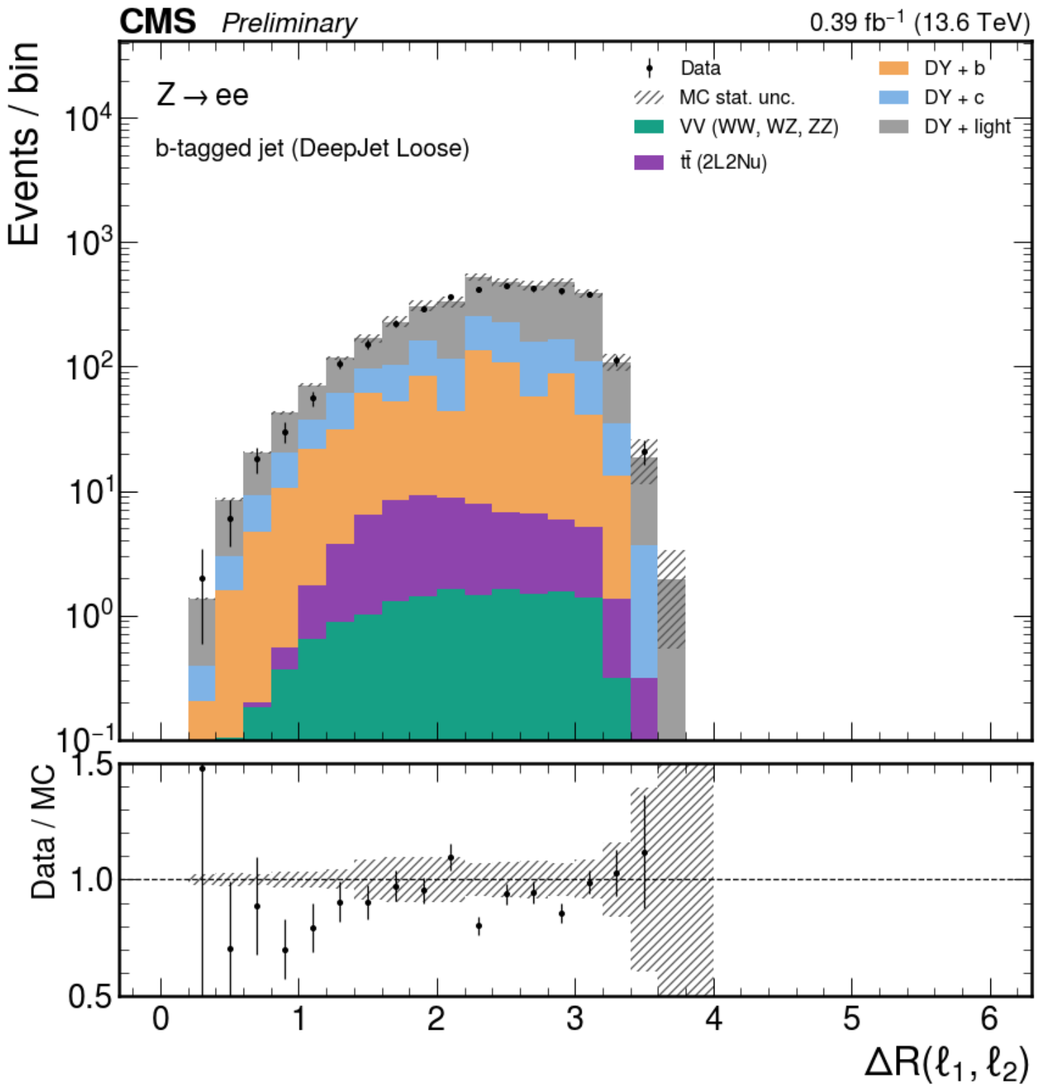
Second-lepton φ

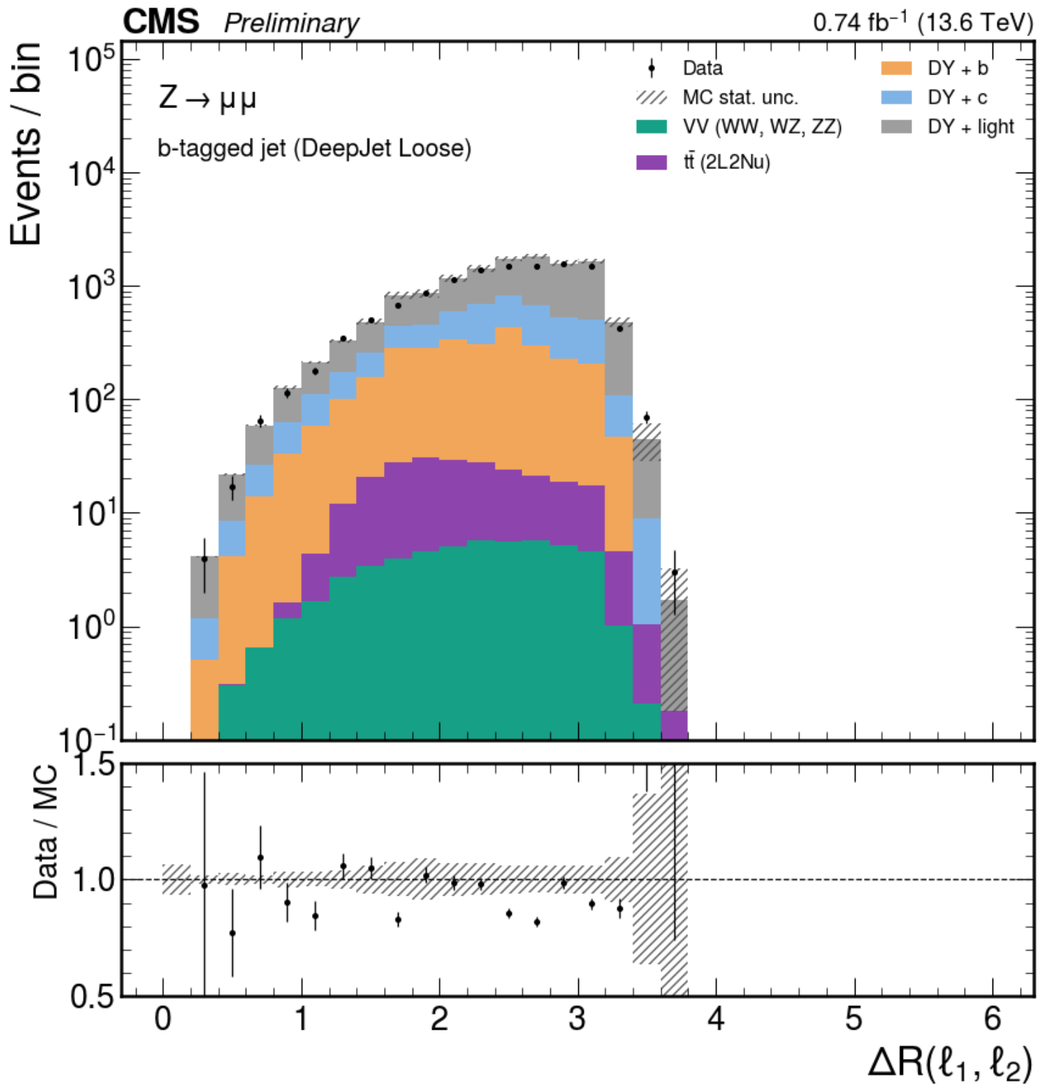




Angular distance between the two leptons, ΔR(ℓℓ)

ΔR = √(Δη² + Δφ²). Small ΔR = boosted Z.

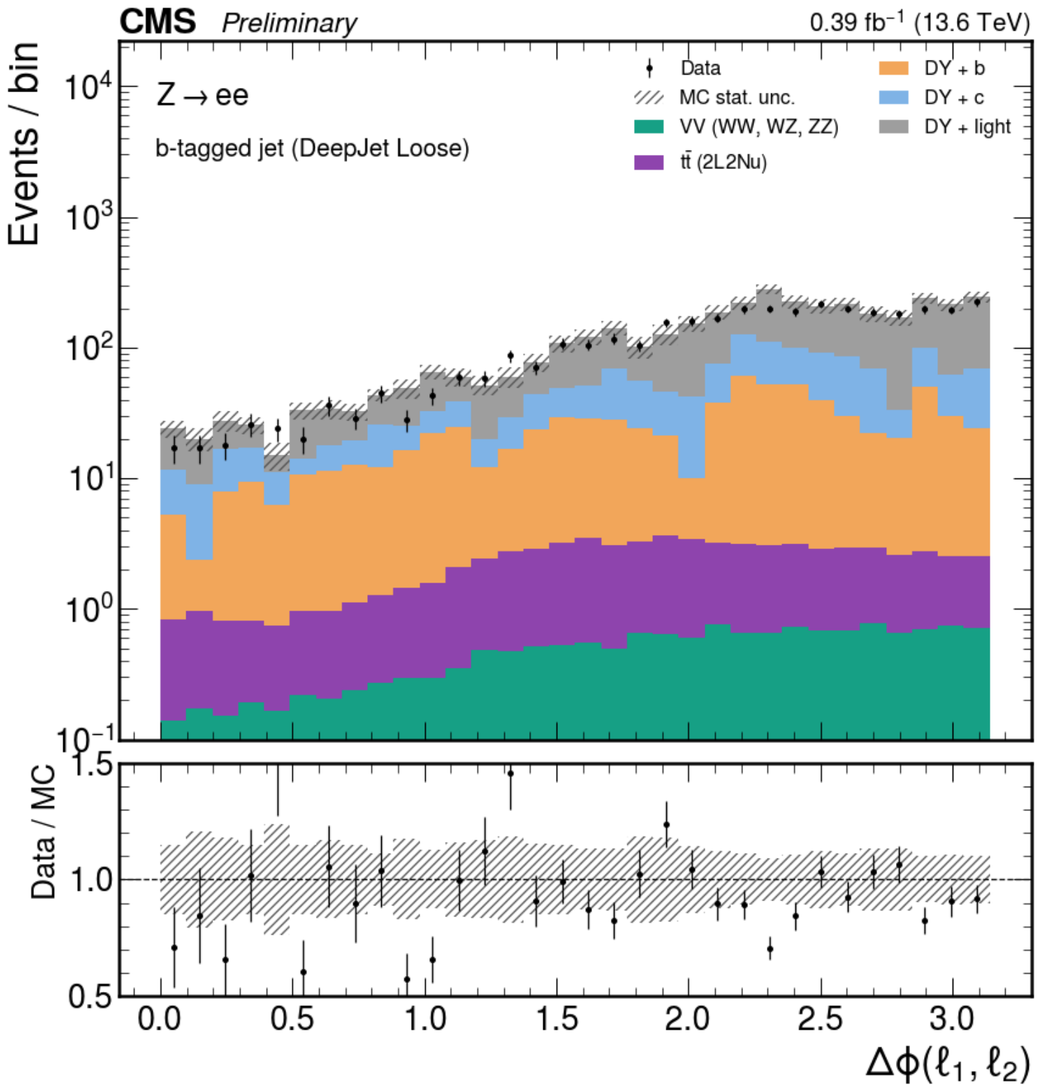


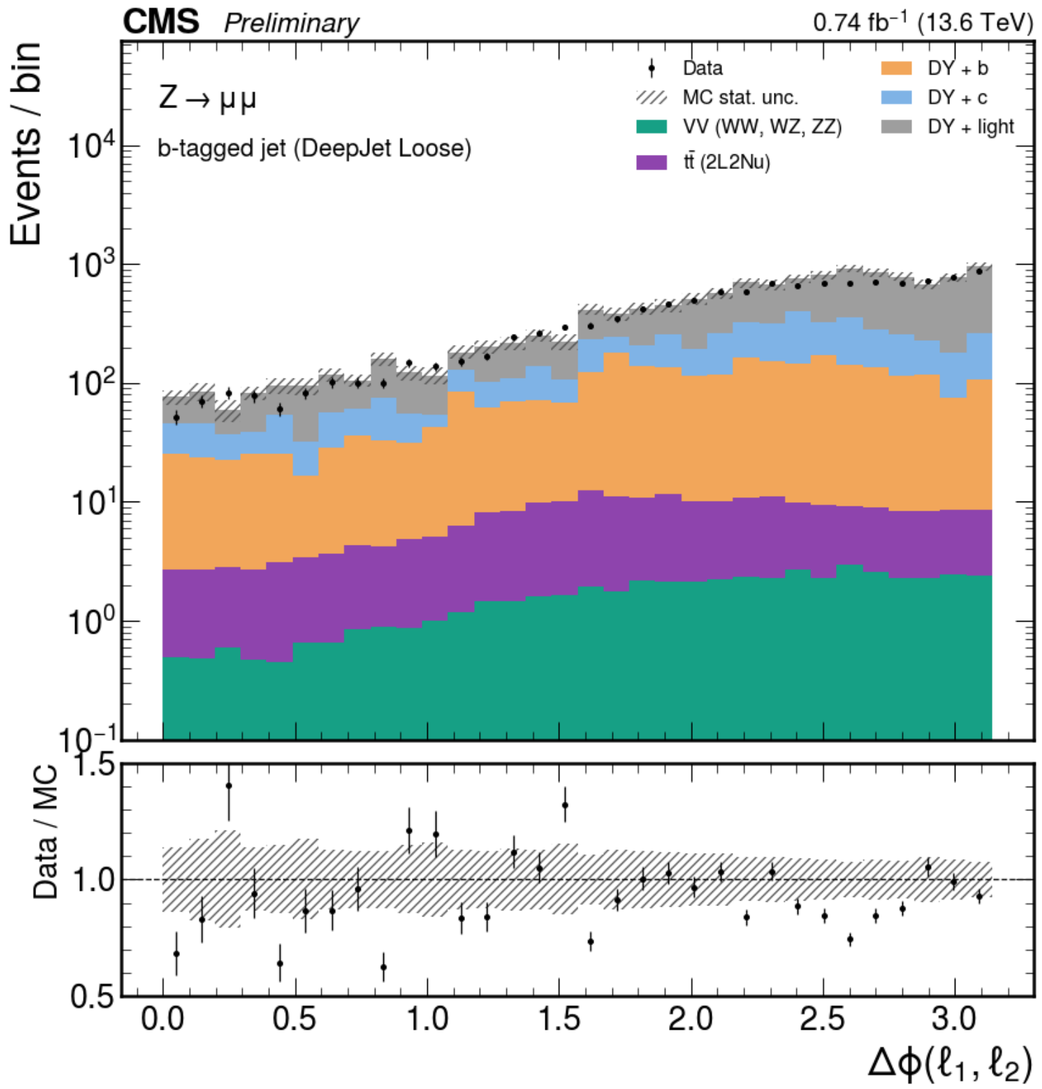


DeepJet · Loose b-tag (B > 0.0583)

Azimuthal angle between the two leptons, Δφ(ℓℓ)

Close to π (back to back) when the Z has low pT.

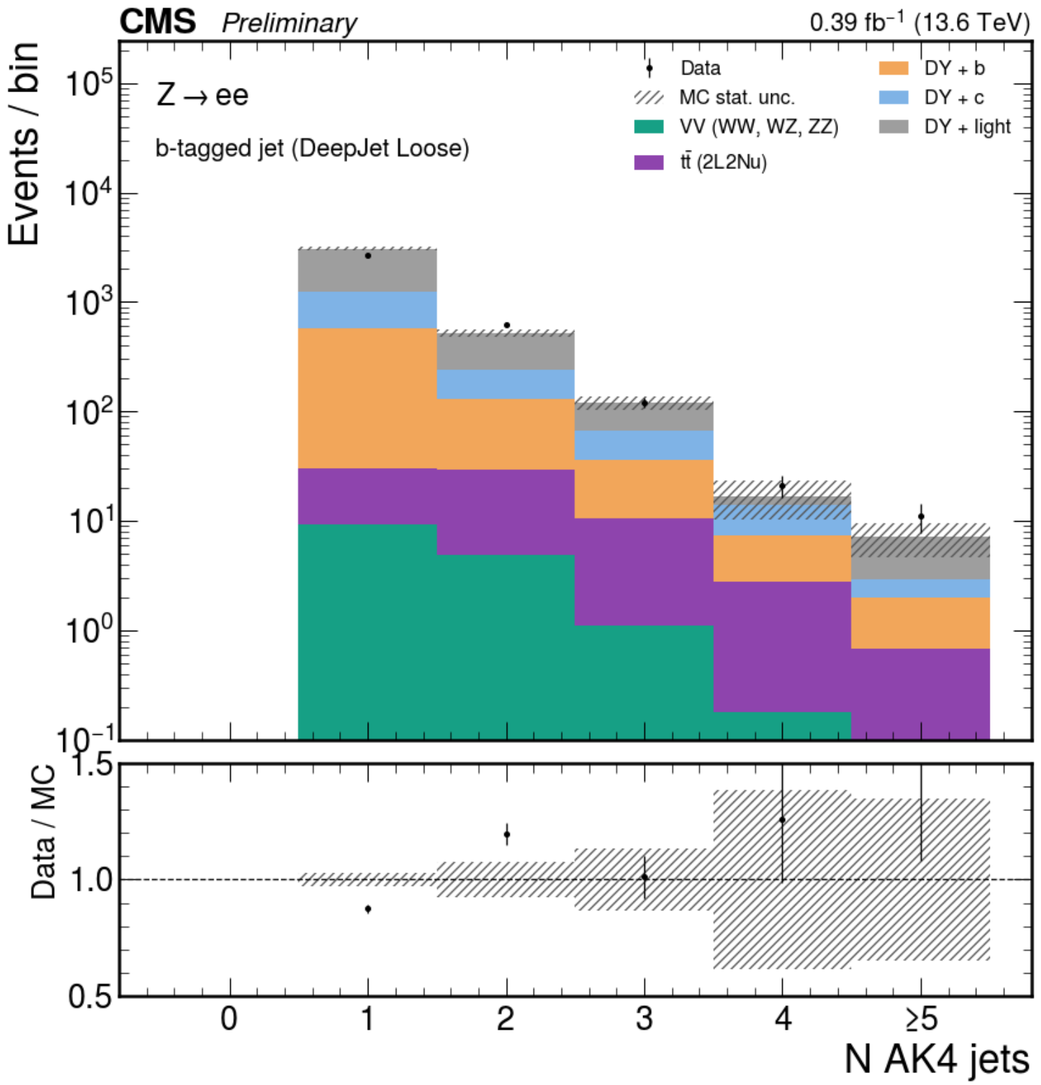


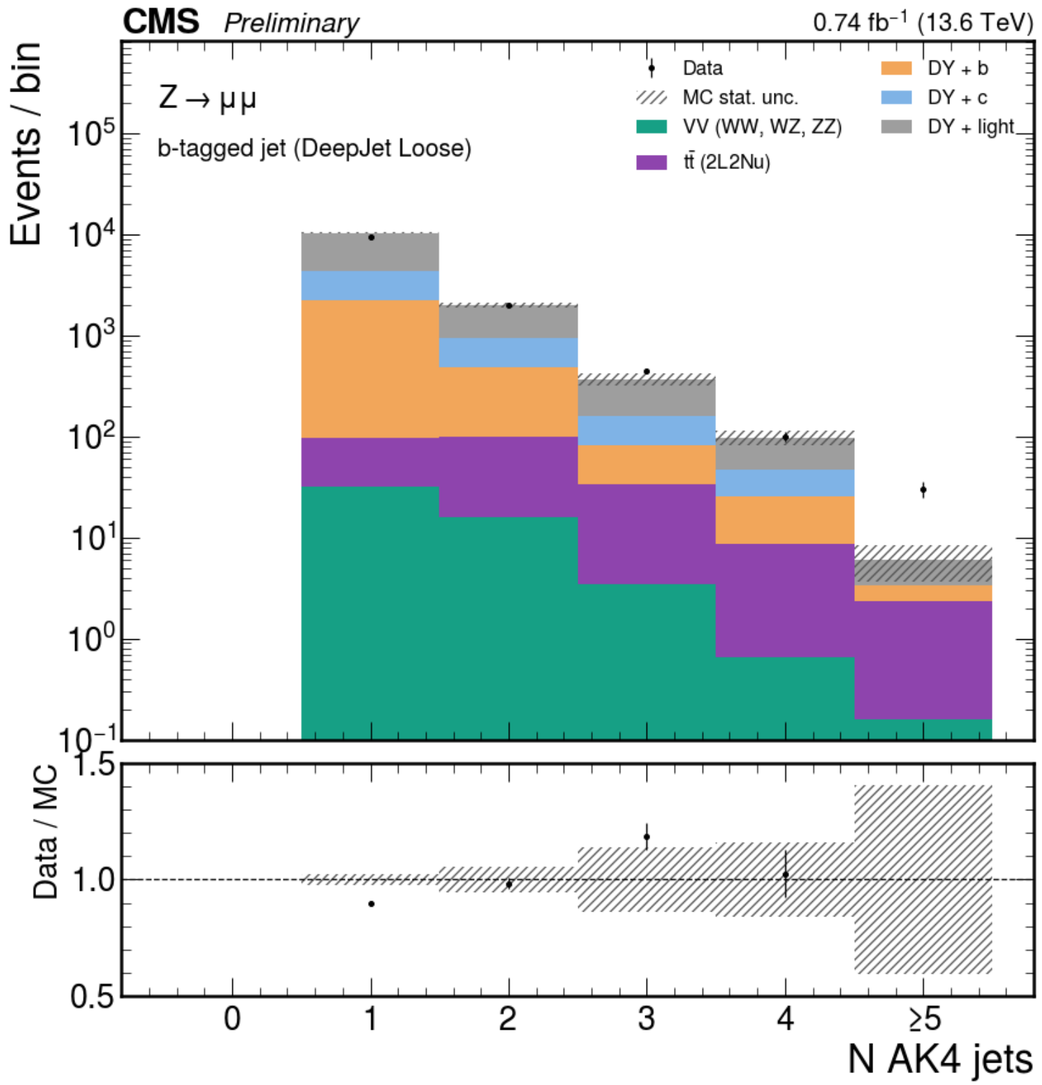


DeepJet · Loose b-tag (B > 0.0583)

Number of selected jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. The leading jet is b-tagged here, so every event has at least one jet.

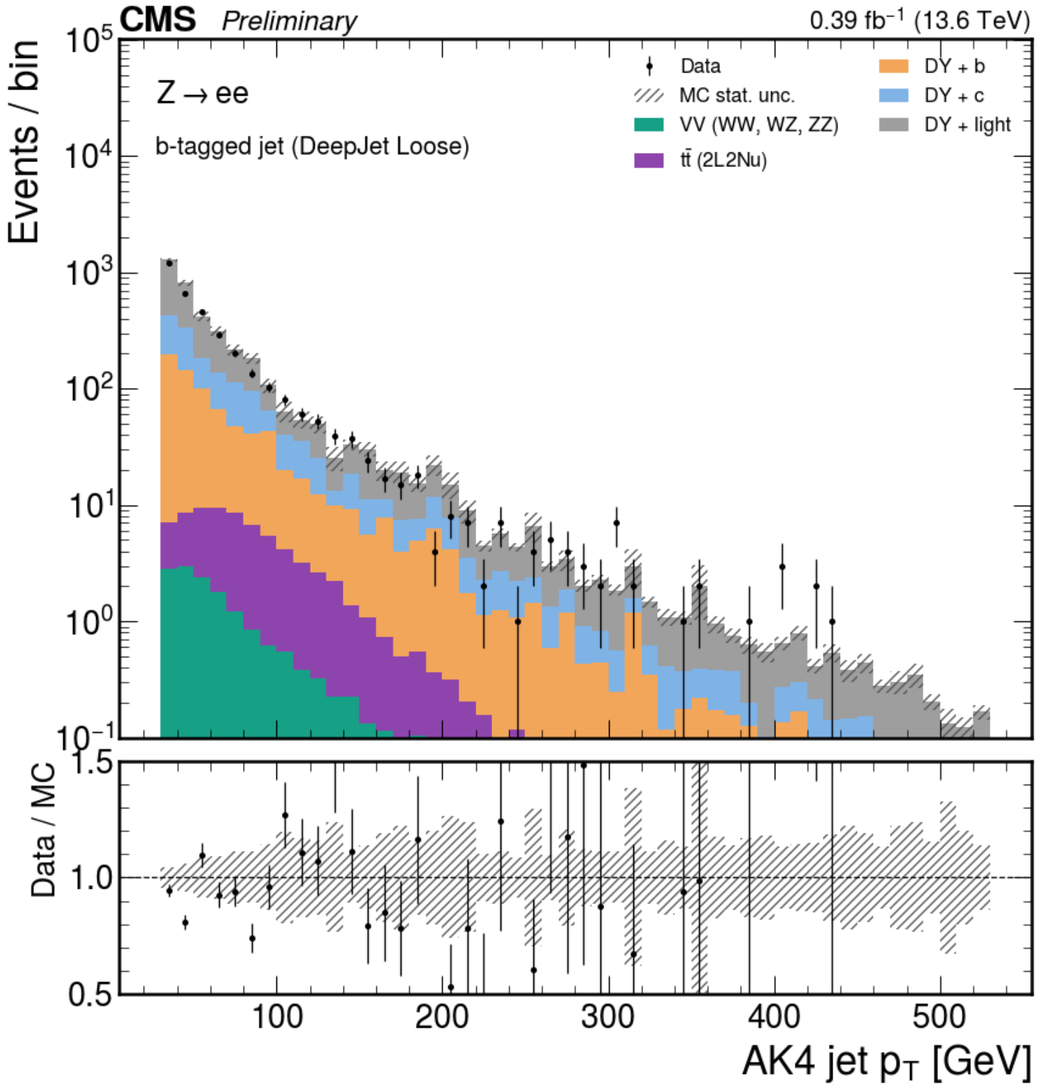


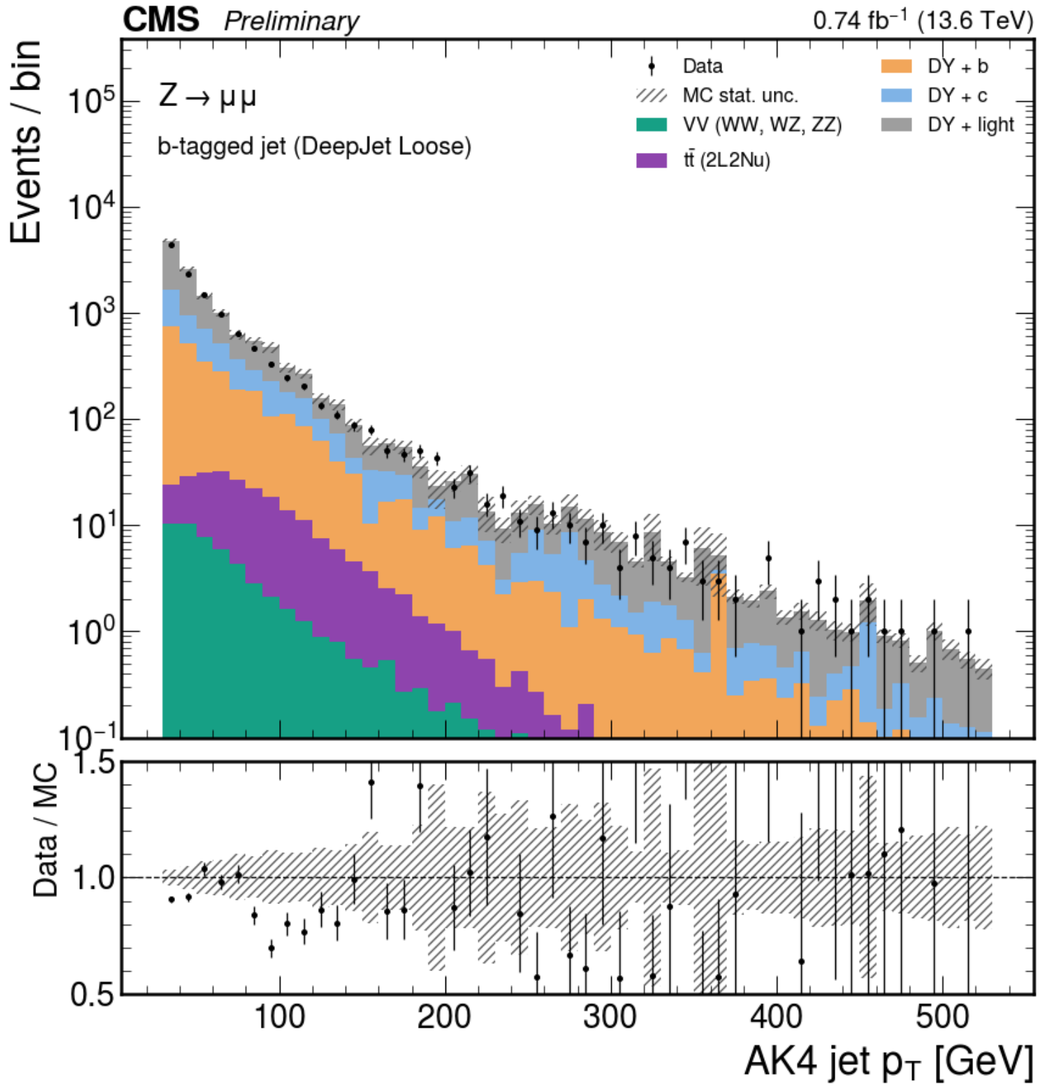


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet pT (this is the b-tagged jet)

pT > 30 GeV.

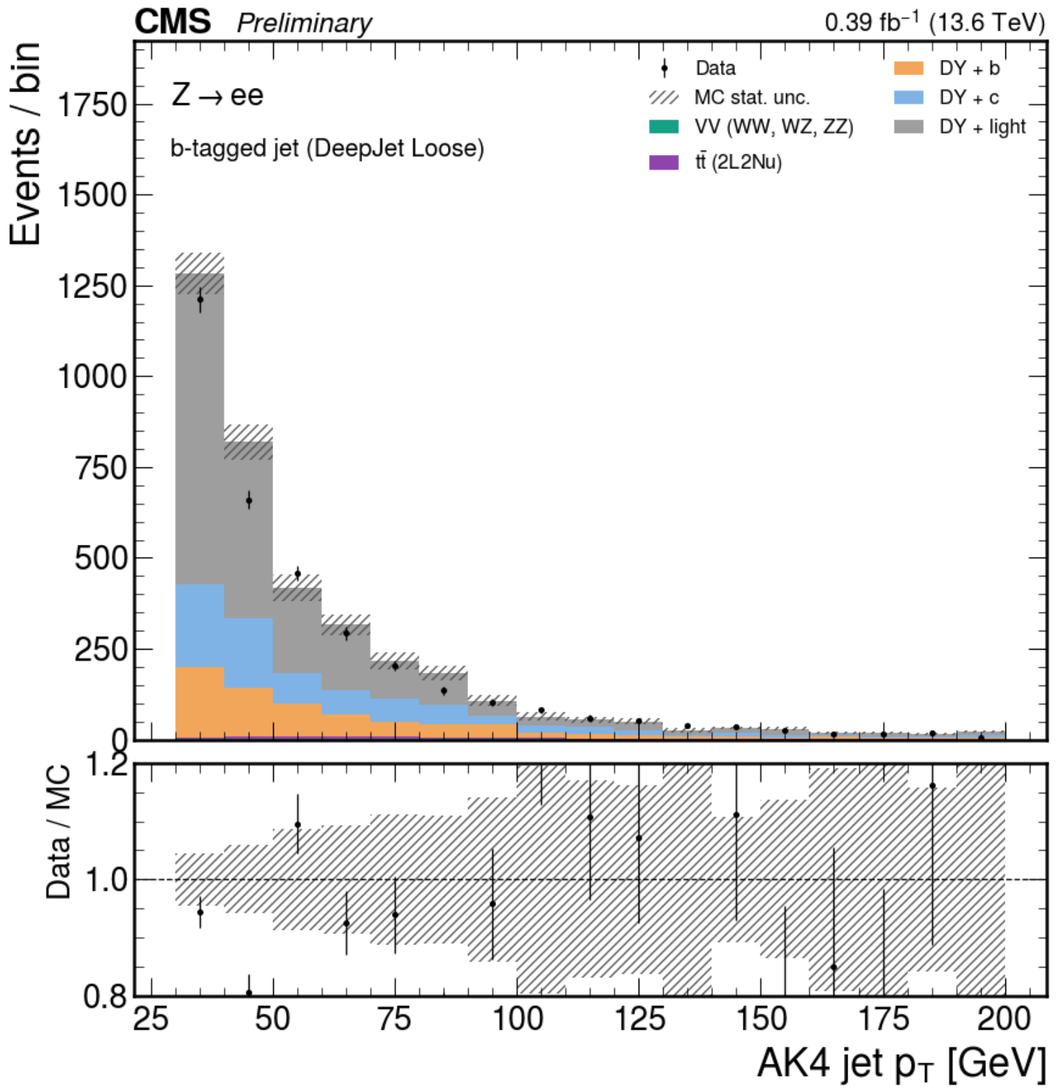


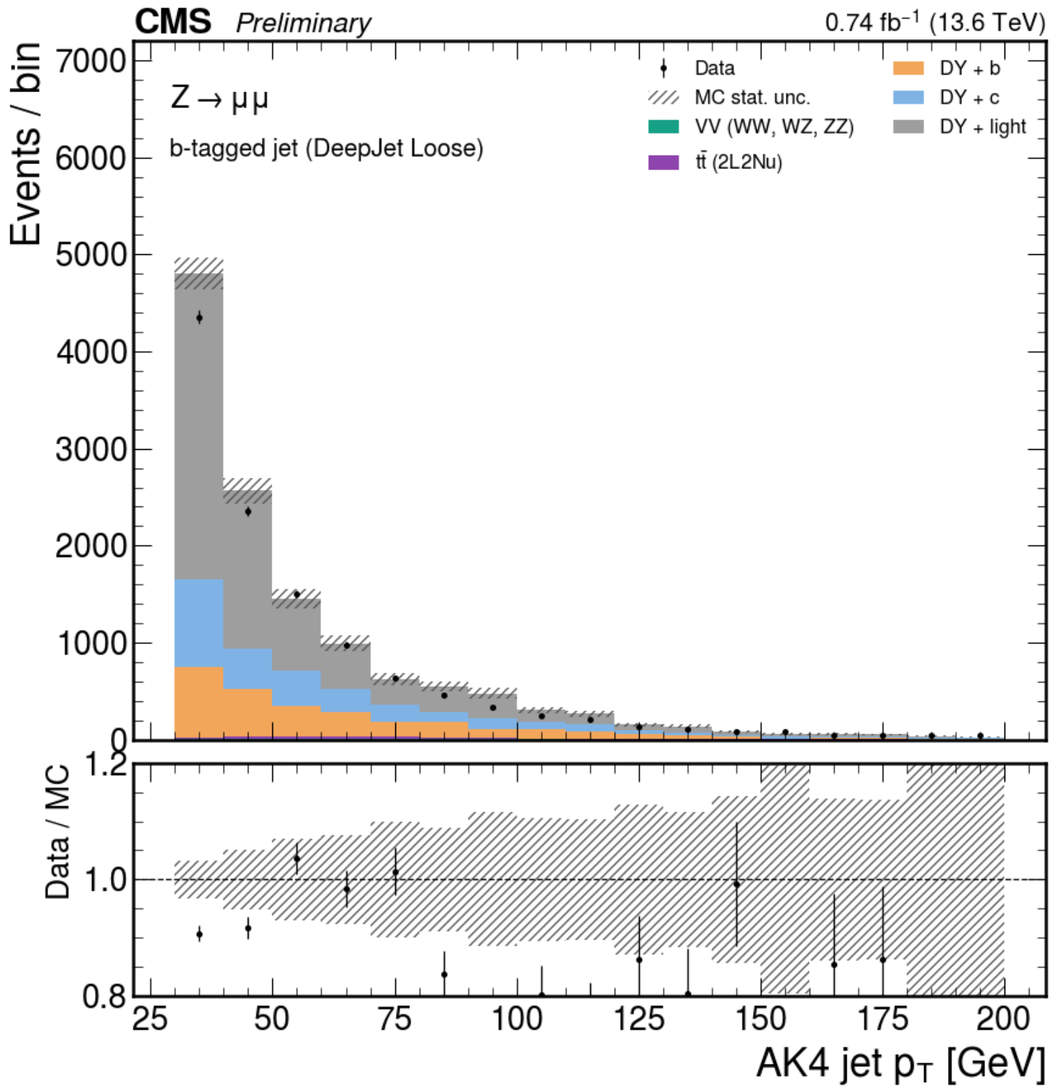


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet pT, zoom on 30–200 GeV

Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.

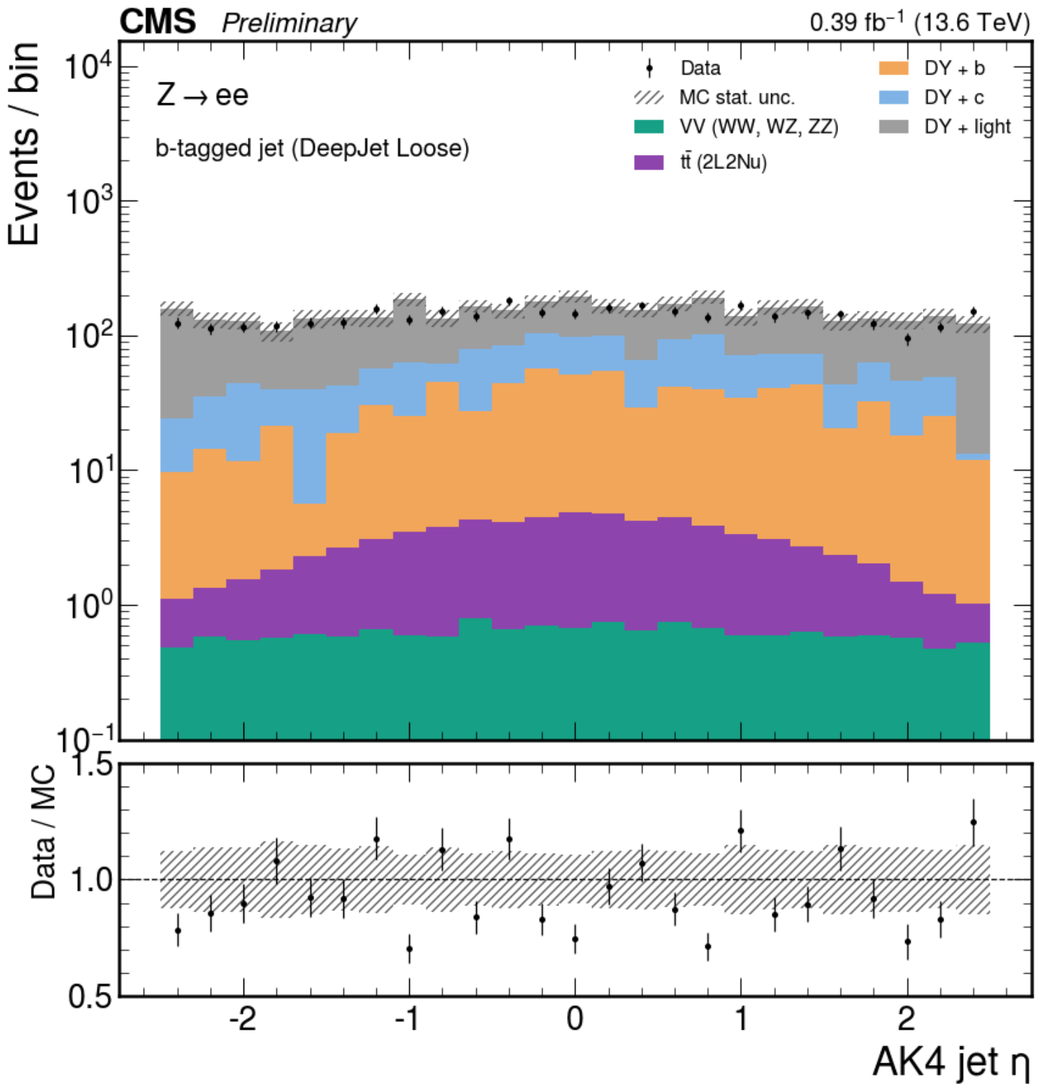


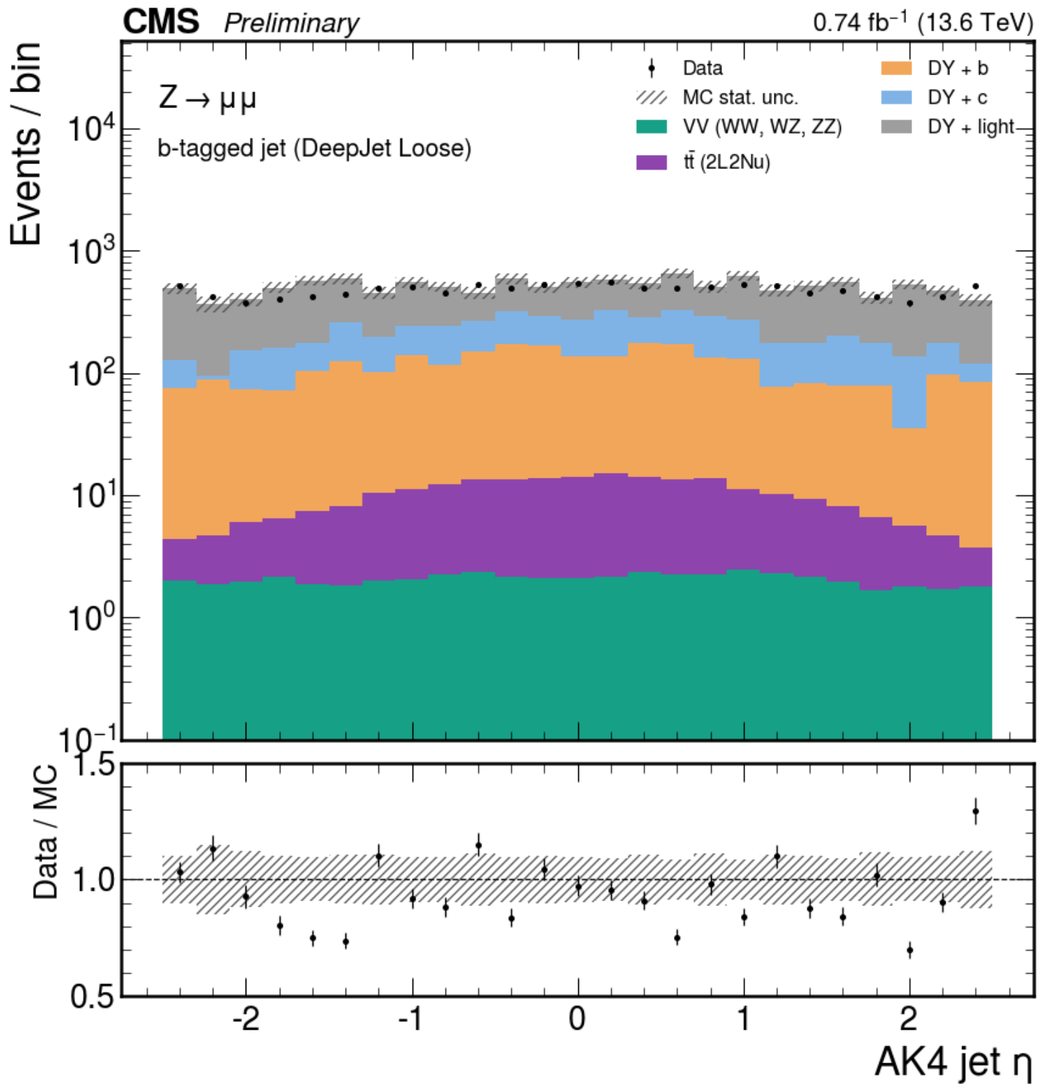


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet η

|η| < 2.5 (tracker coverage, needed for b-tagging).

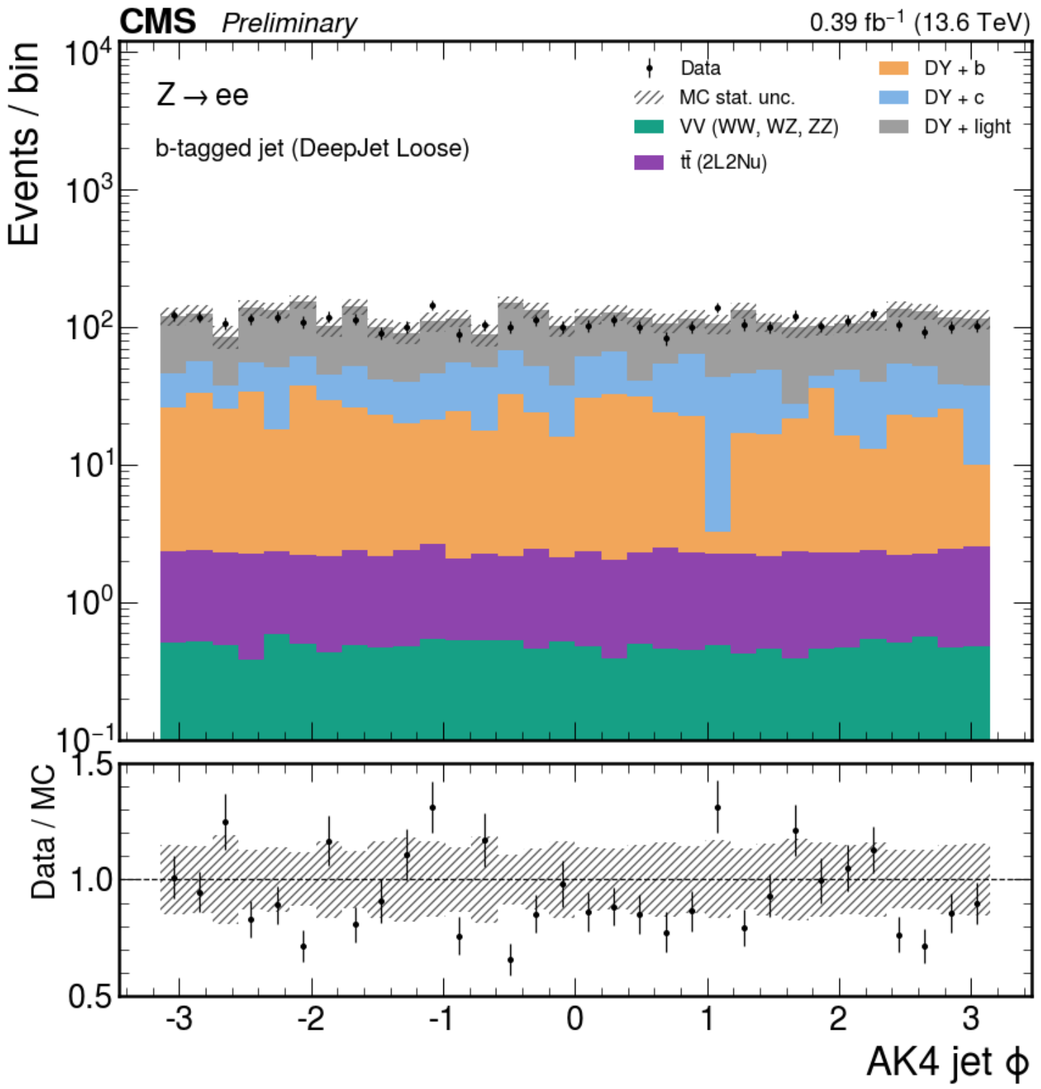


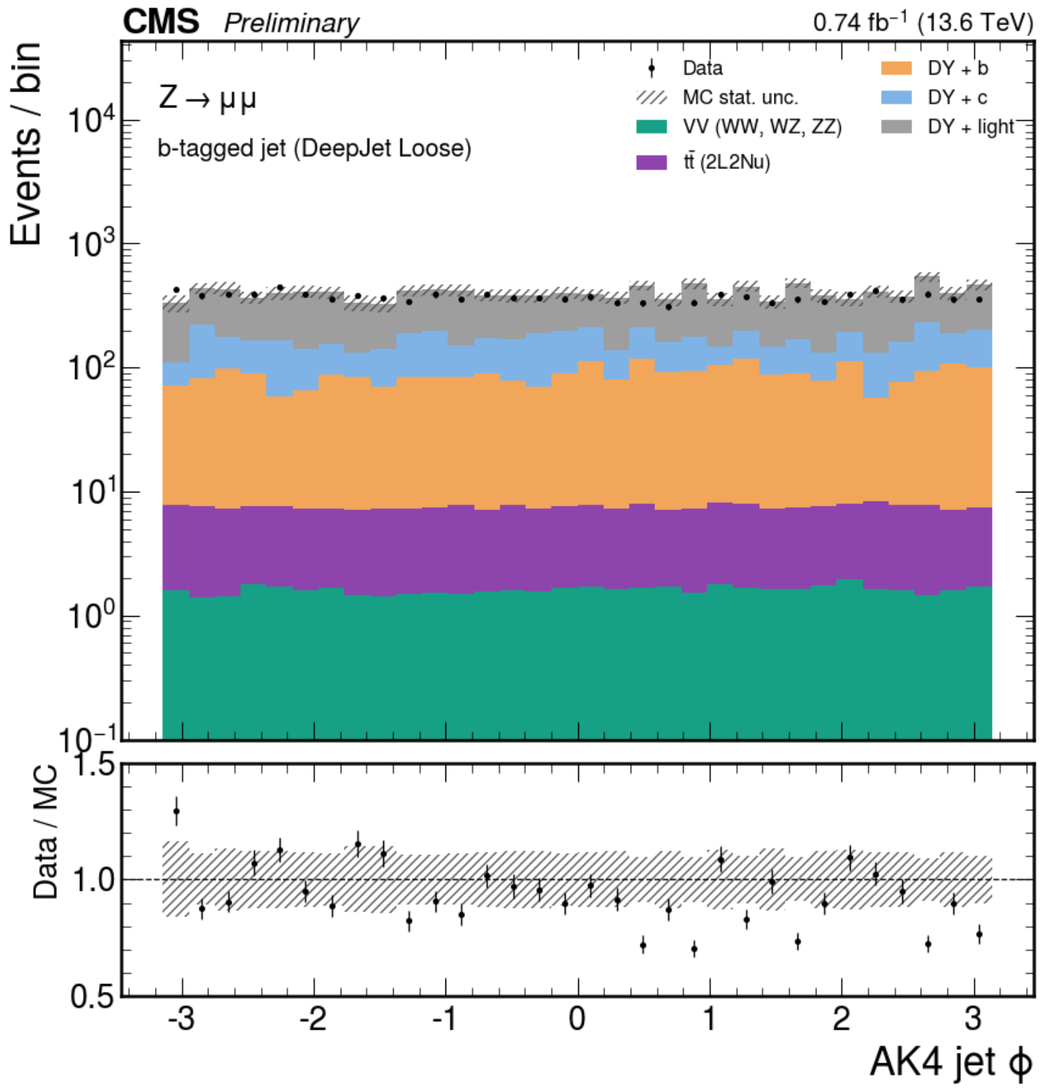


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet φ

Expected to be flat.

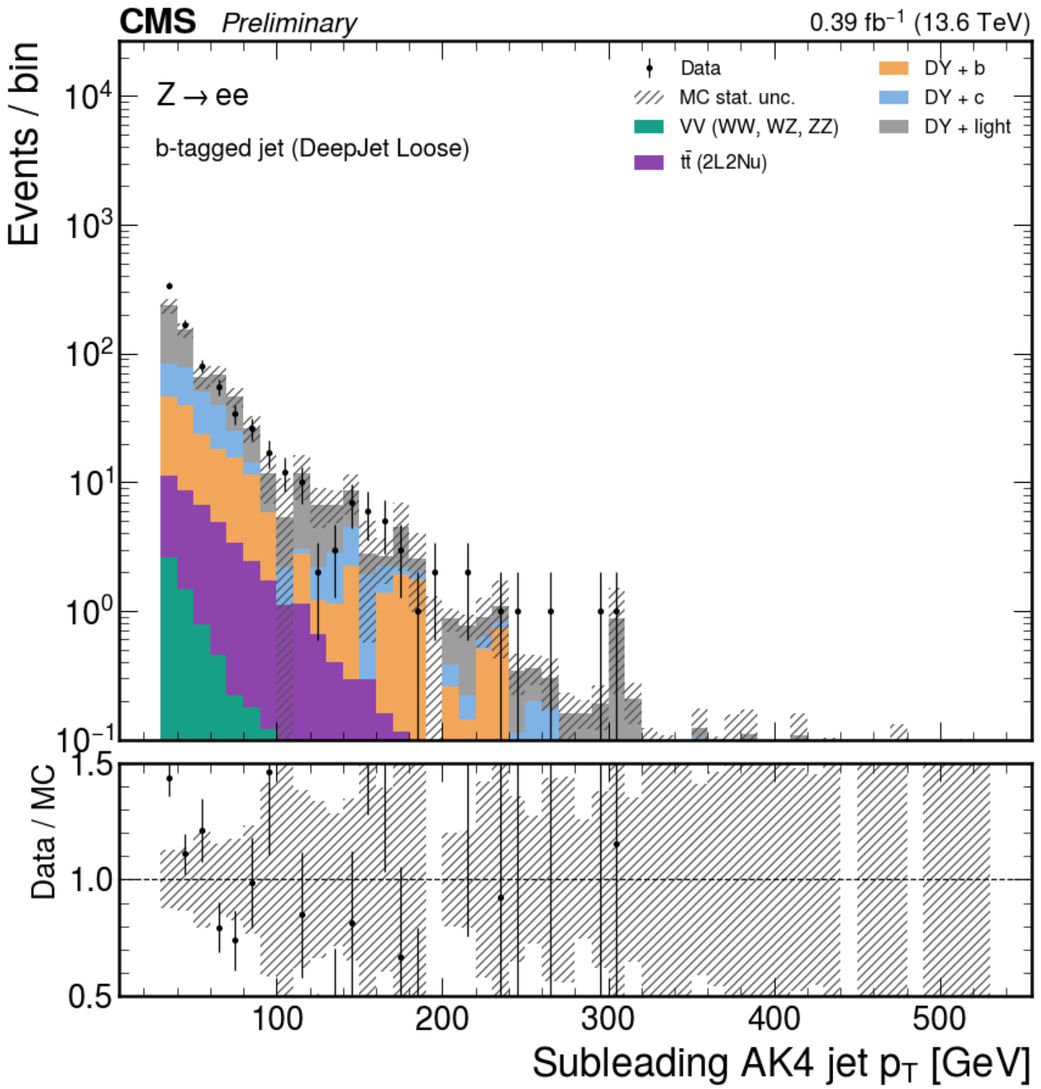


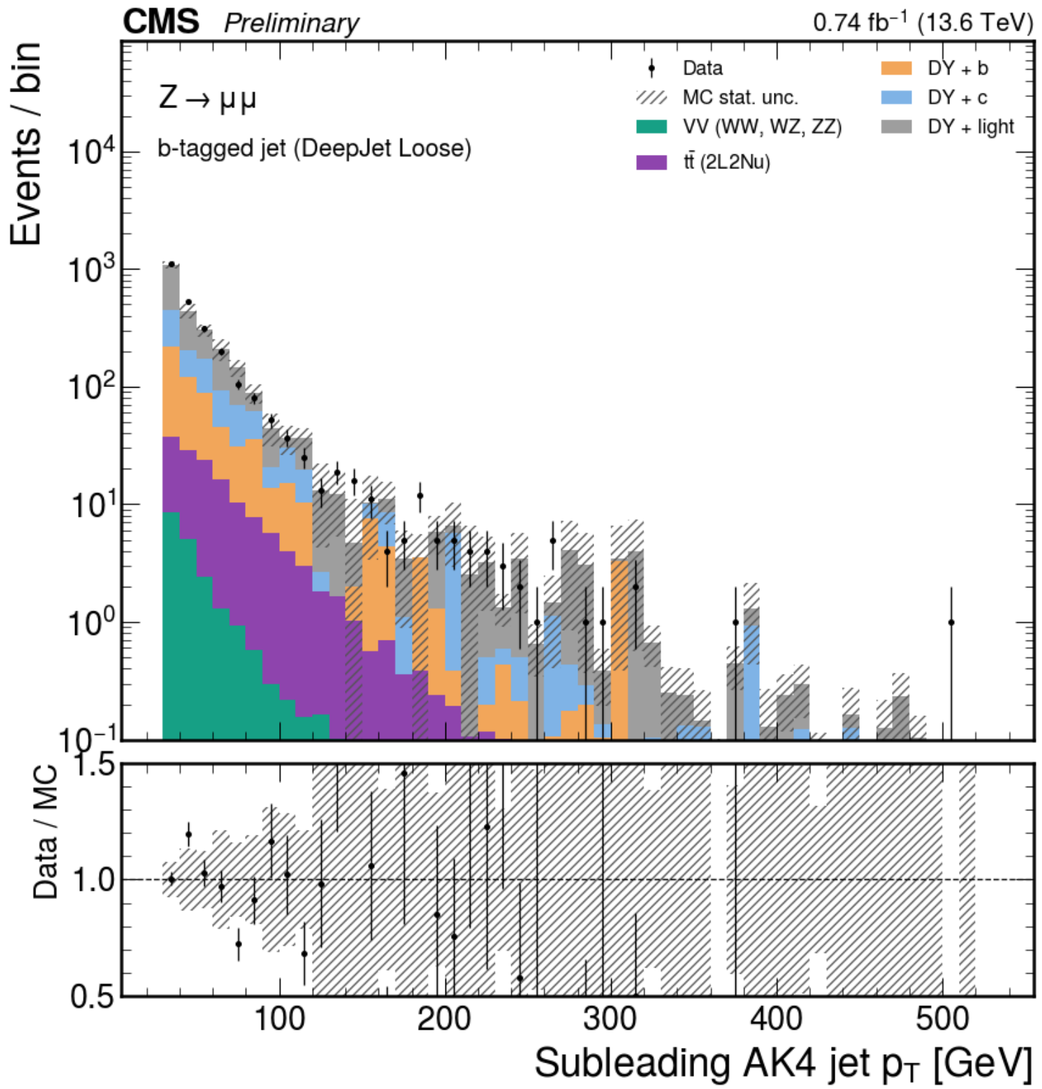


DeepJet · Loose b-tag (B > 0.0583)

Second-jet pT

Only events with at least two jets. The second jet is not tagged.

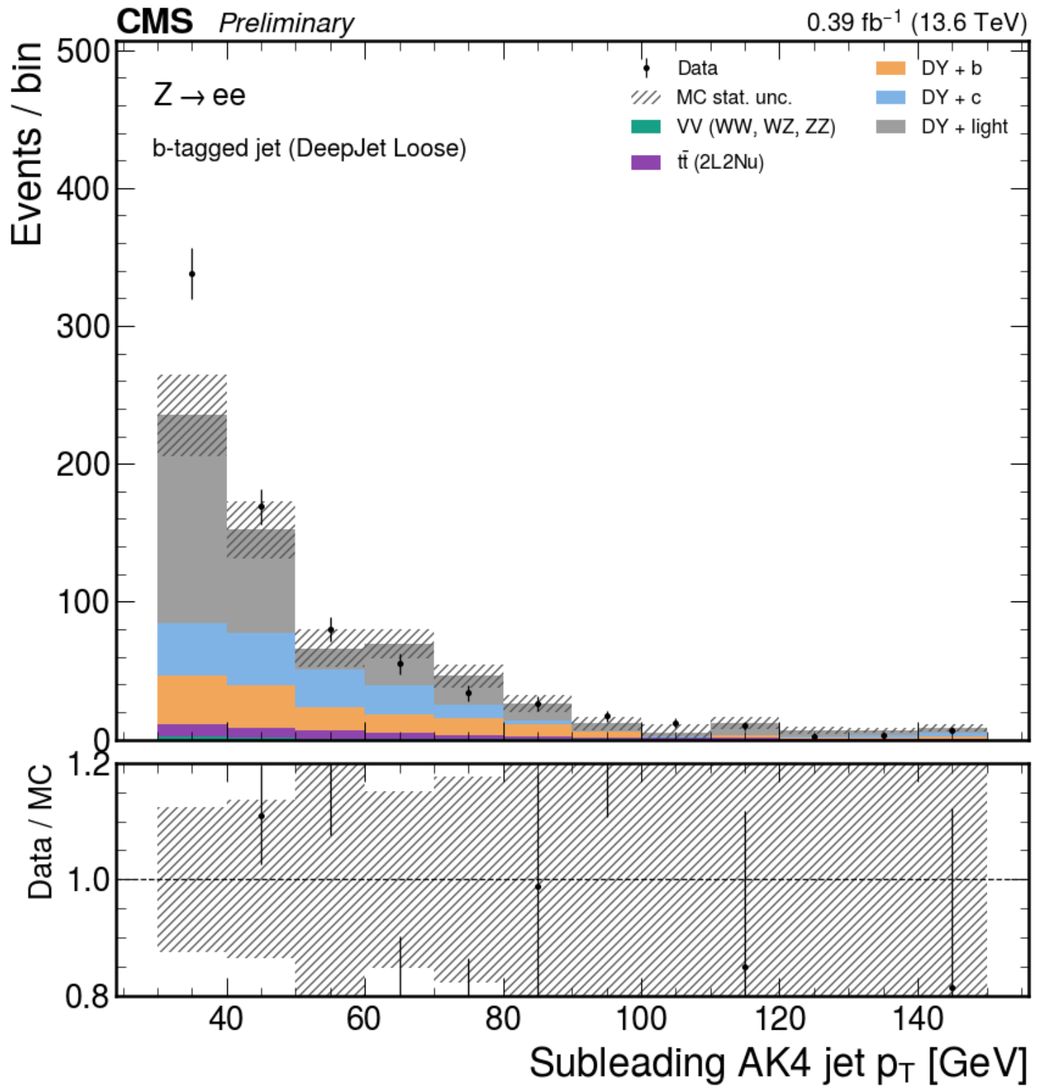


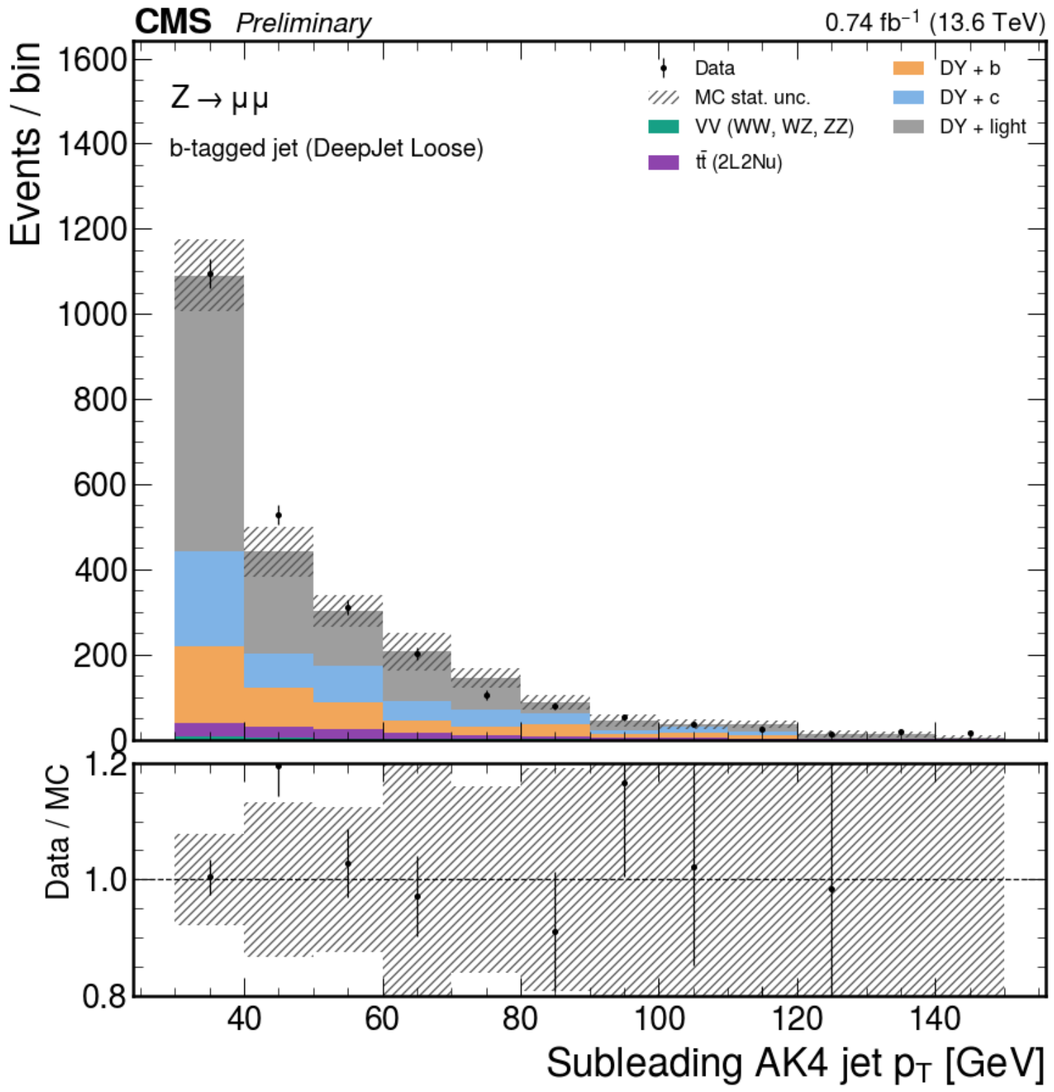


DeepJet · Loose b-tag (B > 0.0583)

Second-jet pT, zoom on 30–150 GeV

Zoom: 30–150, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.

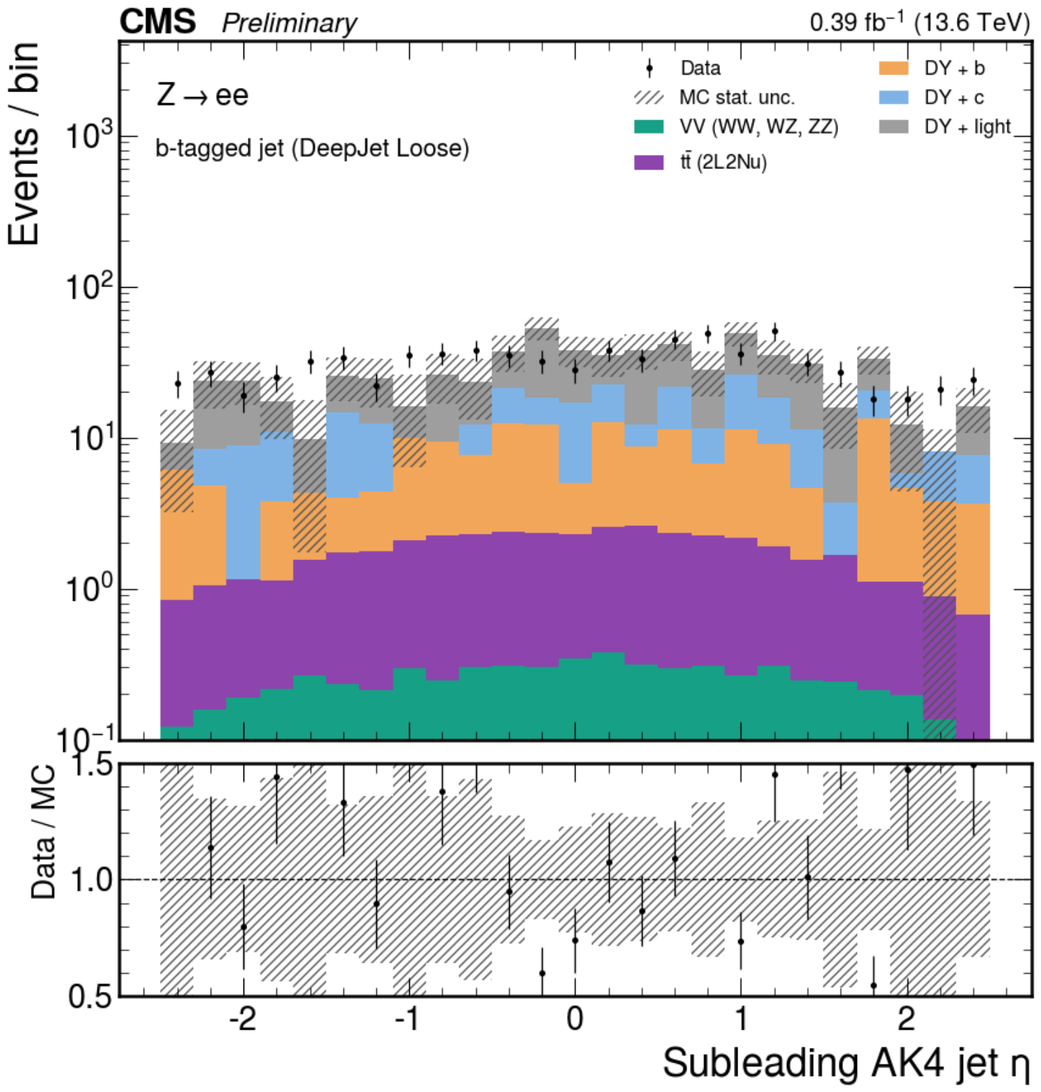


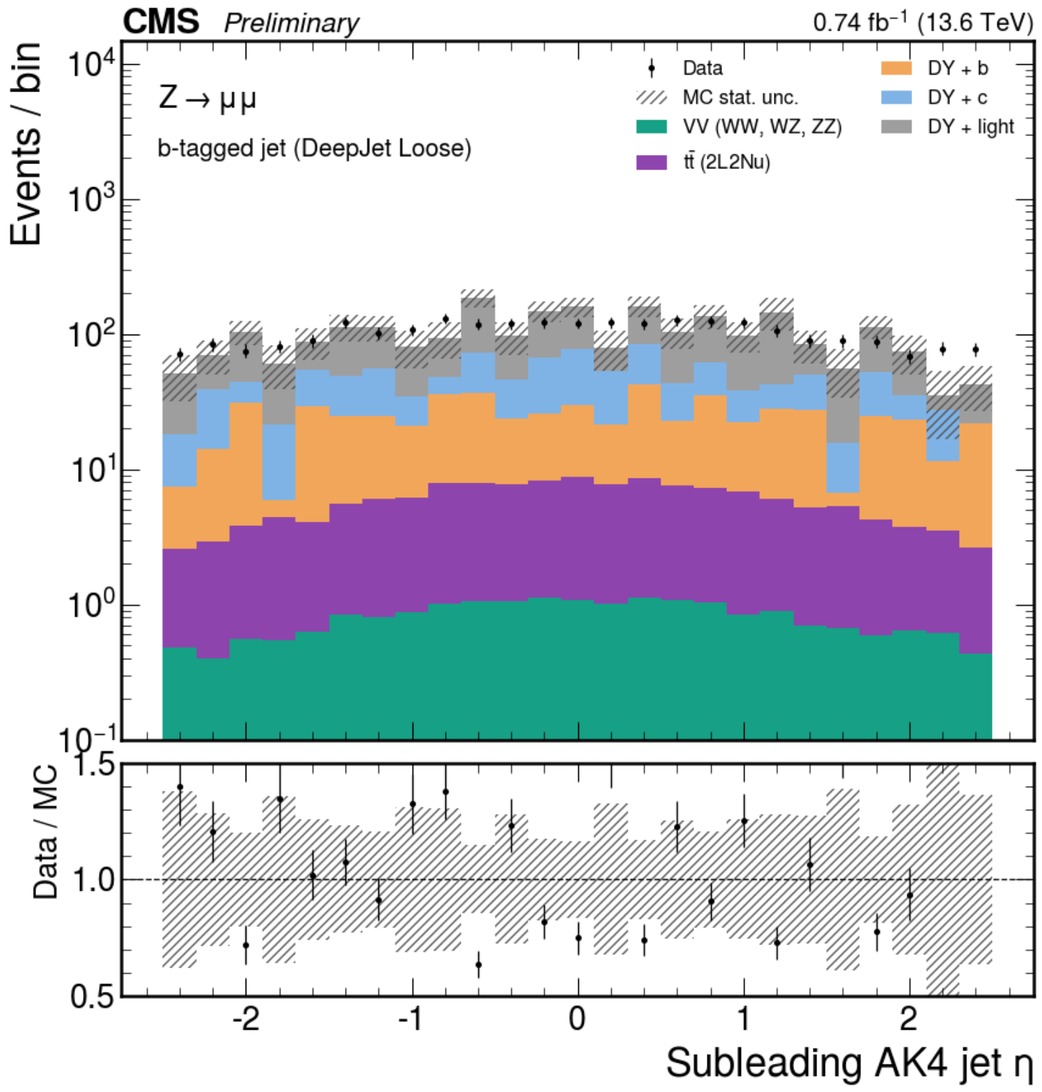


DeepJet · Loose b-tag (B > 0.0583)

Second-jet η

Only events with at least two jets.

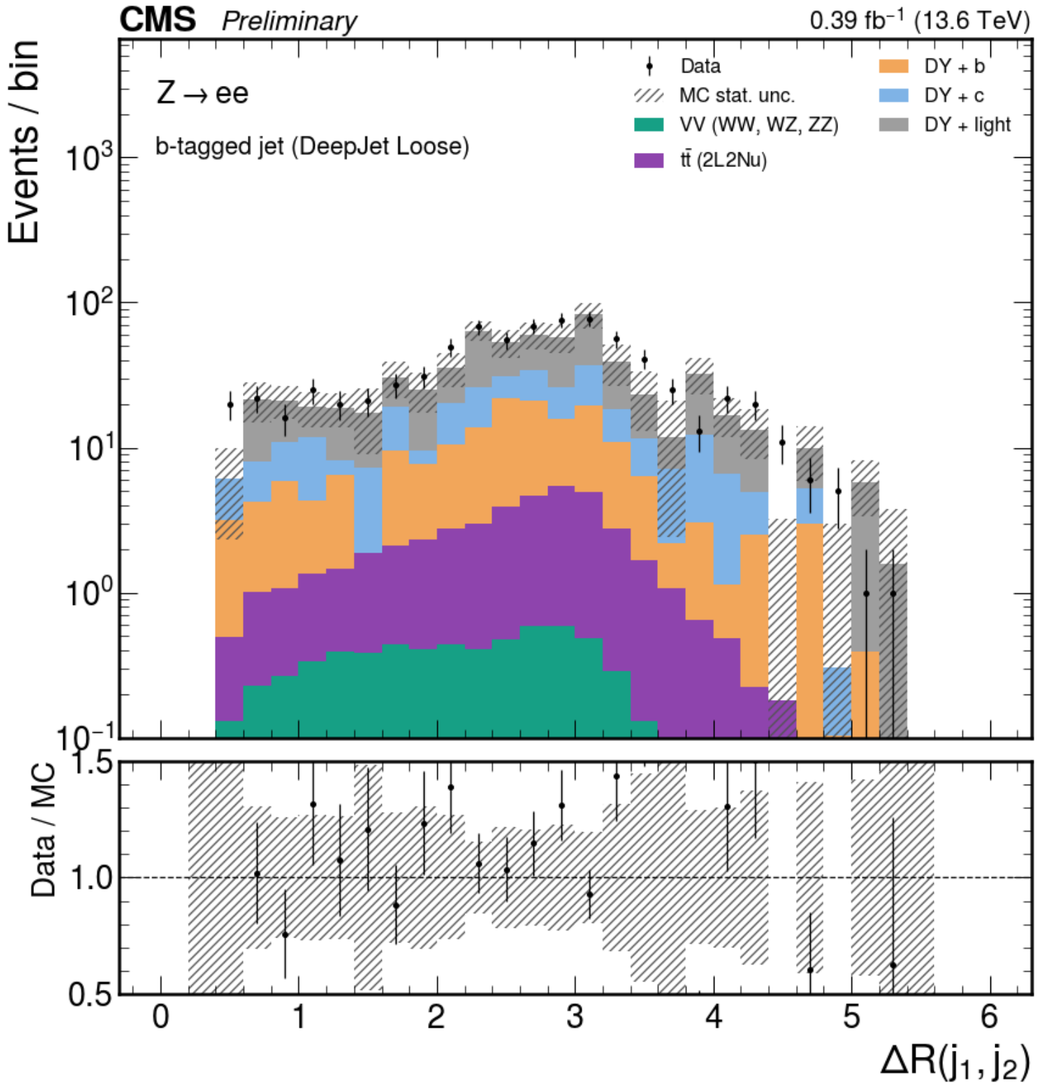


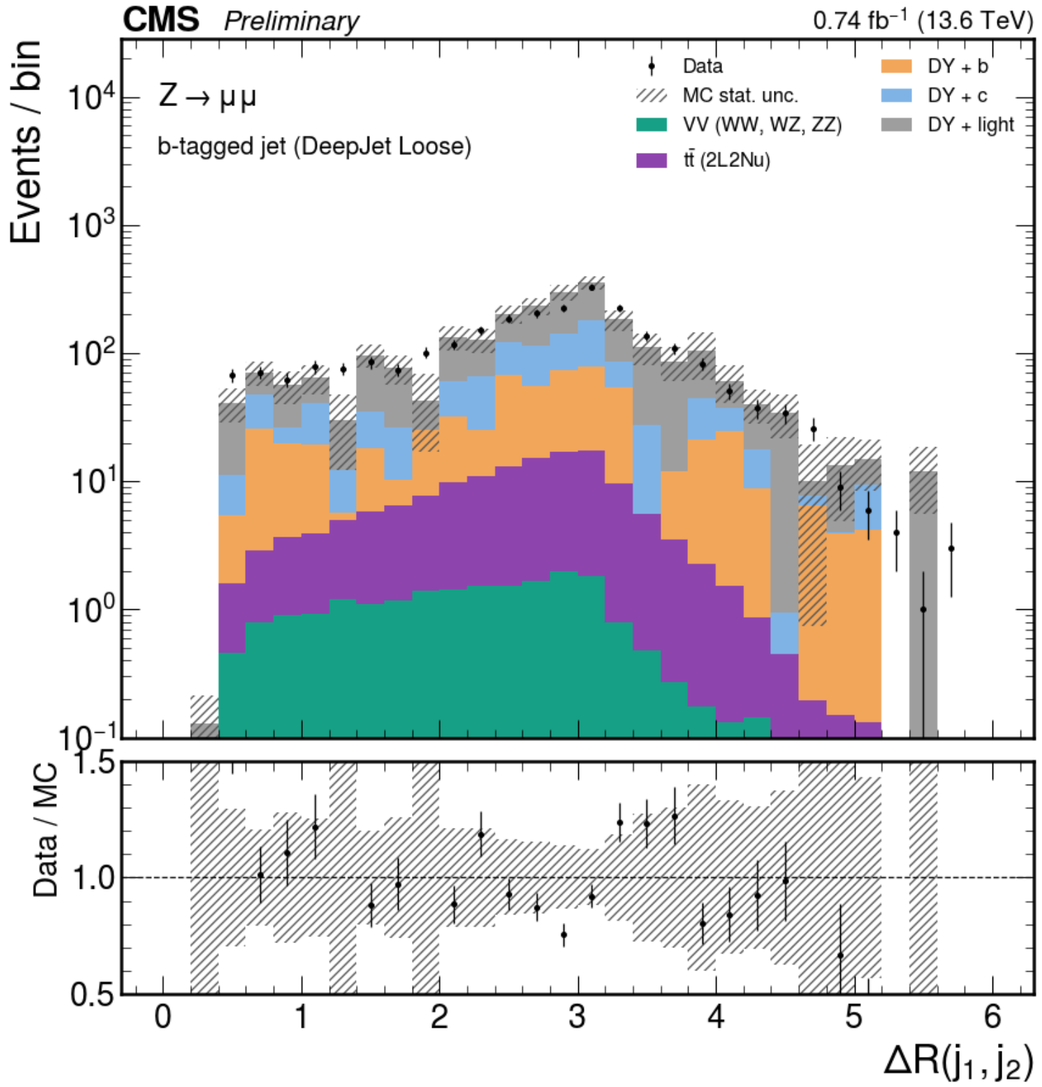


DeepJet · Loose b-tag (B > 0.0583)

Angular distance between the two leading jets, ΔR(jj)

Only events with at least two jets.

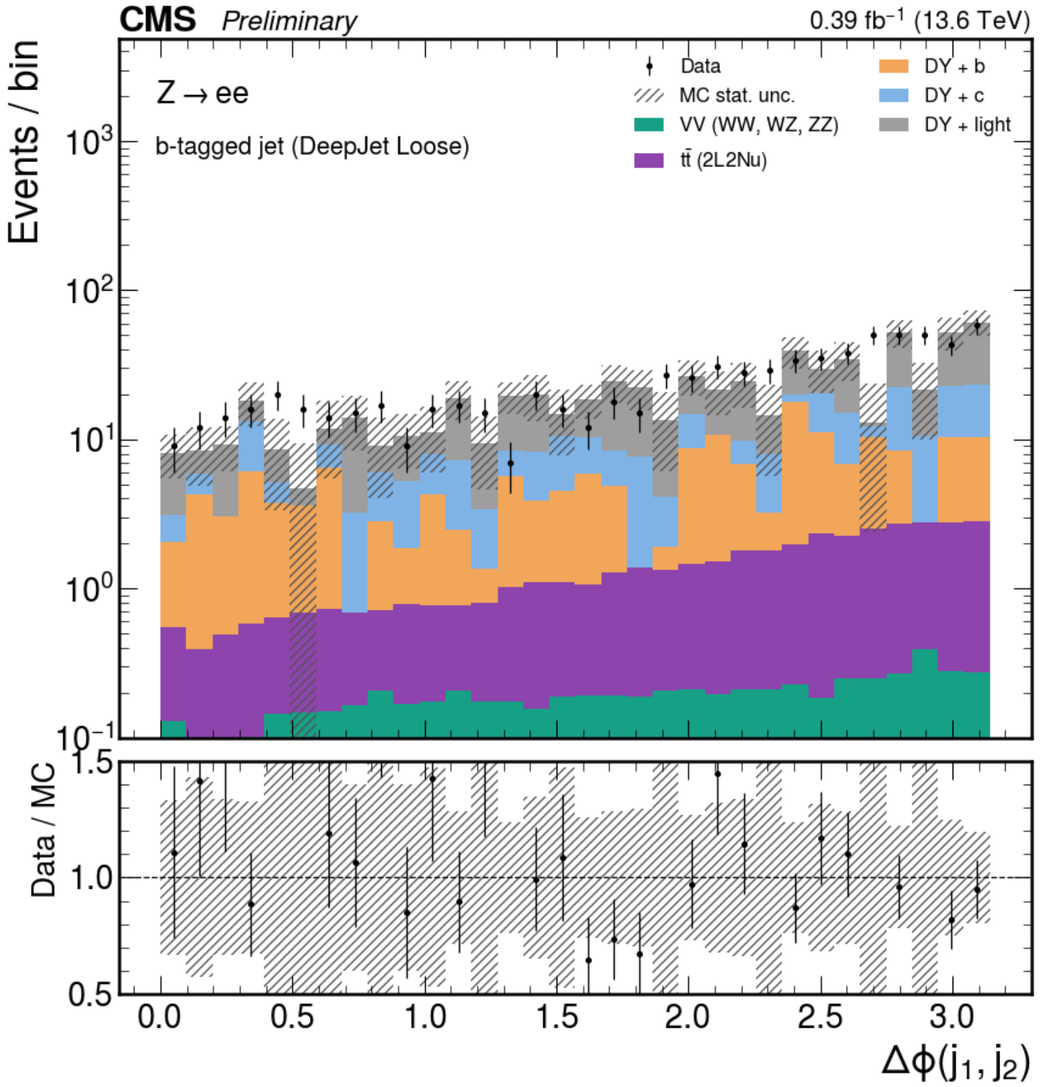


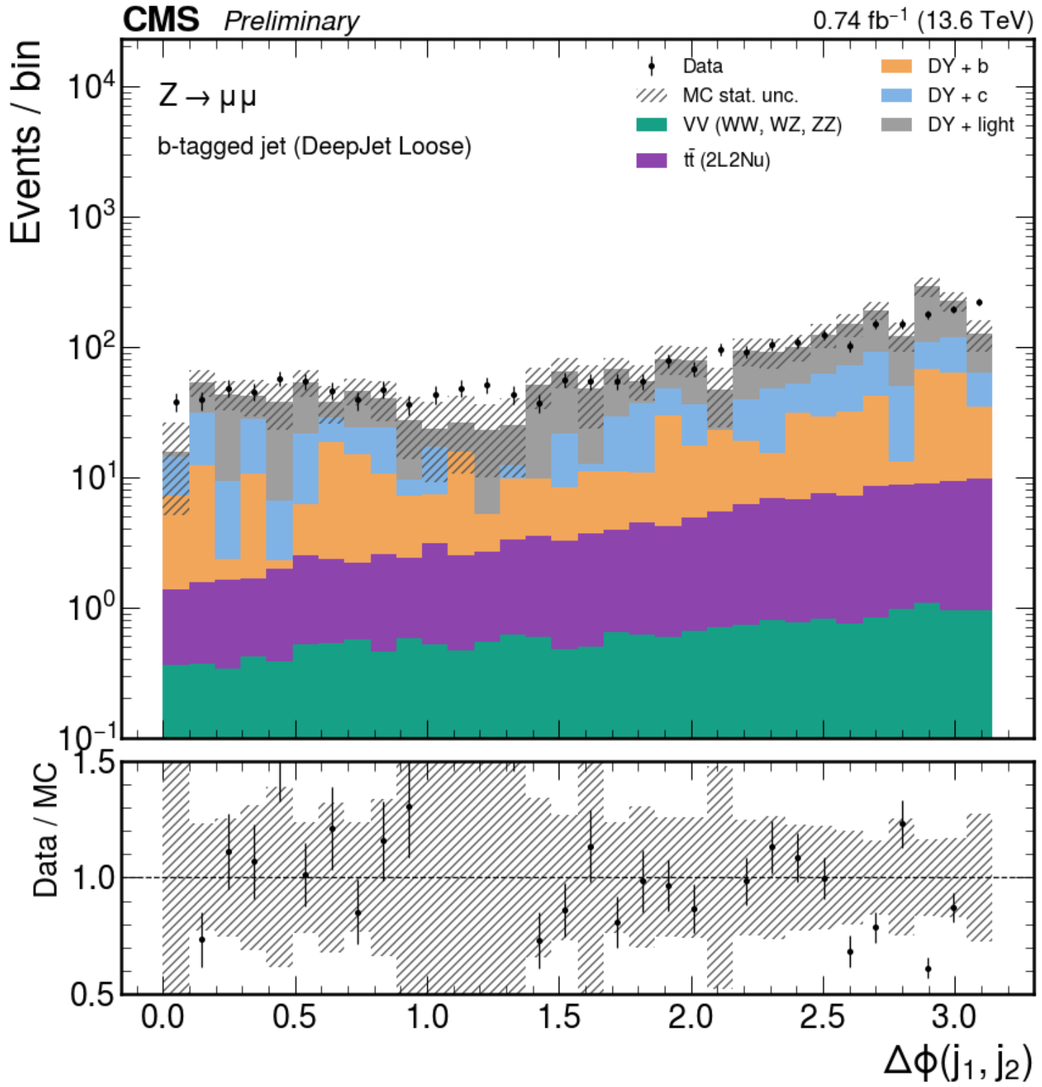


DeepJet · Loose b-tag (B > 0.0583)

Azimuthal angle between the two leading jets, Δφ(jj)

Only events with at least two jets.

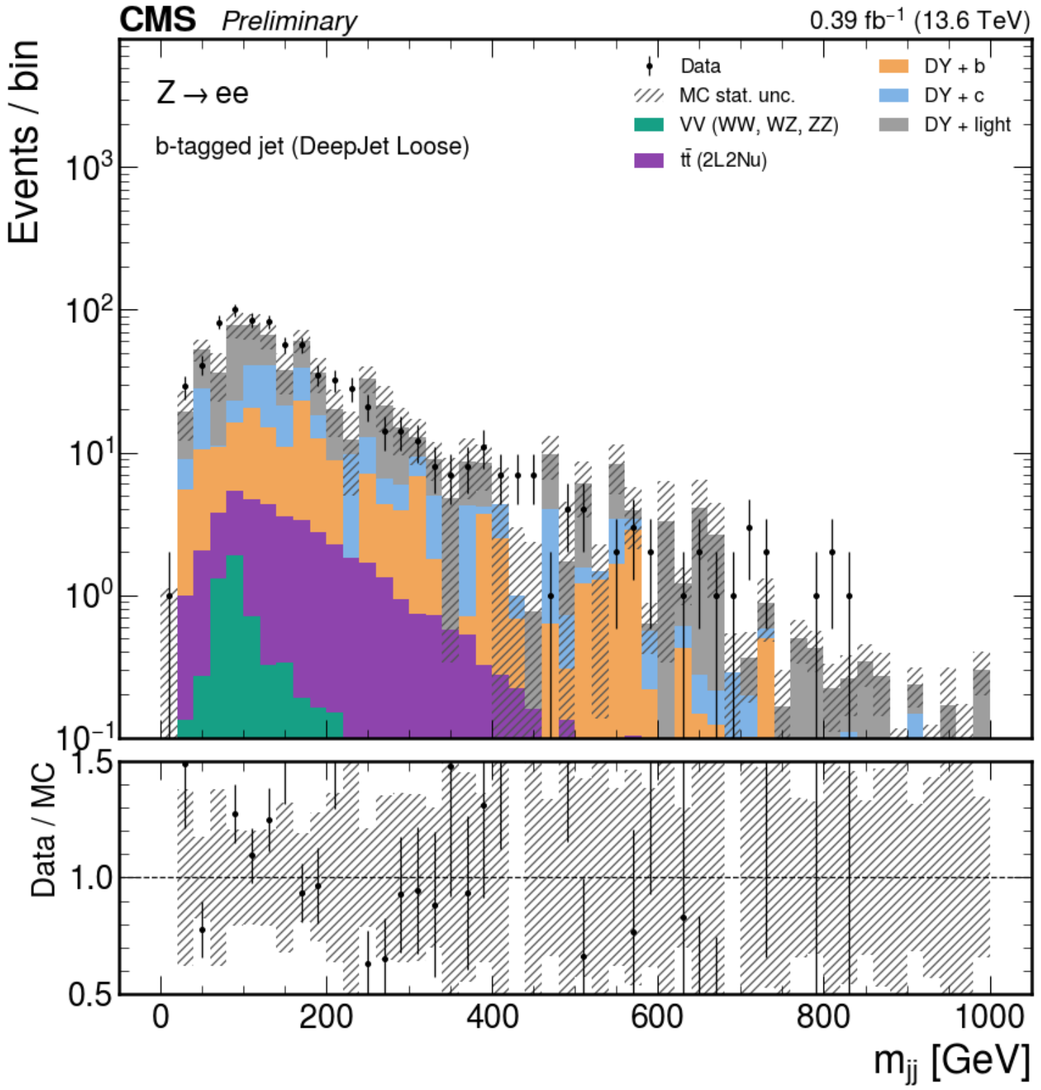


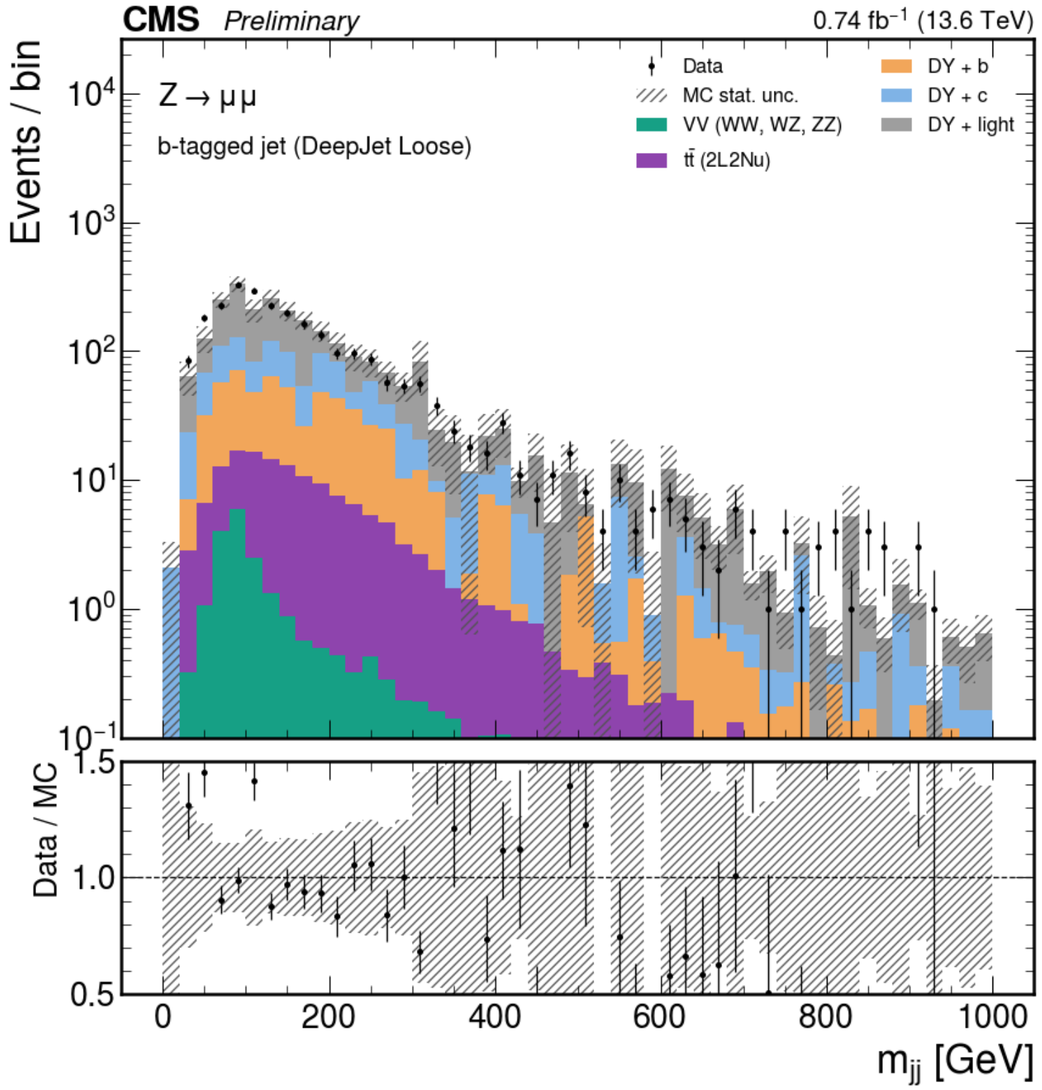


DeepJet · Loose b-tag (B > 0.0583)

Mass of the two leading jets, m(jj)

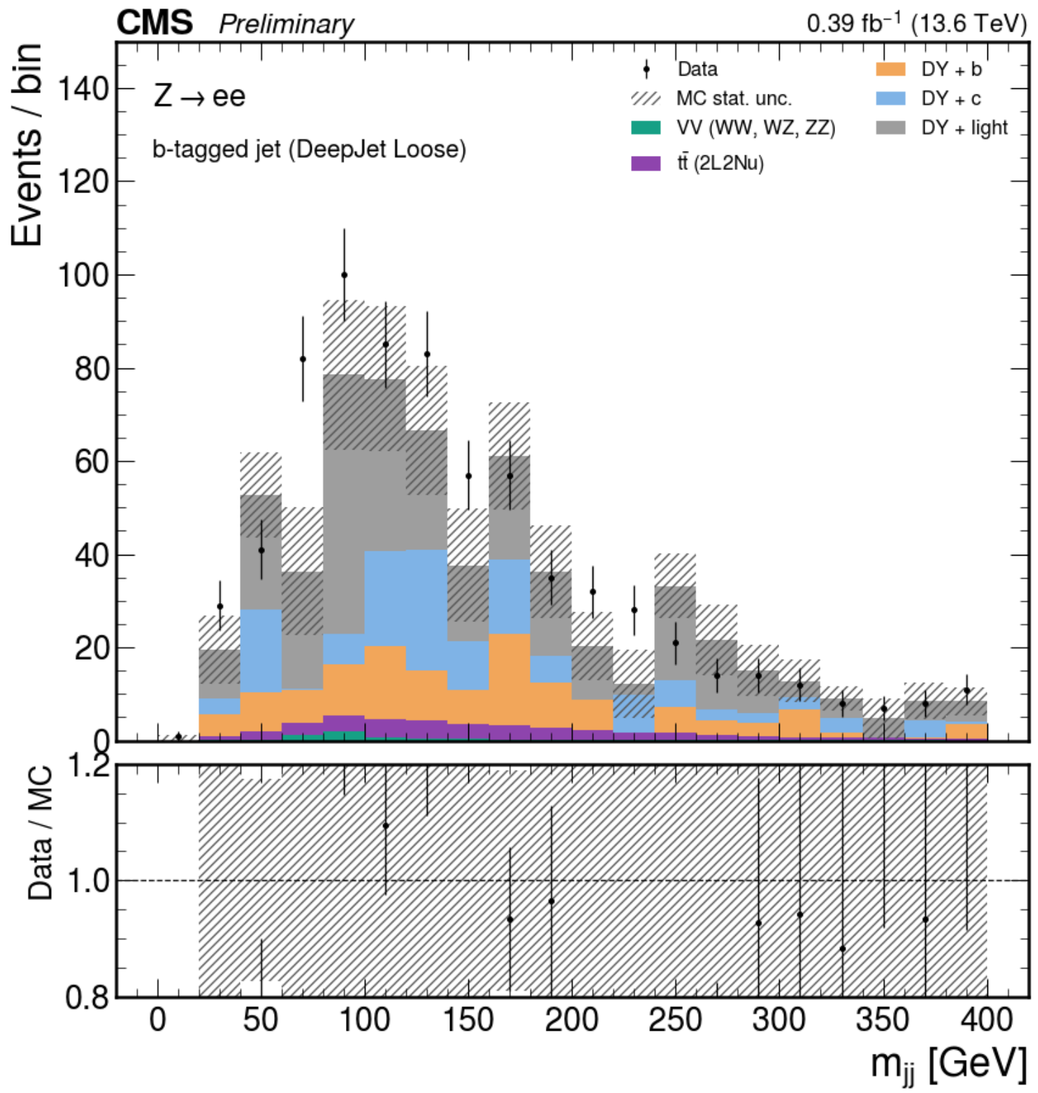
Only events with at least two jets.

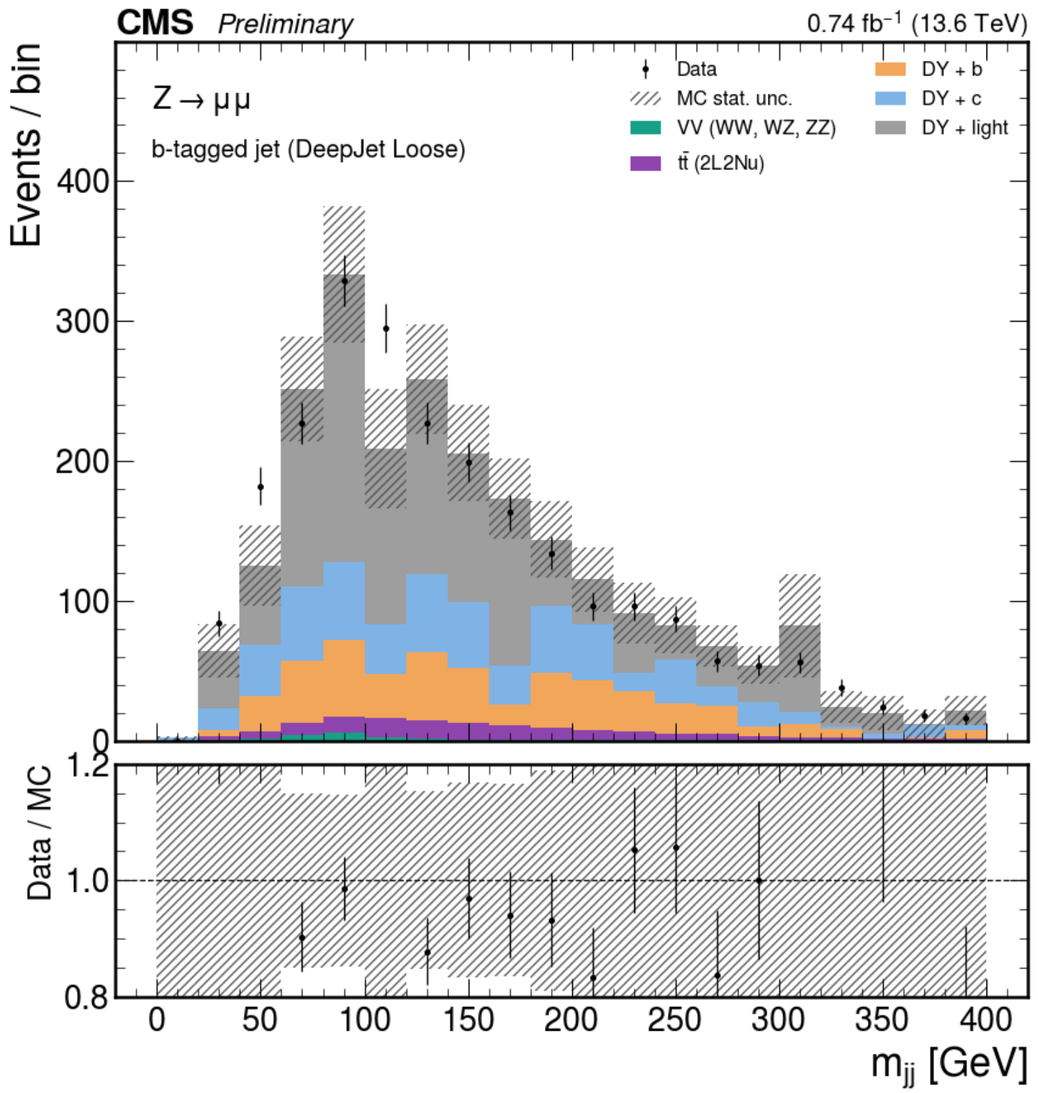




m(jj), zoom on 0–400 GeV

Zoom: 0–400, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.

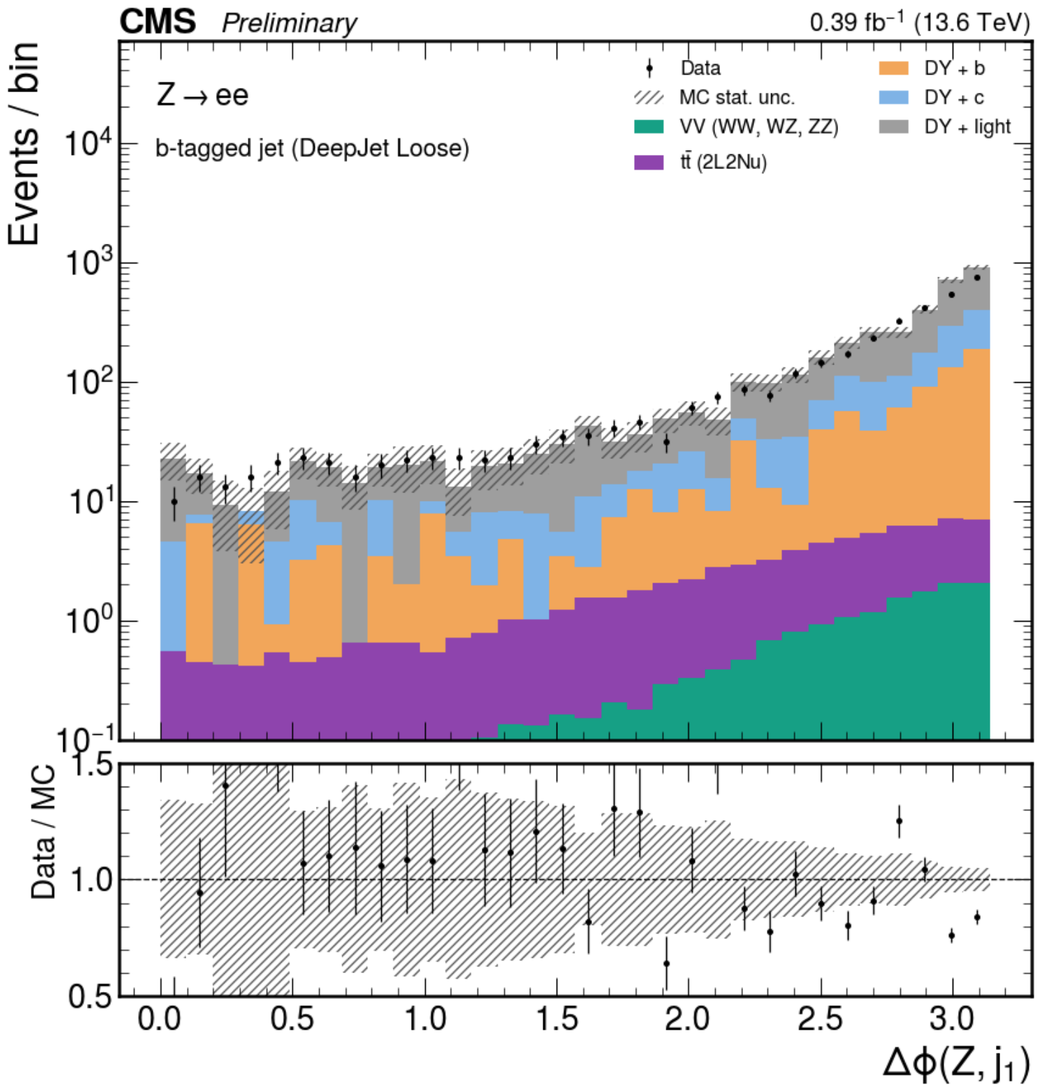


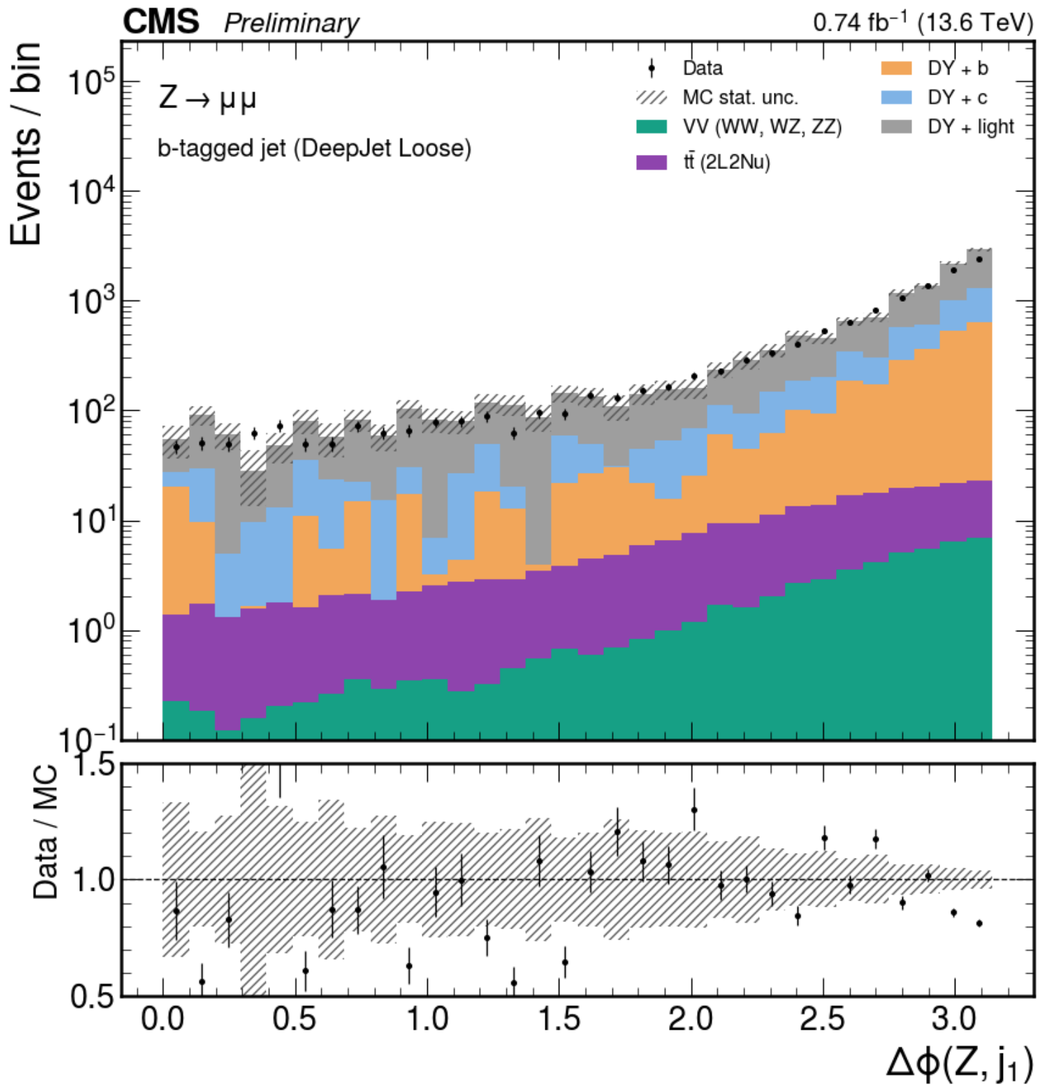


DeepJet · Loose b-tag (B > 0.0583)

Azimuthal angle between the Z and the leading jet, Δφ(Z, j)

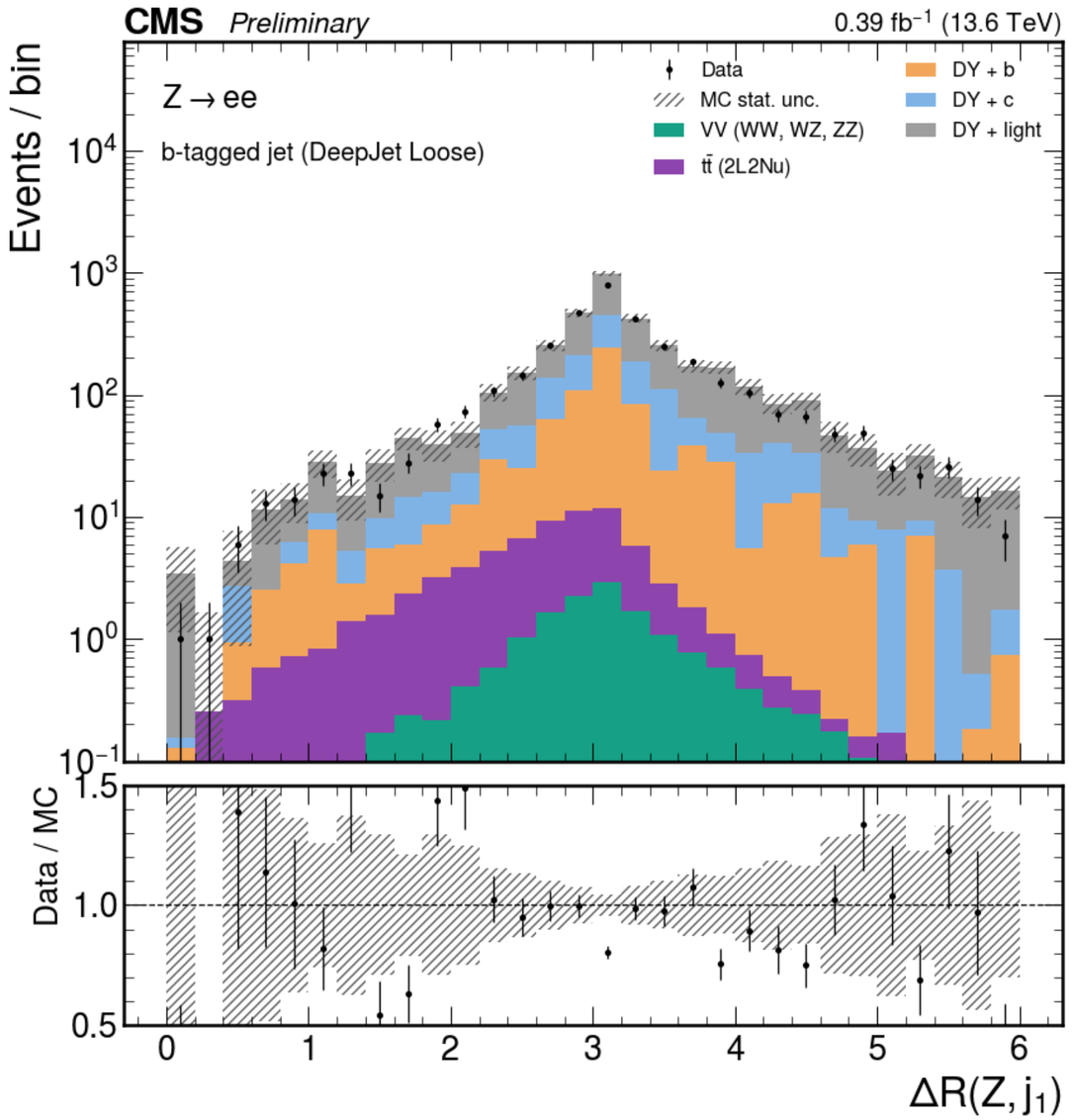
Close to π when the jet recoils against the Z.

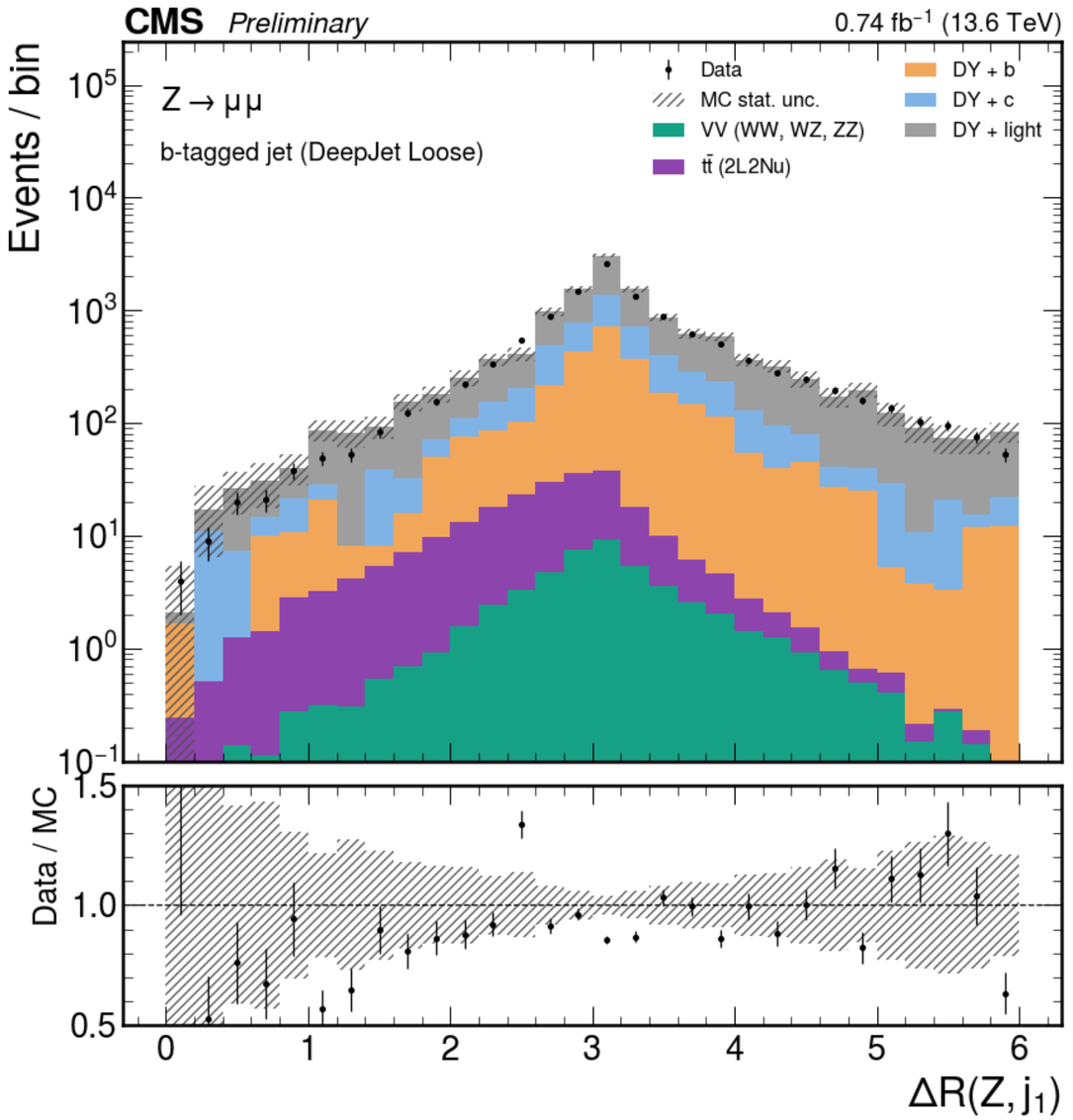




DeepJet · Loose b-tag (B > 0.0583)

Angular distance between the Z and the leading jet, ΔR(Z, j)

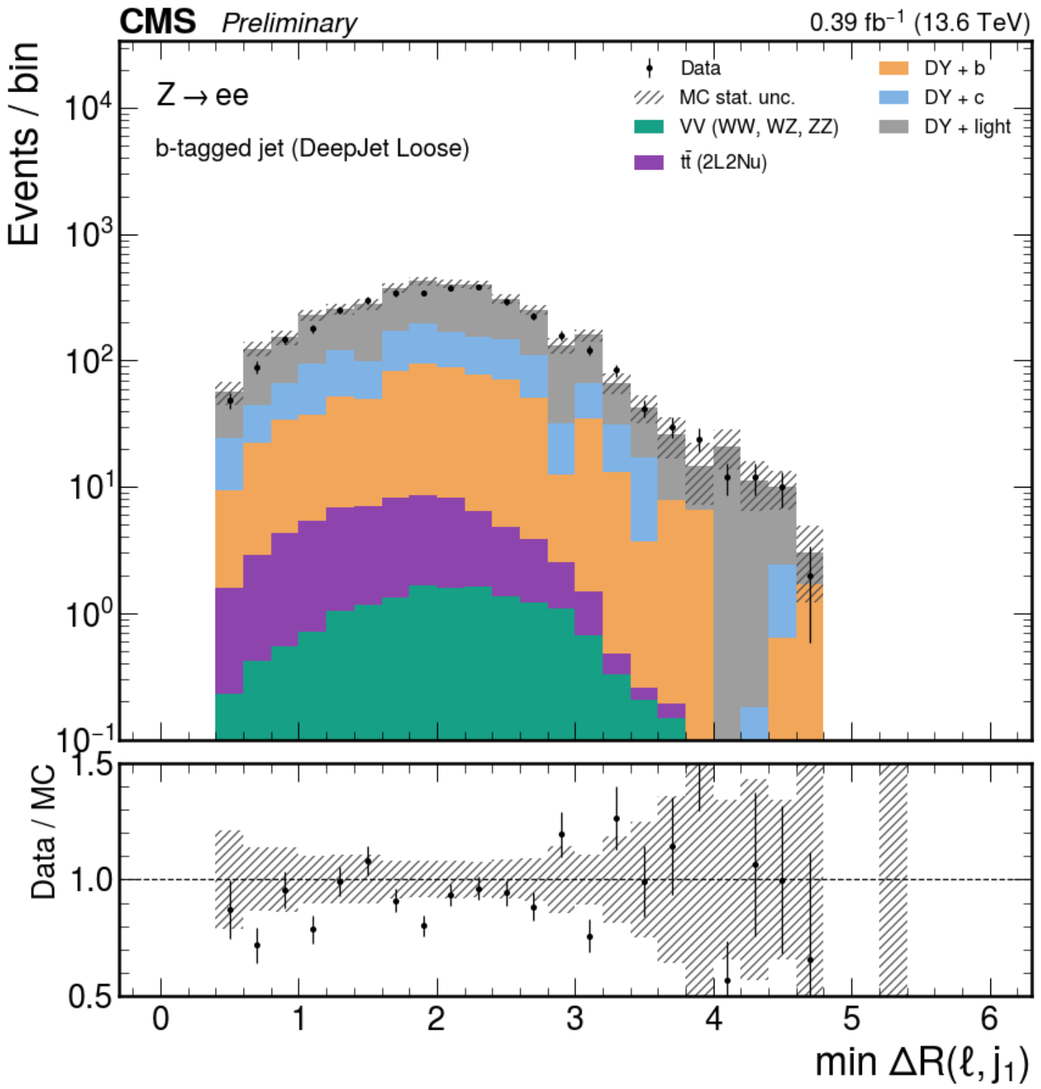


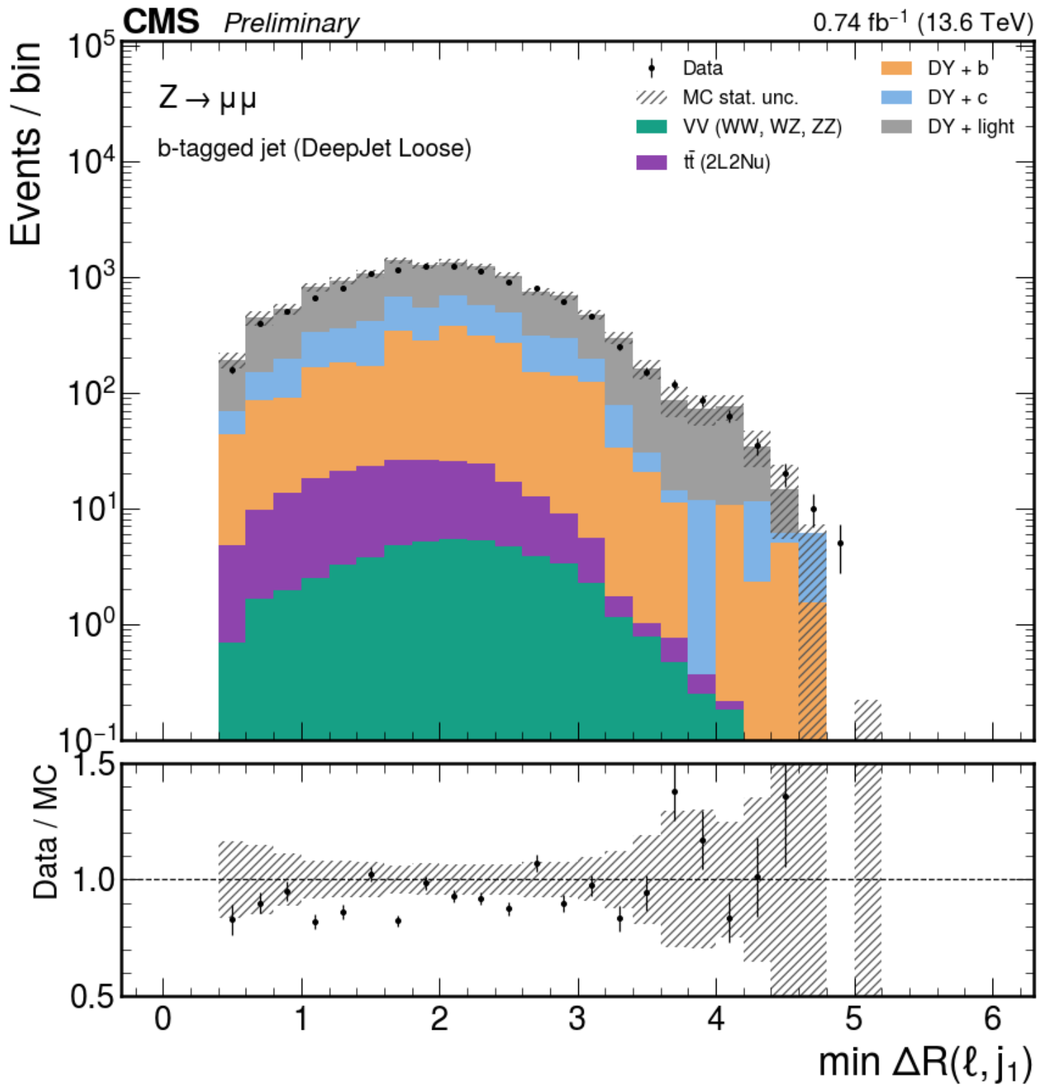


DeepJet · Loose b-tag (B > 0.0583)

Smallest ΔR between a lepton and the leading jet

Must be > 0.4 by selection (jets overlapping a lepton are removed).

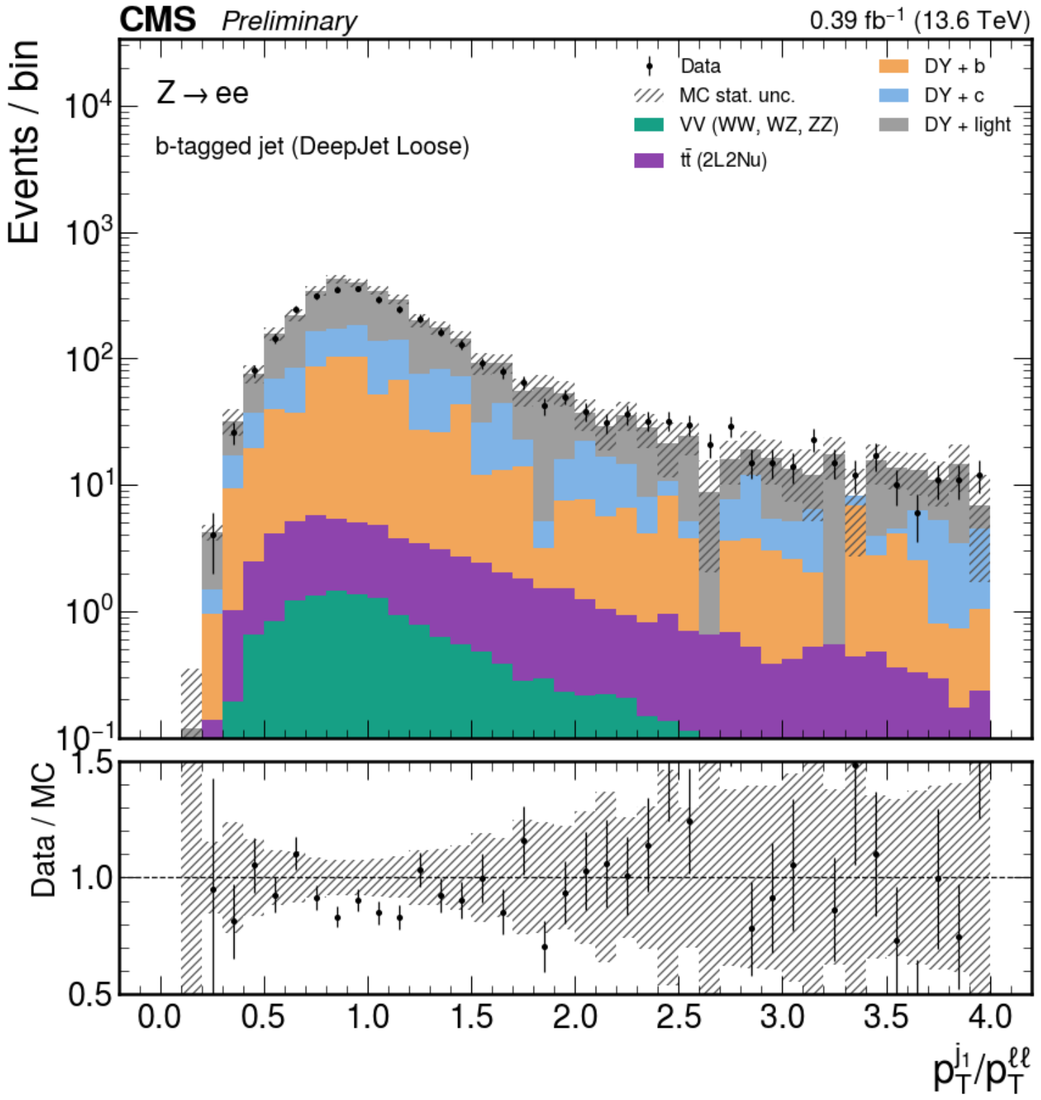


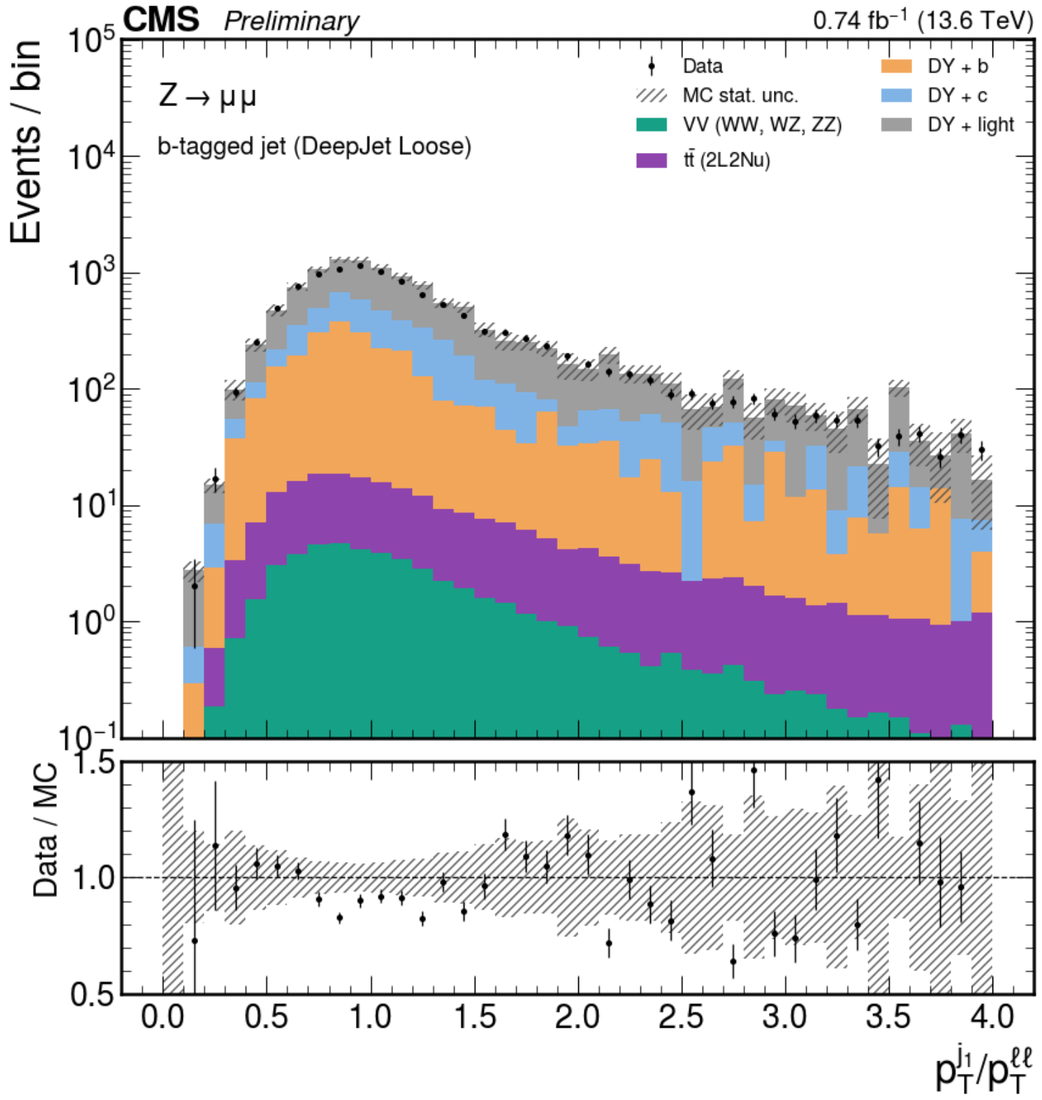


DeepJet · Loose b-tag (B > 0.0583)

pT balance: leading-jet pT / Z pT

Close to 1 when the jet alone balances the Z. Sensitive to the jet energy scale and to extra jets.

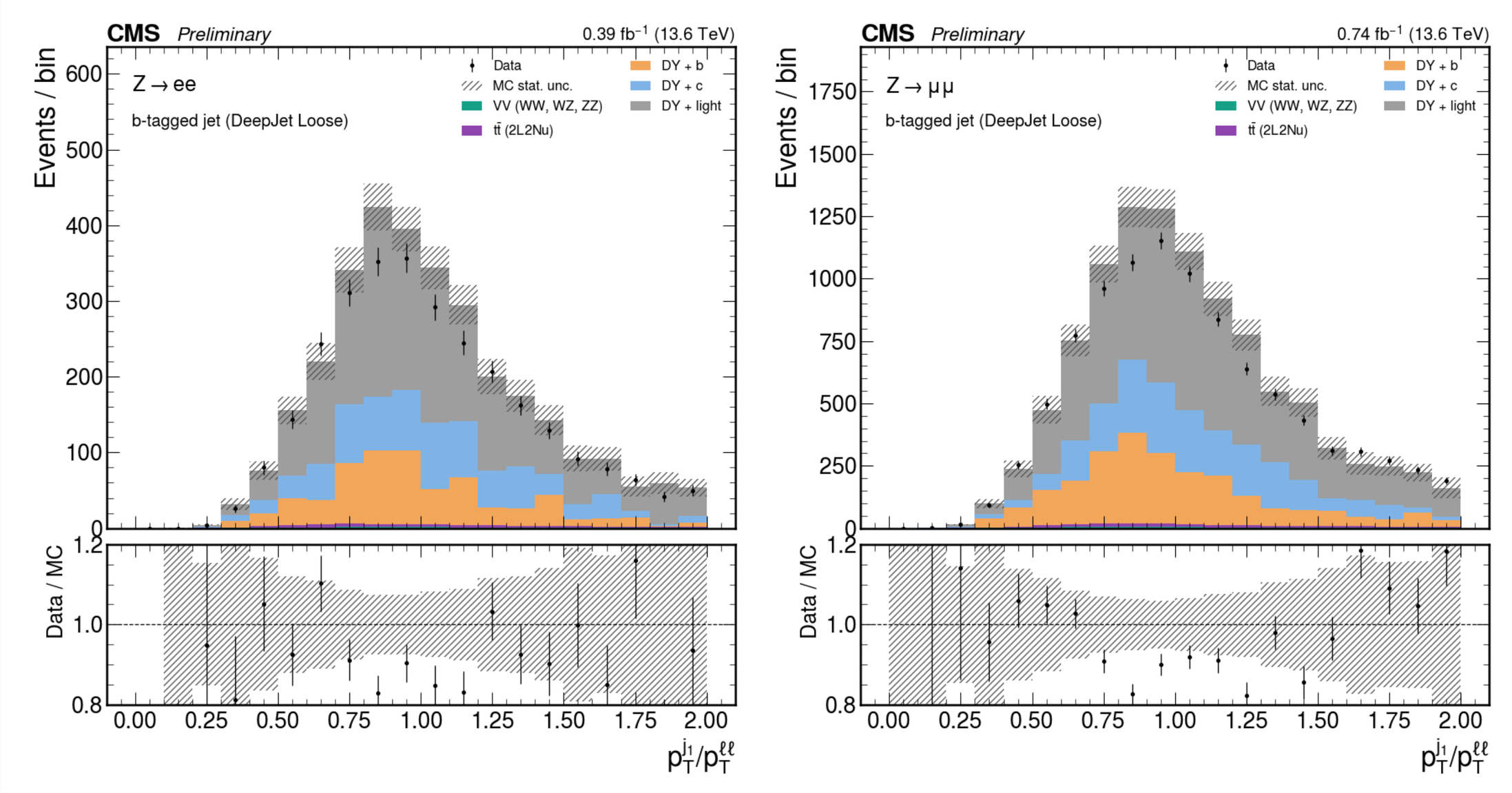




DeepJet · Loose b-tag (B > 0.0583)

pT balance, zoom on 0–2

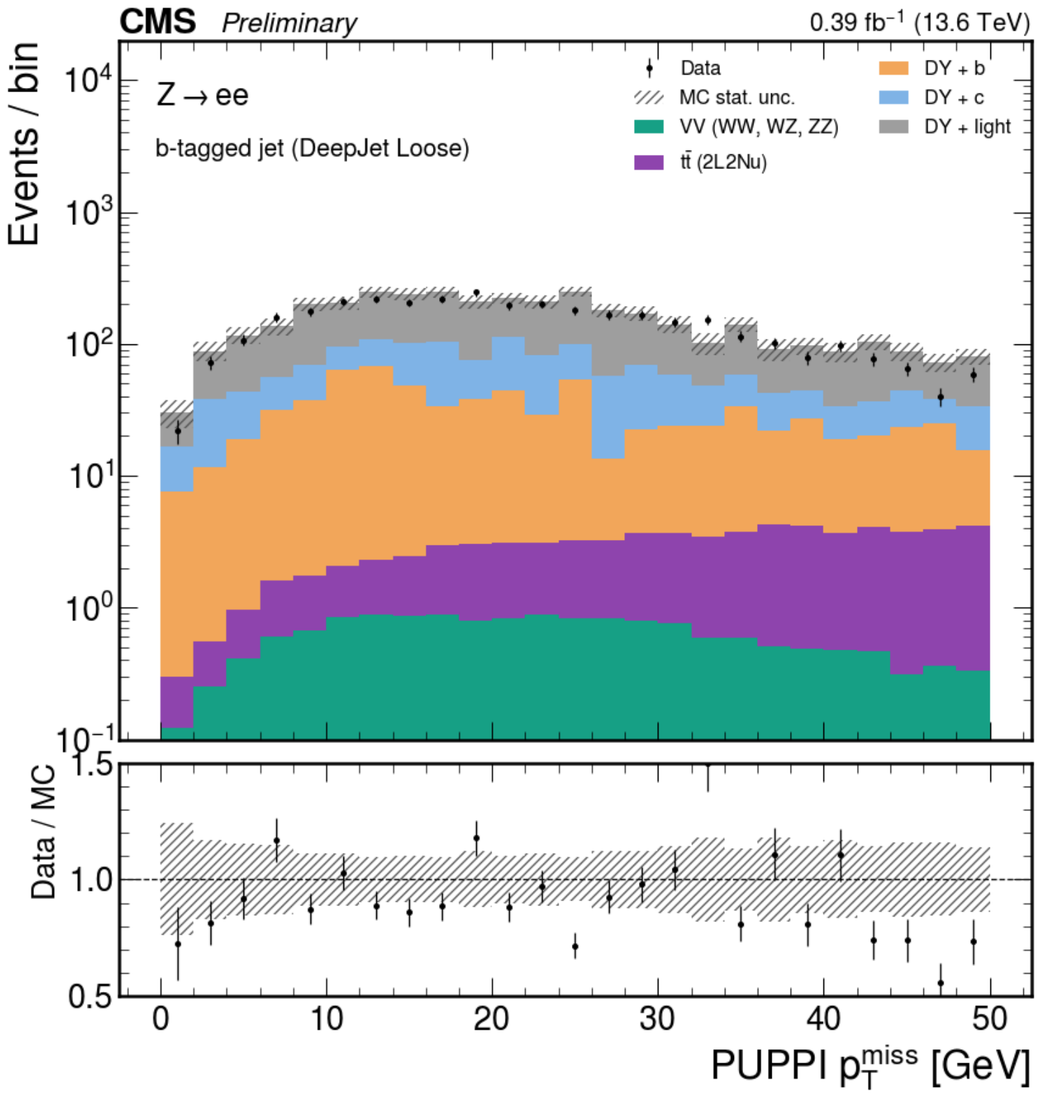
Zoom: 0–2, linear scale, data/MC from 0.8 to 1.2.

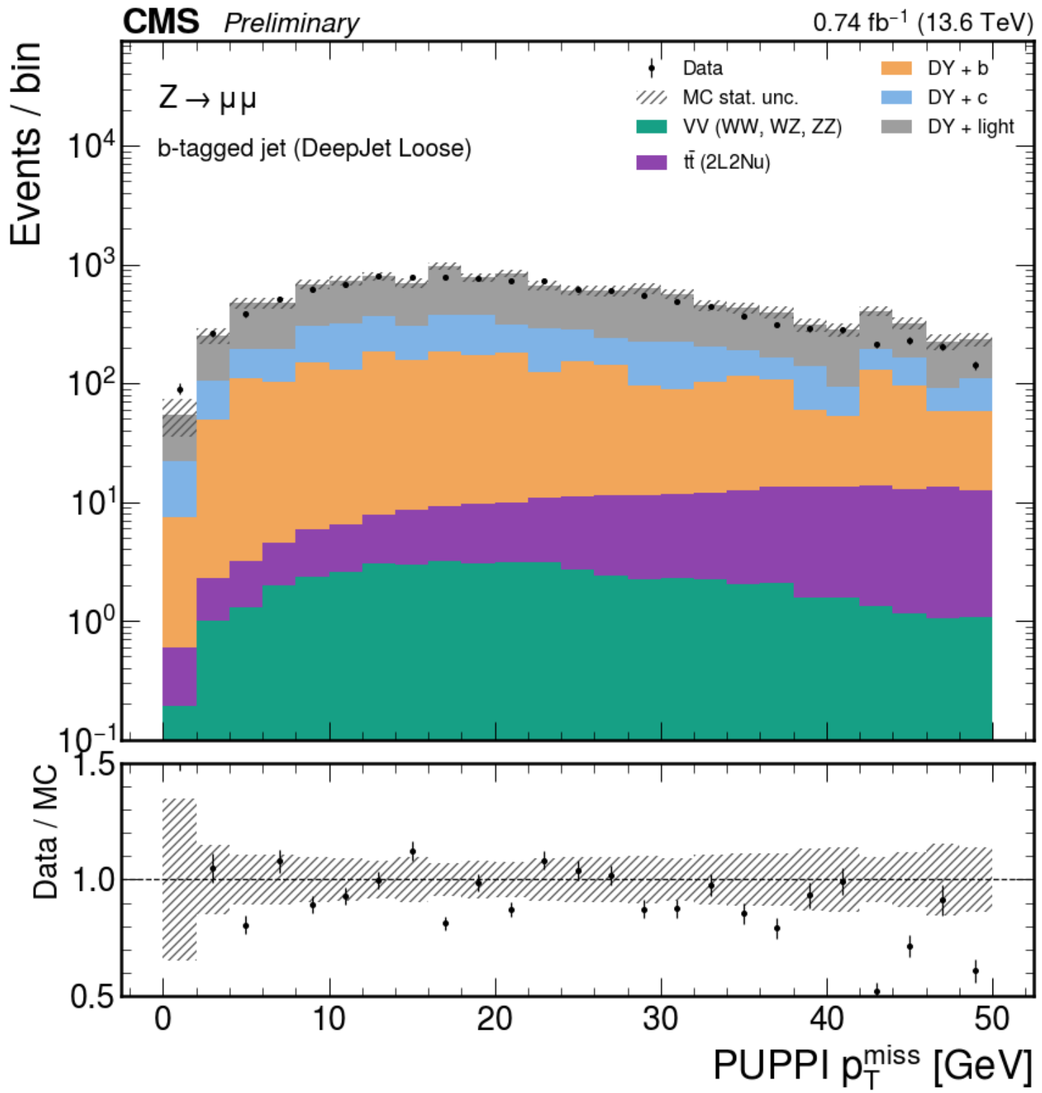


DeepJet · Loose b-tag (B > 0.0583)

Missing transverse momentum (type-1 PUPPI MET), after the MET cut

The selection requires MET < 50 GeV, so the plot stops there.

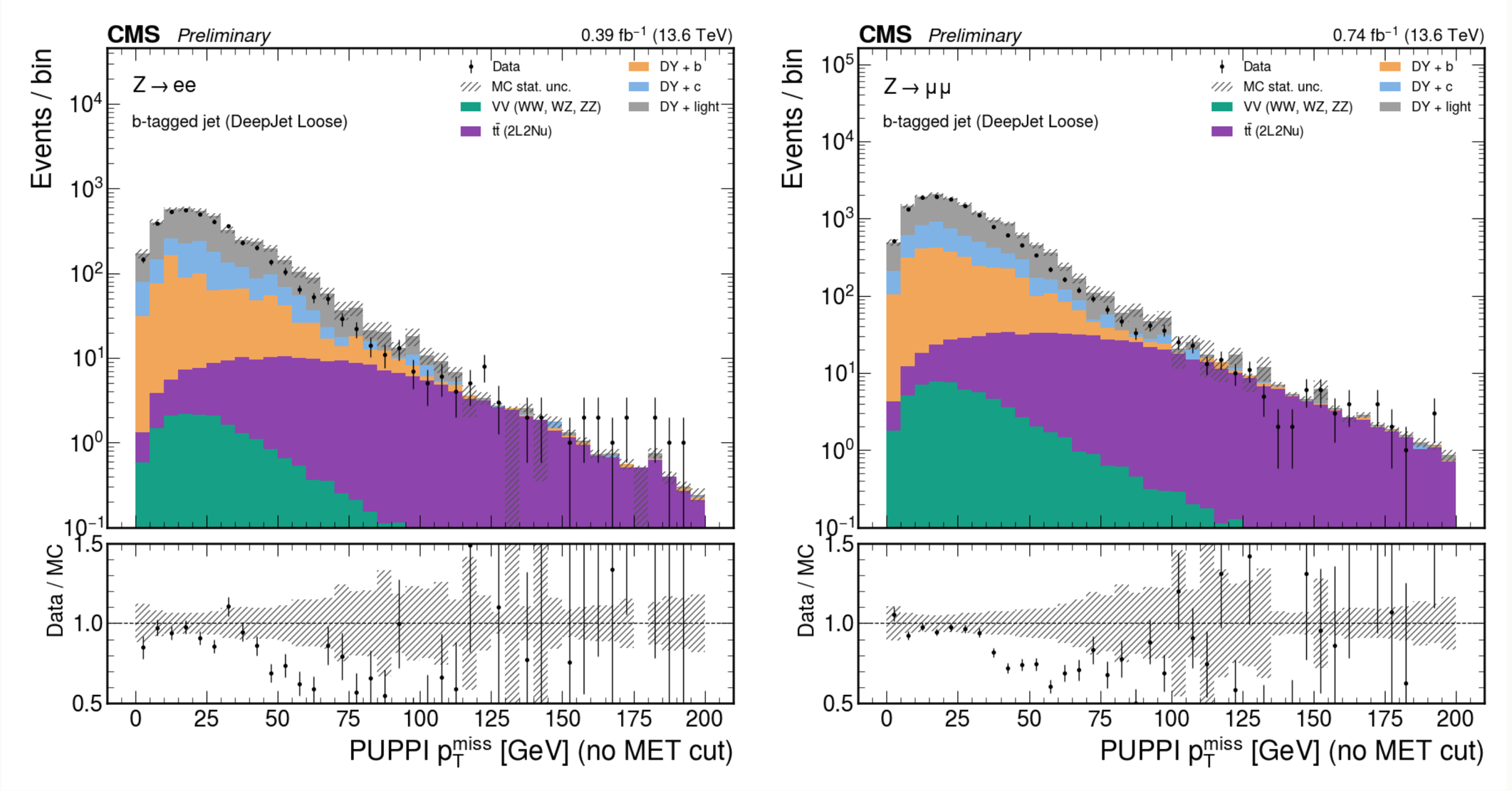




DeepJet · Loose b-tag (B > 0.0583)

MET without the MET cut, up to 200 GeV

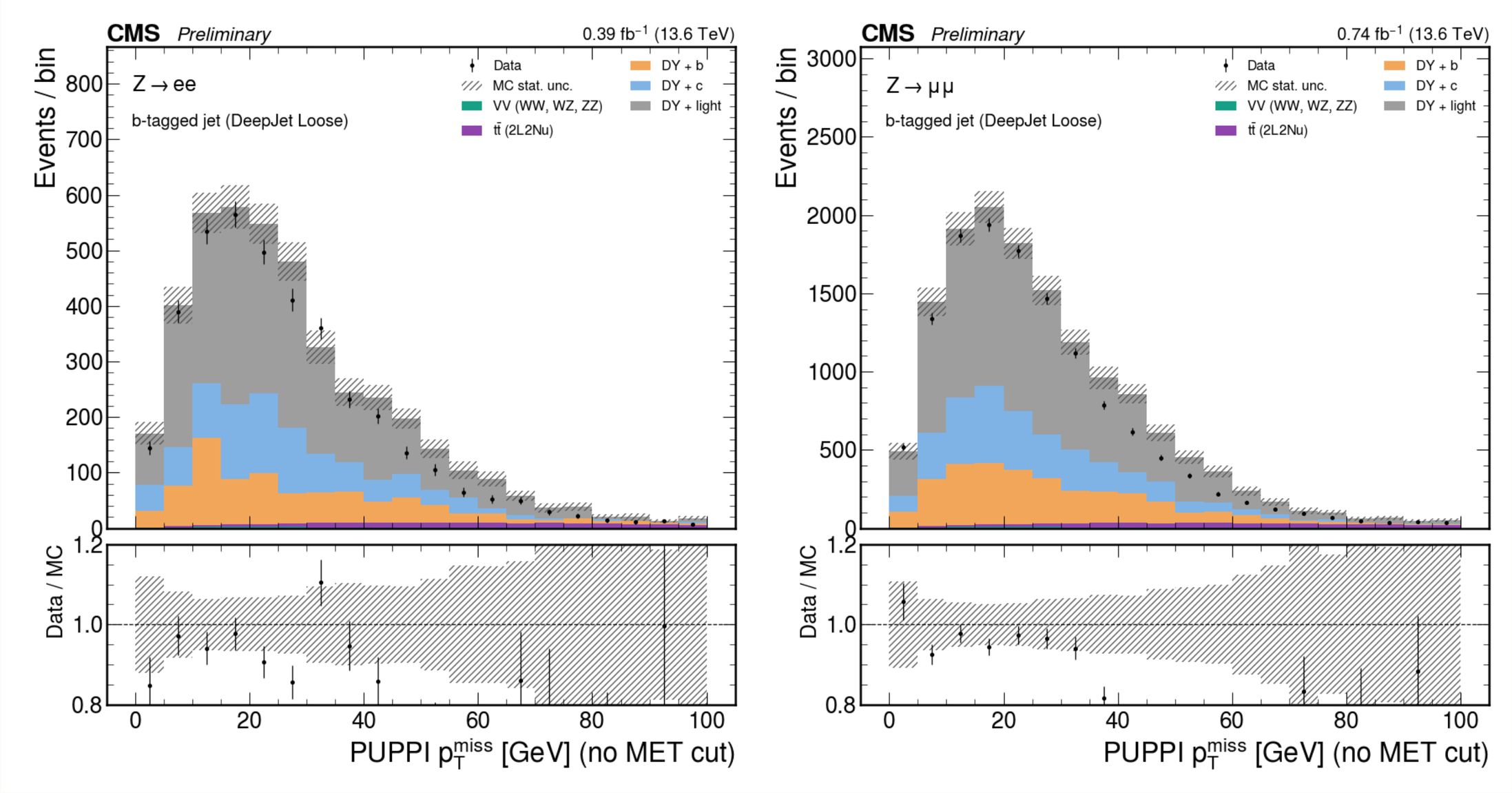
All other cuts applied, so the region removed by MET < 50 GeV is visible. $t\bar{t}$ dominates at high MET. MC is above data at 40–120 GeV, as without tagging (known 2022 jet/MET issue).



DeepJet · Loose b-tag (B > 0.0583)

MET without the MET cut, zoom on 0–100 GeV

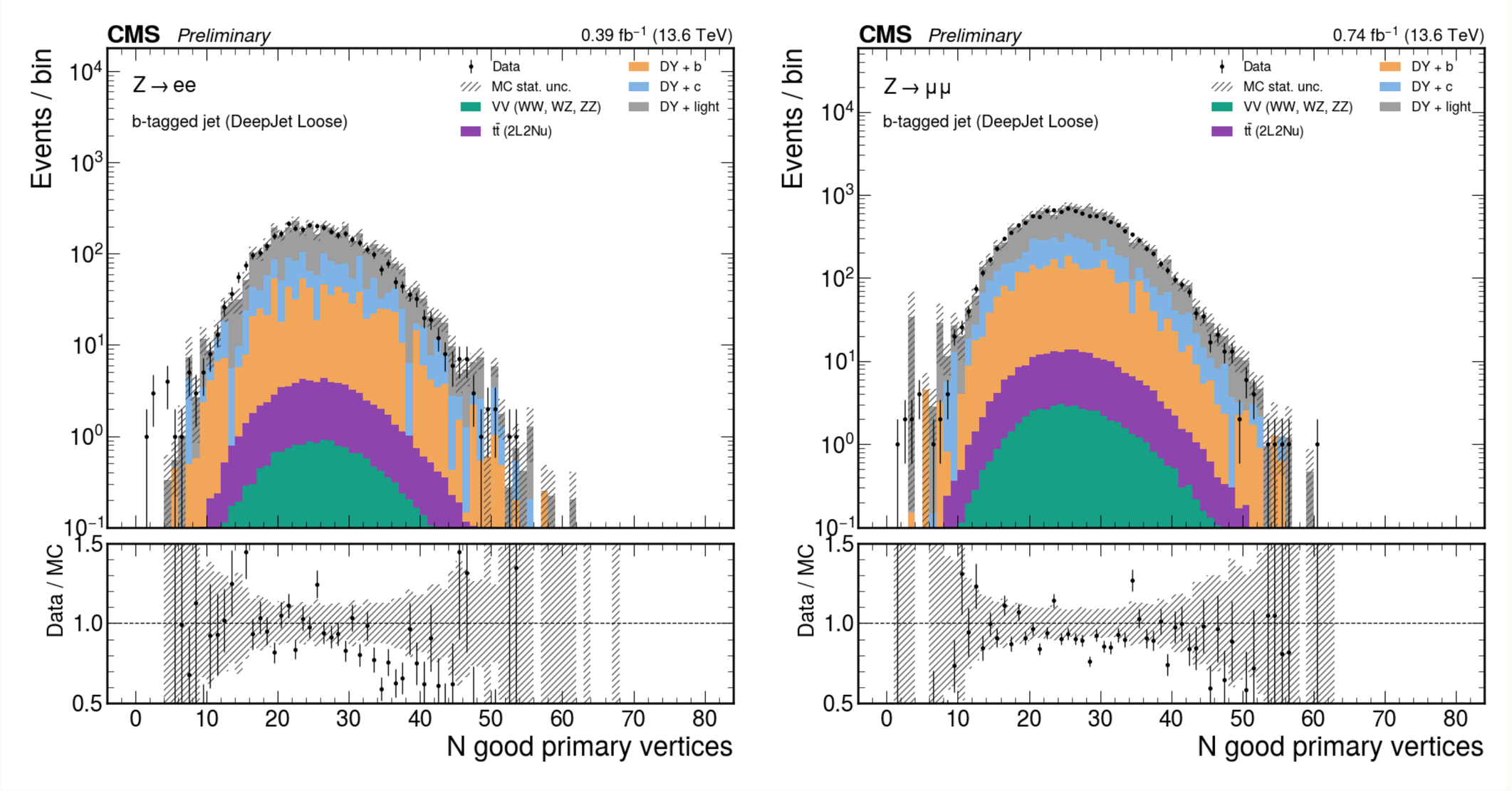
Zoom: 0–100, linear scale, data/MC from 0.8 to 1.2.



DeepJet · Loose b-tag (B > 0.0583)

Number of good primary vertices

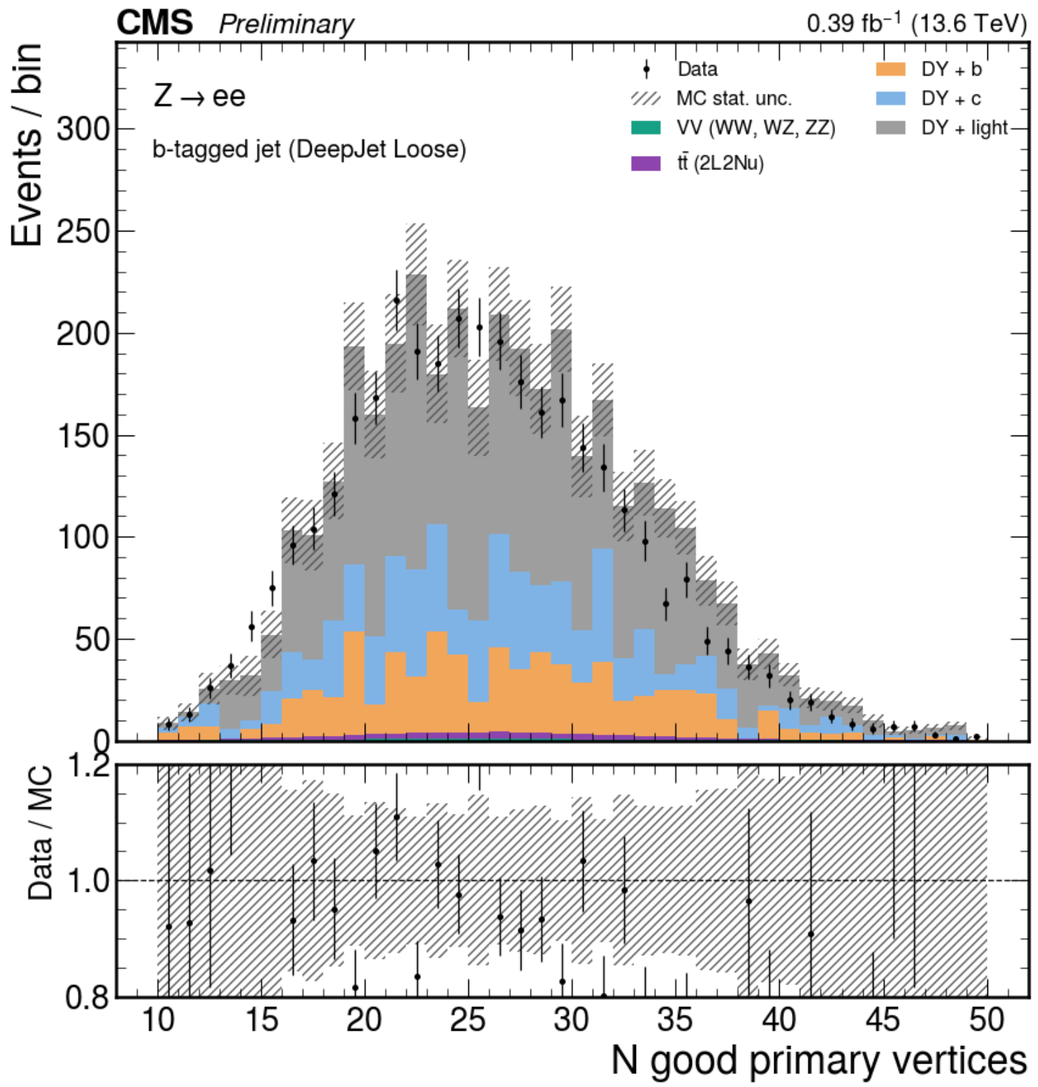
After the pileup weights. Data and MC differ slightly because our 20 test data files cover runs with less (ee) or about the same (μμ) pileup as all of 2022 C–D, which the MC is weighted to.

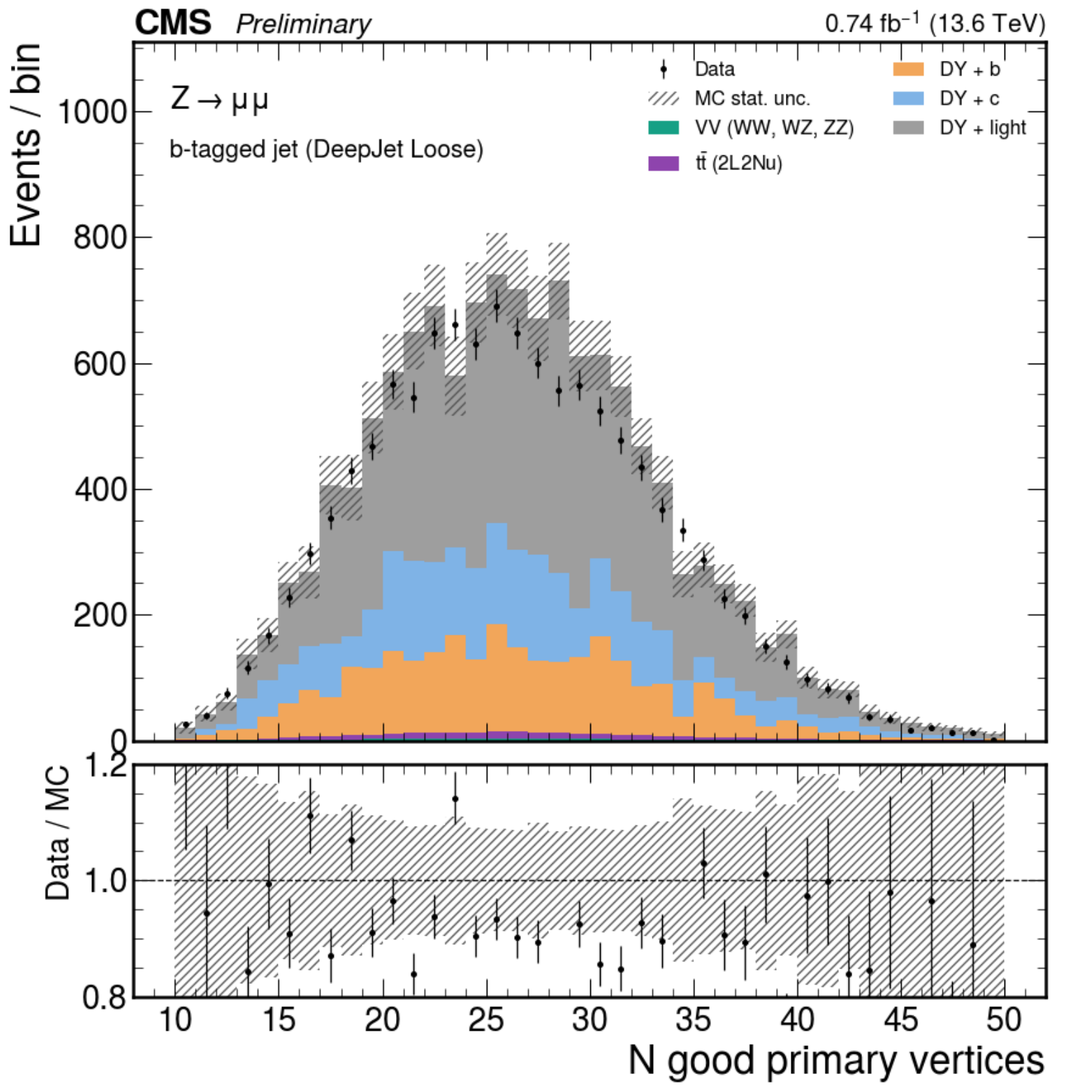


DeepJet · Loose b-tag (B > 0.0583)

Number of good primary vertices, zoom on 10–50

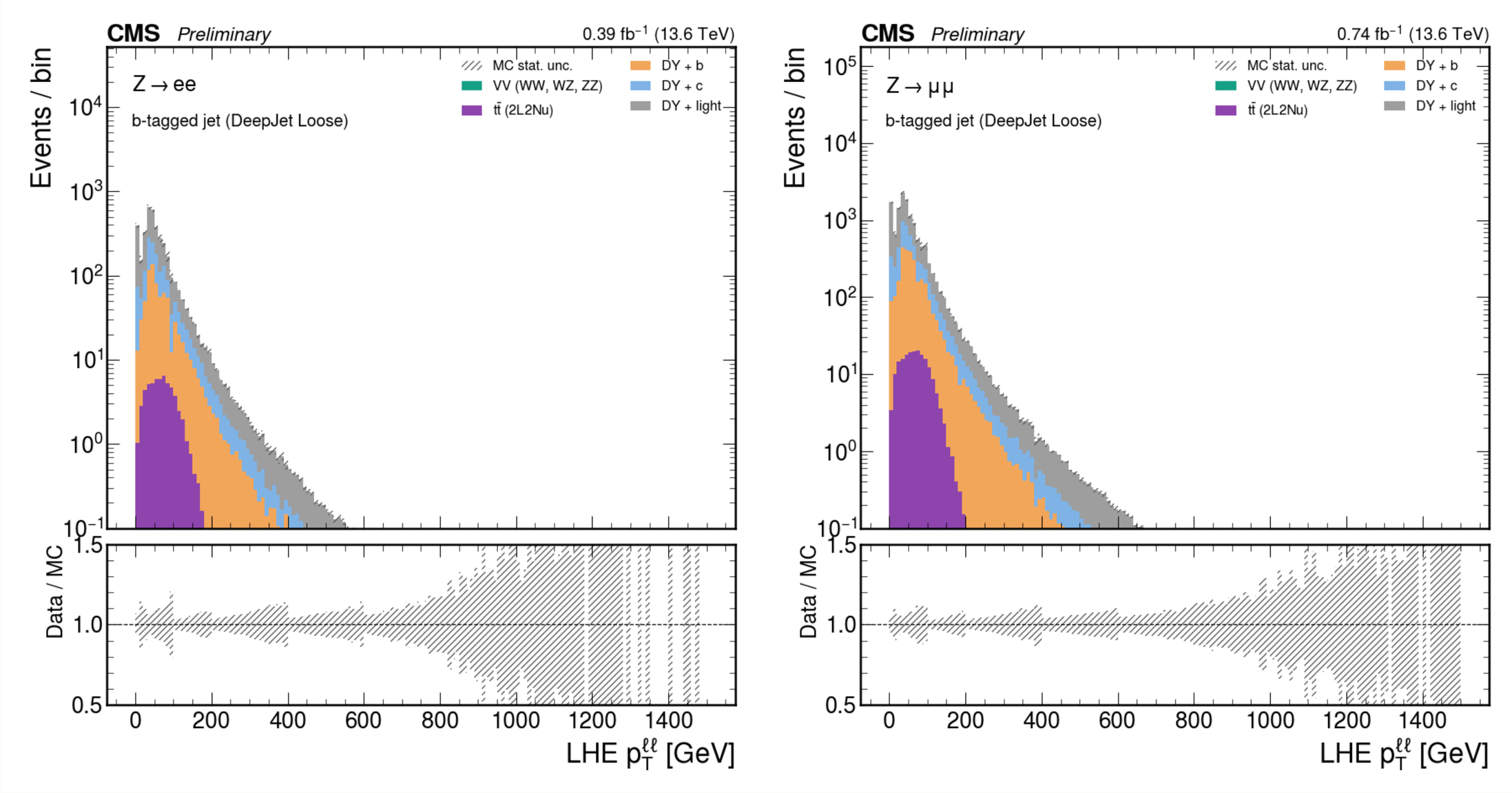
Zoom: 10–50, linear scale, data/MC from 0.8 to 1.2.





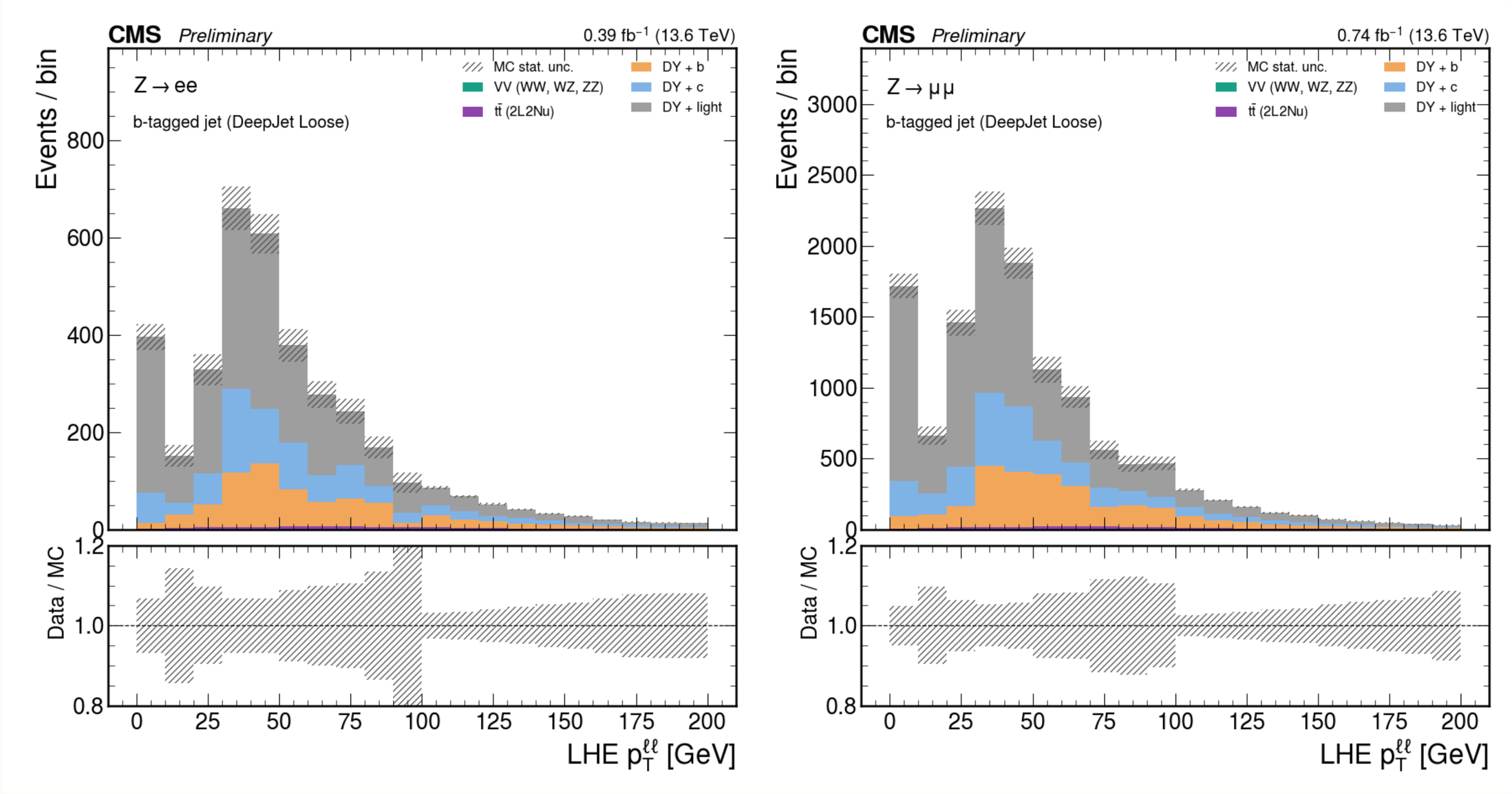
Generator-level Z pT (simulation only)

Z pT from the event generator, before any detector effect; no data. Shows which pT(ℓℓ)-binned DY sample contributes where.



Generator-level Z pT, zoom on 0–200 GeV

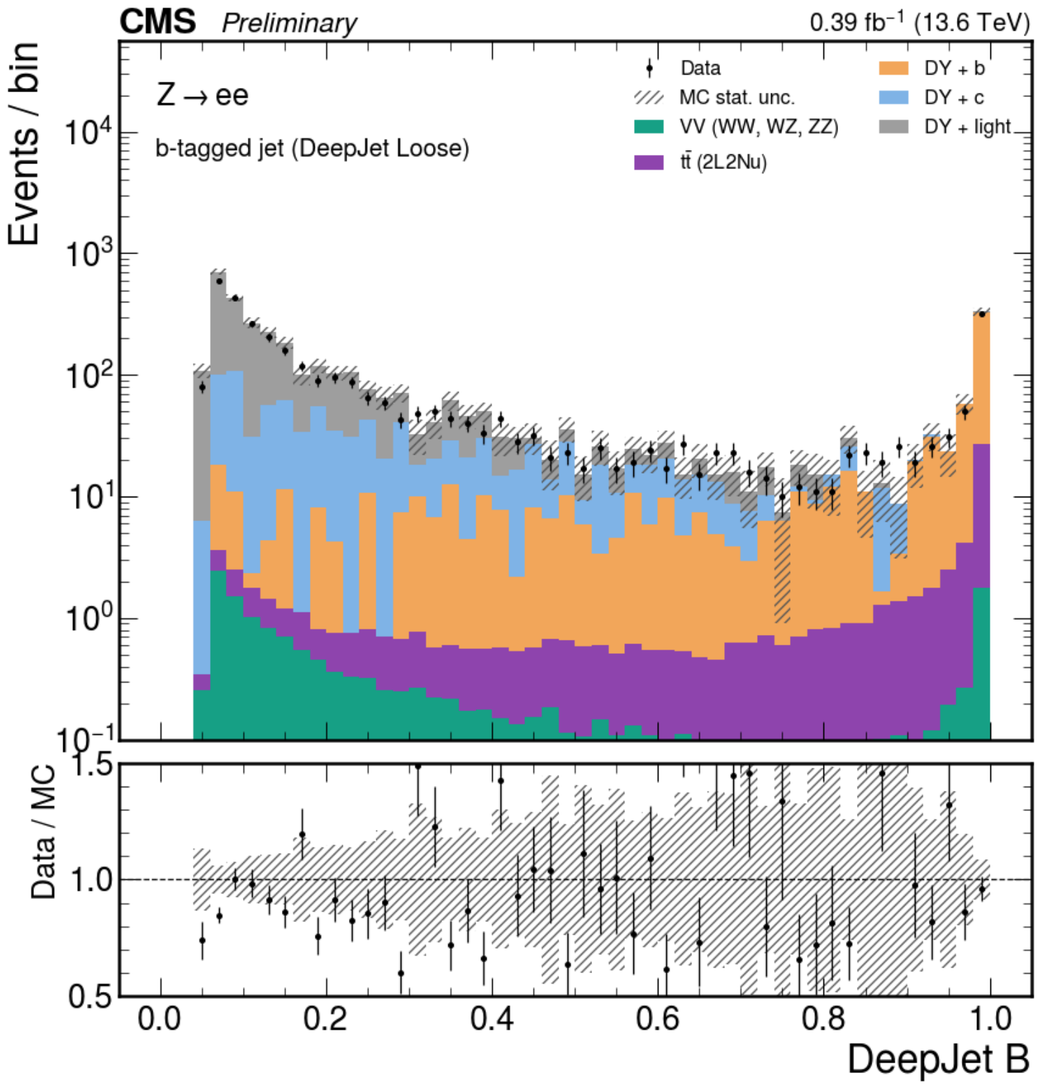
Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.

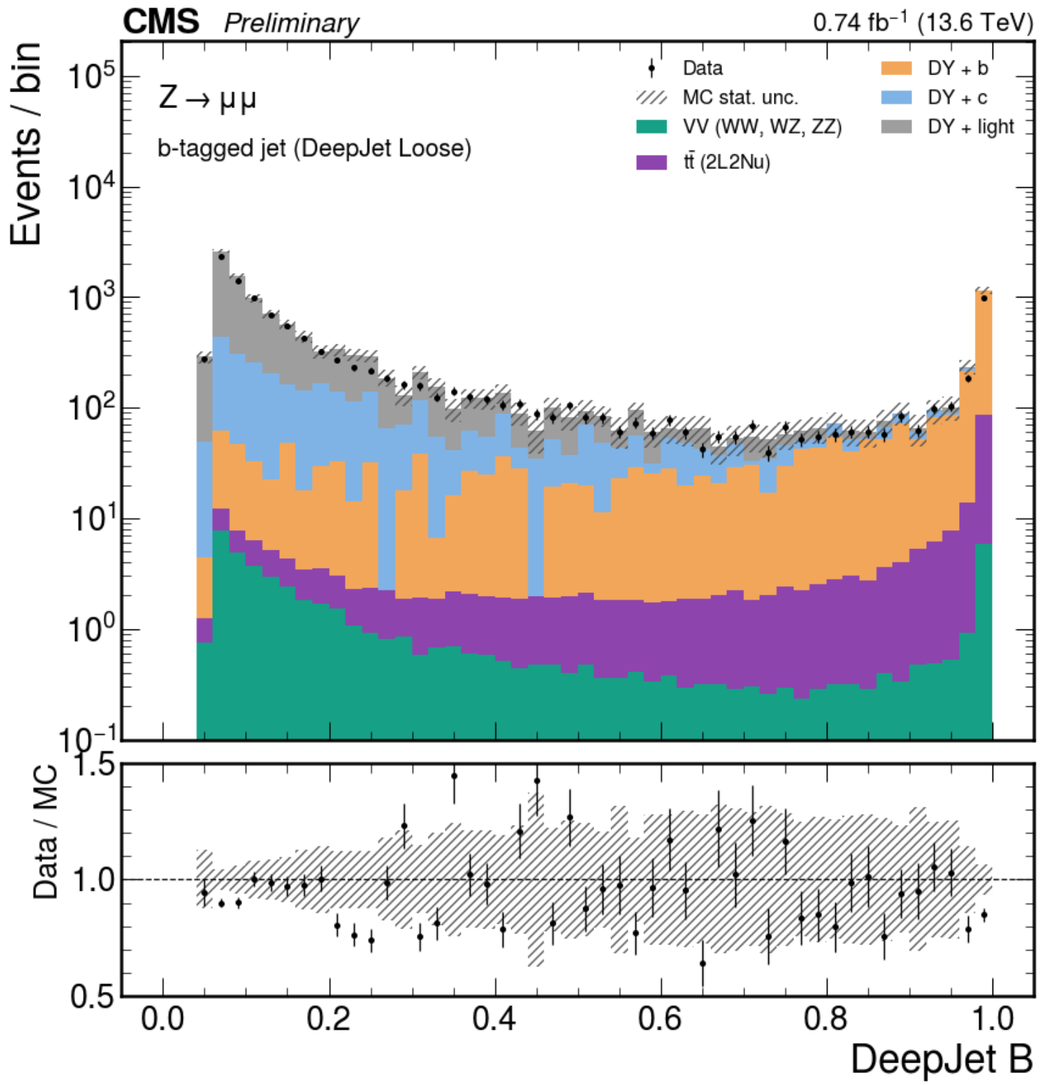


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet DeepJet B score (b vs everything)

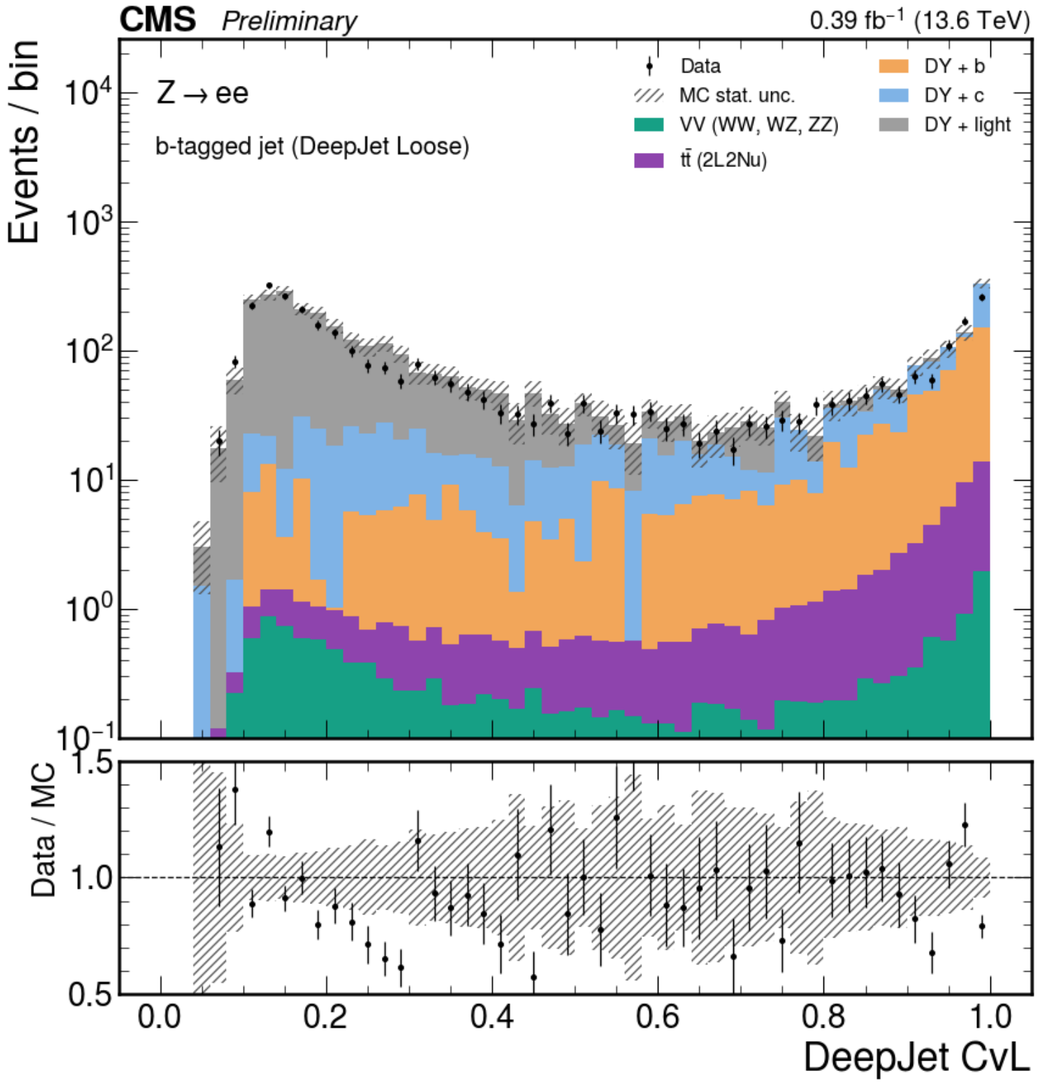
The score used for the selection: only jets above 0.0583 enter this region.

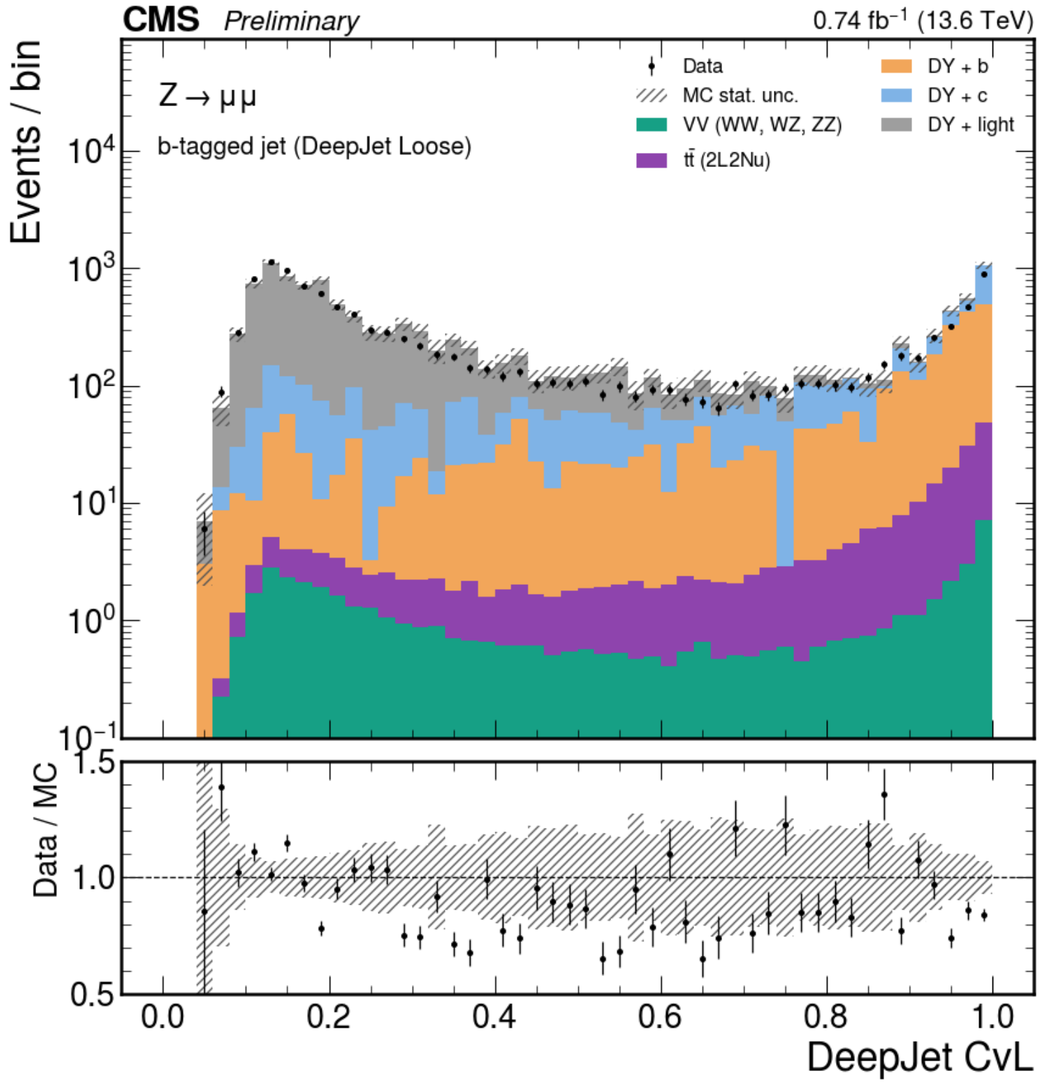




Leading-jet DeepJet CvL score (c vs light)

Not used in this b-tag selection; shown for comparison.

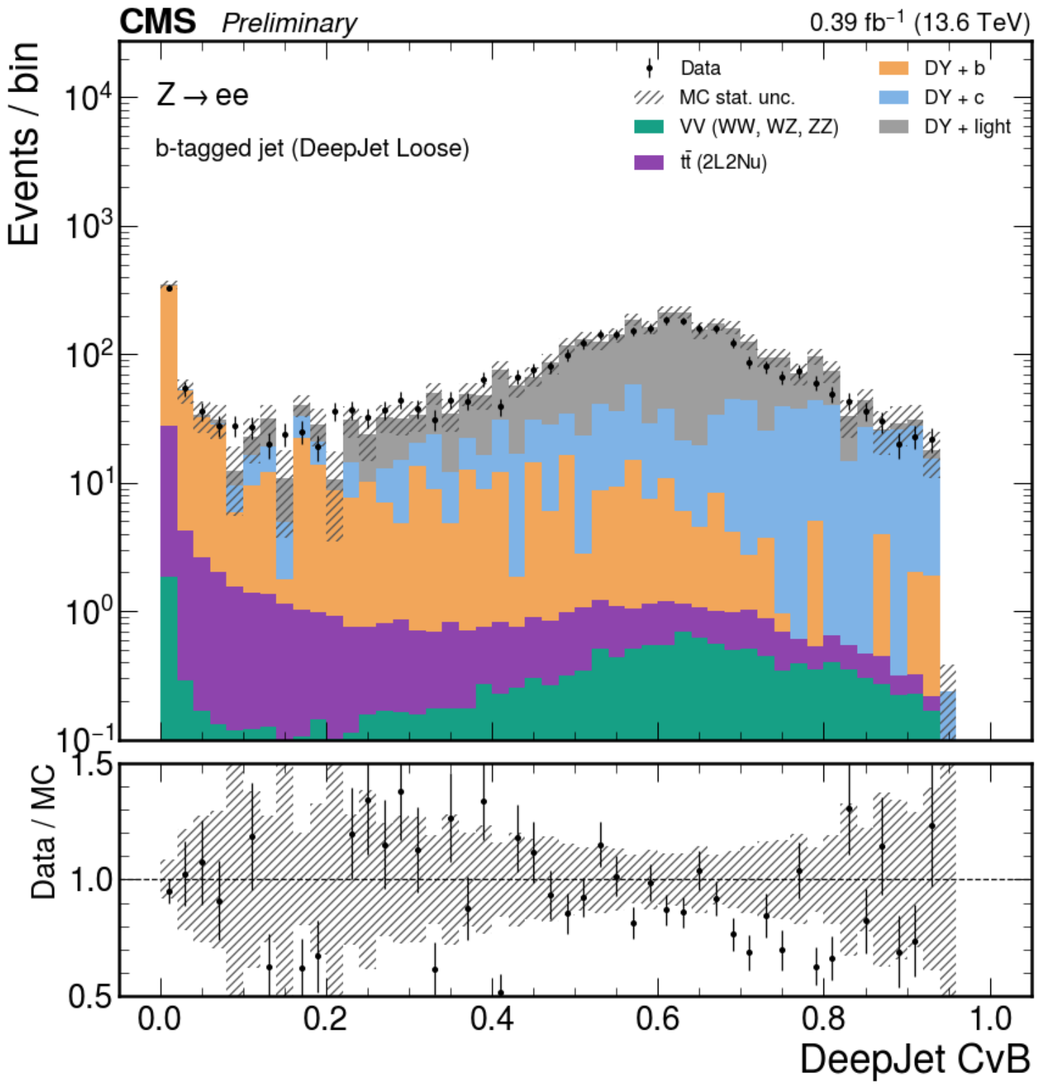


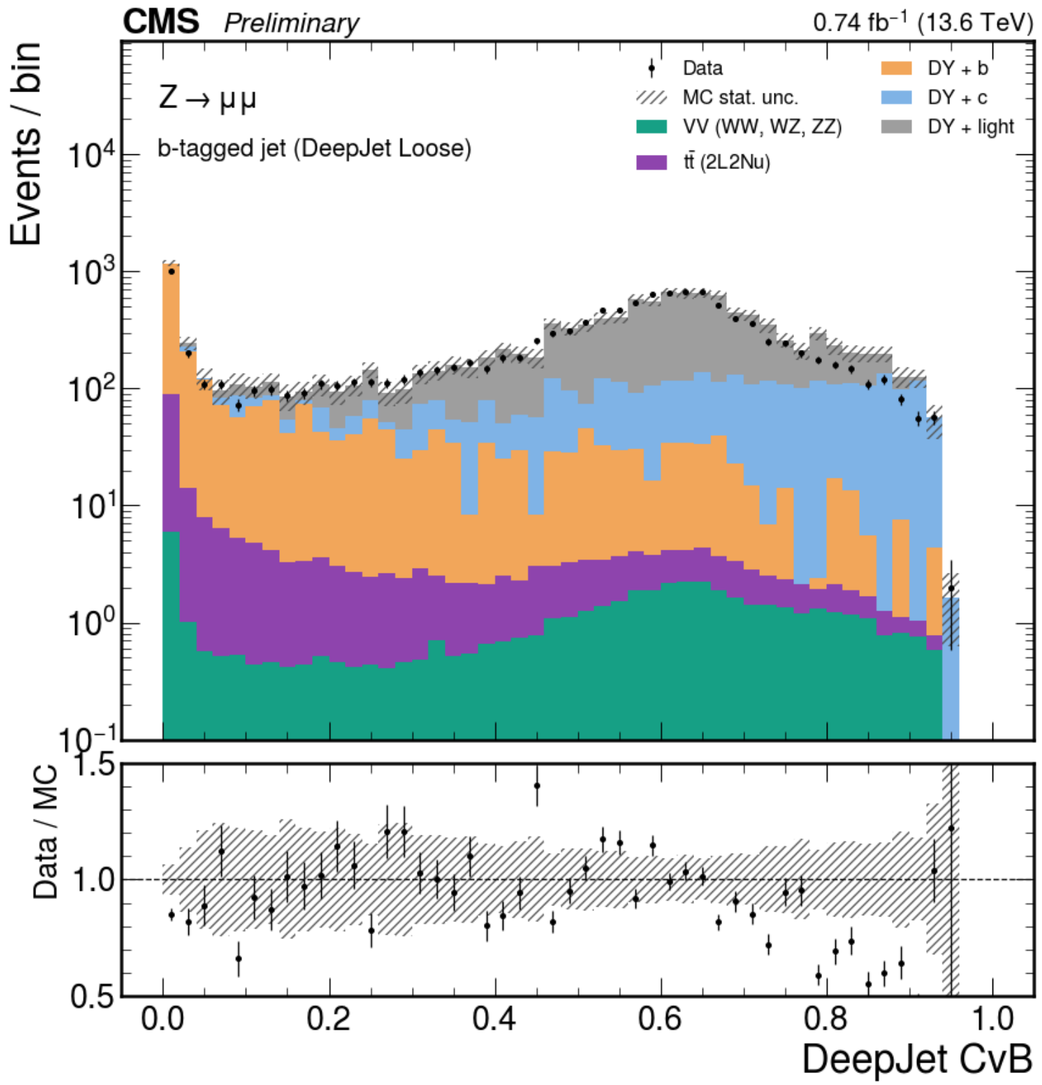


DeepJet · Loose b-tag (B > 0.0583)

Leading-jet DeepJet CvB score (c vs b)

Not used in this b-tag selection. Low values = b-like.

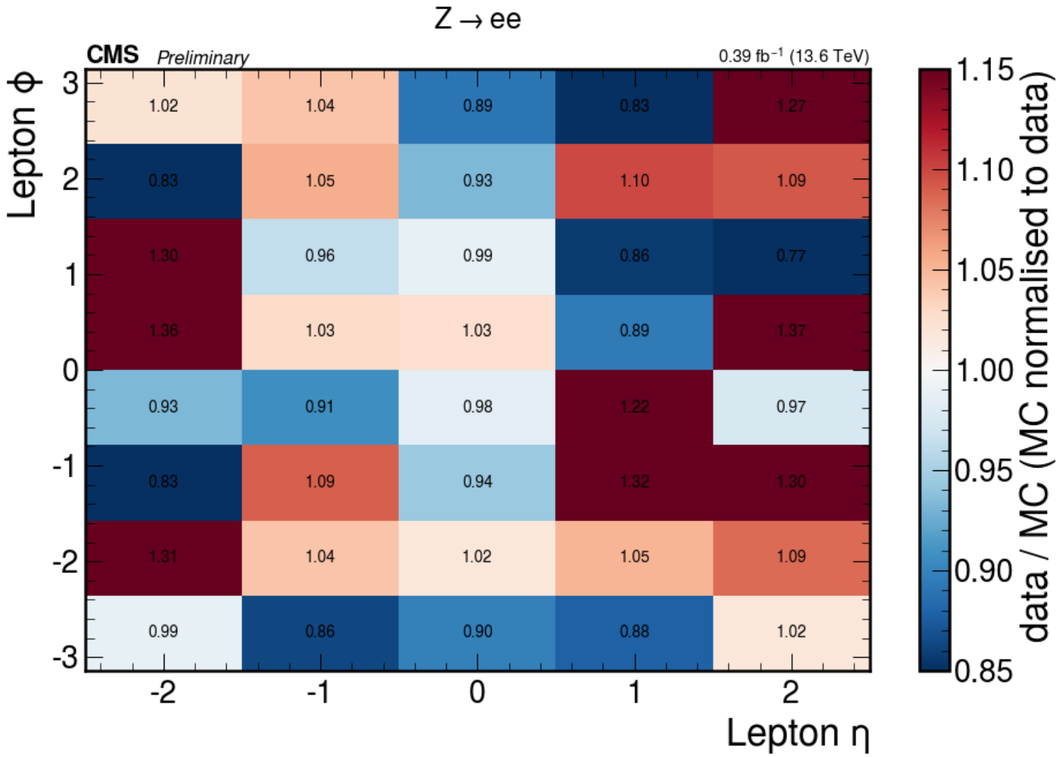


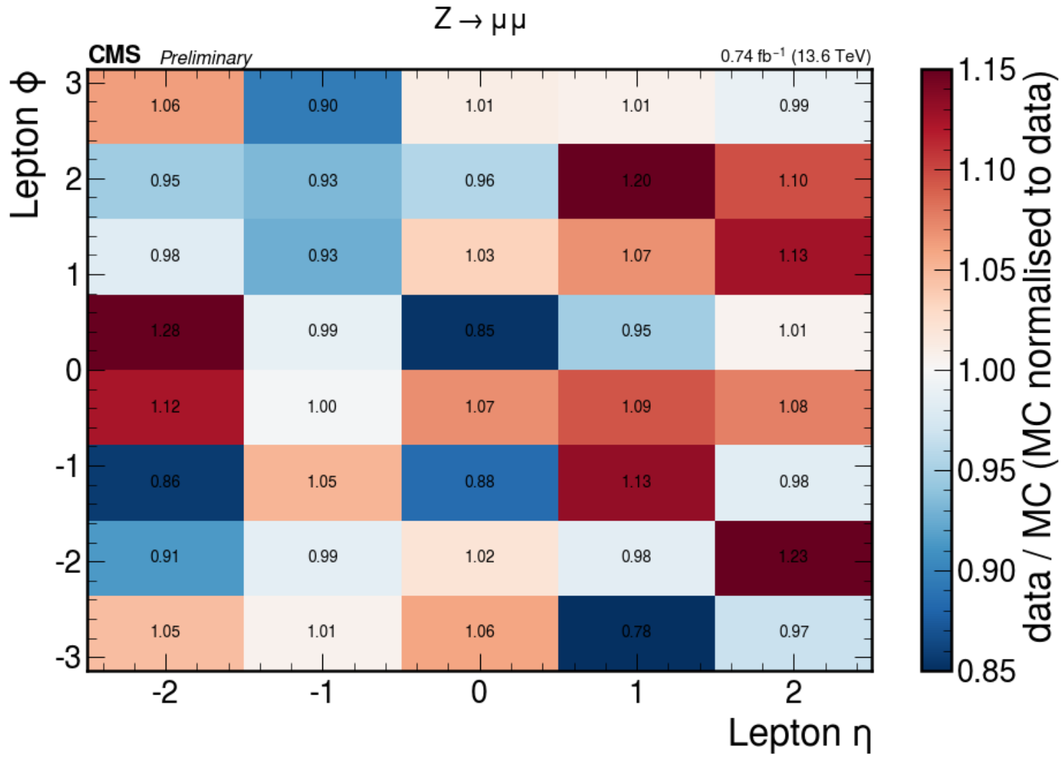


DeepJet · Loose b-tag (B > 0.0583)

Lepton η–φ map: data / MC

Both Z leptons, MC normalised to data in each map, cells of 1.0 in η × π/4 in φ. Colour scale from 0.85 to 1.15; the number in each cell is data/MC. Few events per cell in a tagged region.





RobustParT (Robust Particle Transformer), Loose b-tag

Leading jet RobustParT B score > 0.0849; MC weighted with the robustParticleTransformer BTV scale factors.

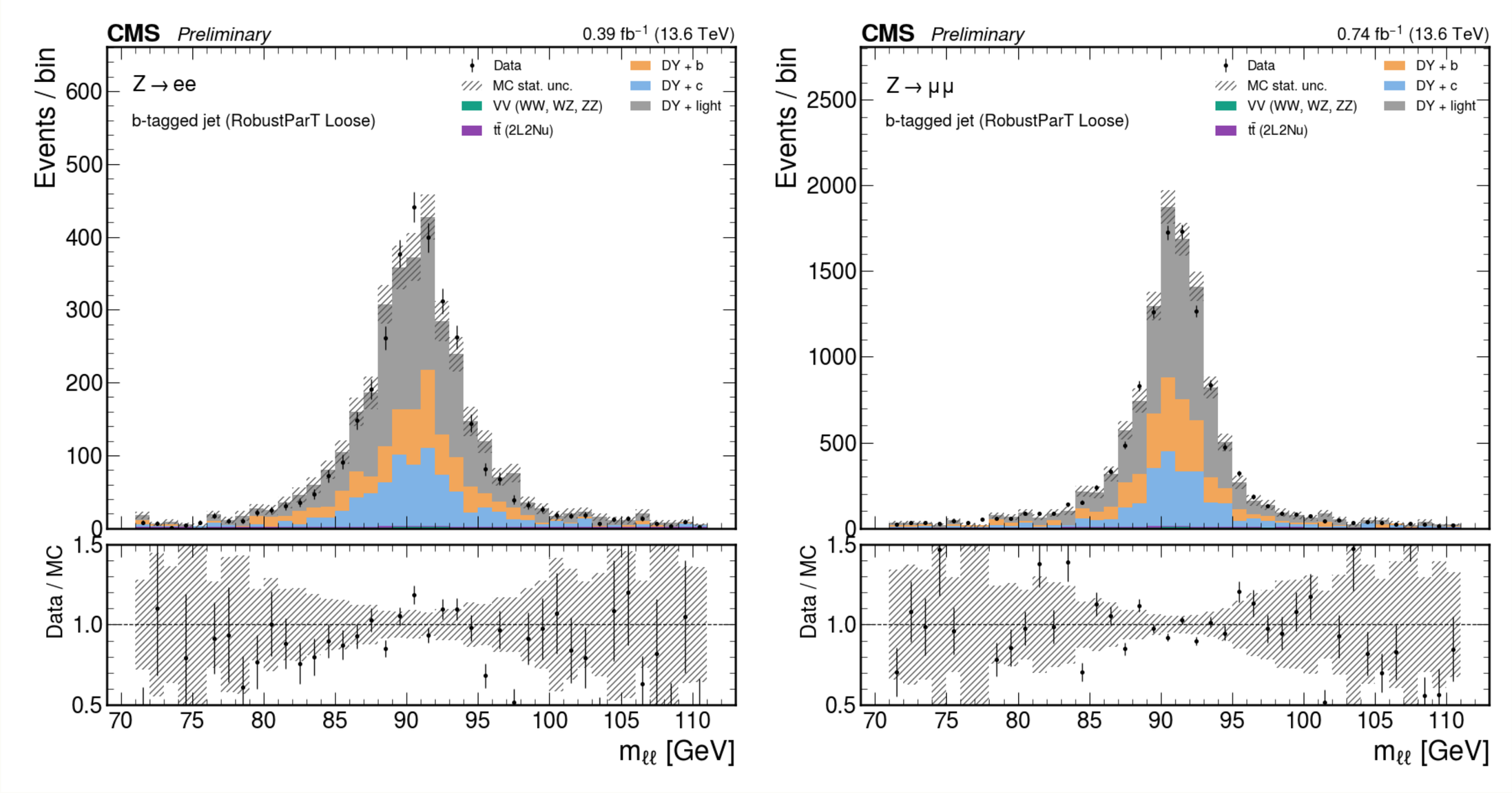
	data/MC	DY+b	DY+c	DY+light	tt̄	tagged MC	tagged data
Z→ee	0.962 ± 0.031	20%	22%	55%	2%	17.2%	17.4%
Z→μμ	0.981 ± 0.023	22%	21%	54%	2%	17.2%	17.9%

Flavour shares are of all MC. Tagged = fraction of Z + jet events whose leading jet passes the tag. Statistical errors only.

42 plots follow, in this order: Z and leptons, jets, Z–jet, MET and vertices, generator Z pT, the RobustParT scores, the lepton η–φ map. Left Z→ee, right Z→μμ.

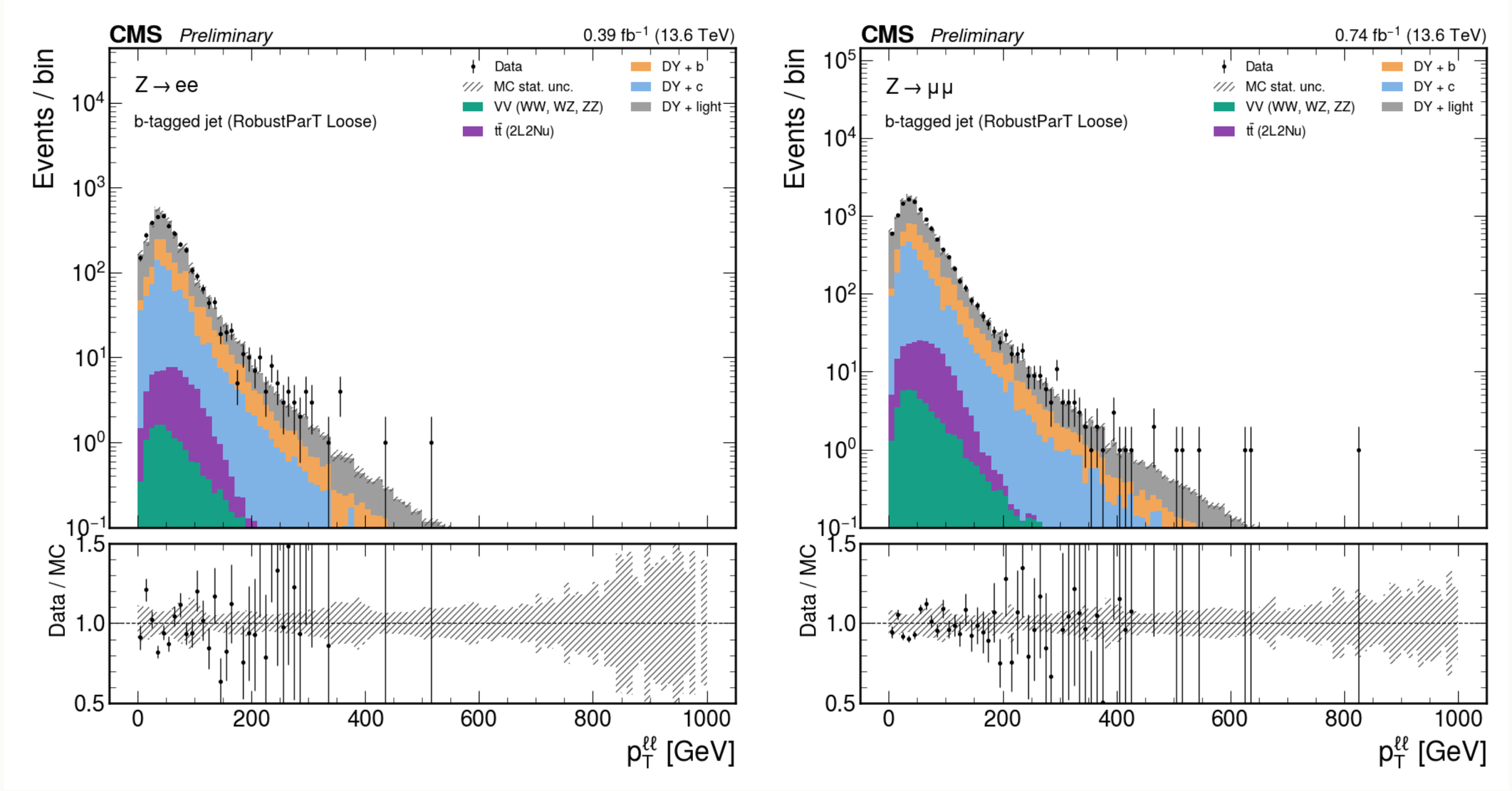
Mass of the lepton pair, m(ℓℓ)

The Z peak, linear scale. Selection: 71–111 GeV. Tests the lepton energy scale and resolution.



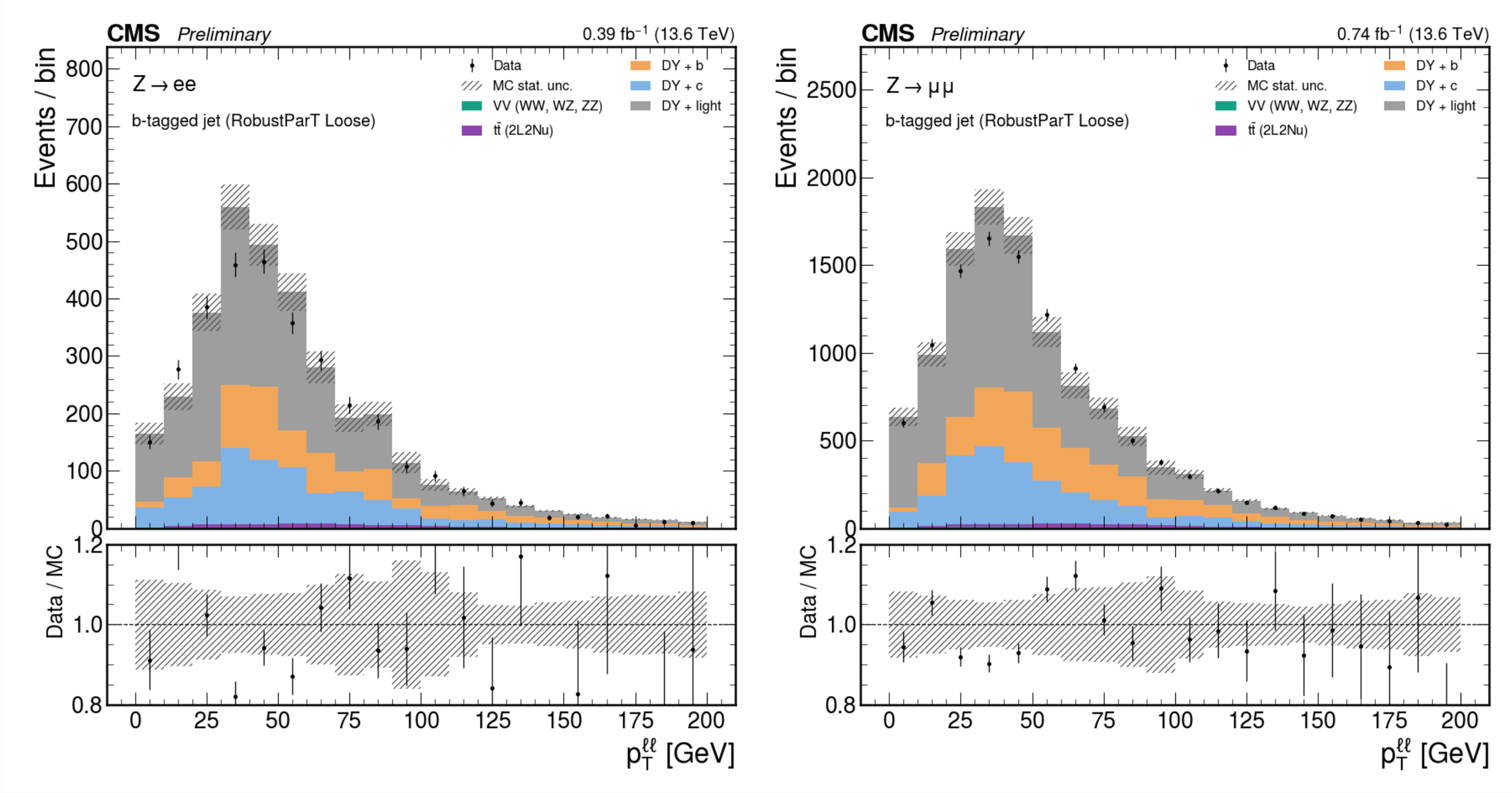
Transverse momentum of the lepton pair, pT(ℓℓ) = Z pT

Log scale up to 1 TeV. This is the distribution with the pT trend discussed in the follow-up slides (low at 30–60 GeV, high at 100–150 GeV).

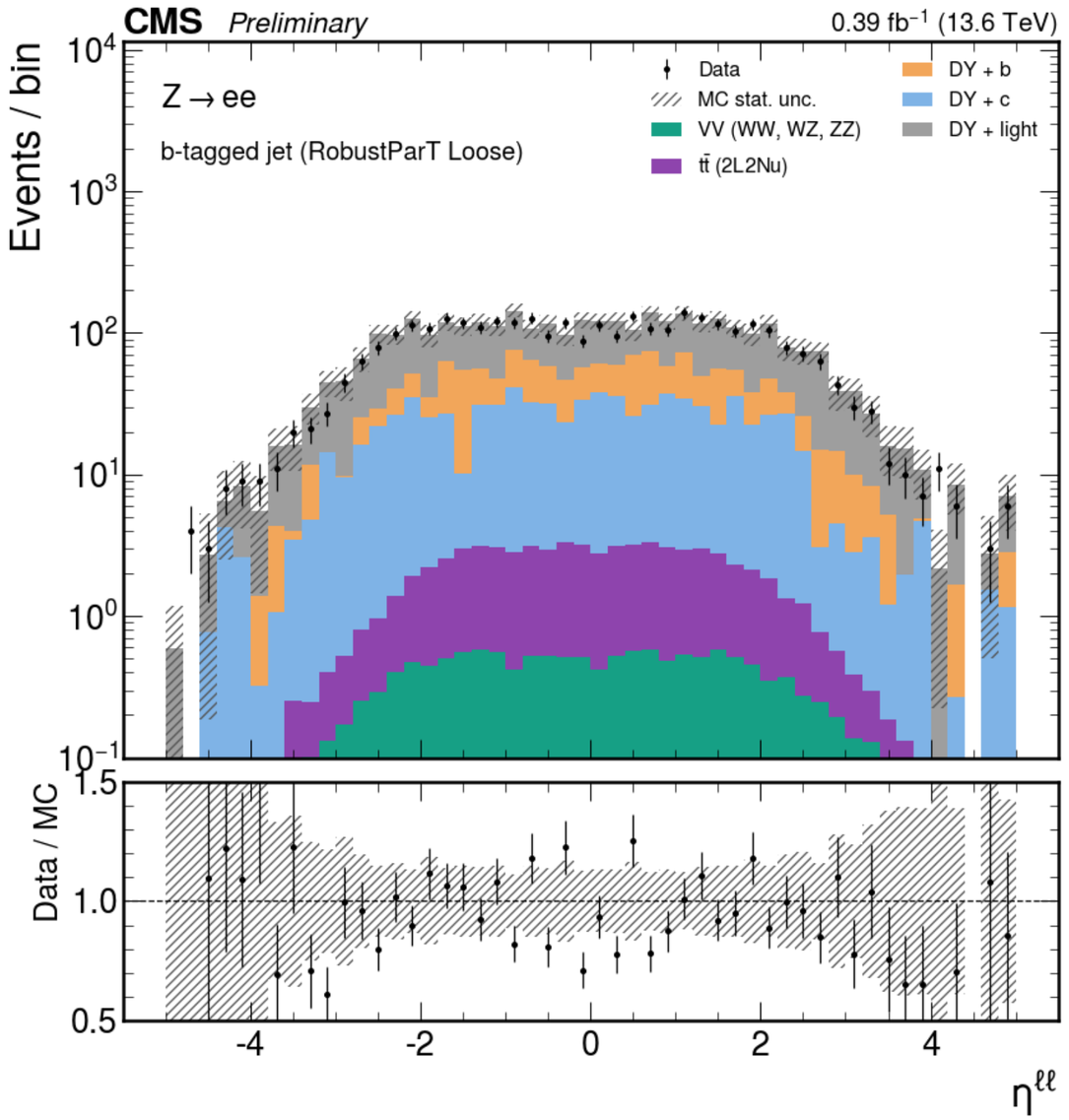


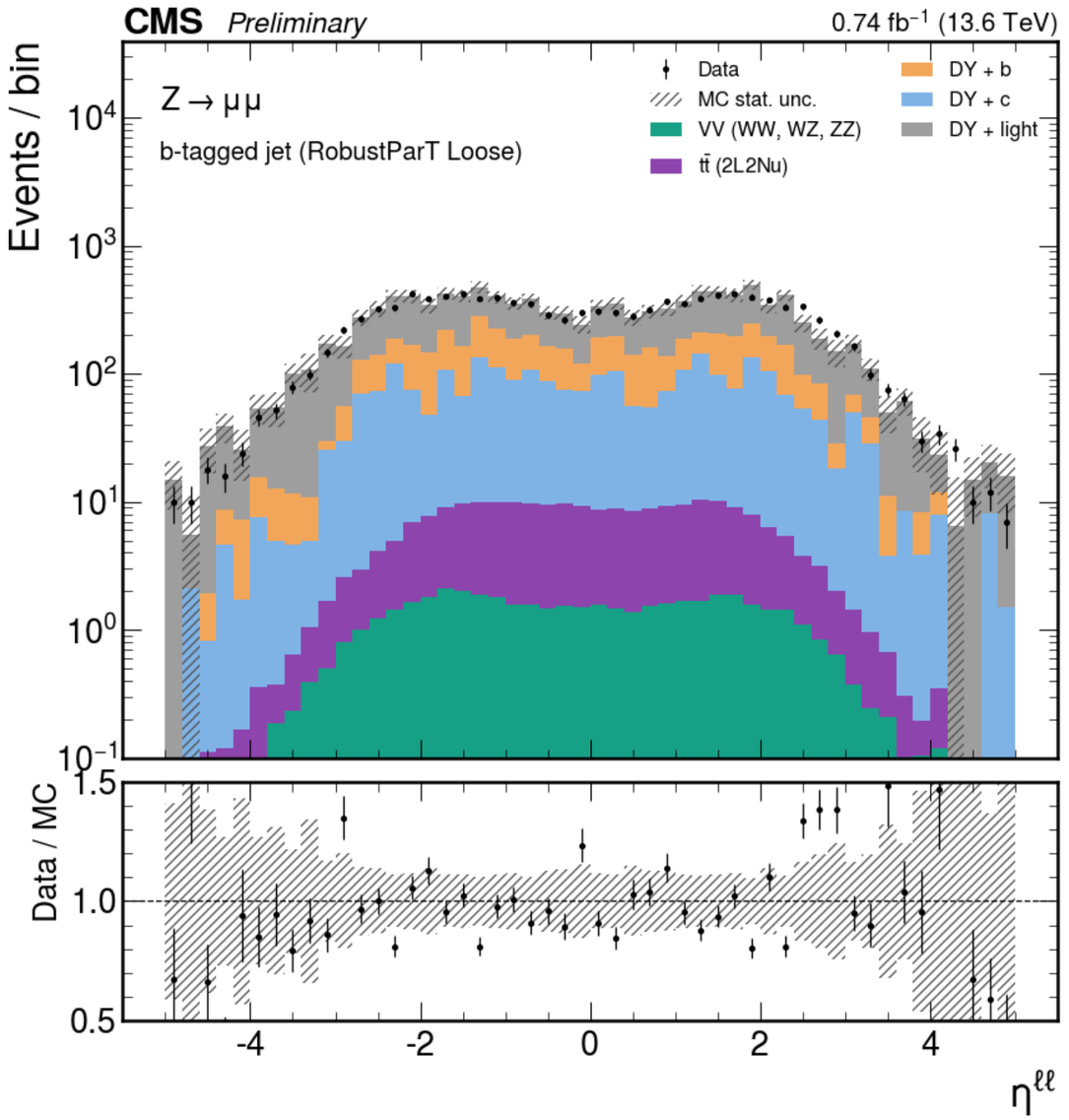
Z pT, zoom on 0–200 GeV

Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.



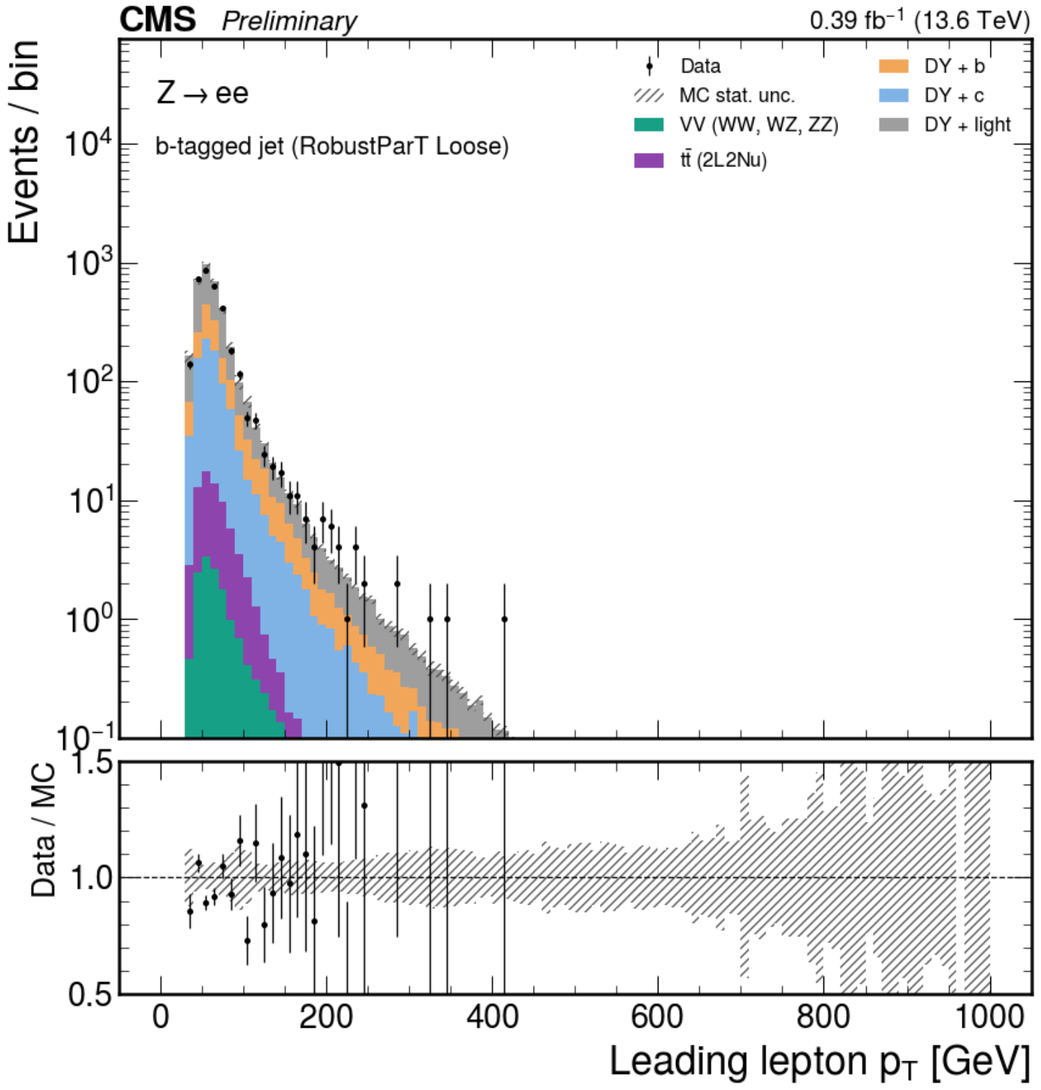
Pseudorapidity of the lepton pair, η(ℓℓ)

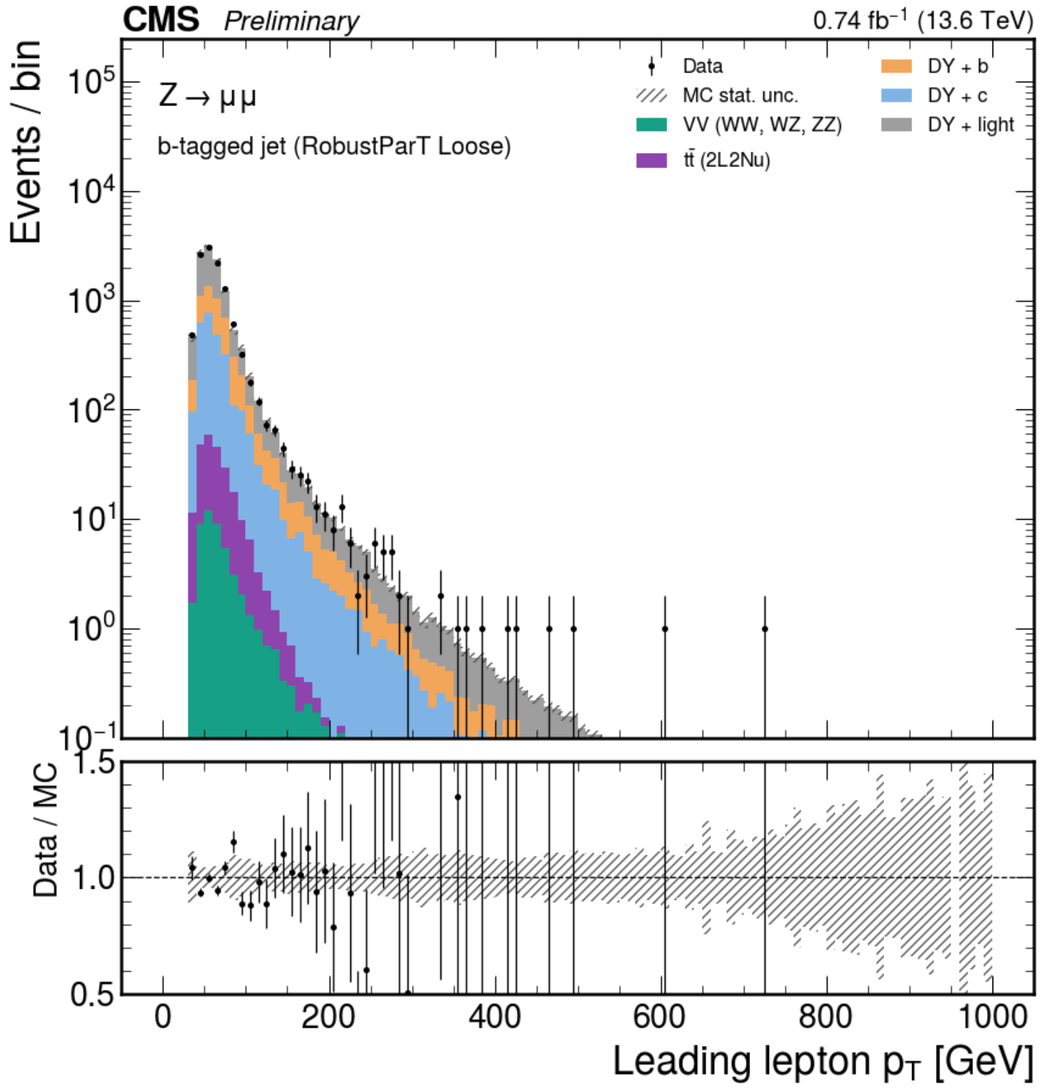




Leading-lepton pT

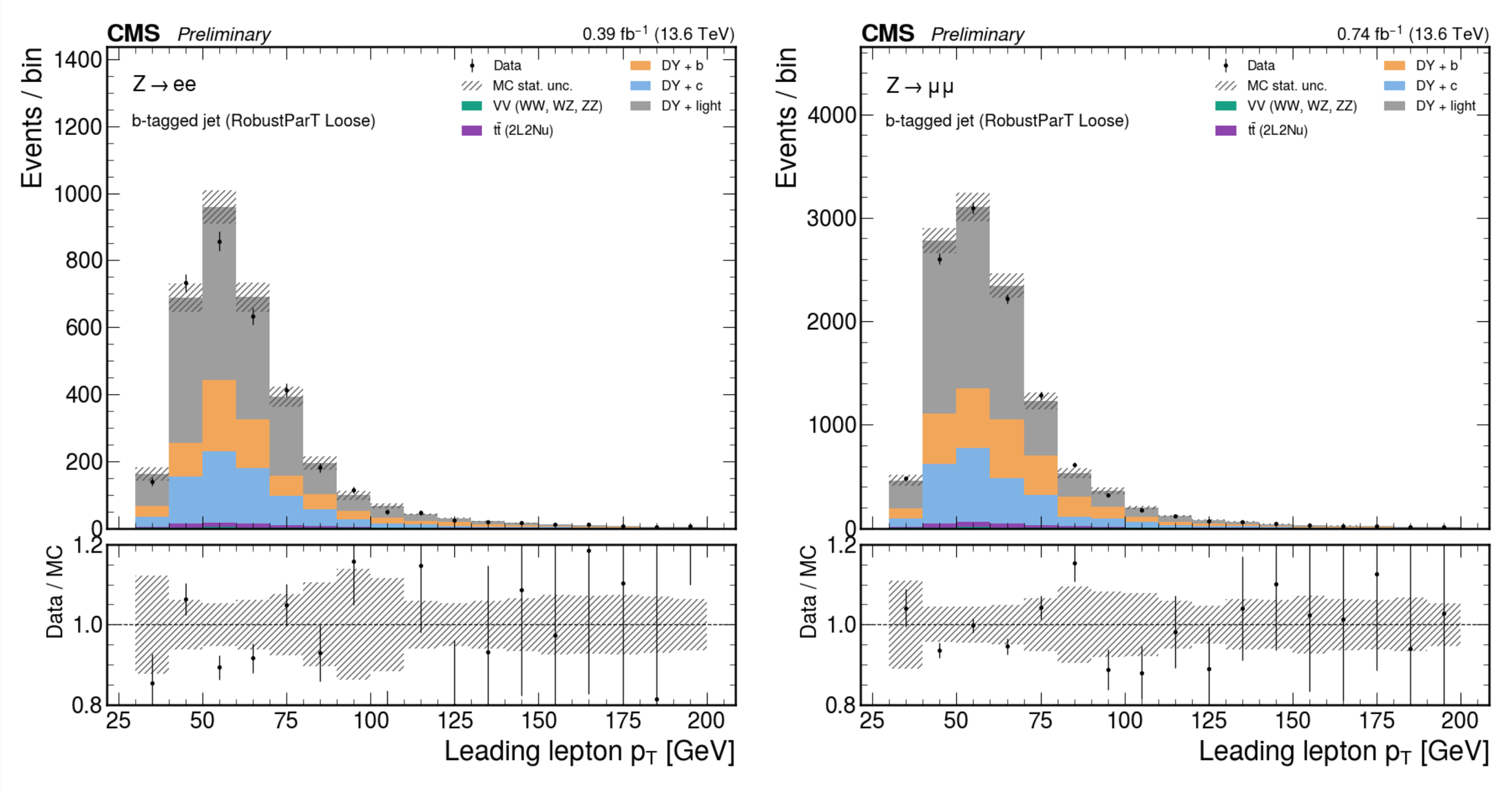
The selection requires pT > 35 GeV for the leading lepton (on the trigger plateau).





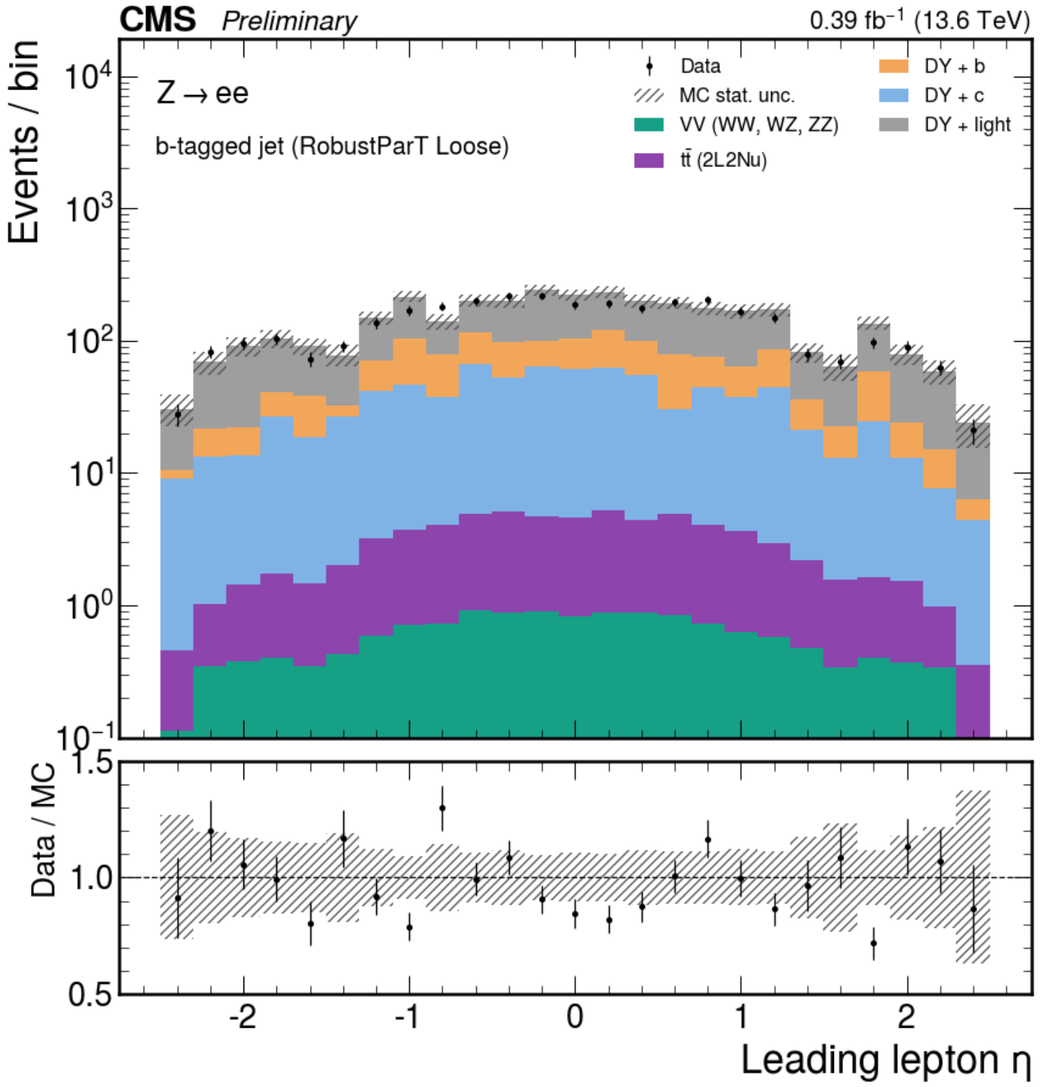
Leading-lepton pT, zoom on 30–200 GeV

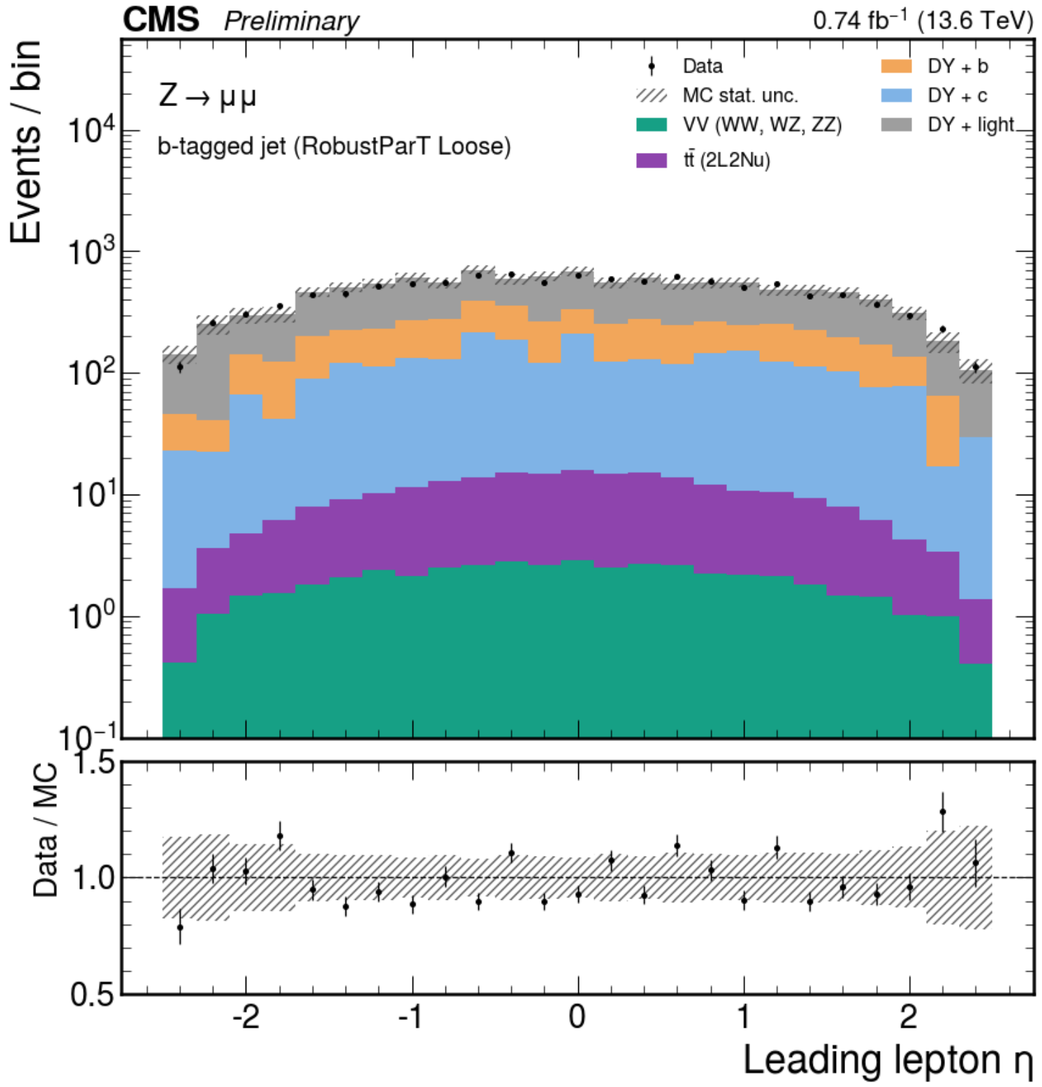
Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.



Leading-lepton η

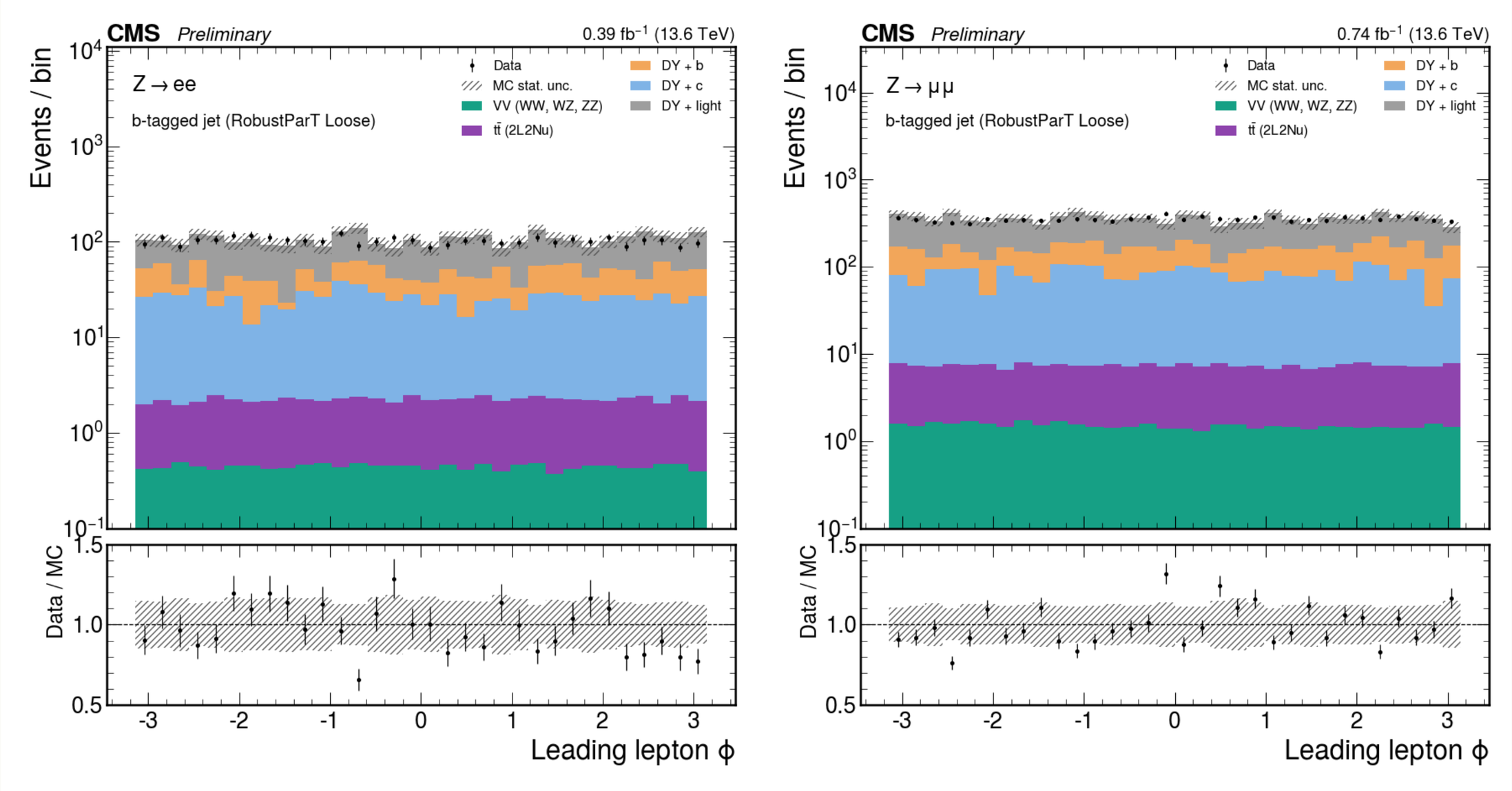
|η| < 2.4. For electrons the barrel–endcap gap (1.44 < |η| < 1.57) is removed, so that bin is low.





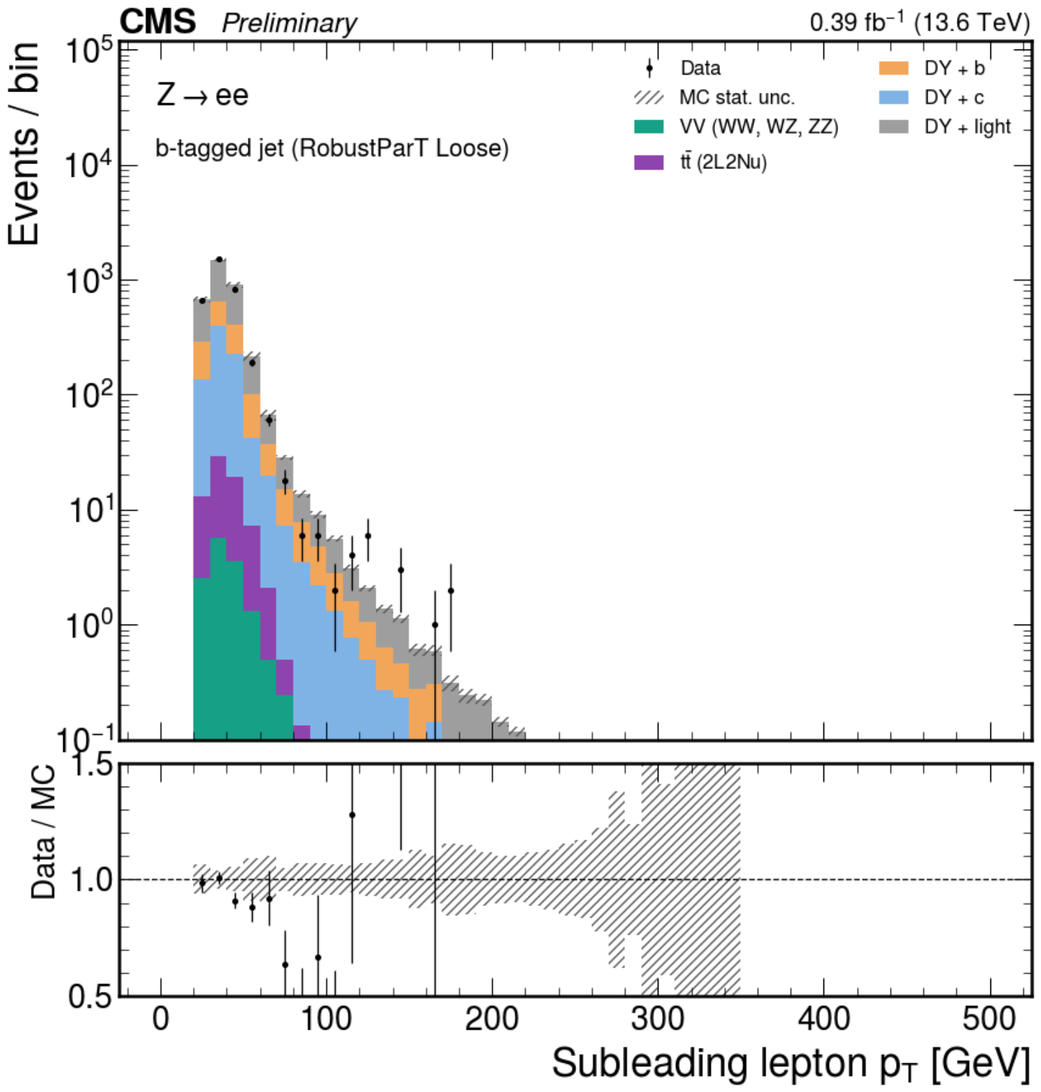
Leading-lepton φ

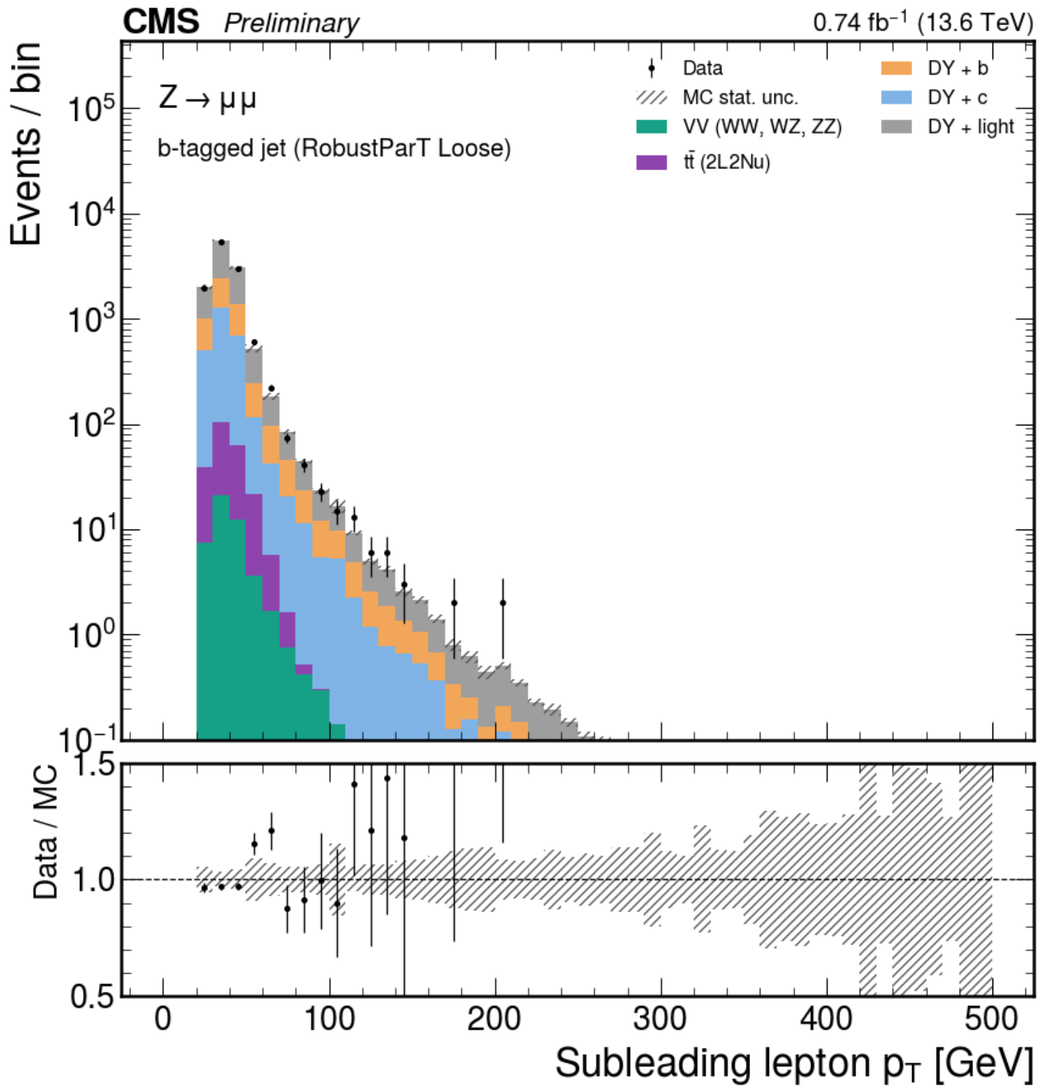
Expected to be flat; dips would point to detector problems.



Second-lepton pT

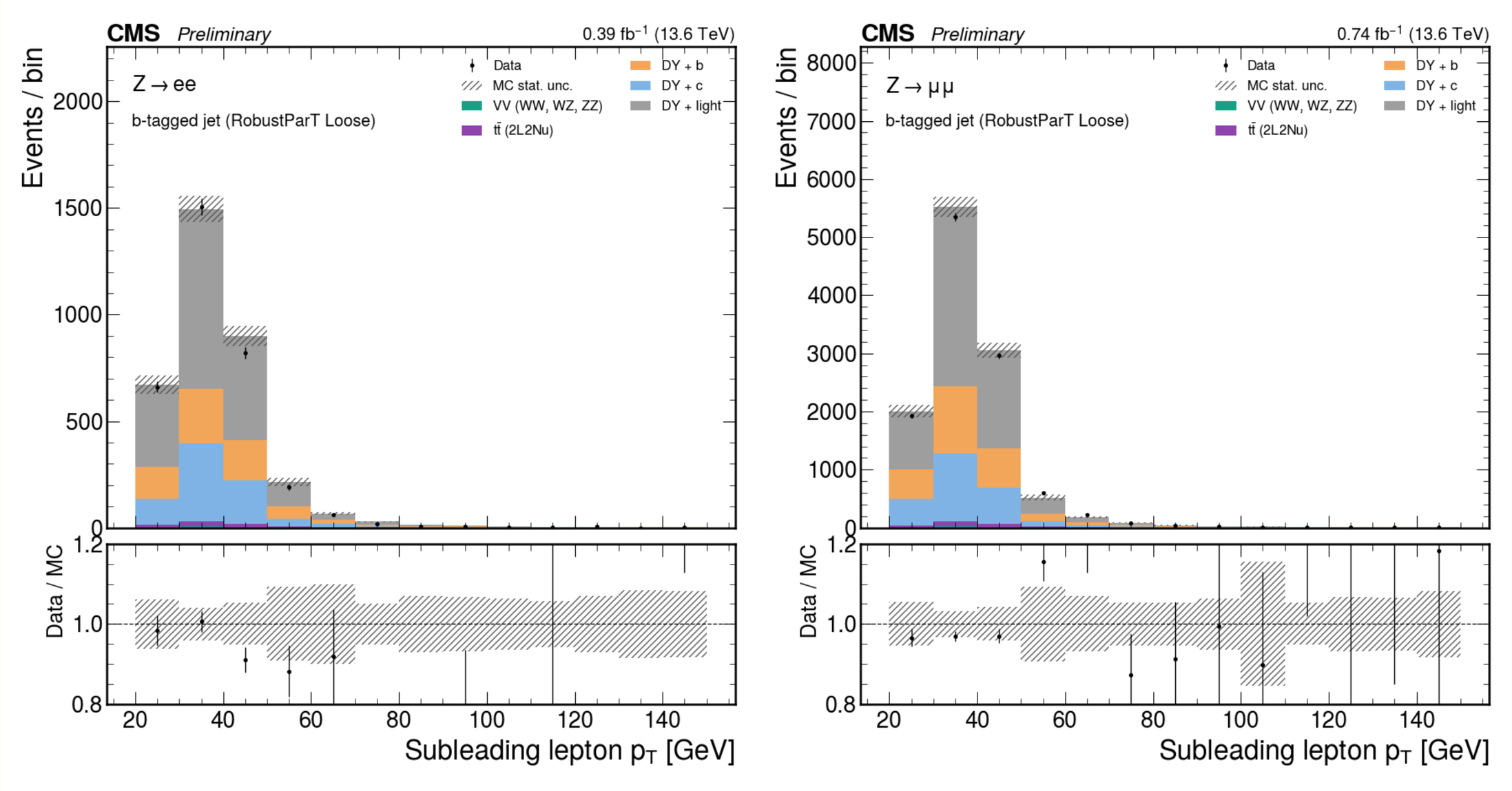
The selection requires pT > 25 GeV for electrons and > 26 GeV for muons.



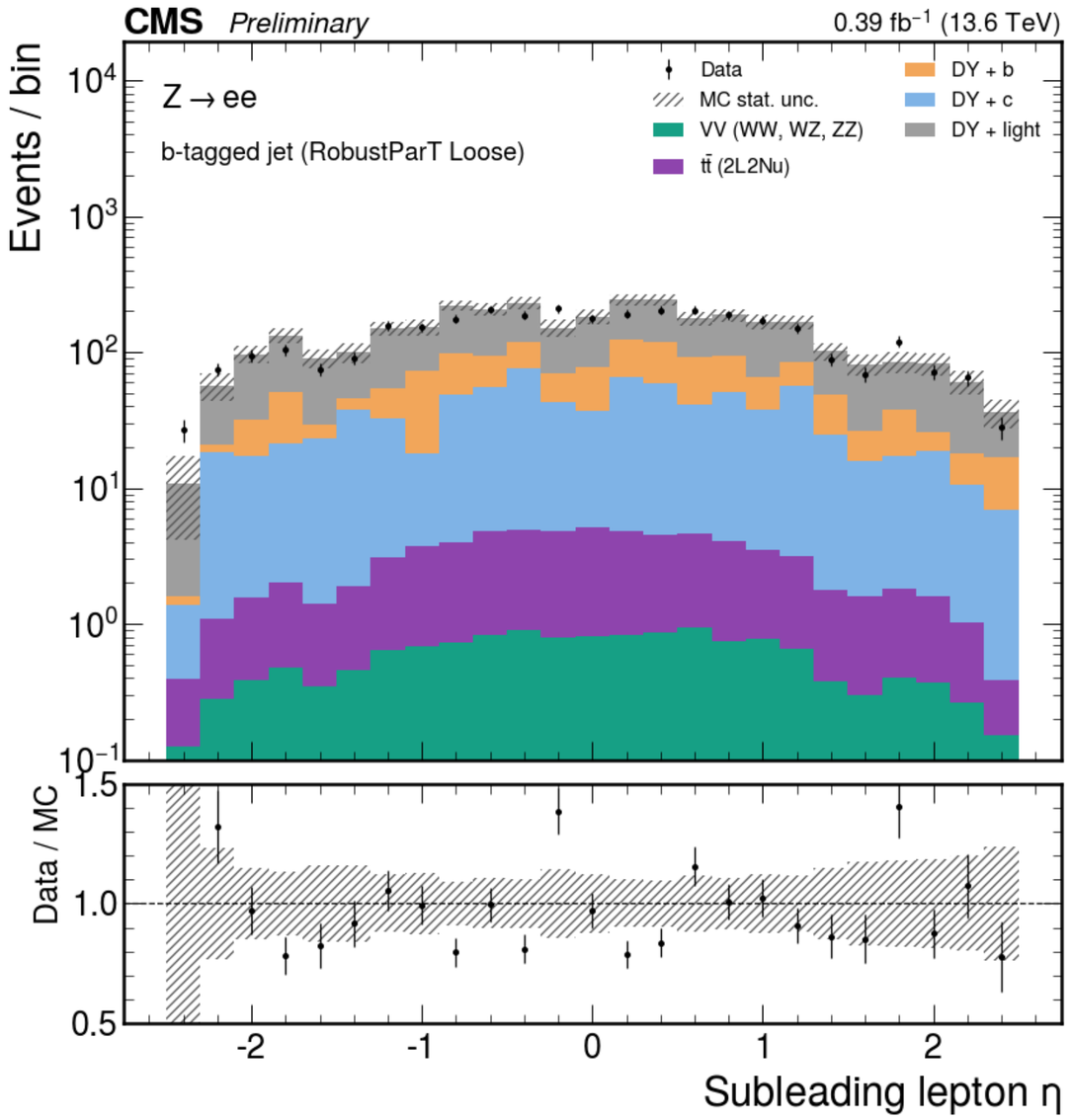


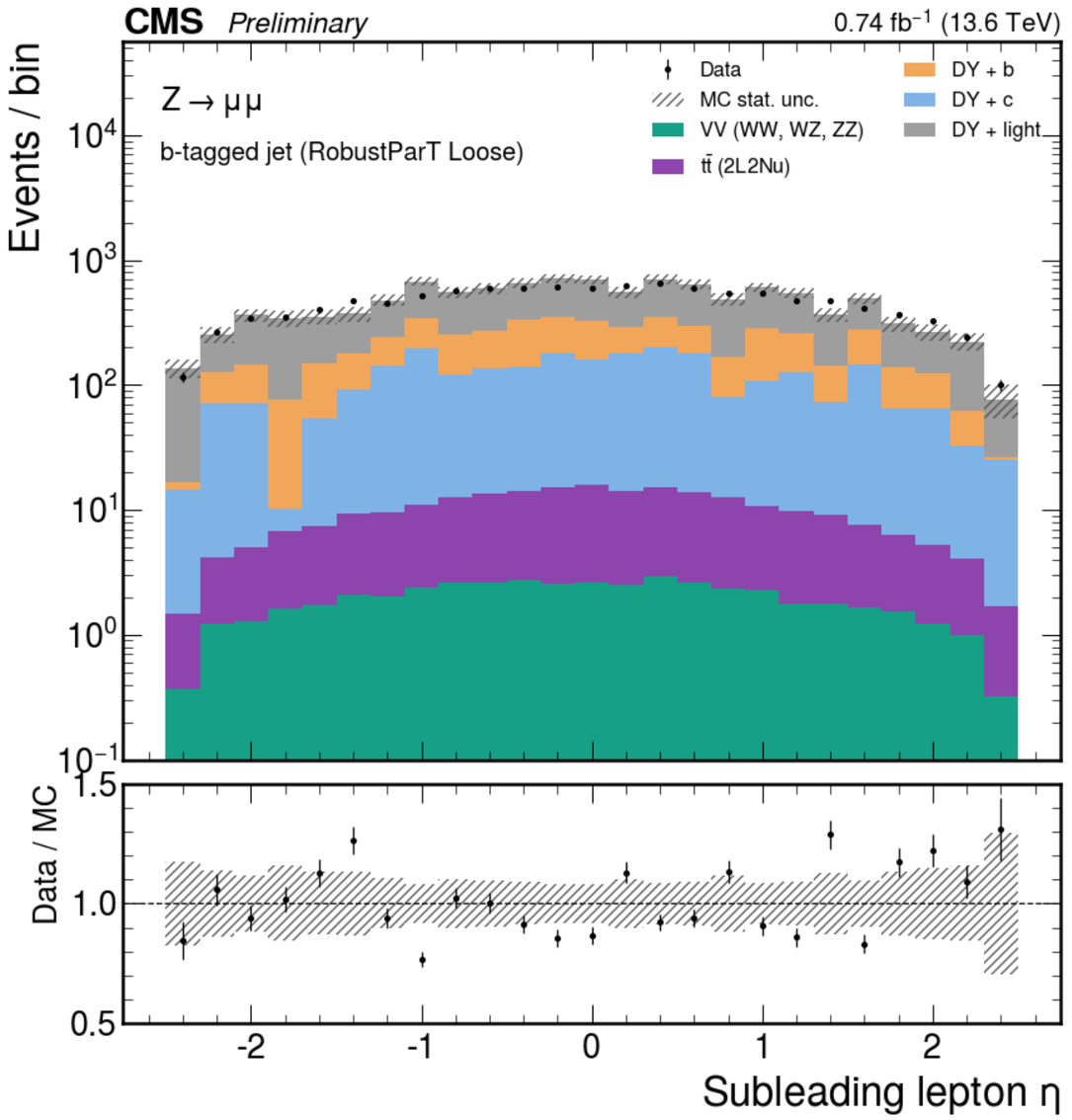
Second-lepton pT, zoom on 20–150 GeV

Zoom: 20–150, linear scale, data/MC from 0.8 to 1.2.

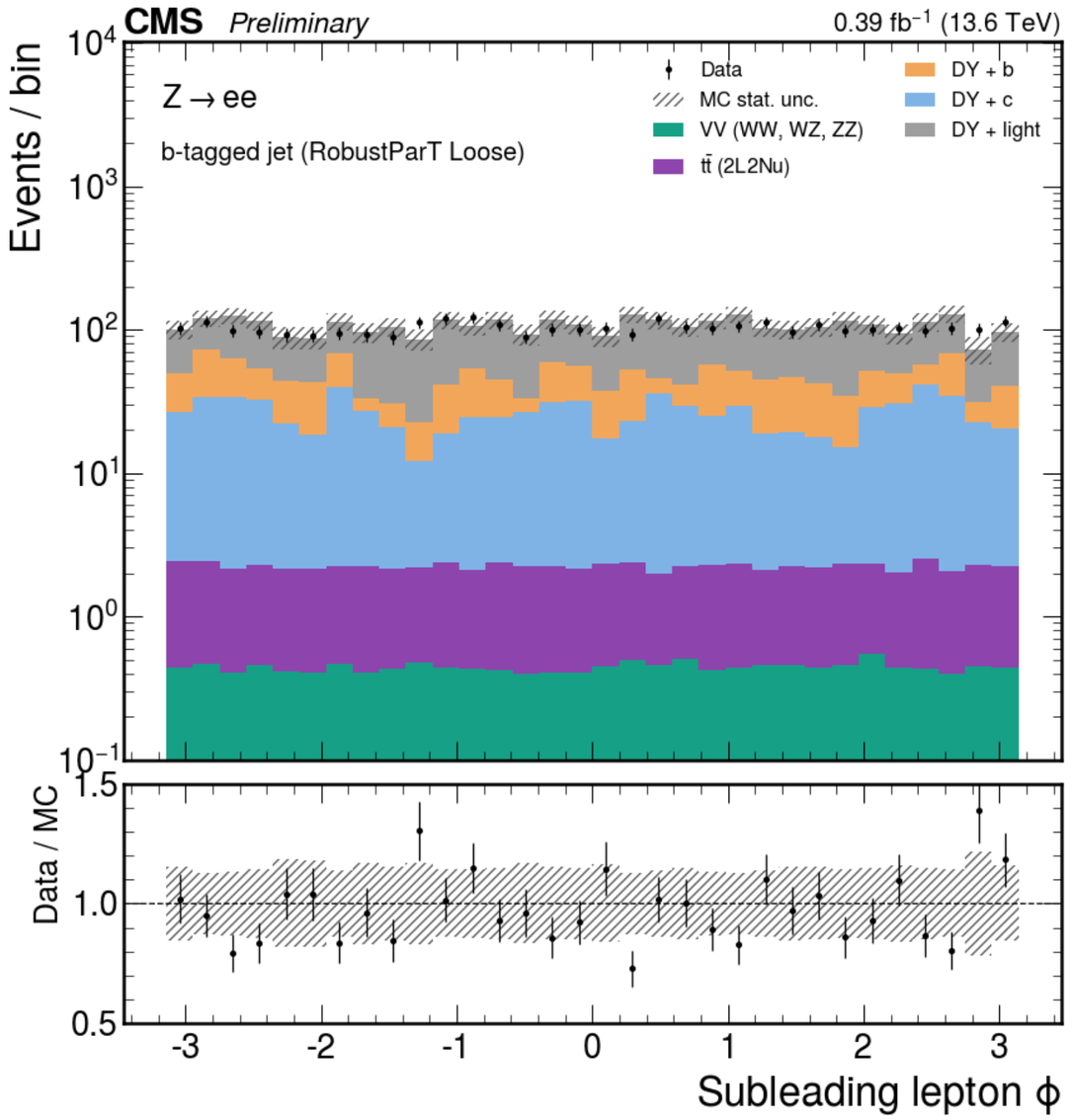


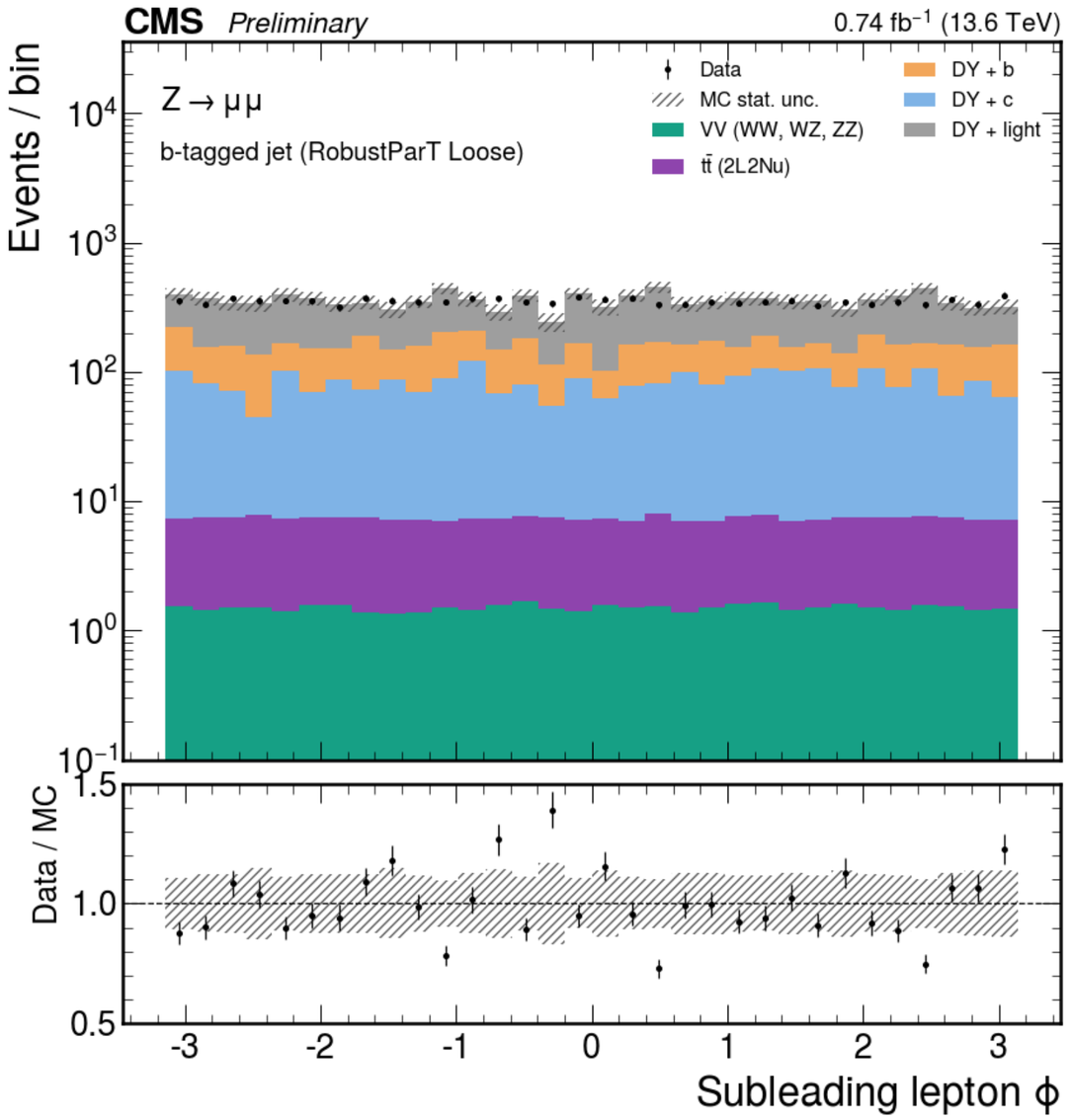
Second-lepton η





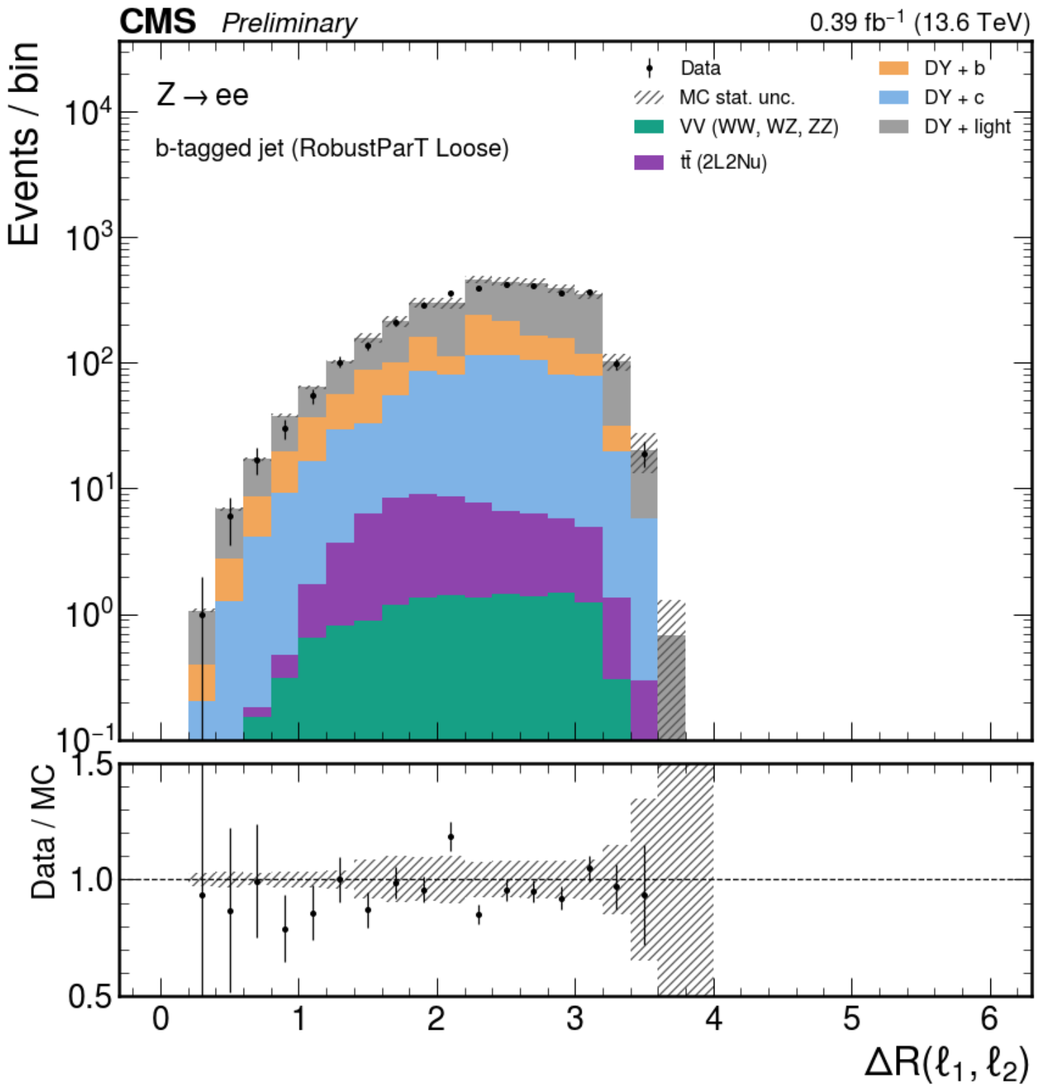
Second-lepton φ

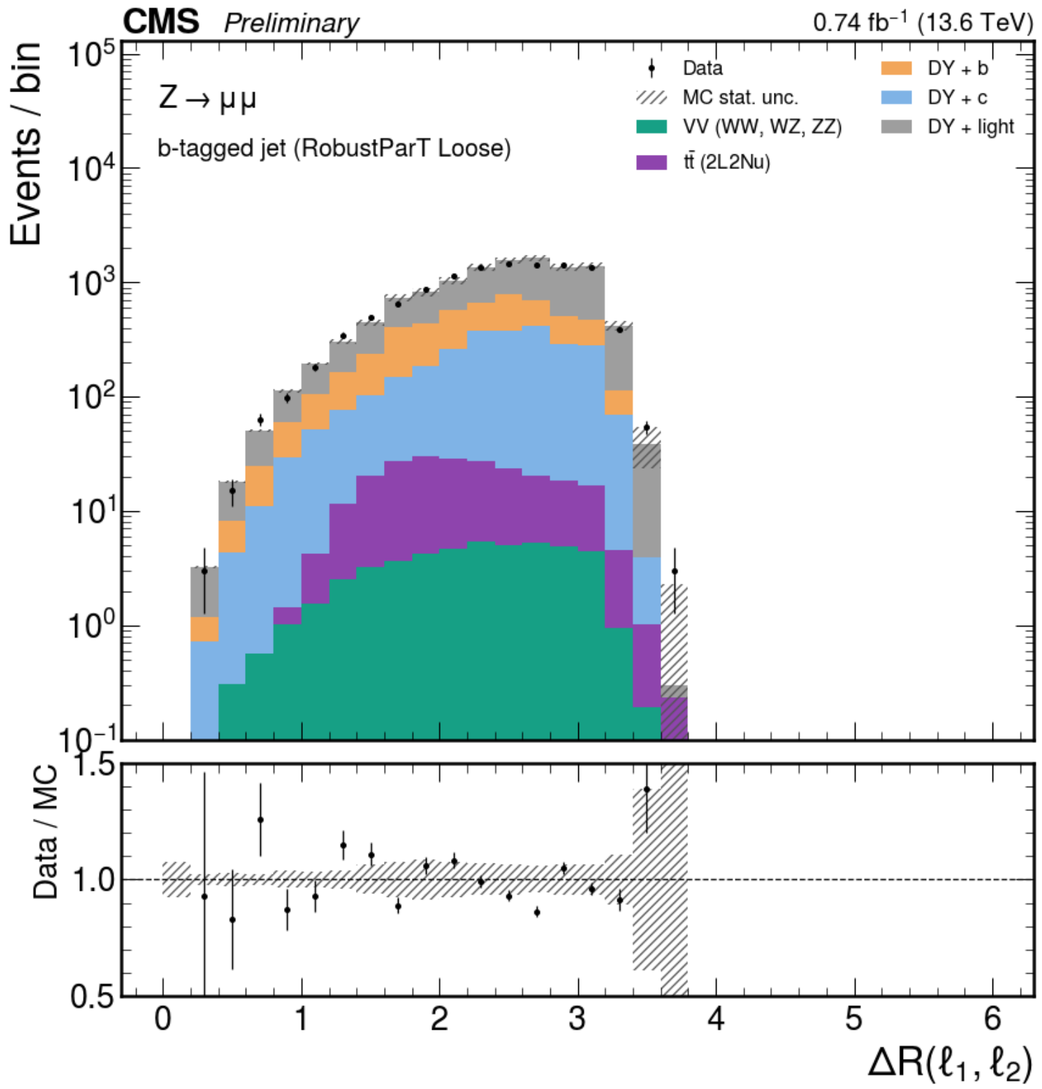




Angular distance between the two leptons, ΔR(ℓℓ)

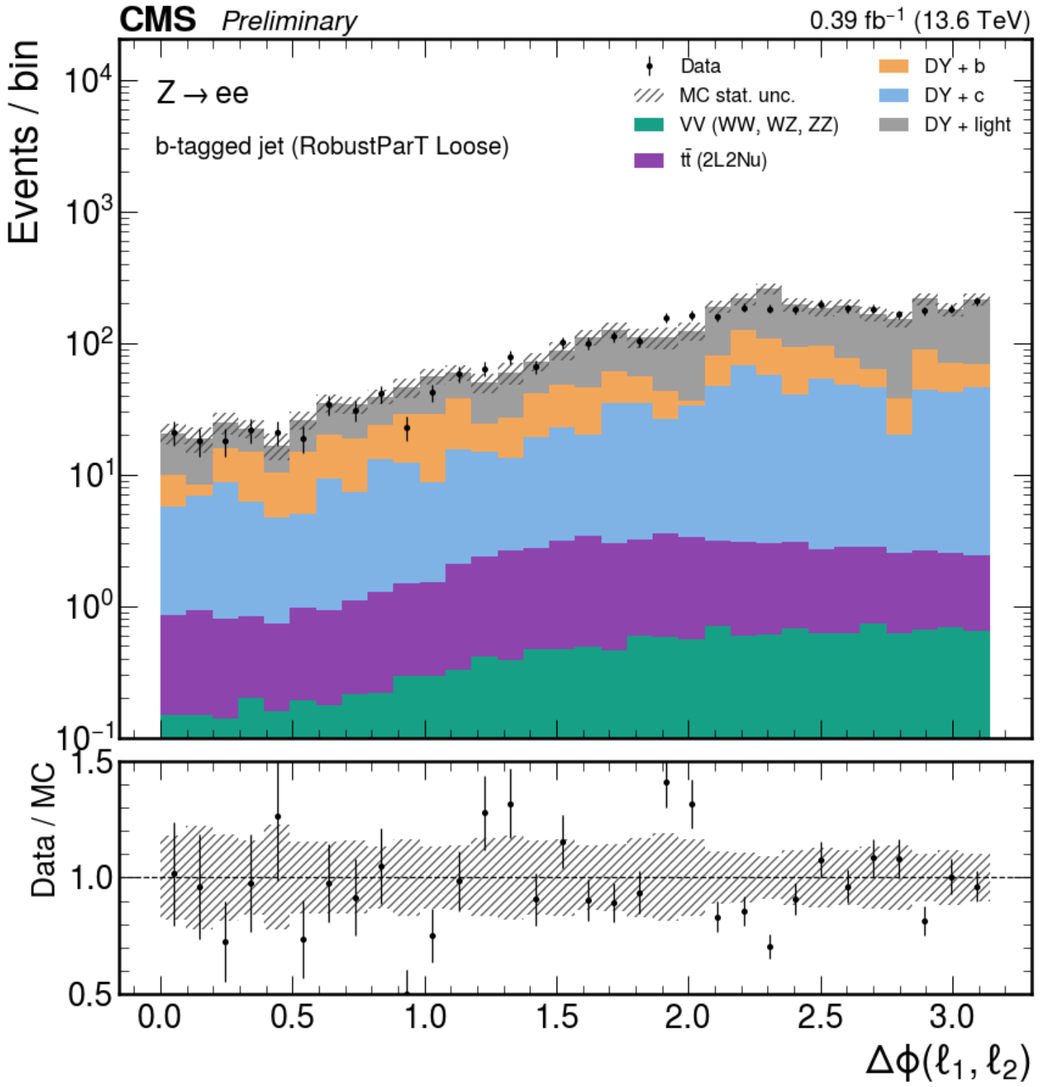
ΔR = √(Δη² + Δφ²). Small ΔR = boosted Z.

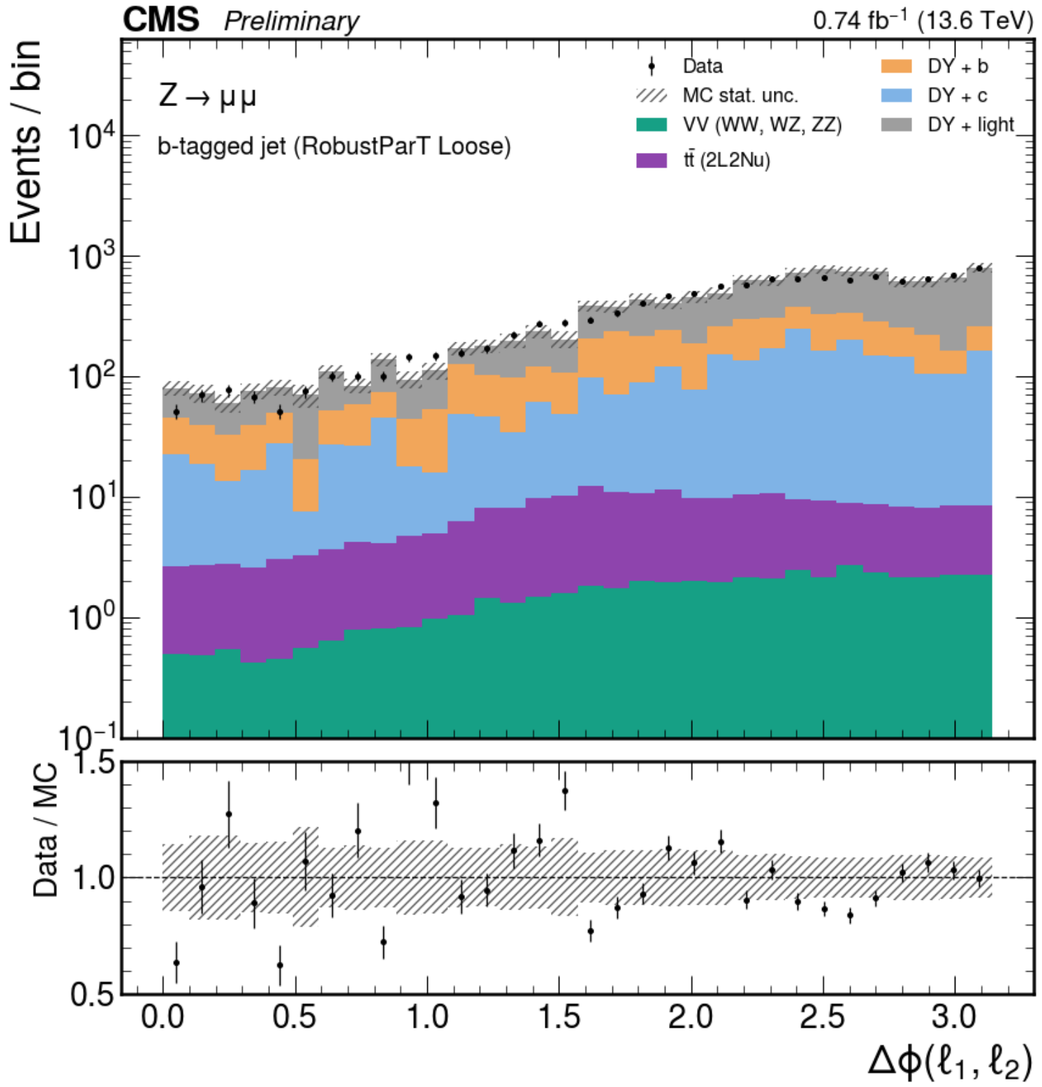




Azimuthal angle between the two leptons, Δφ(ℓℓ)

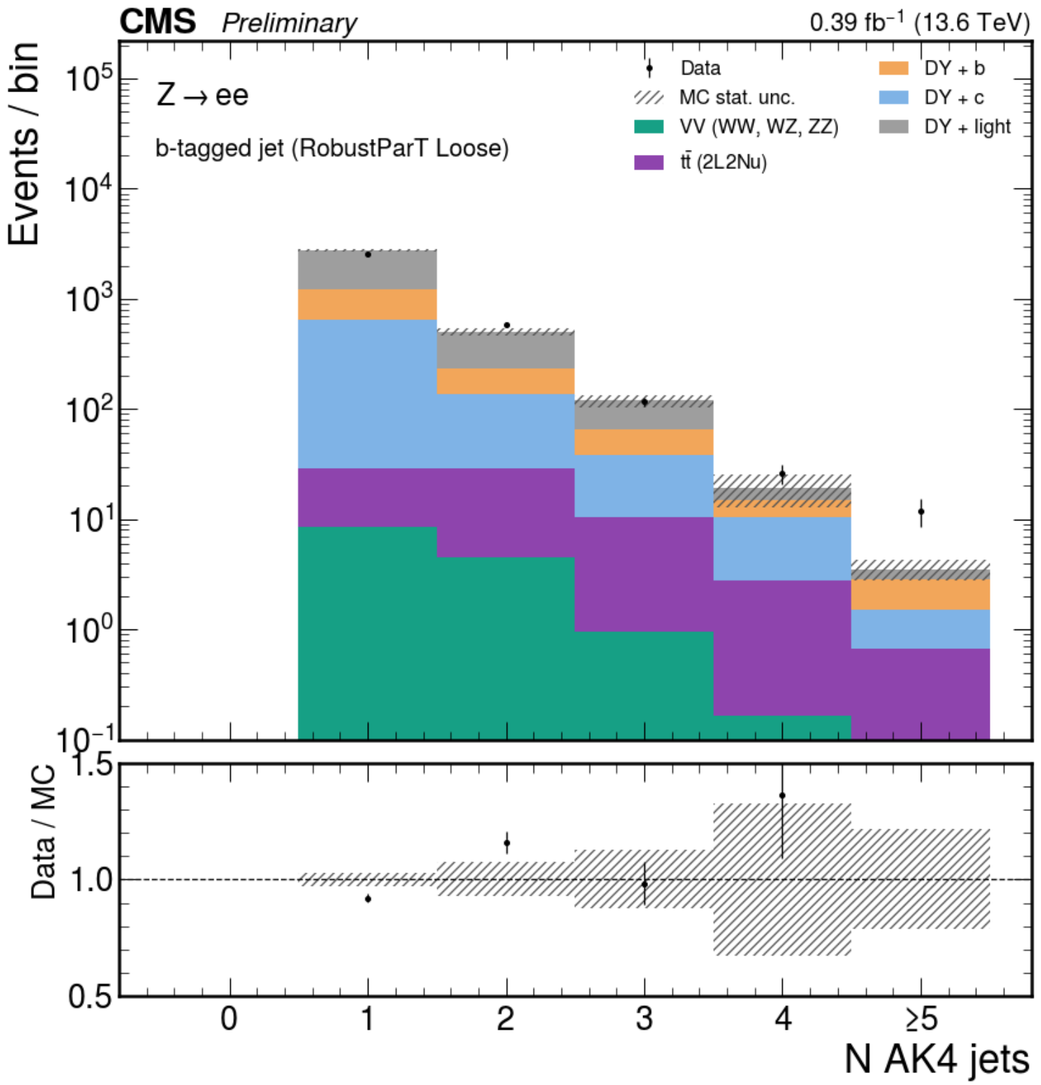
Close to π (back to back) when the Z has low pT.

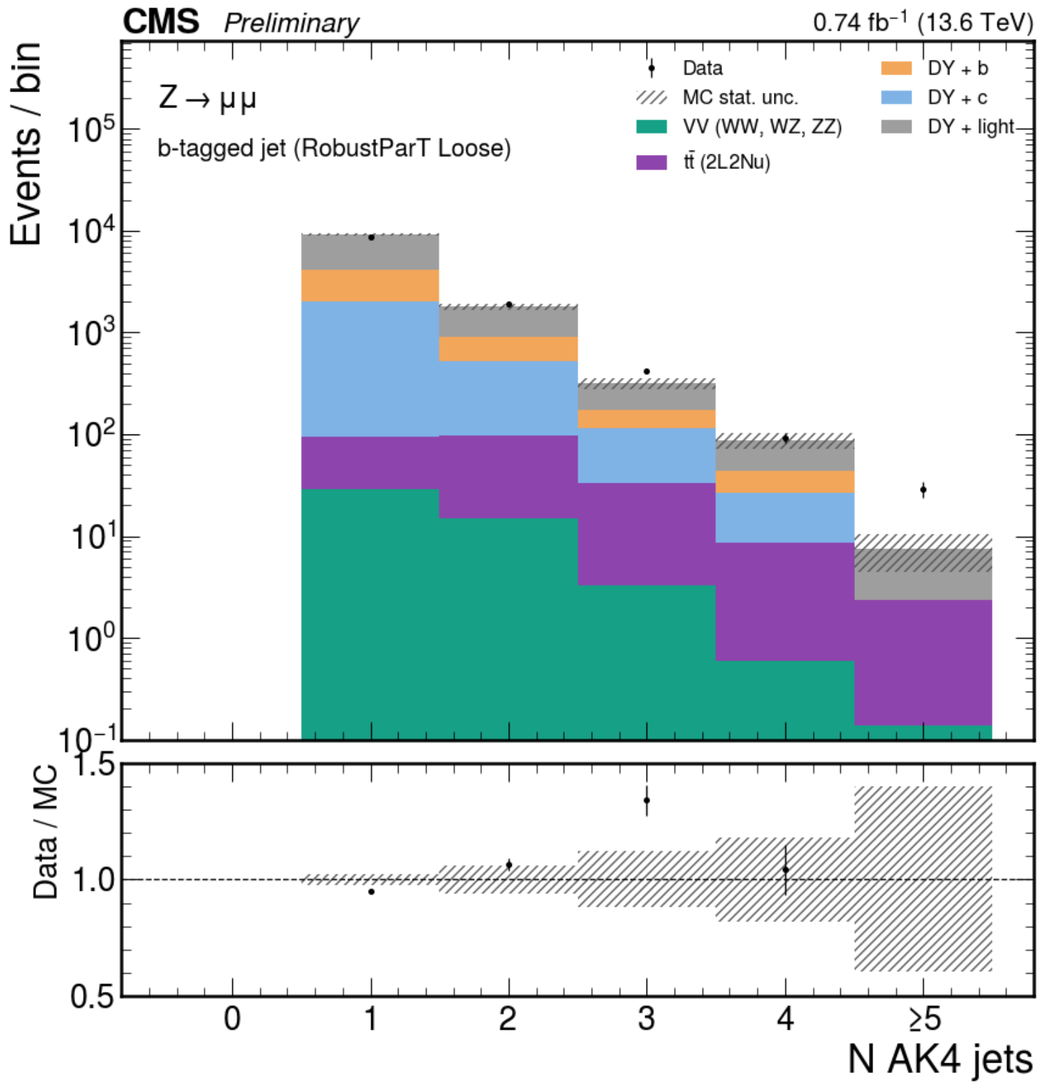




Number of selected jets

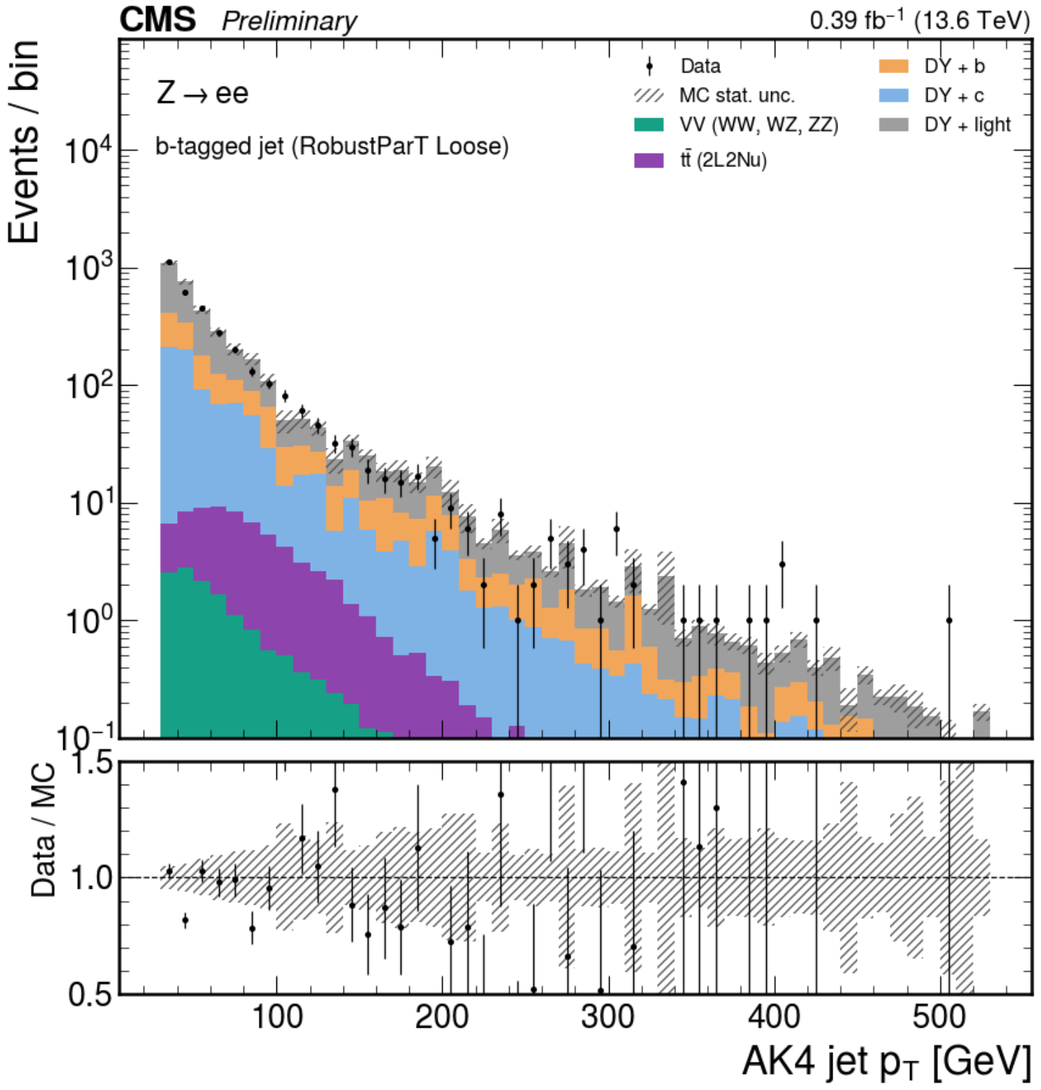
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. The leading jet is b-tagged here, so every event has at least one jet.

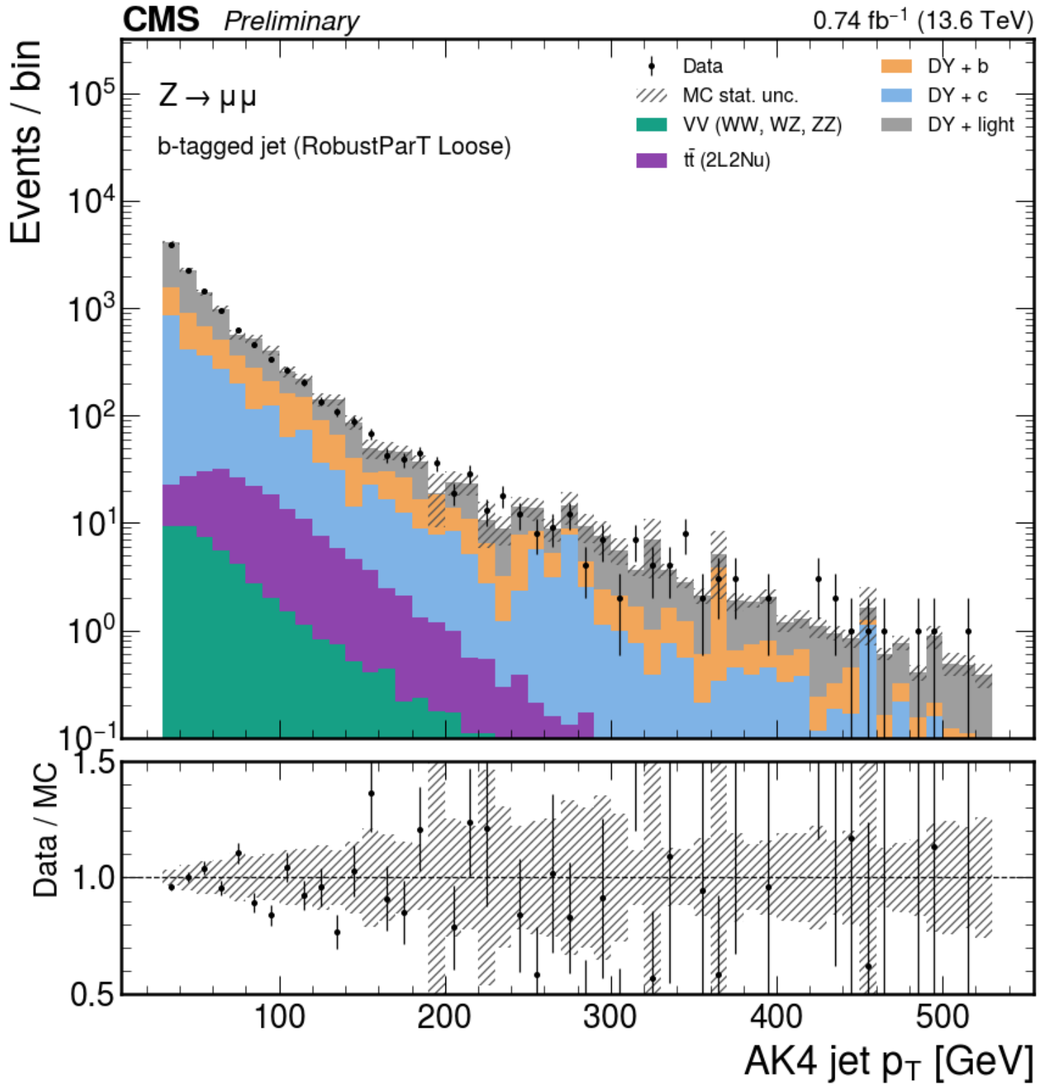




Leading-jet pT (this is the b-tagged jet)

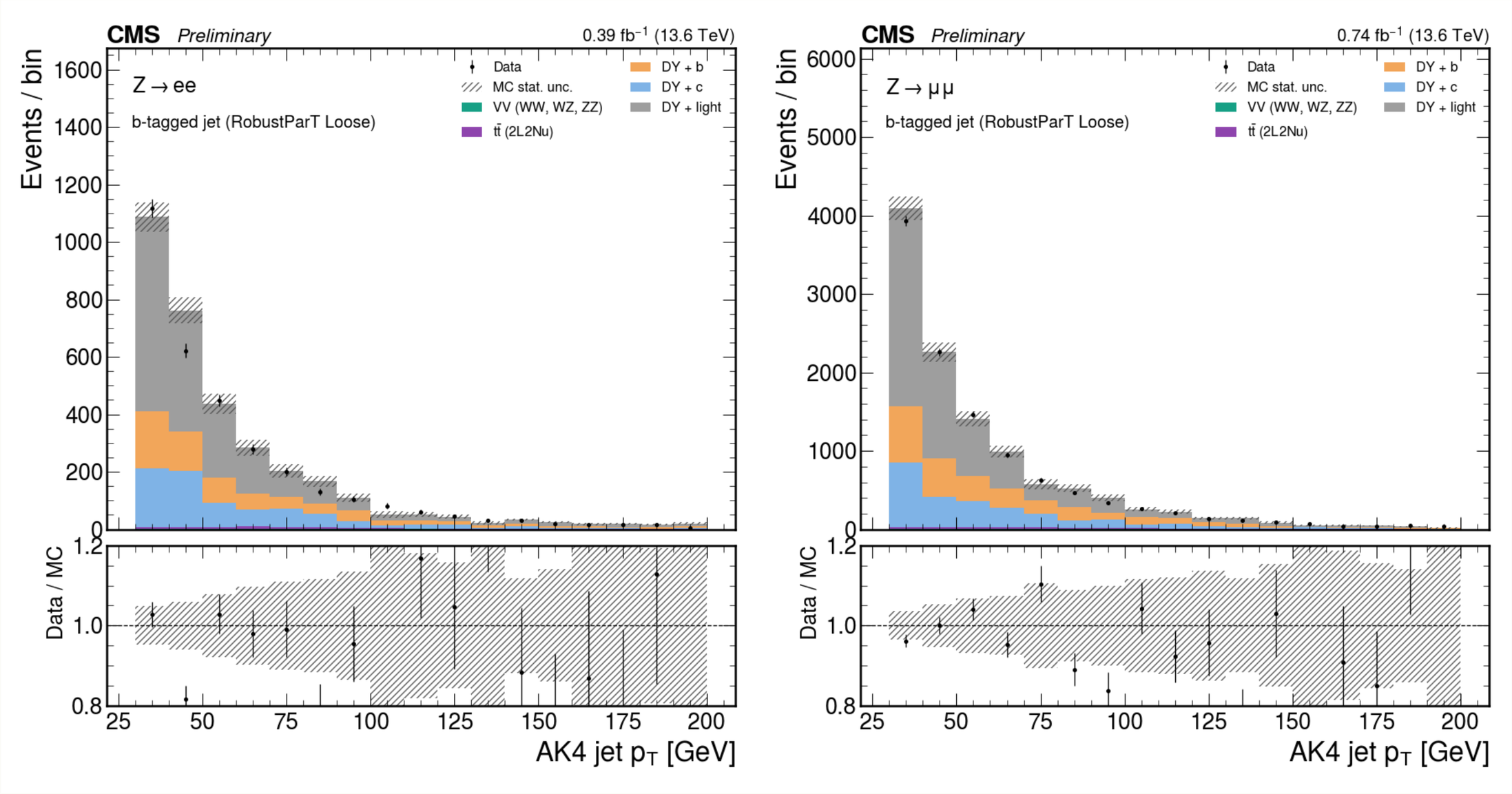
pT > 30 GeV.





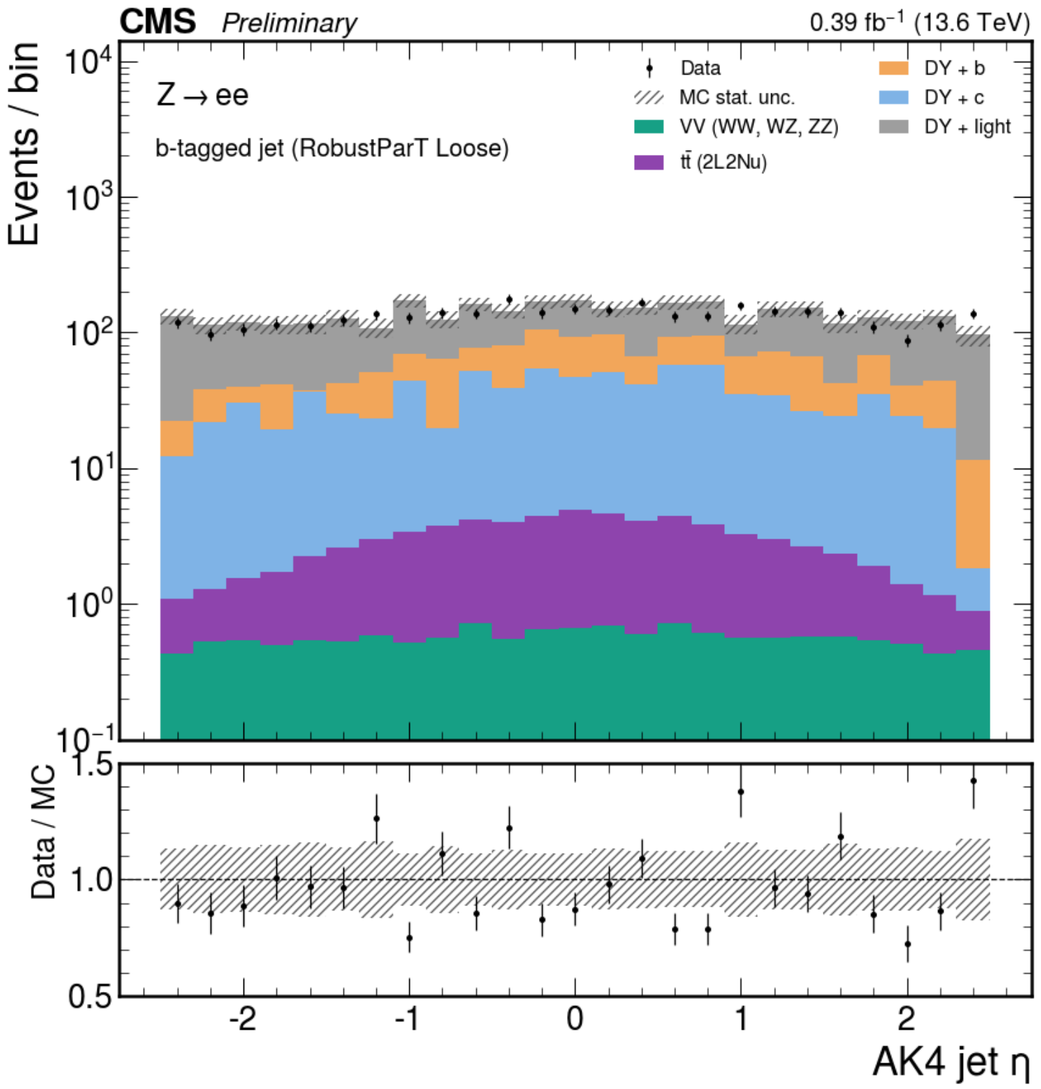
Leading-jet pT, zoom on 30–200 GeV

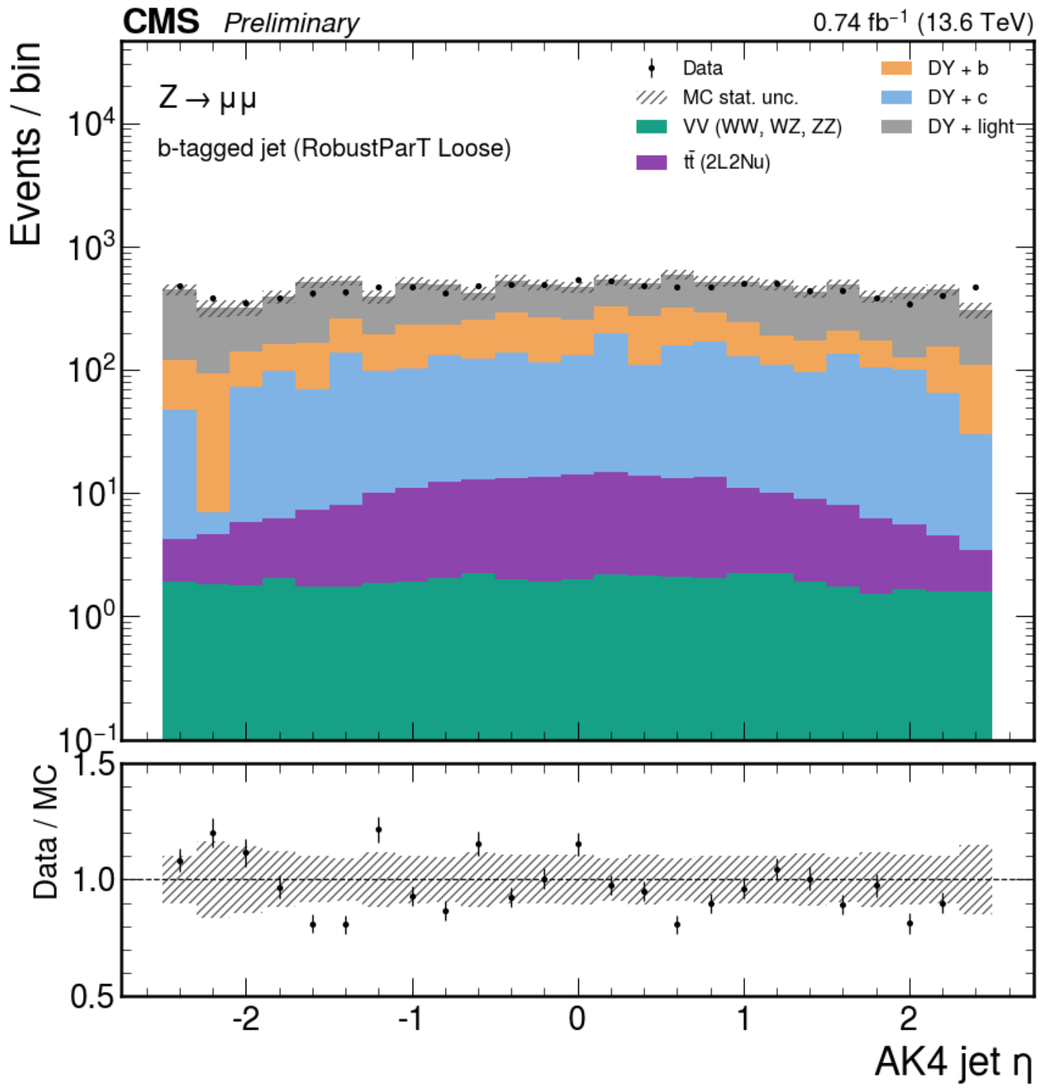
Zoom: 30–200, linear scale, data/MC from 0.8 to 1.2.



Leading-jet η

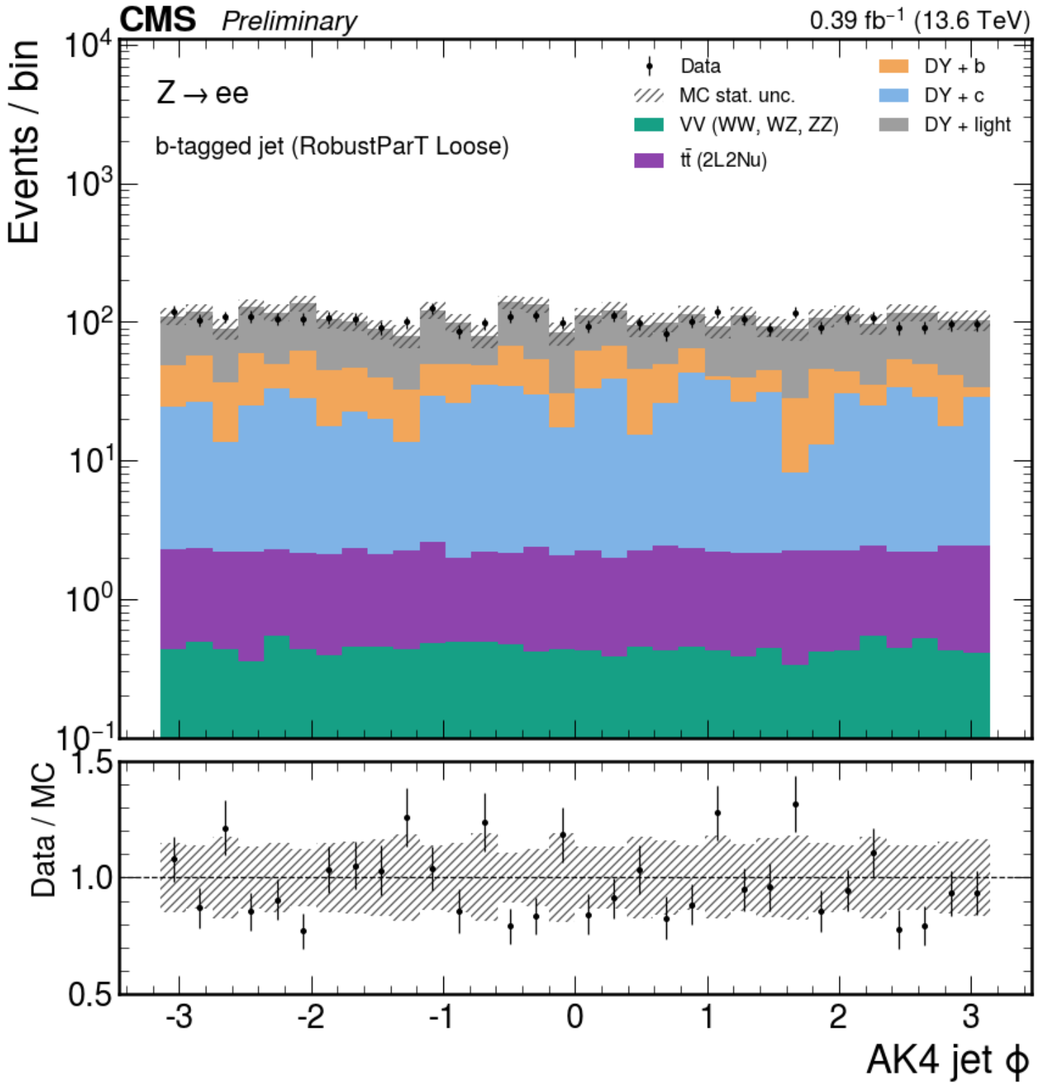
|η| < 2.5 (tracker coverage, needed for b-tagging).

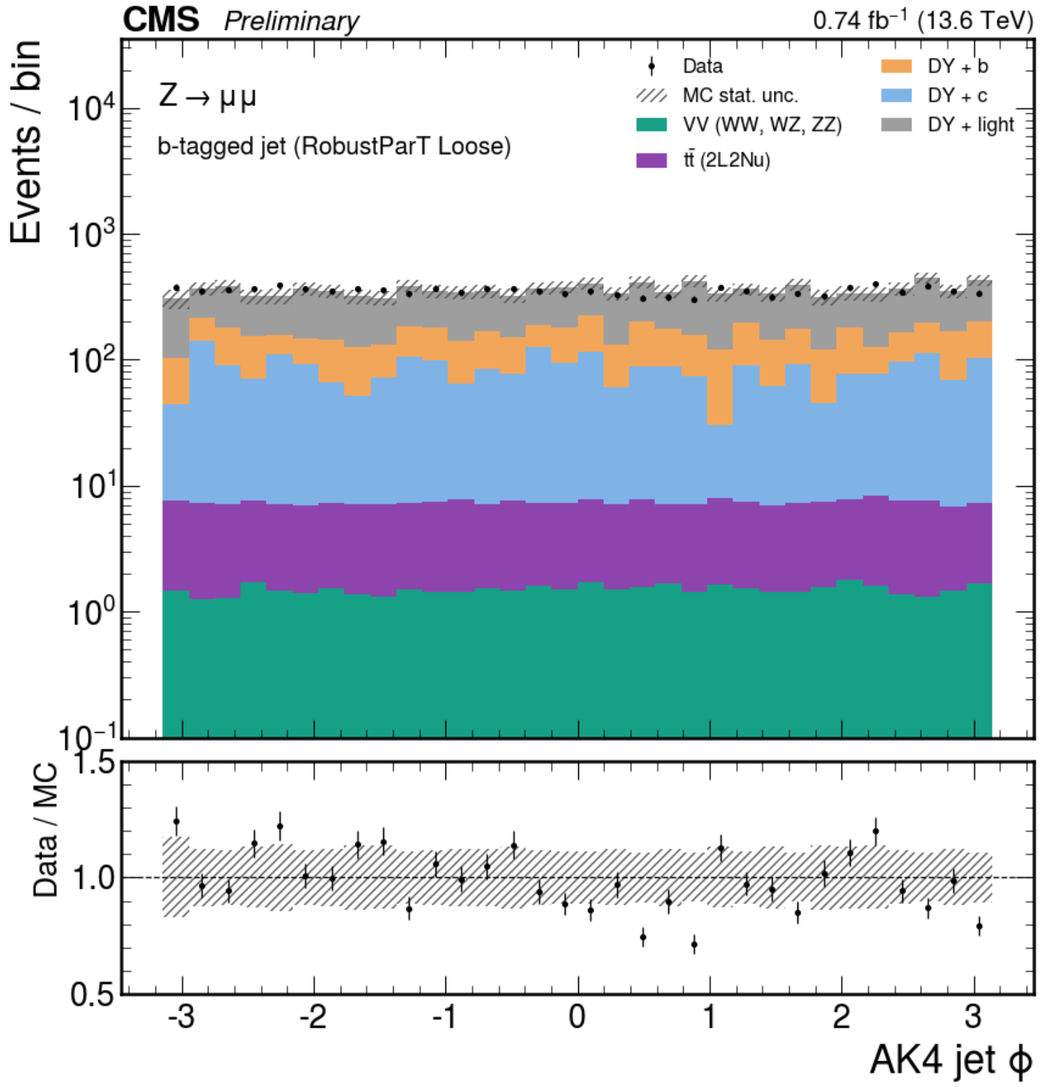




Leading-jet φ

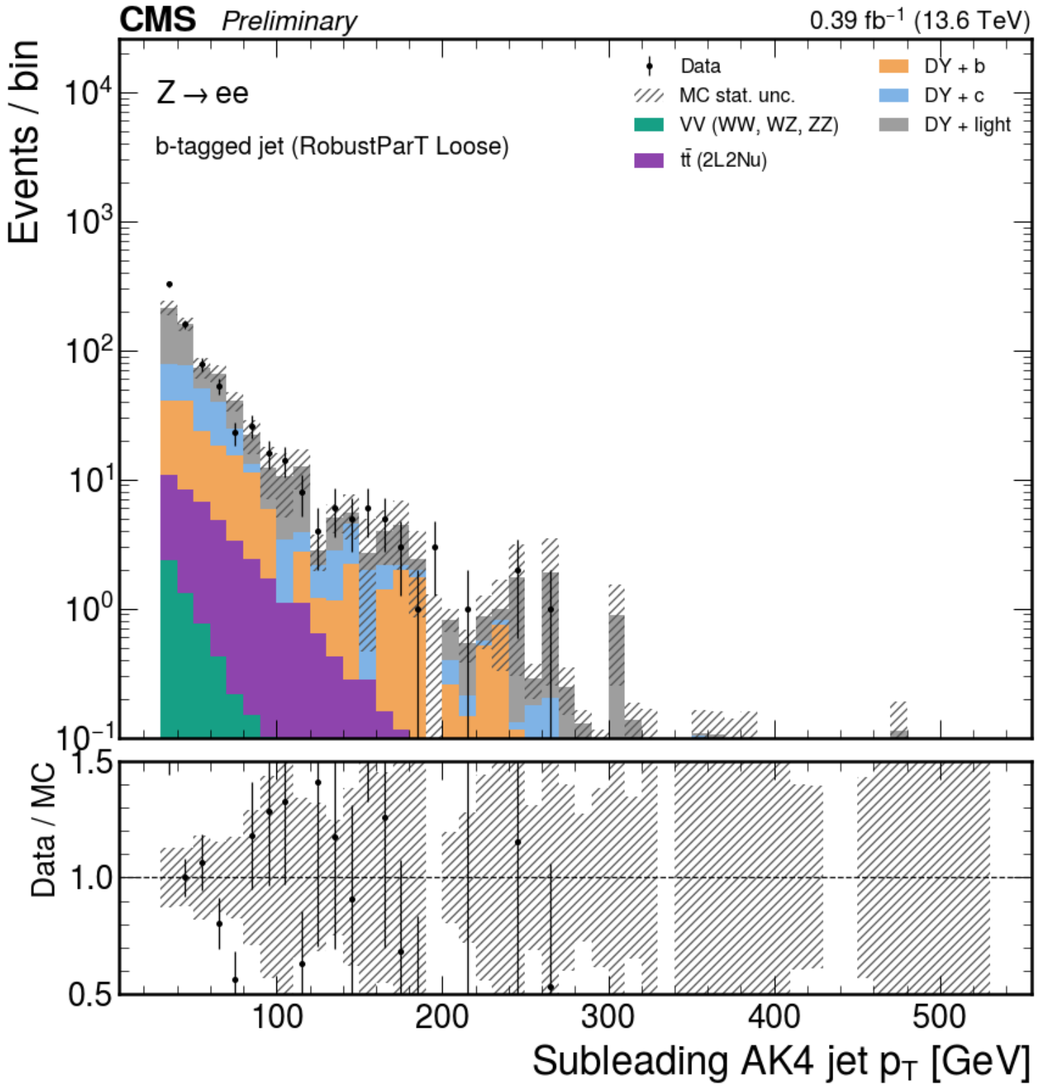
Expected to be flat.

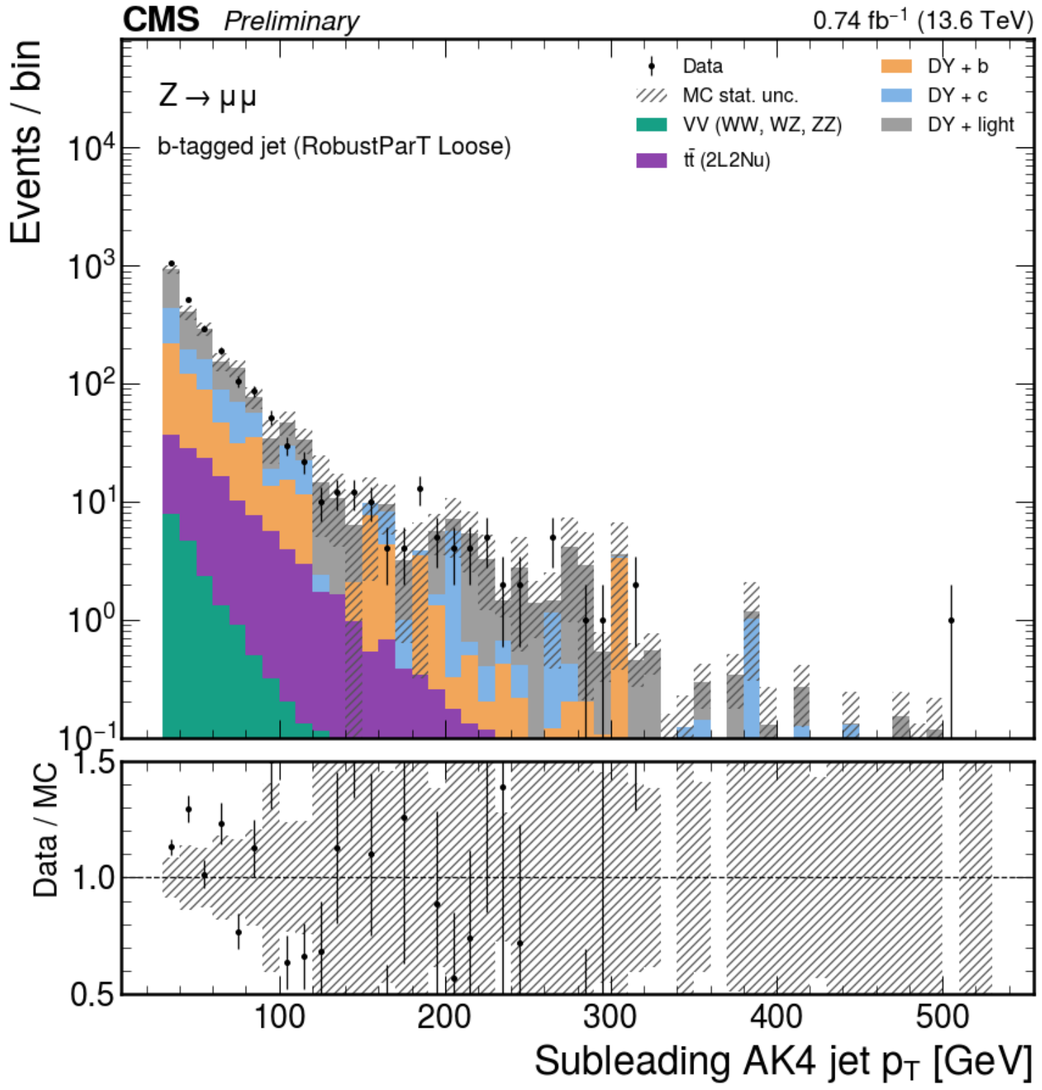




Second-jet pT

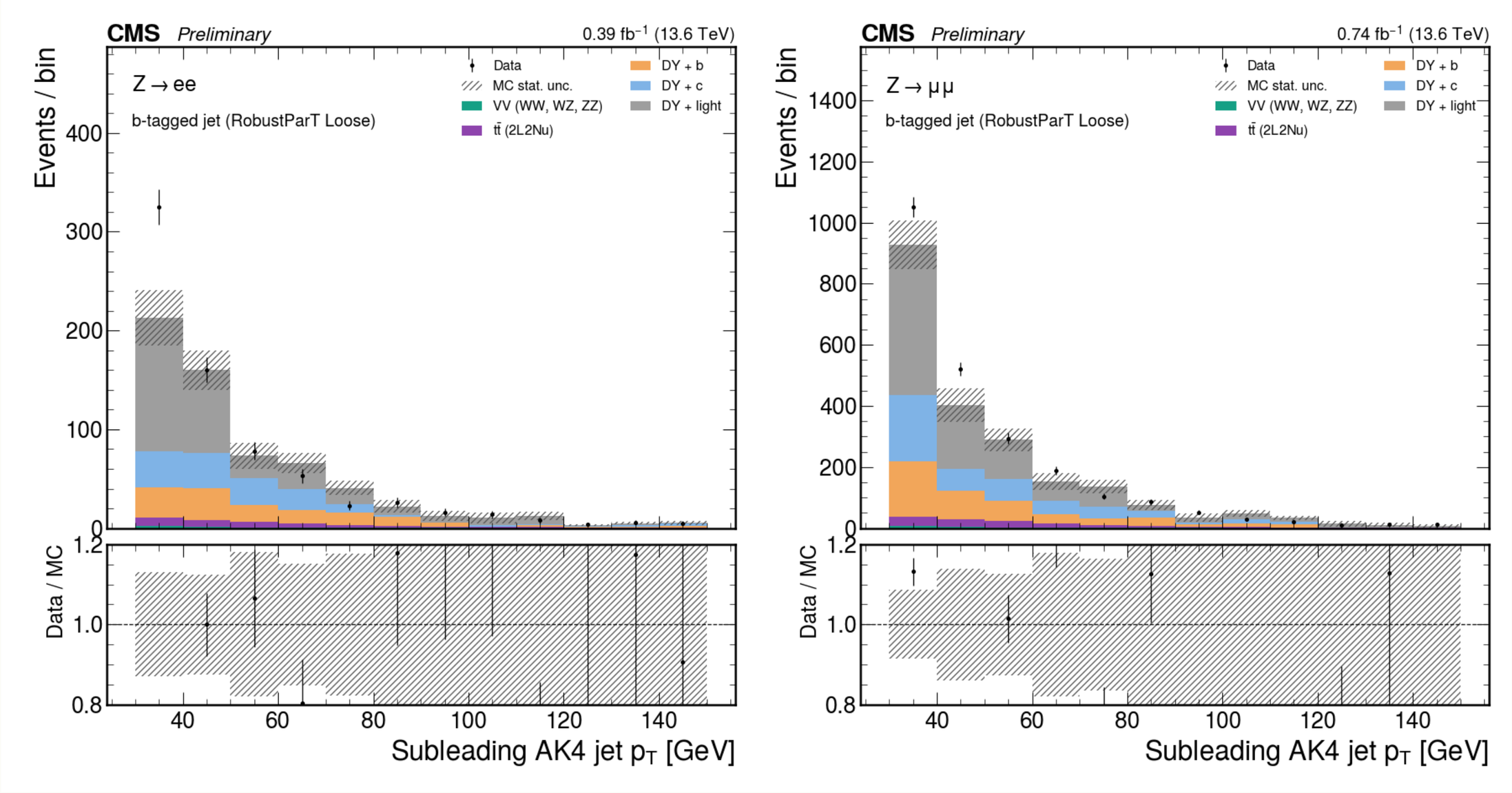
Only events with at least two jets. The second jet is not tagged.





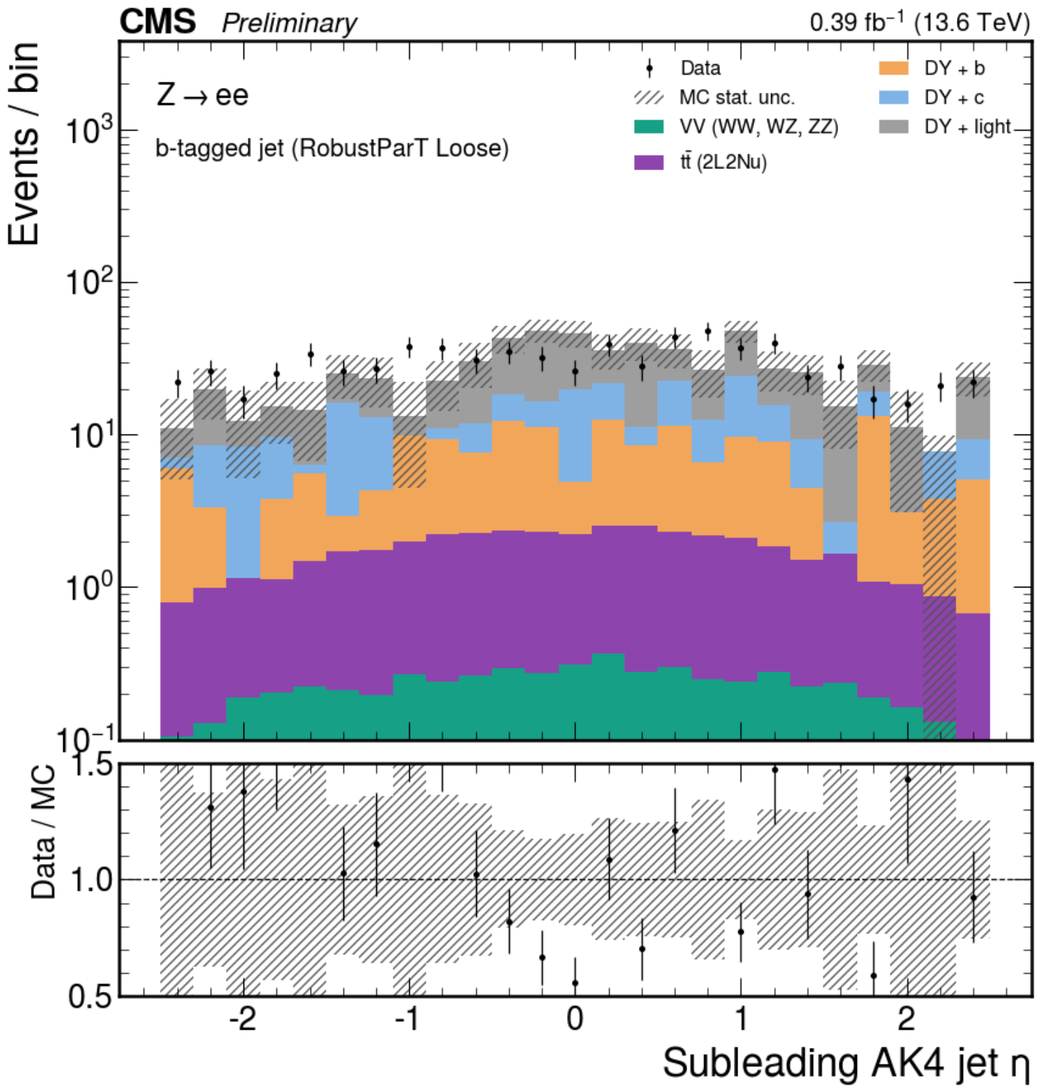
Second-jet pT, zoom on 30–150 GeV

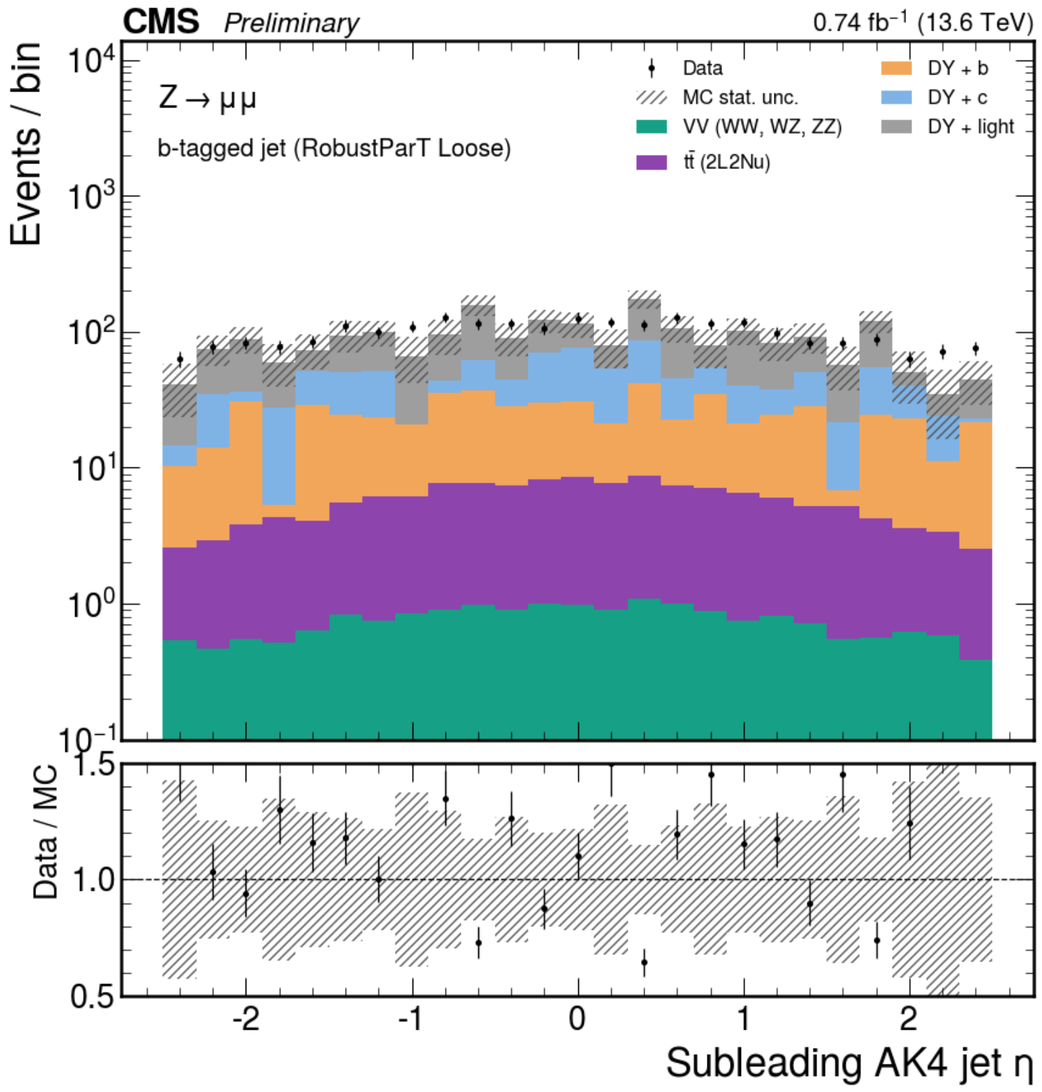
Zoom: 30–150, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.



Second-jet η

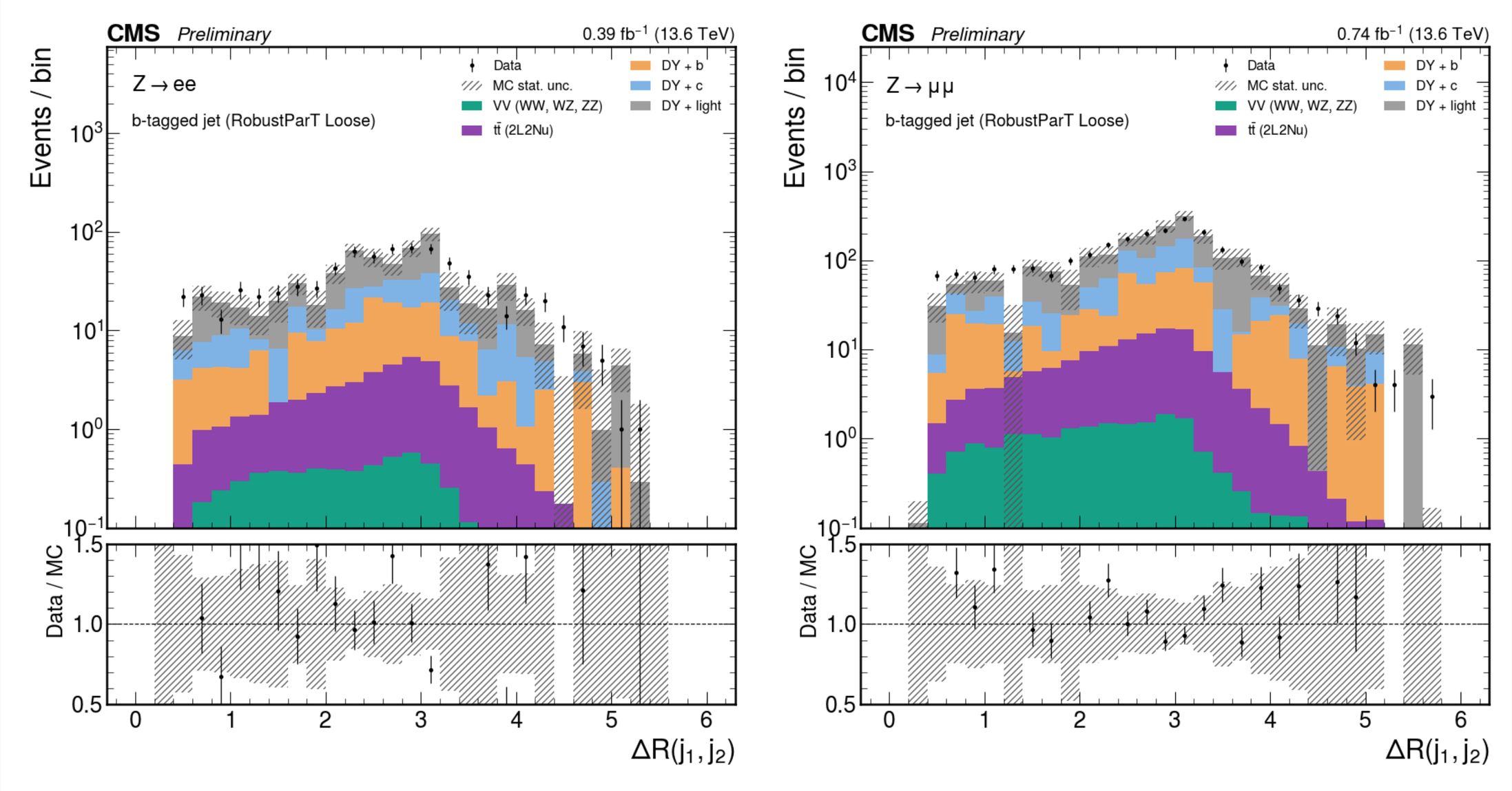
Only events with at least two jets.





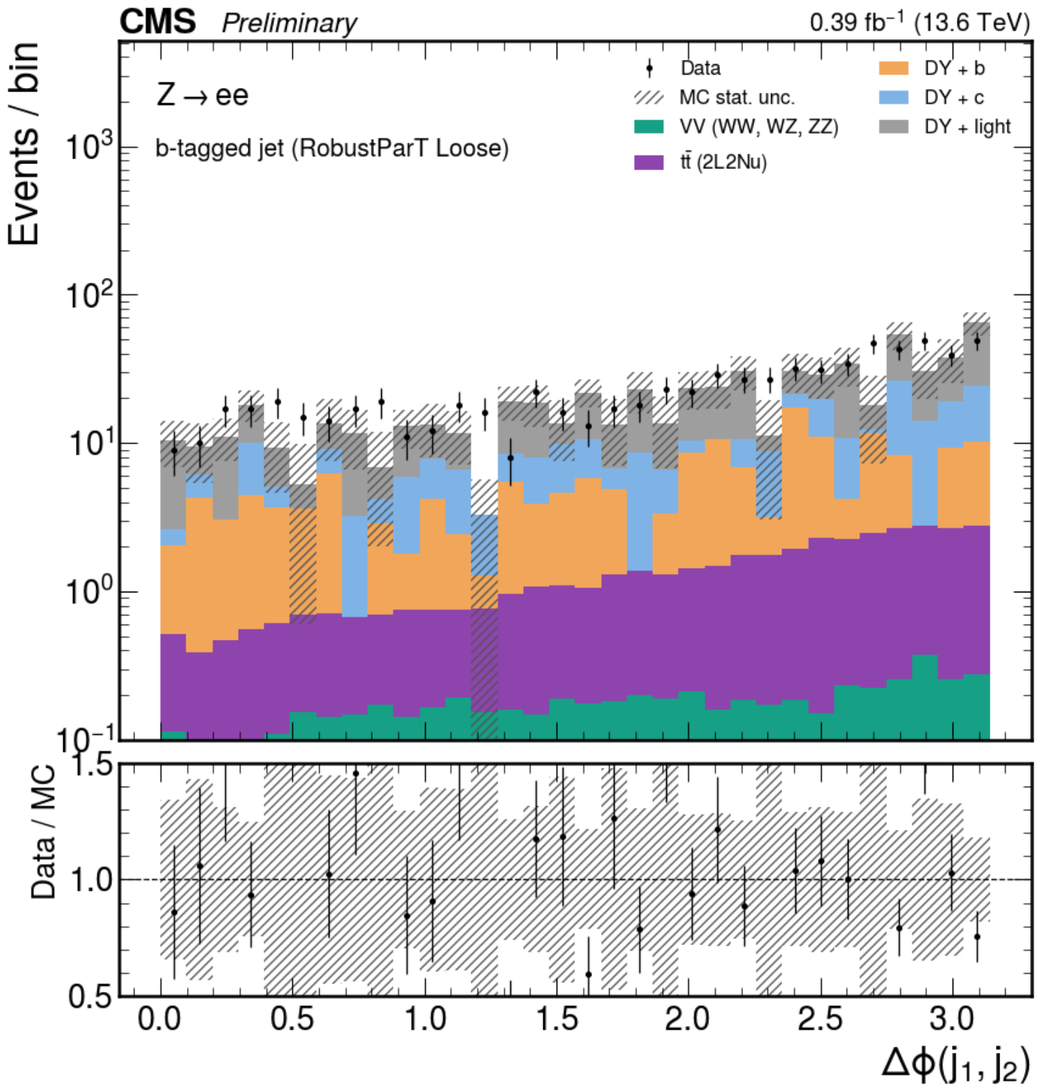
Angular distance between the two leading jets, ΔR(jj)

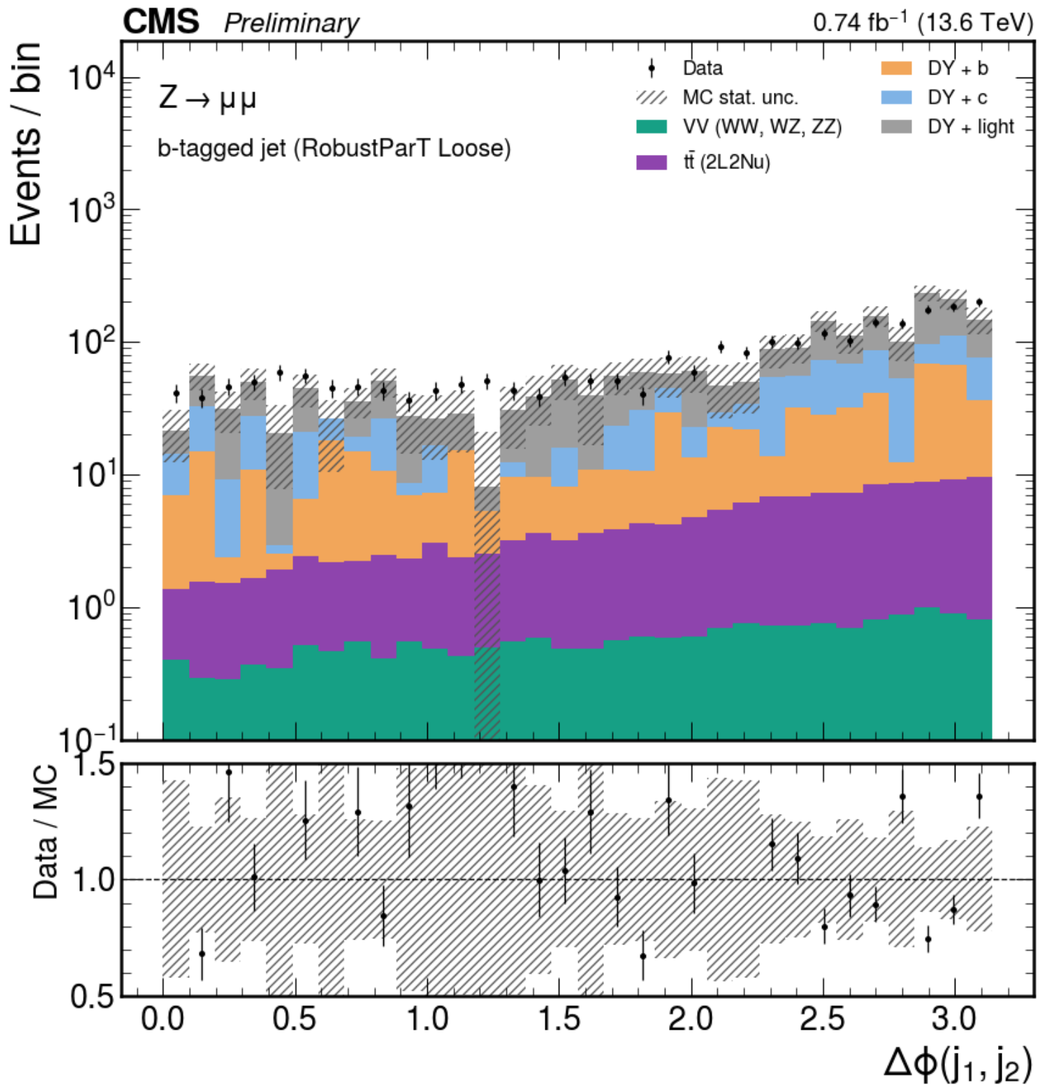
Only events with at least two jets.



Azimuthal angle between the two leading jets, Δφ(jj)

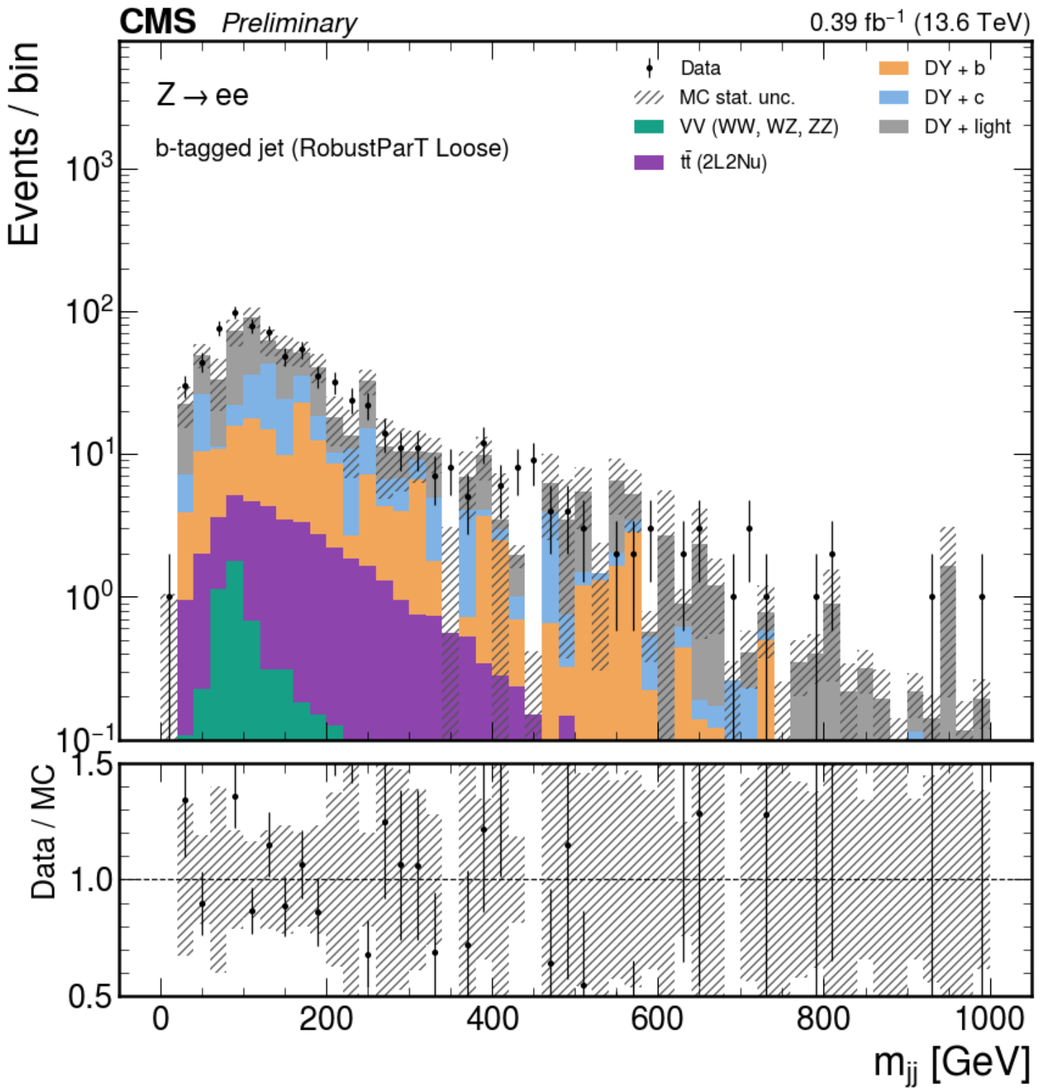
Only events with at least two jets.

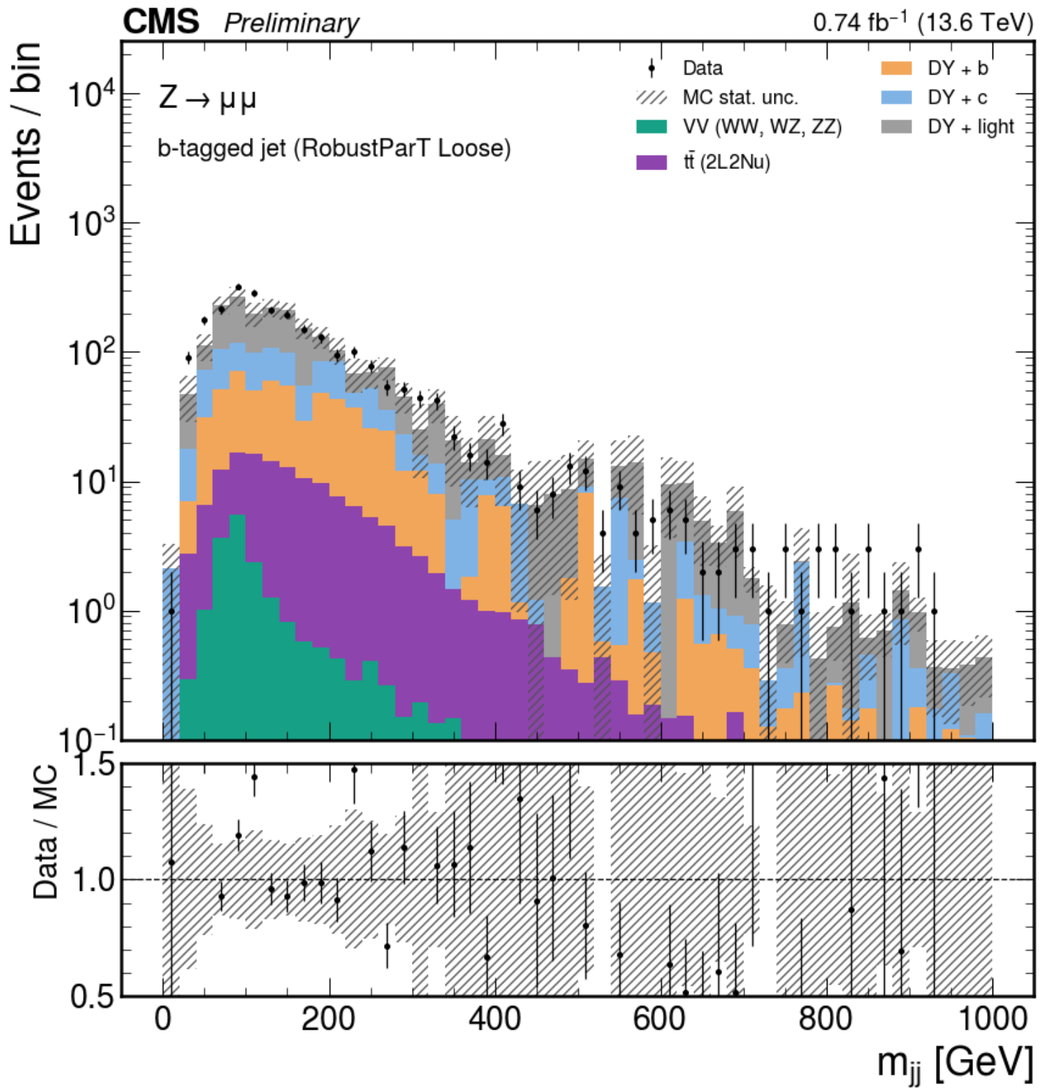




Mass of the two leading jets, m(jj)

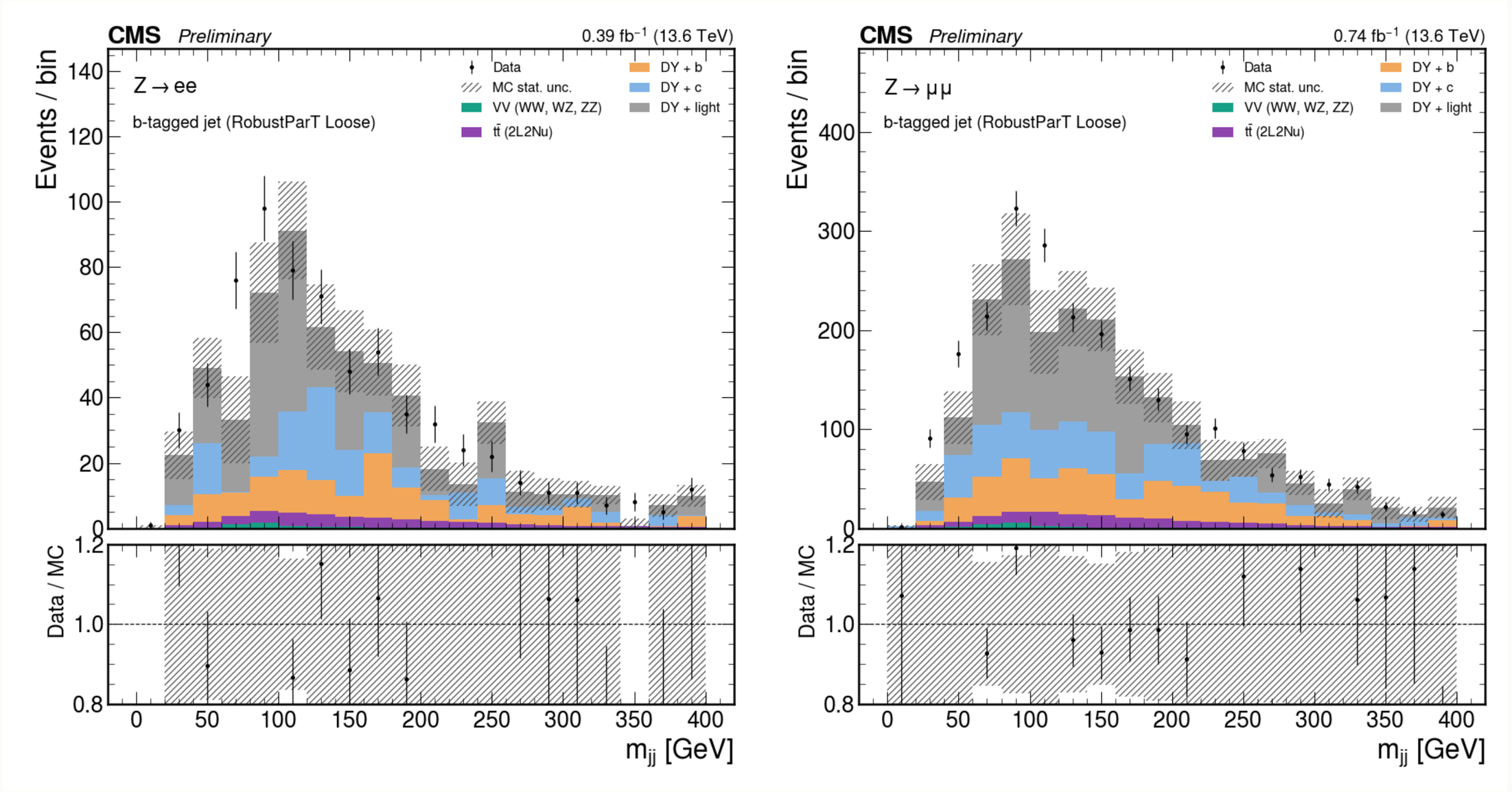
Only events with at least two jets.





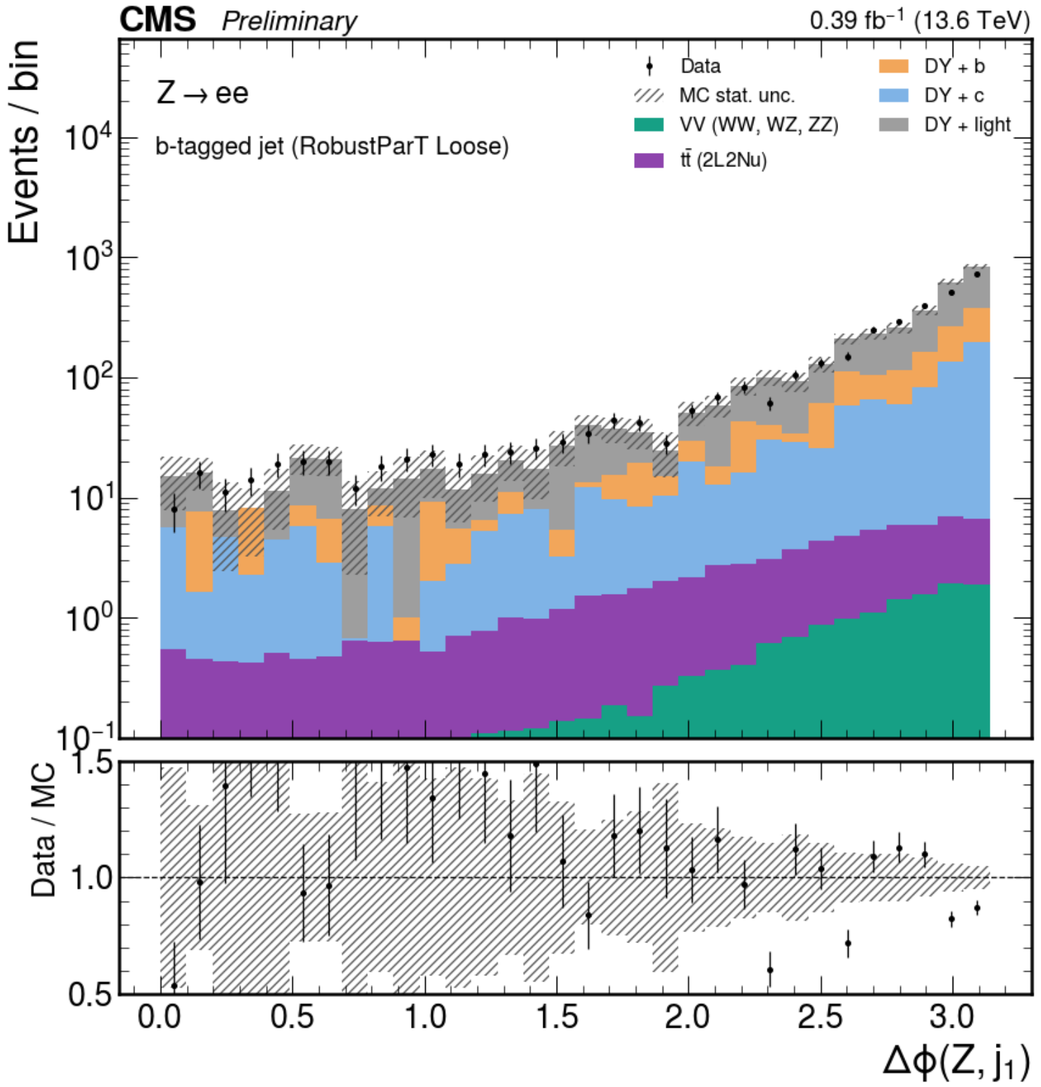
m(jj), zoom on 0–400 GeV

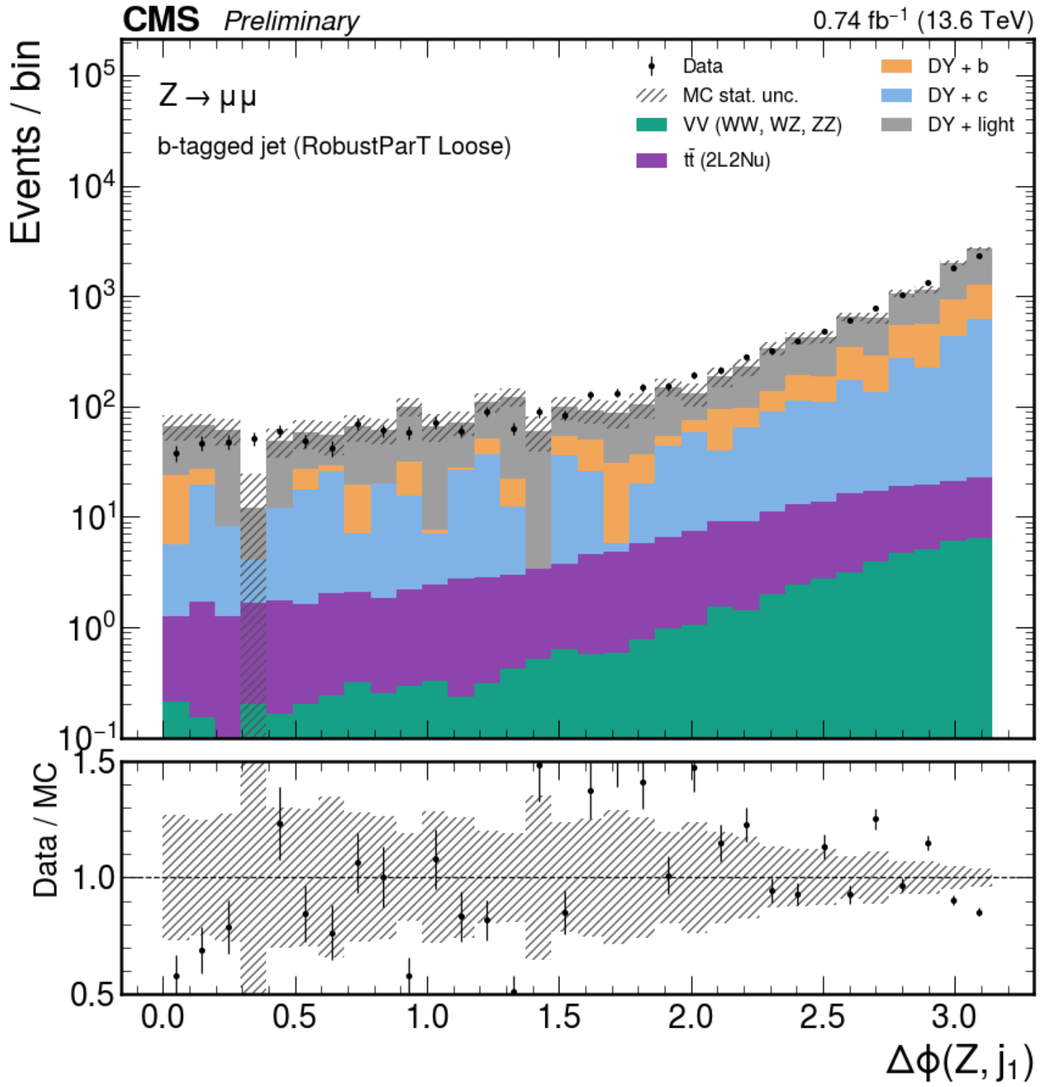
Zoom: 0–400, linear scale, data/MC from 0.8 to 1.2. Only events with at least two jets.



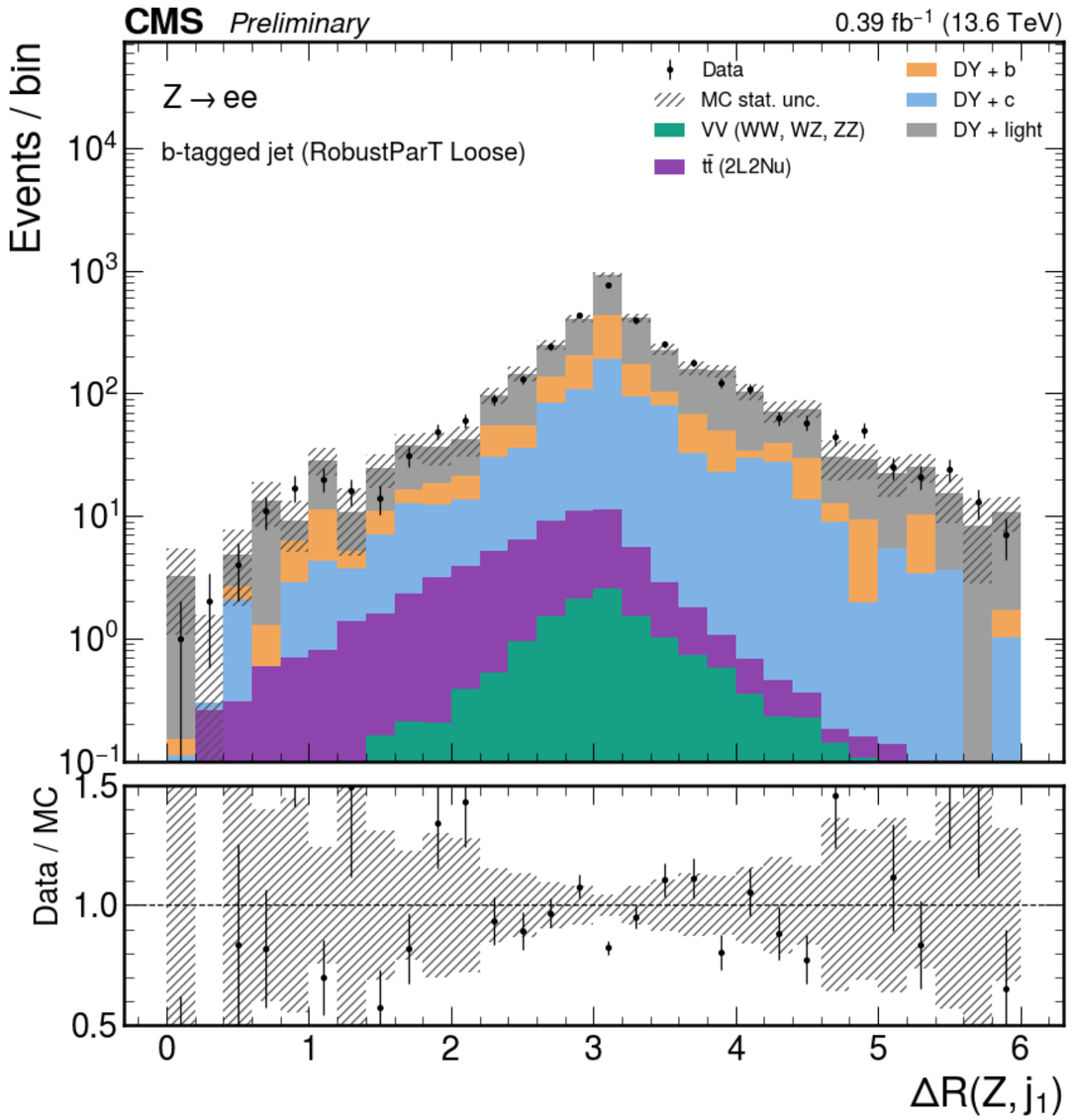
Azimuthal angle between the Z and the leading jet, Δφ(Z, j)

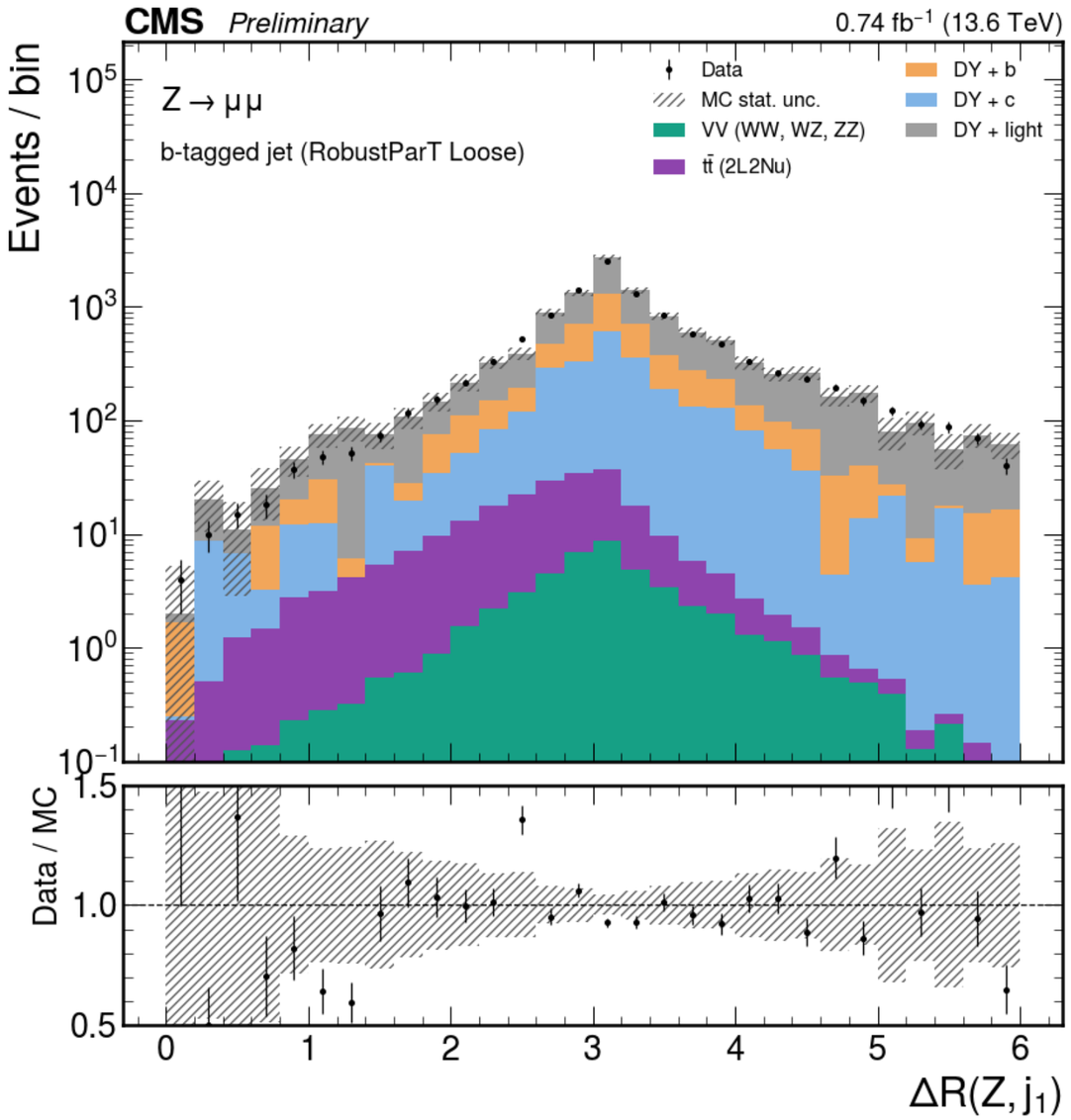
Close to π when the jet recoils against the Z.





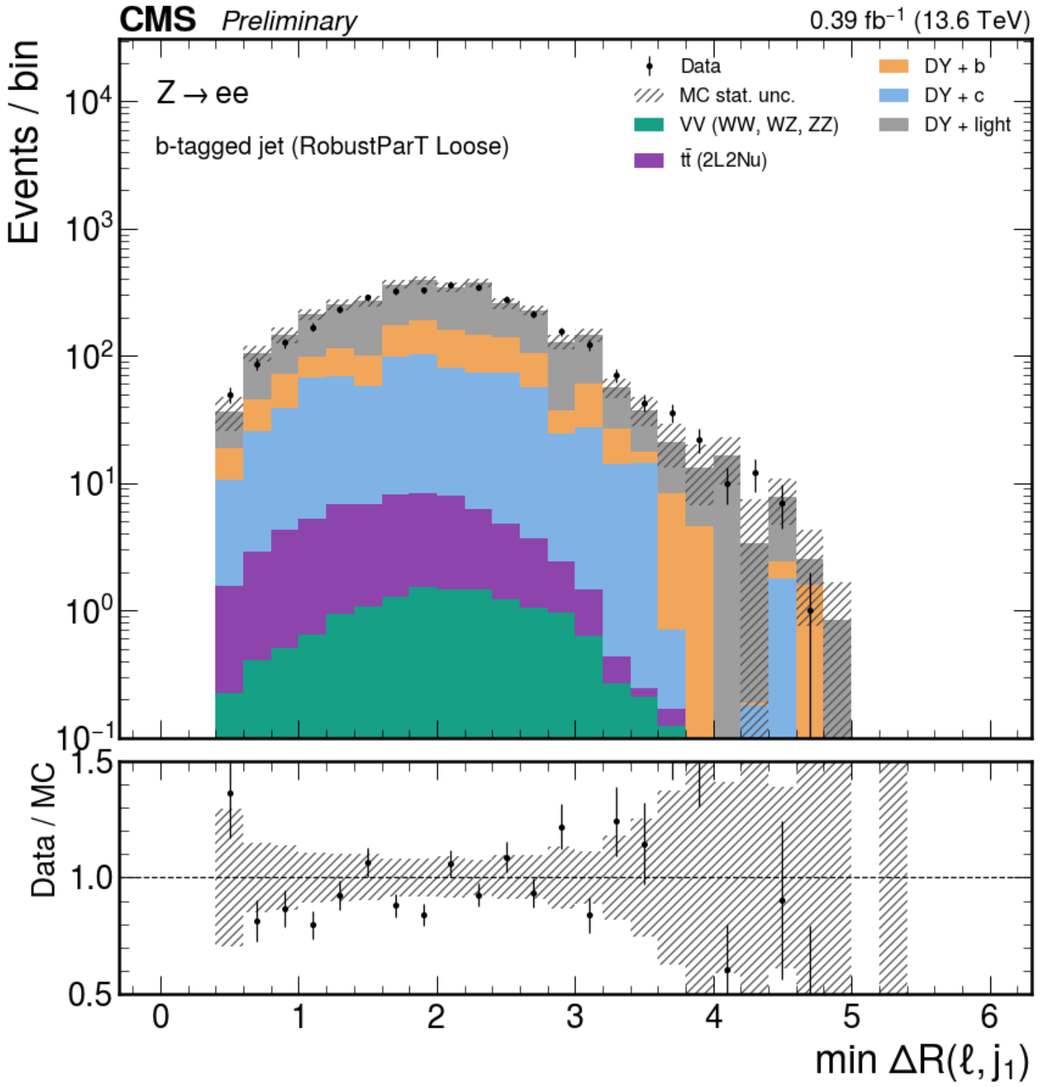
Angular distance between the Z and the leading jet, ΔR(Z, j)

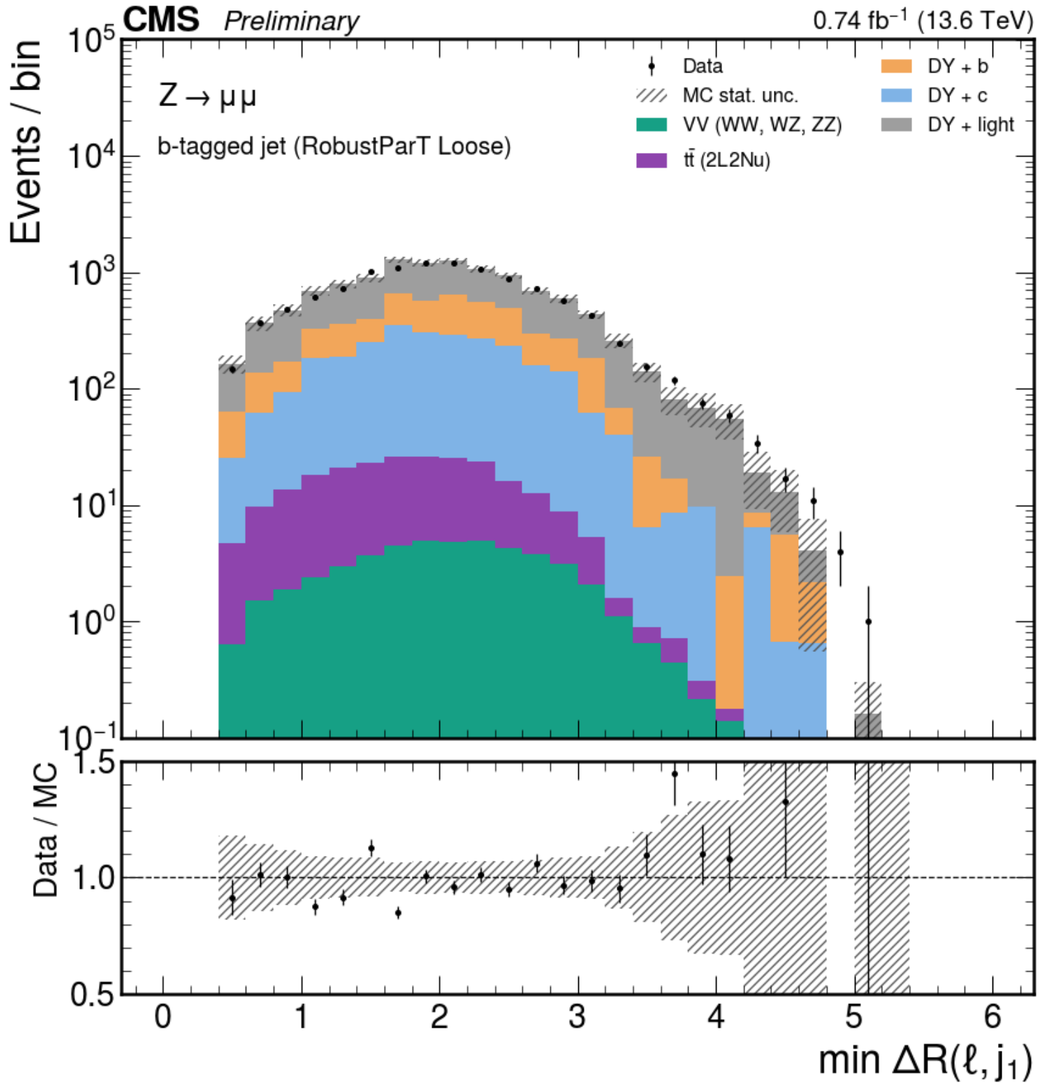




Smallest ΔR between a lepton and the leading jet

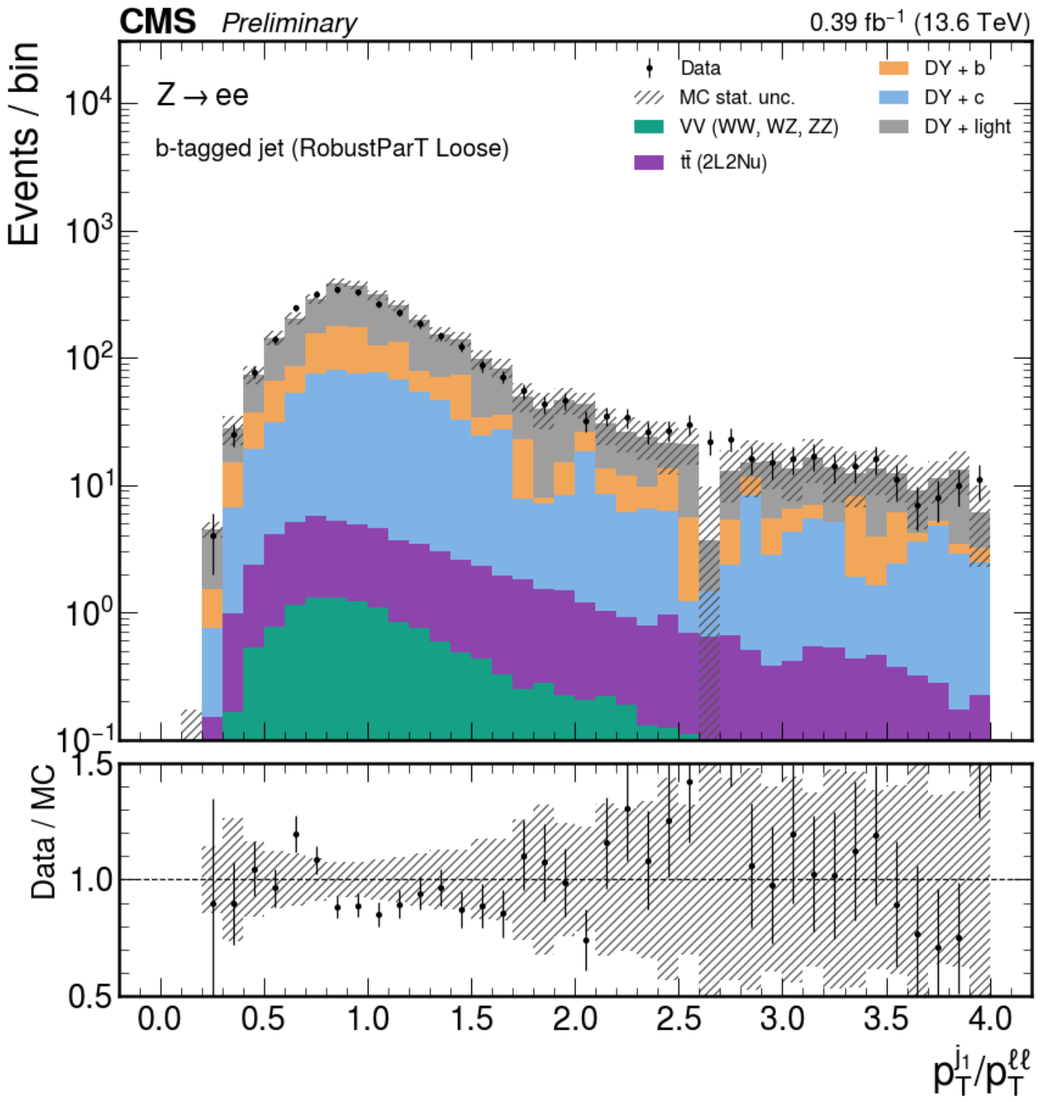
Must be > 0.4 by selection (jets overlapping a lepton are removed).

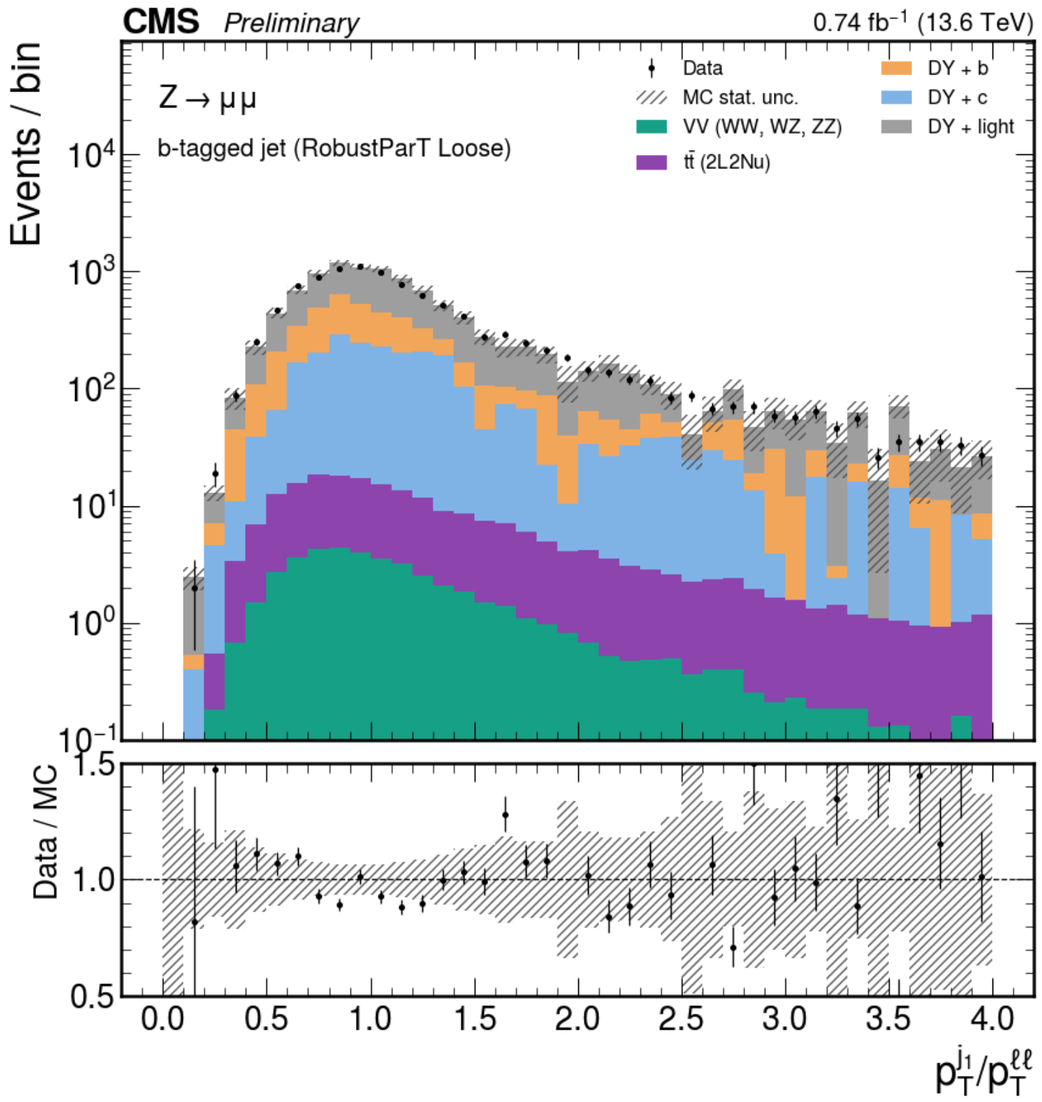




pT balance: leading-jet pT / Z pT

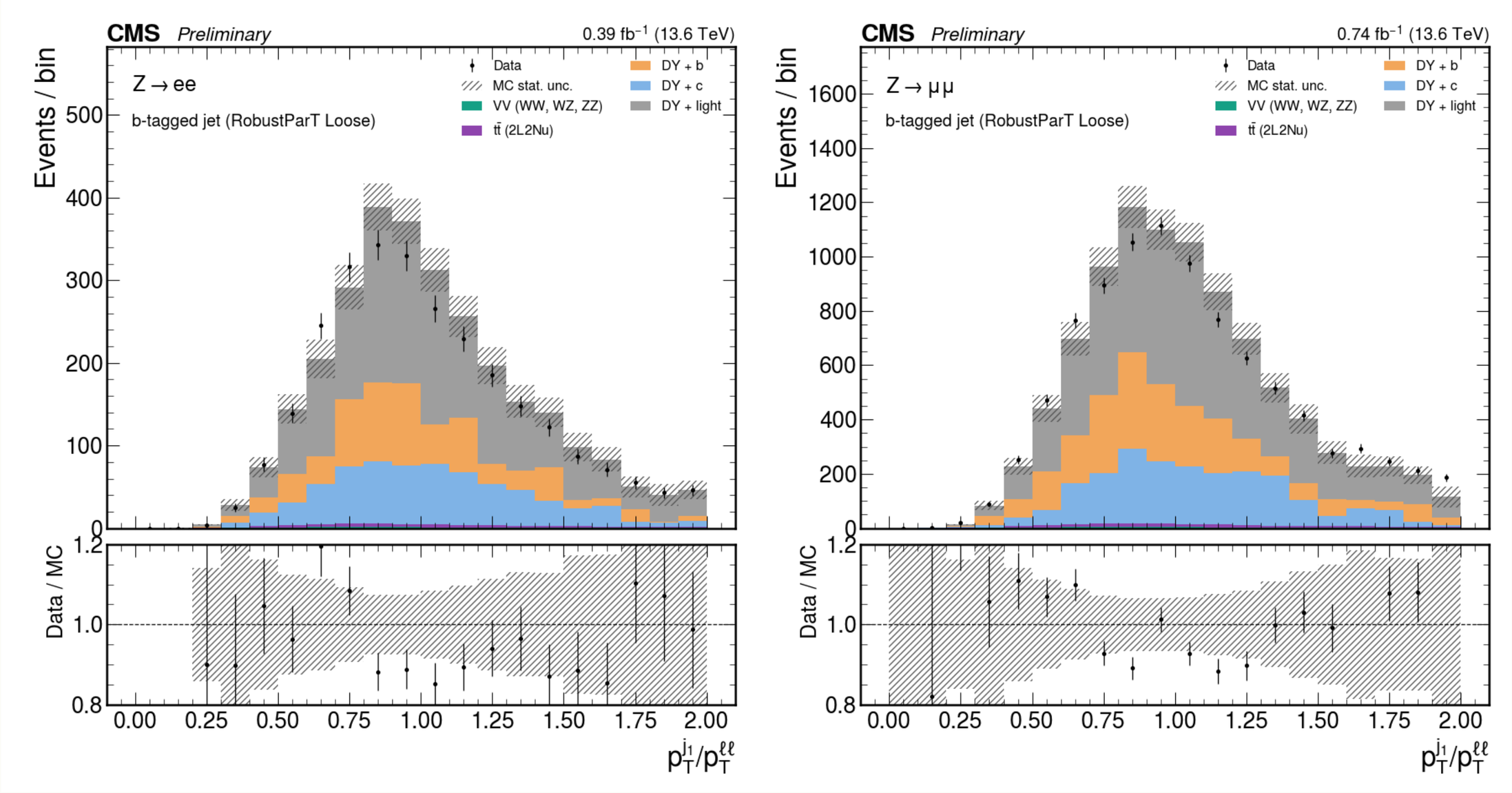
Close to 1 when the jet alone balances the Z. Sensitive to the jet energy scale and to extra jets.





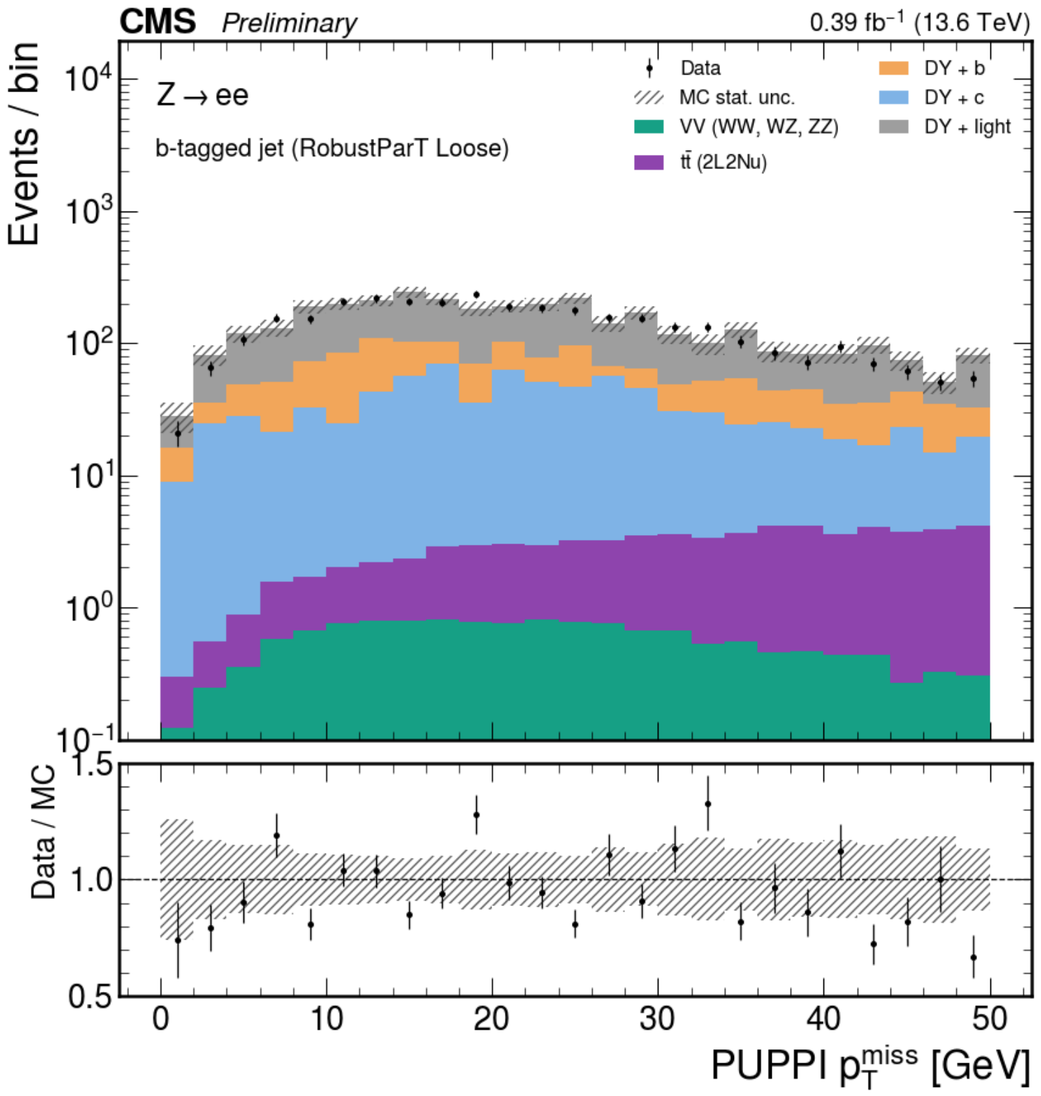
pT balance, zoom on 0–2

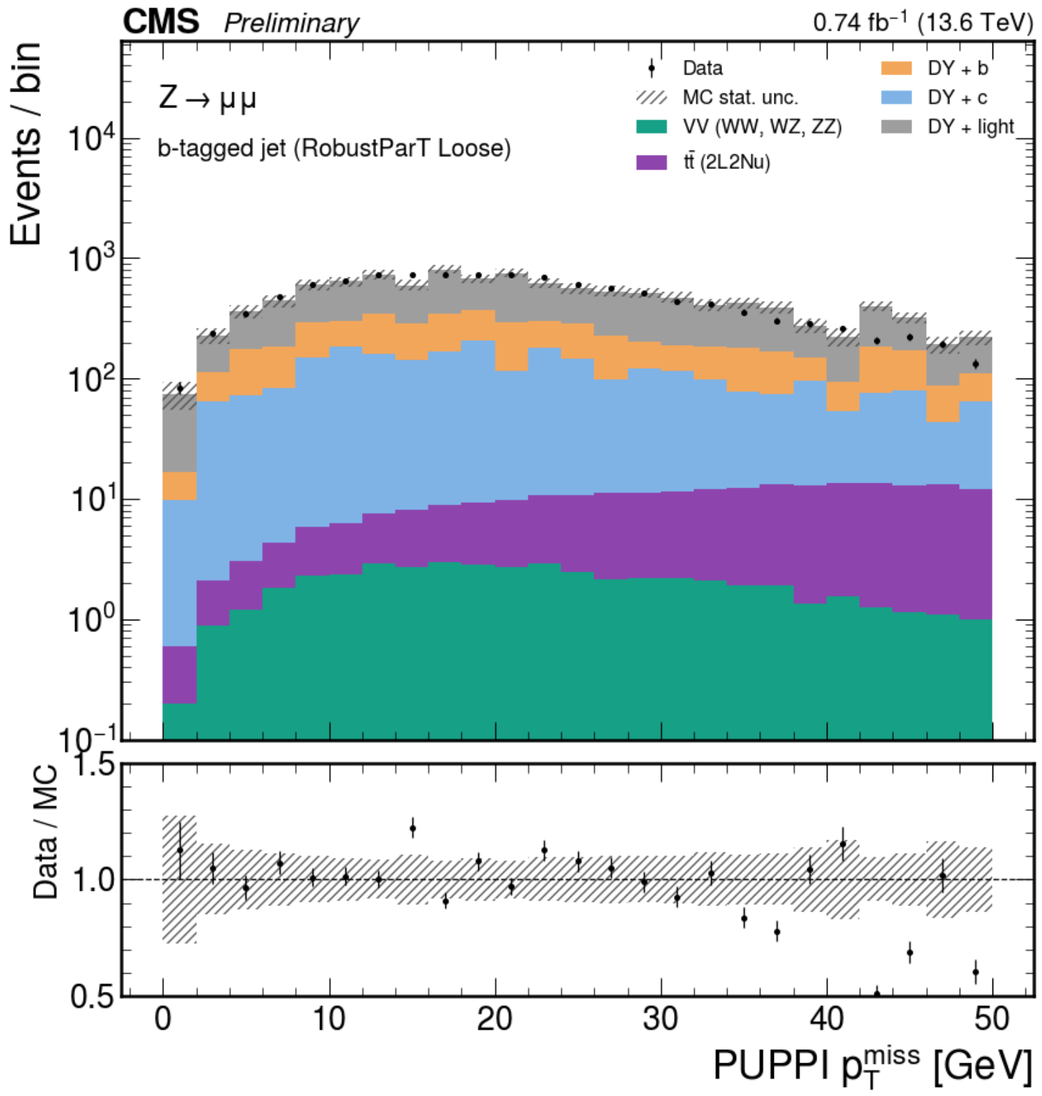
Zoom: 0–2, linear scale, data/MC from 0.8 to 1.2.



Missing transverse momentum (type-1 PUPPI MET), after the MET cut

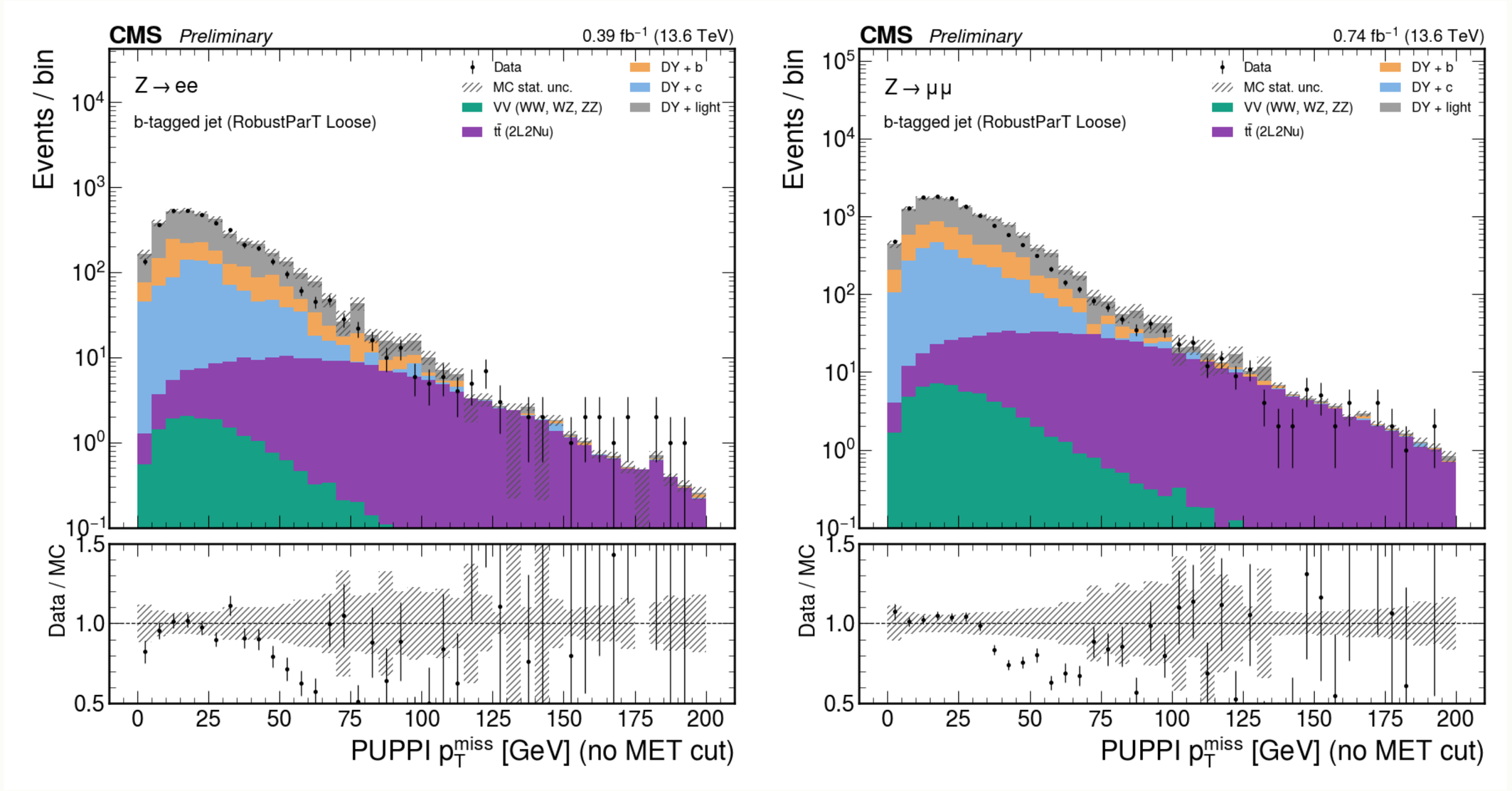
The selection requires MET < 50 GeV, so the plot stops there.





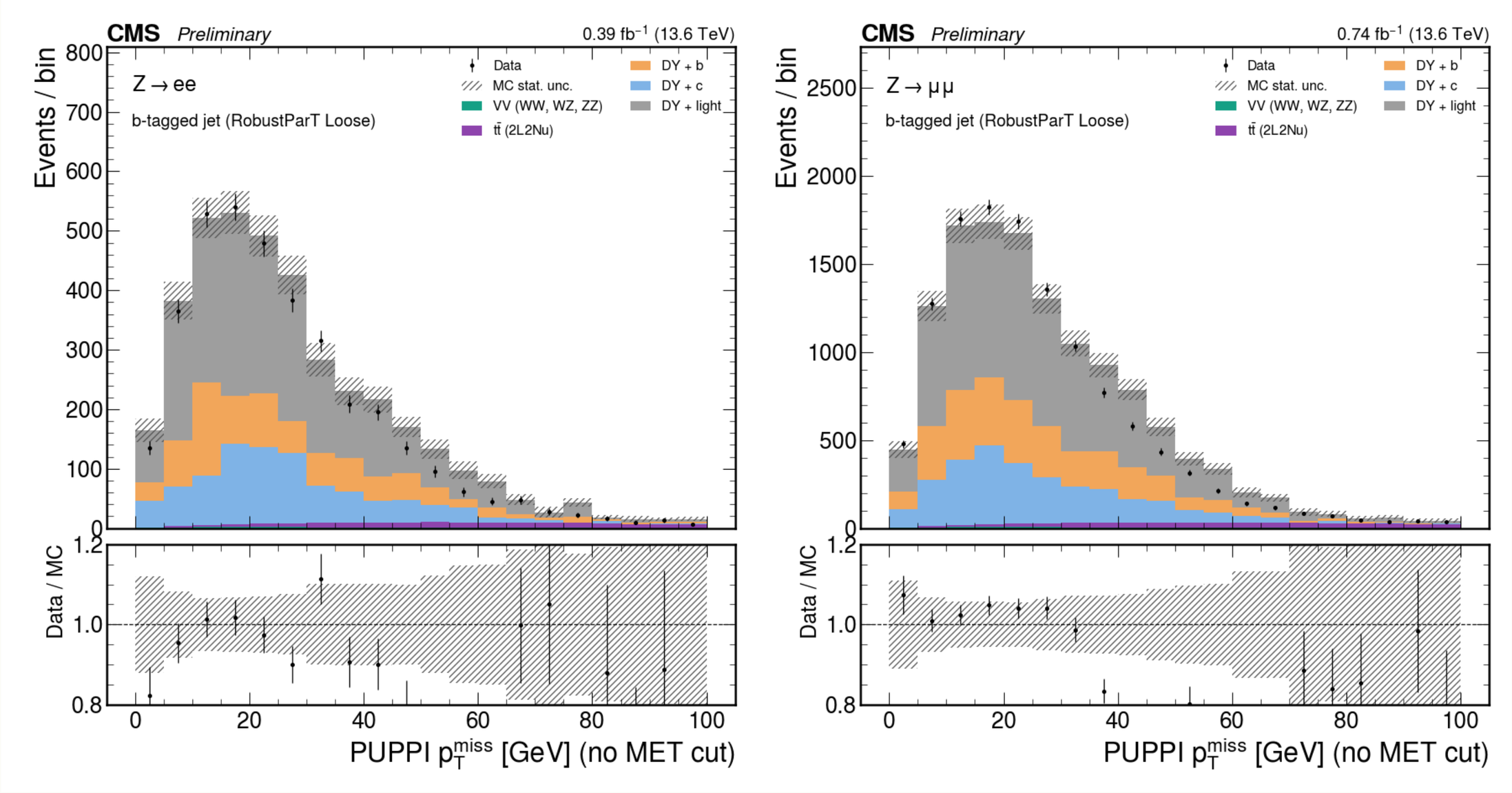
MET without the MET cut, up to 200 GeV

All other cuts applied, so the region removed by MET < 50 GeV is visible. $t\bar{t}$ dominates at high MET. MC is above data at 40–120 GeV, as without tagging (known 2022 jet/MET issue).



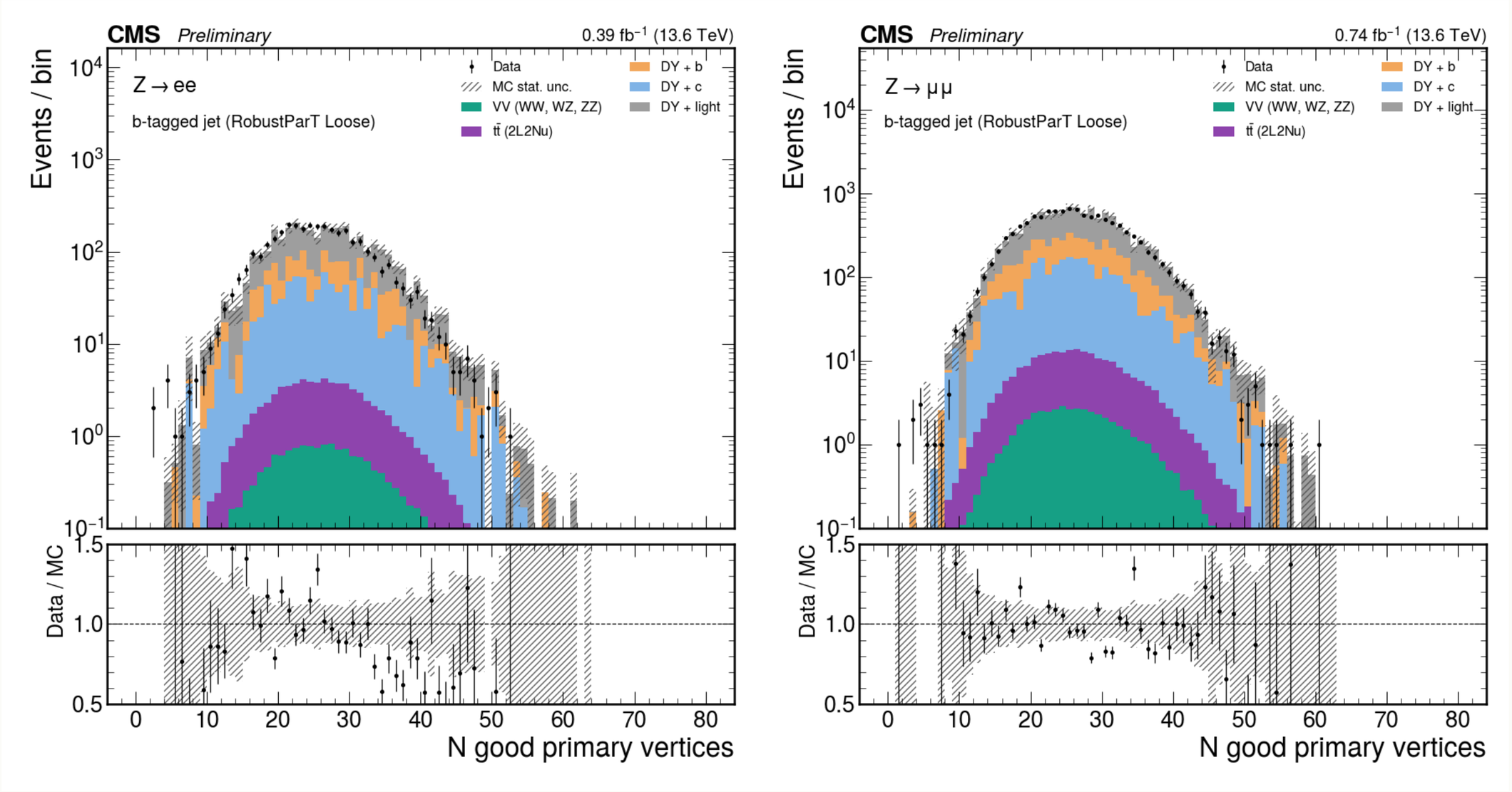
MET without the MET cut, zoom on 0–100 GeV

Zoom: 0–100, linear scale, data/MC from 0.8 to 1.2.



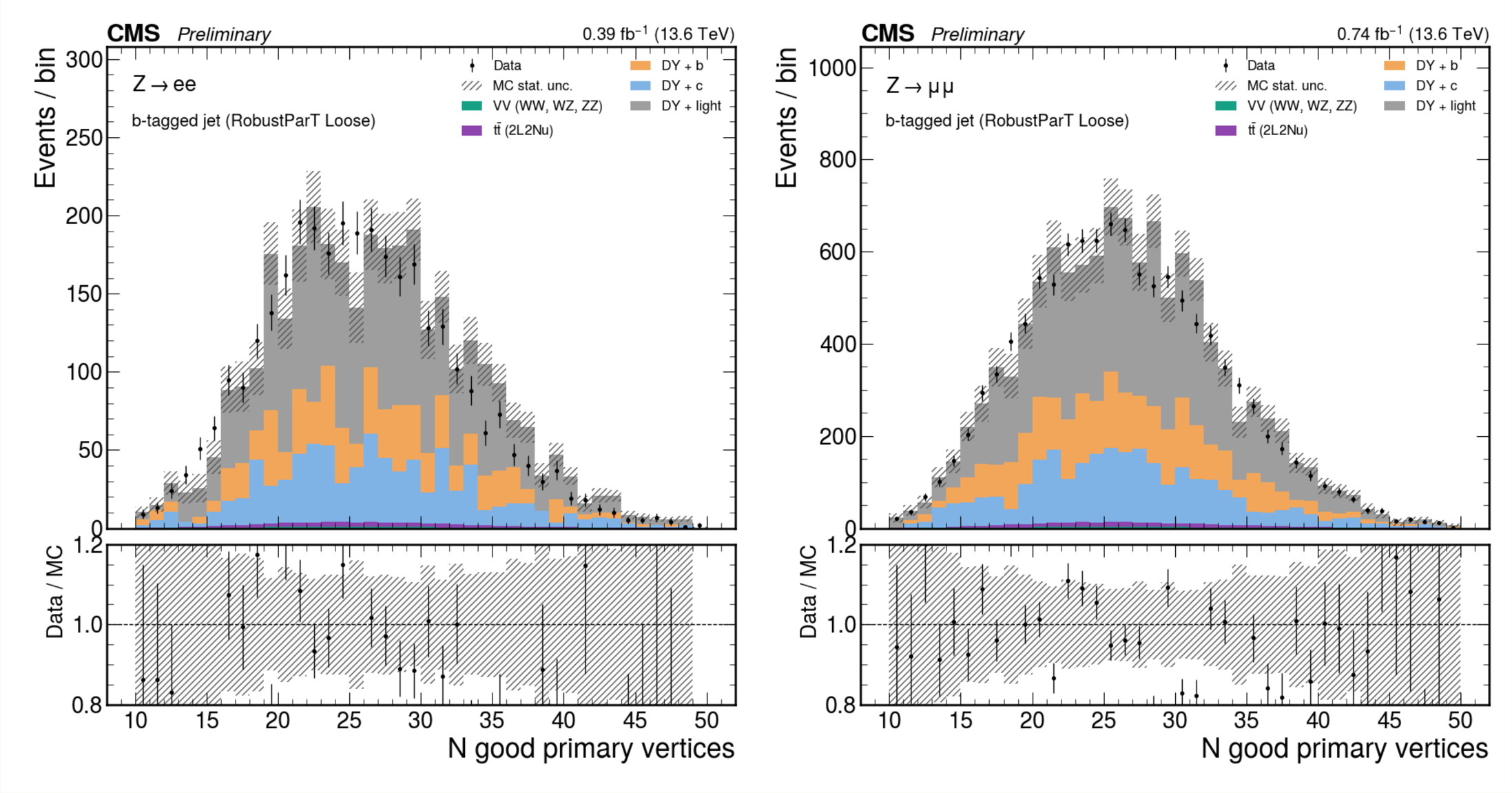
Number of good primary vertices

After the pileup weights. Data and MC differ slightly because our 20 test data files cover runs with less (ee) or about the same (μμ) pileup as all of 2022 C–D, which the MC is weighted to.



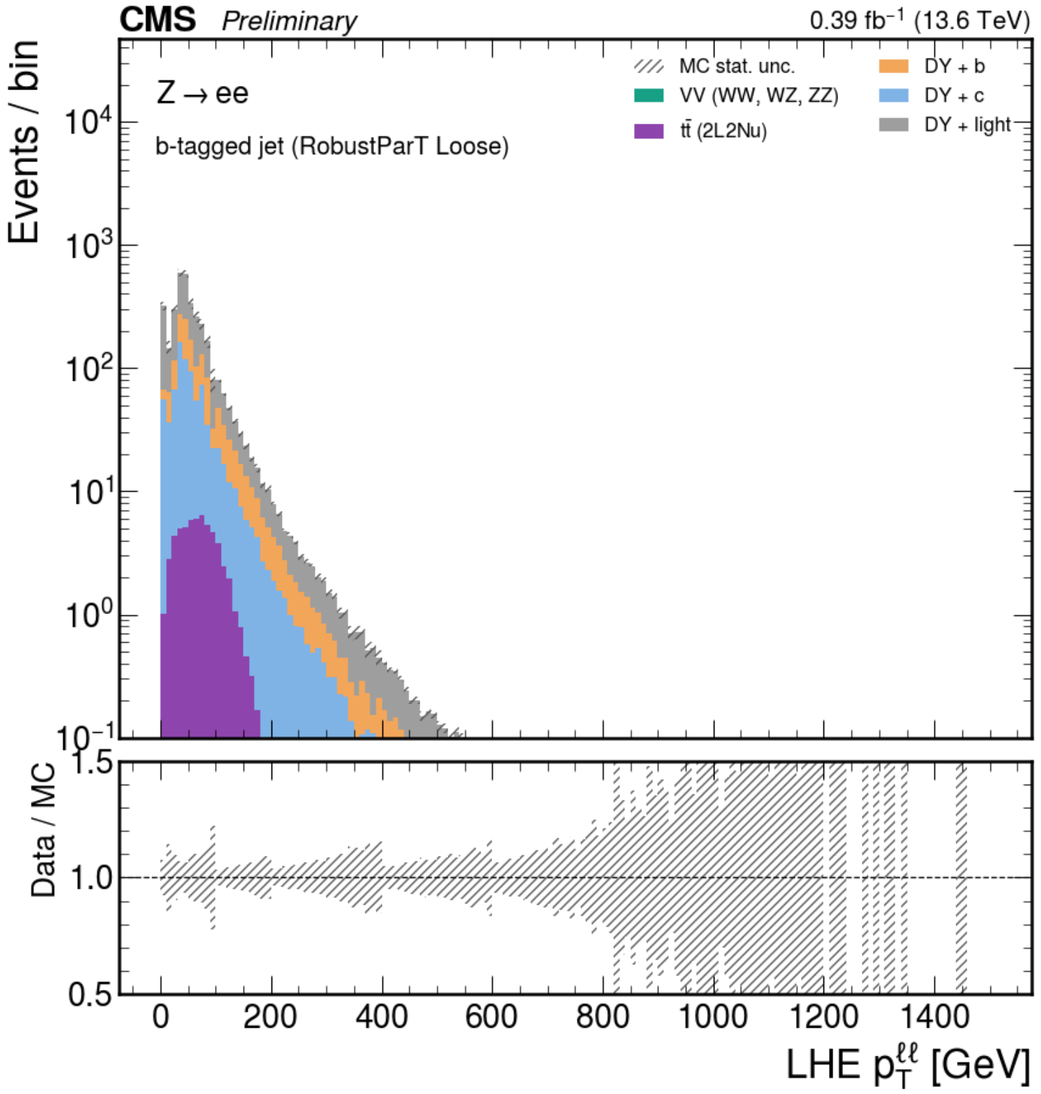
Number of good primary vertices, zoom on 10–50

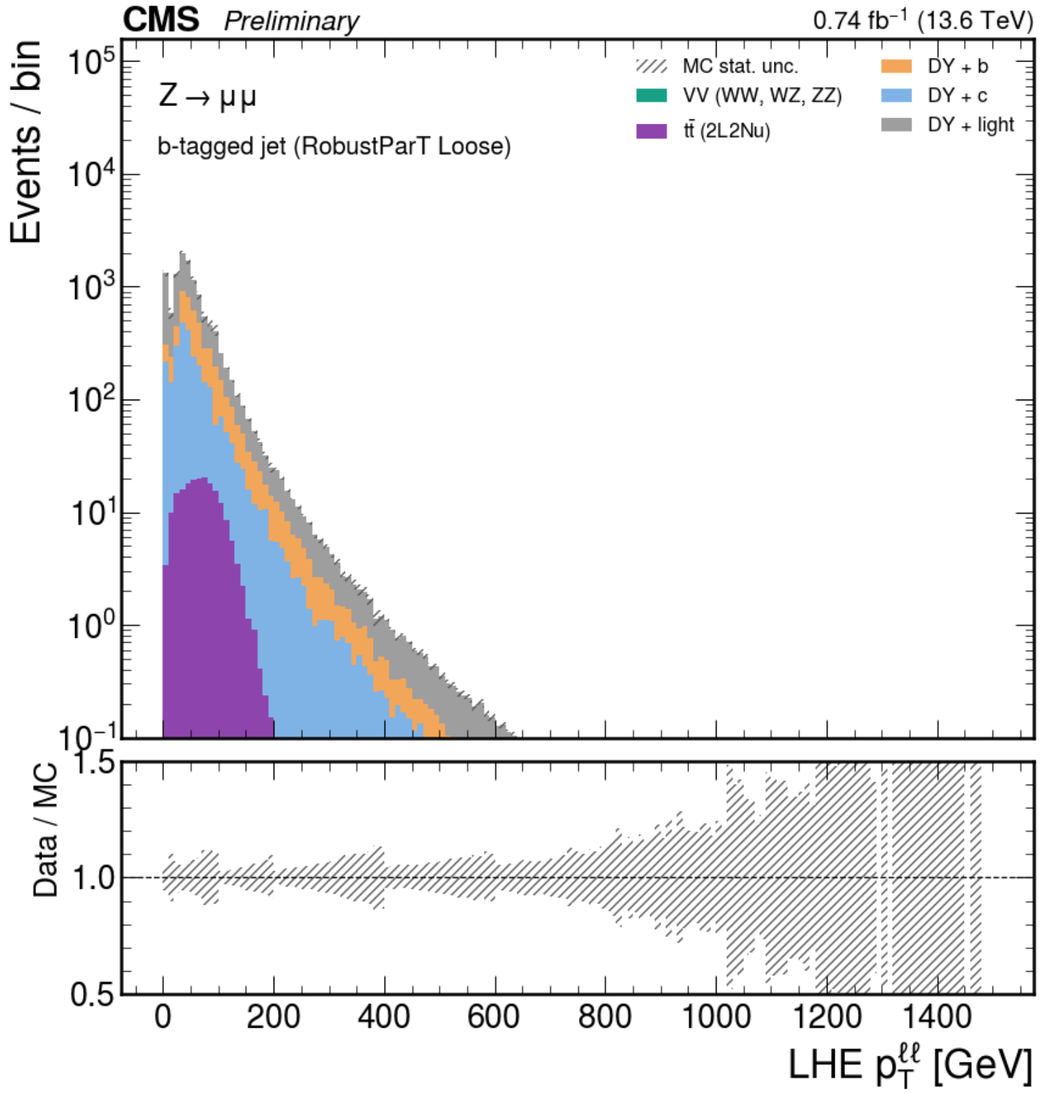
Zoom: 10–50, linear scale, data/MC from 0.8 to 1.2.



Generator-level Z pT (simulation only)

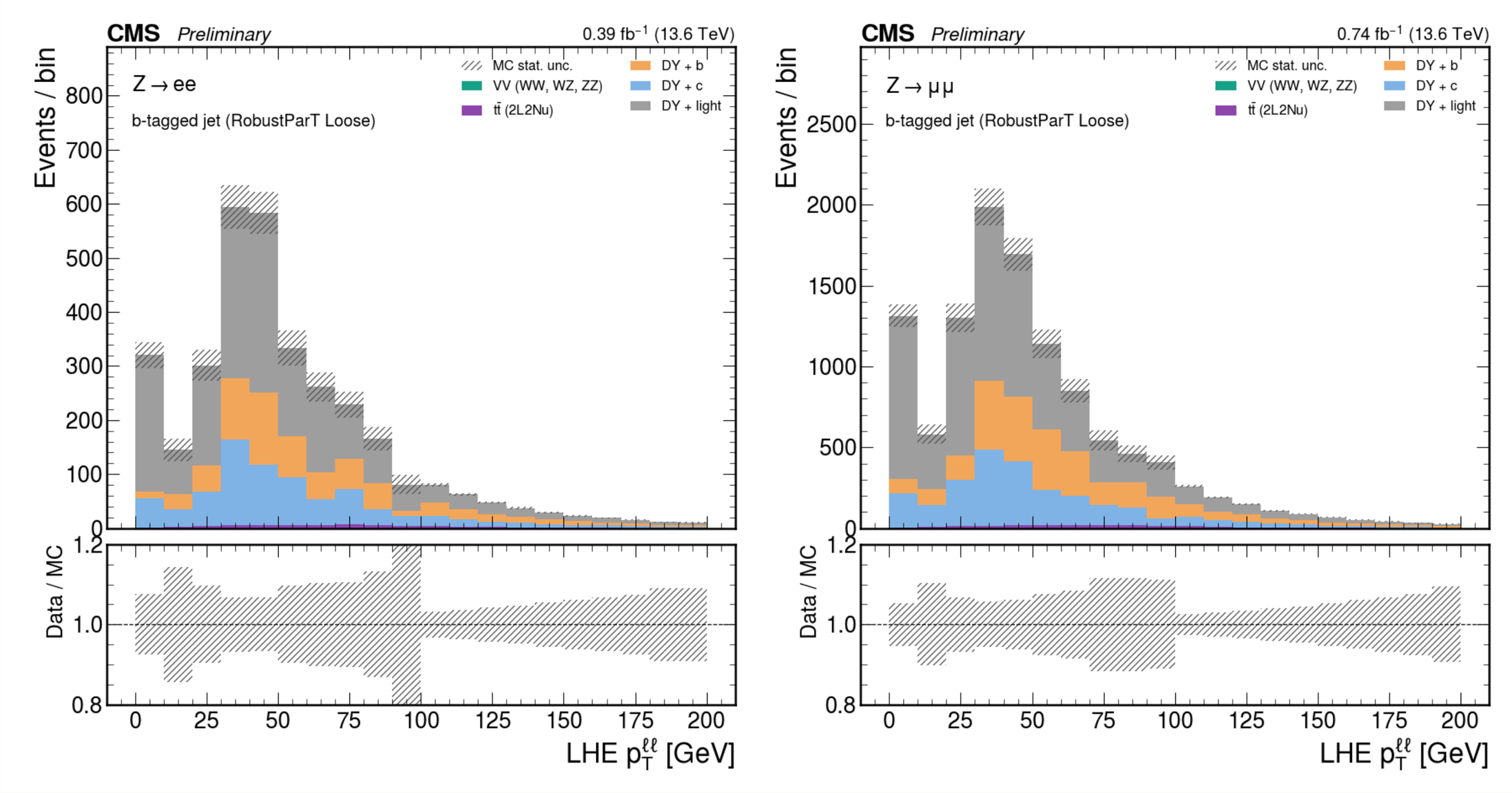
Z pT from the event generator, before any detector effect; no data. Shows which pT(ℓℓ)-binned DY sample contributes where.





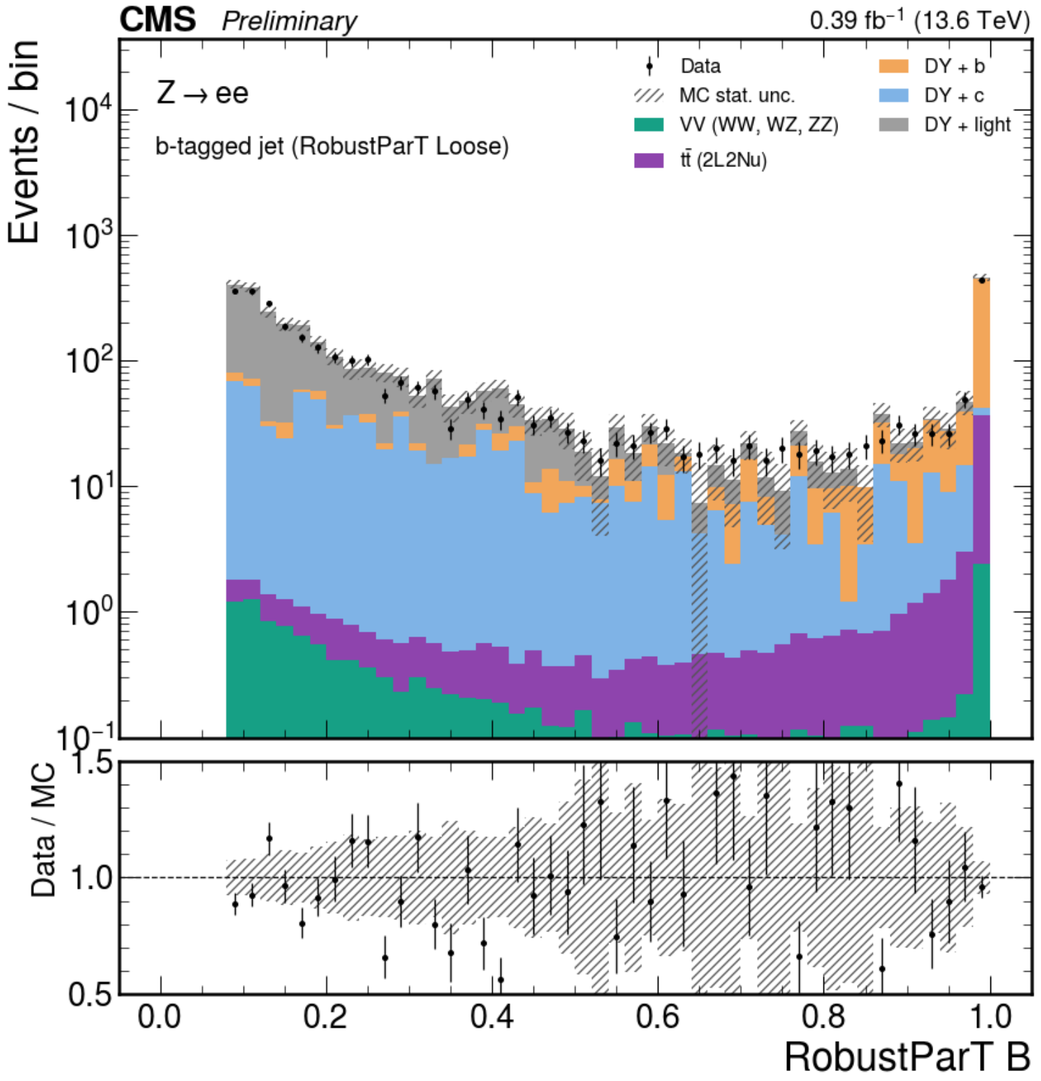
Generator-level Z pT, zoom on 0–200 GeV

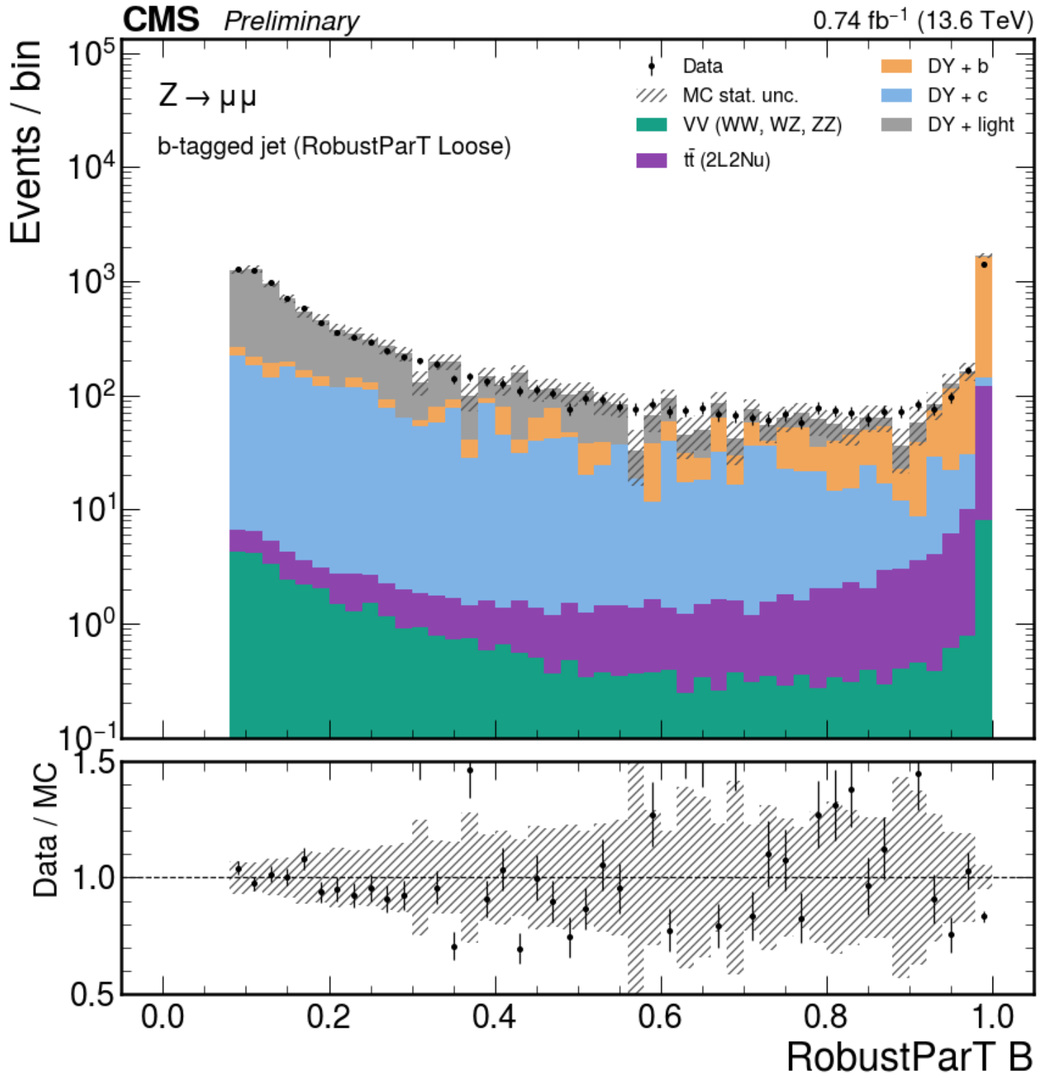
Zoom: 0–200, linear scale, data/MC from 0.8 to 1.2.



Leading-jet RobustParT B score (b vs everything)

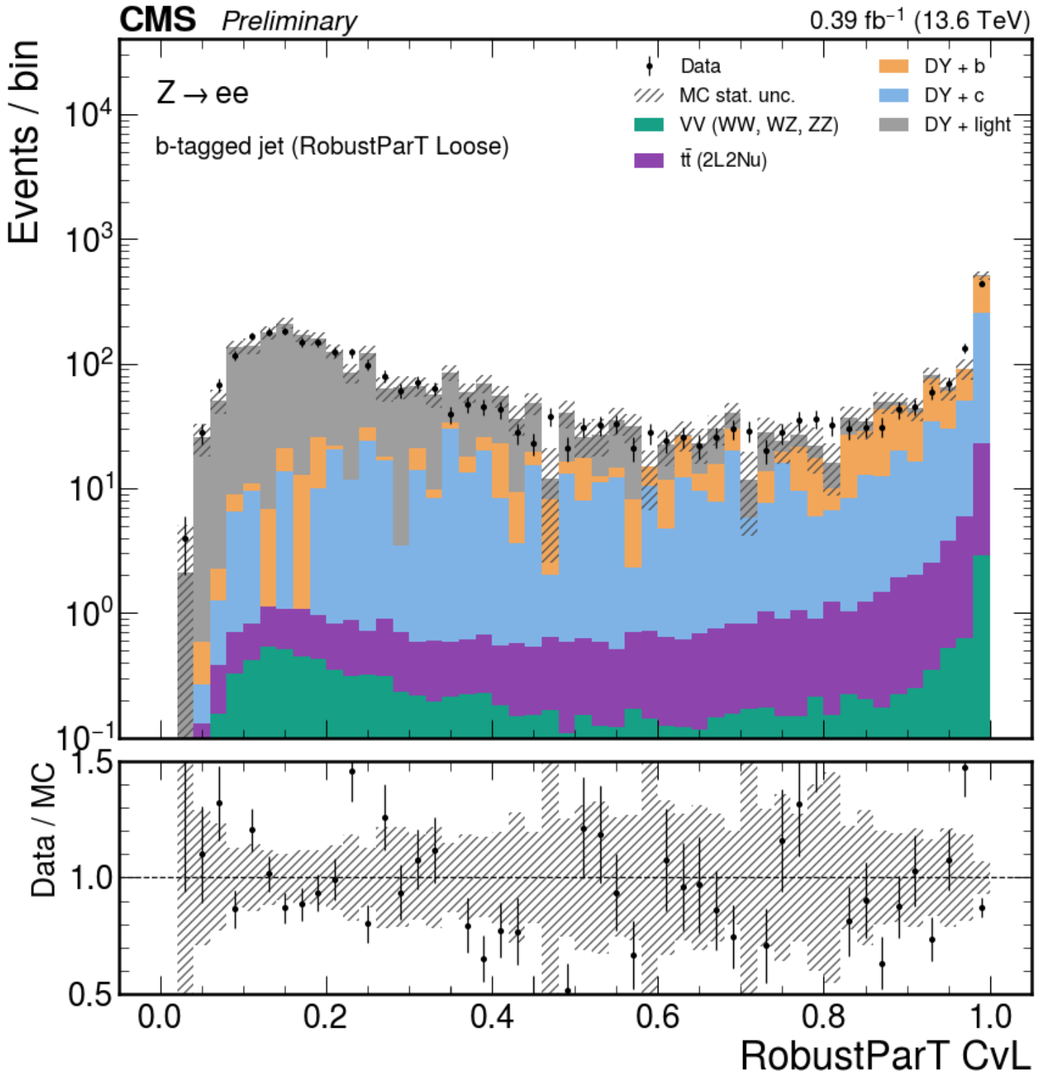
The score used for the selection: only jets above 0.0849 enter this region.

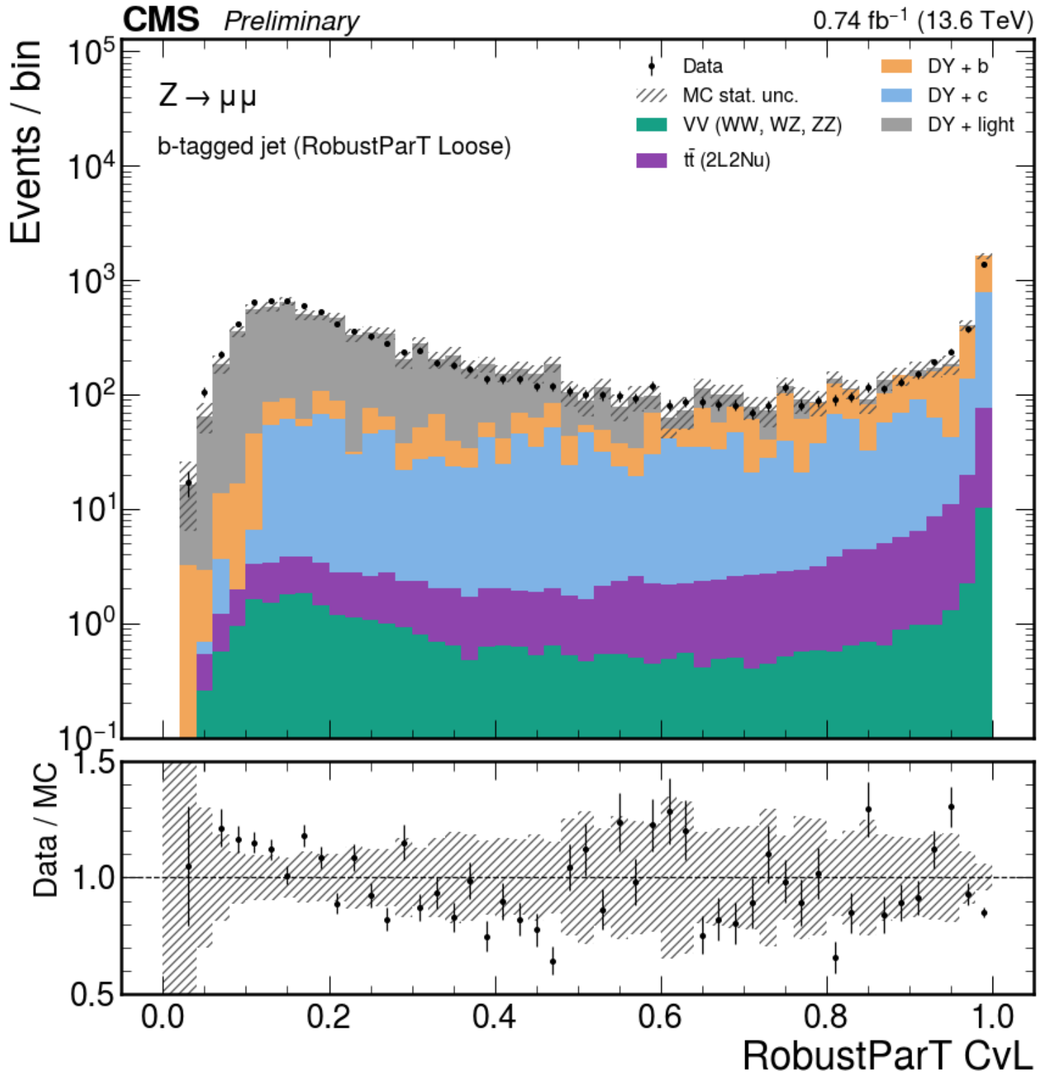




Leading-jet RobustParT CvL score (c vs light)

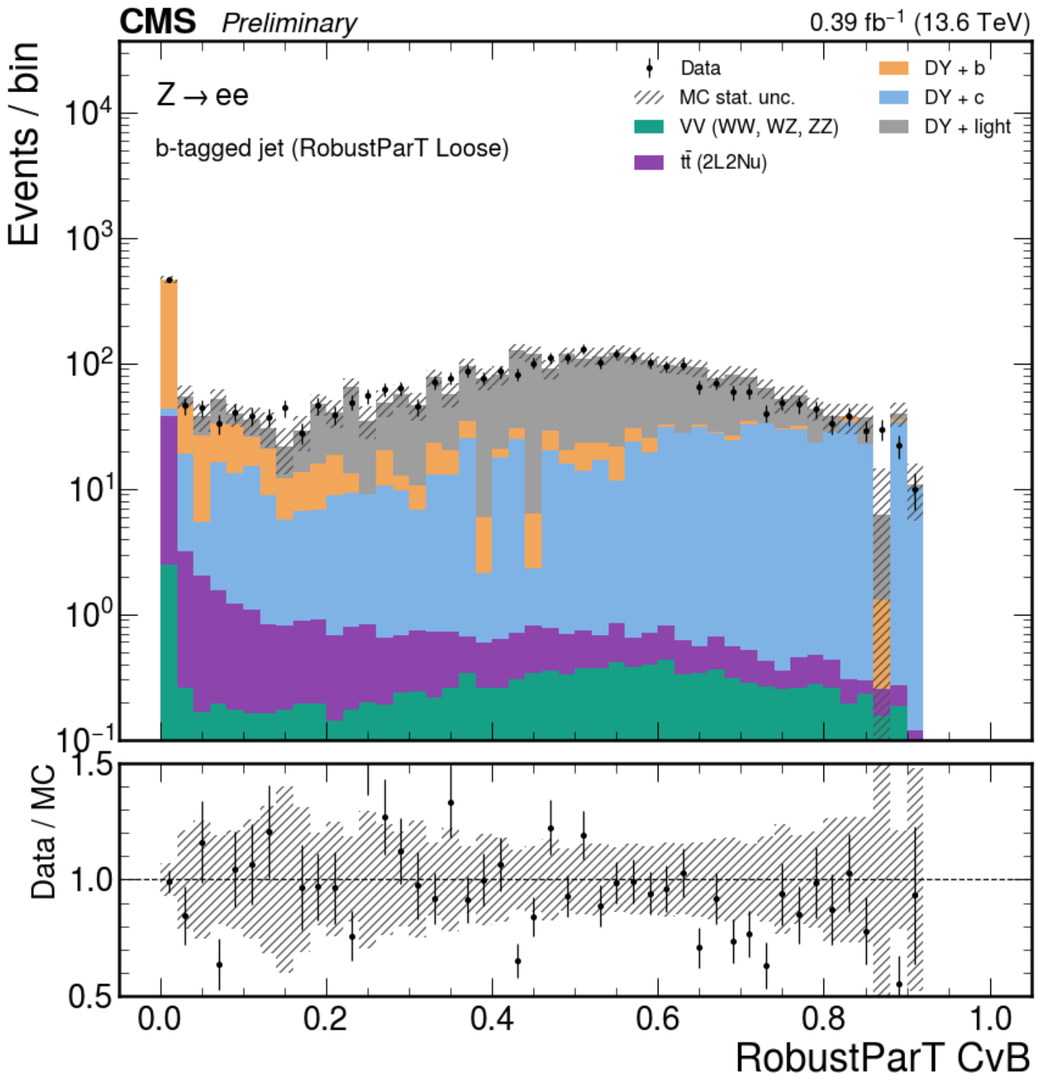
Not used in this b-tag selection; shown for comparison.

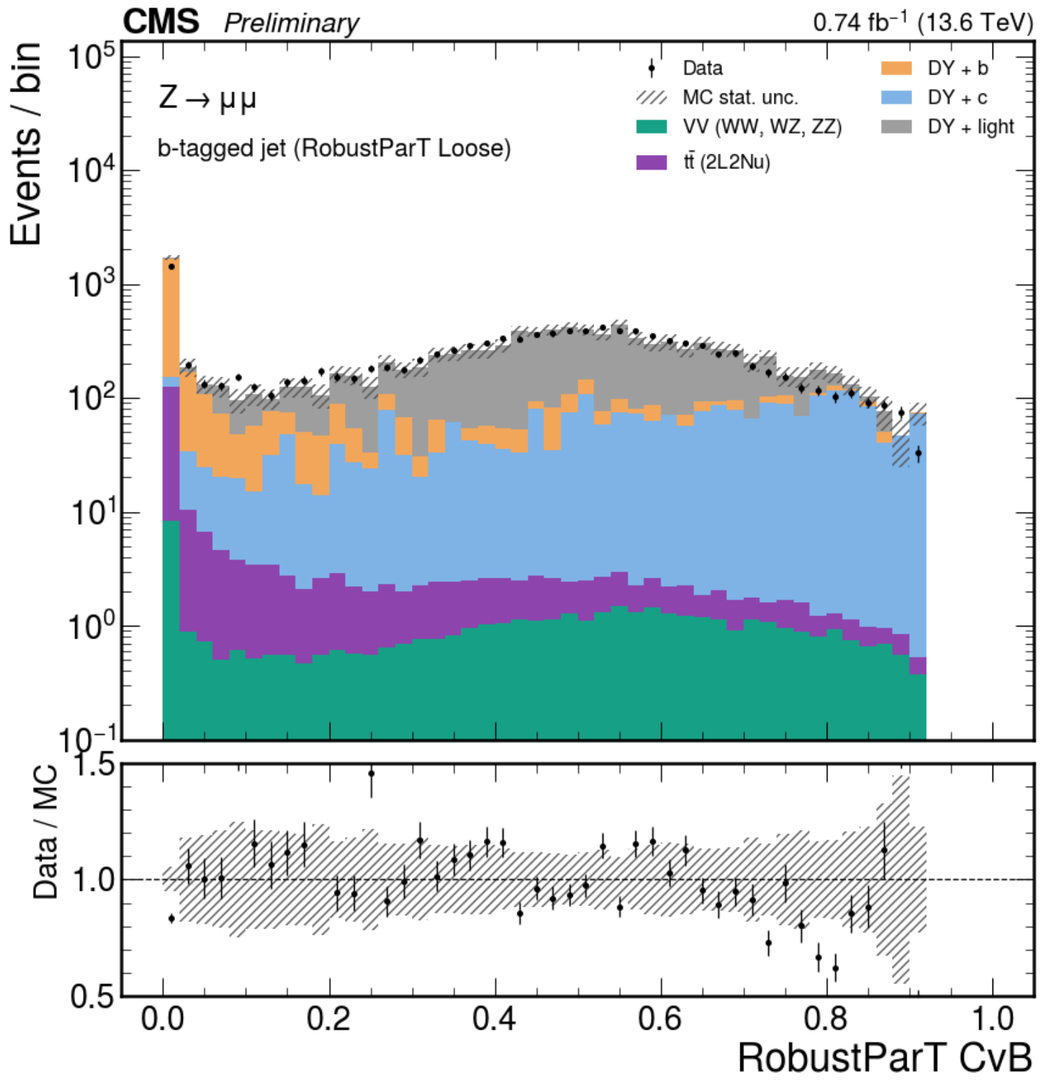




Leading-jet RobustParT CvB score (c vs b)

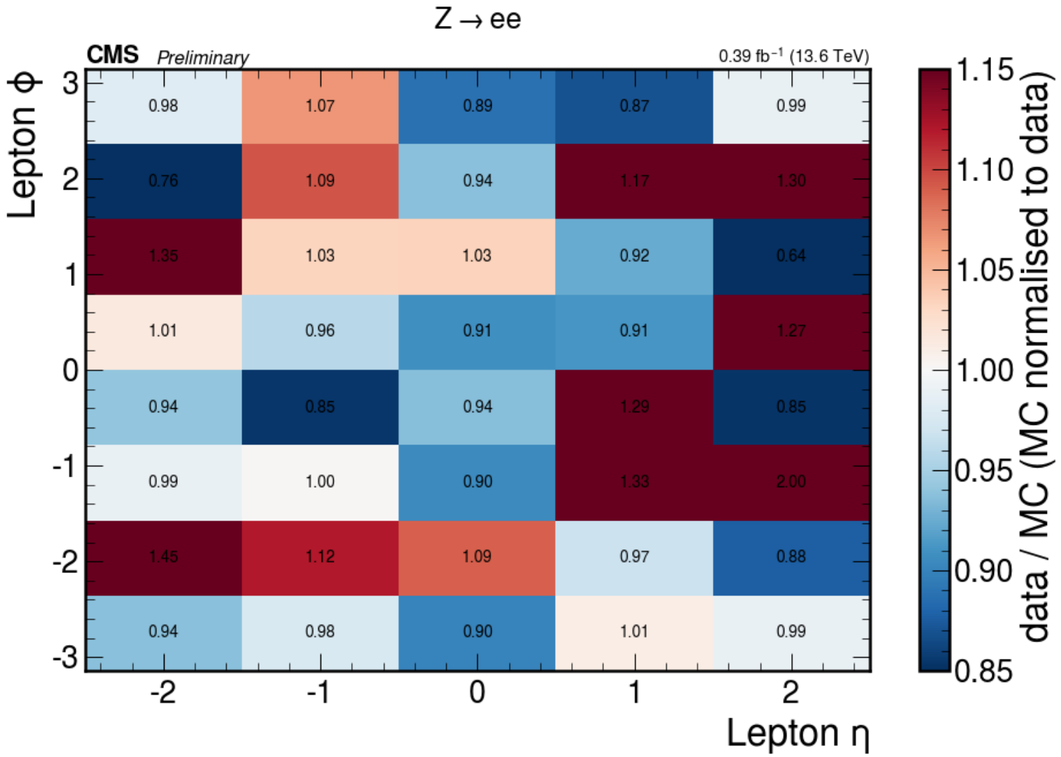
Not used in this b-tag selection. Low values = b-like.

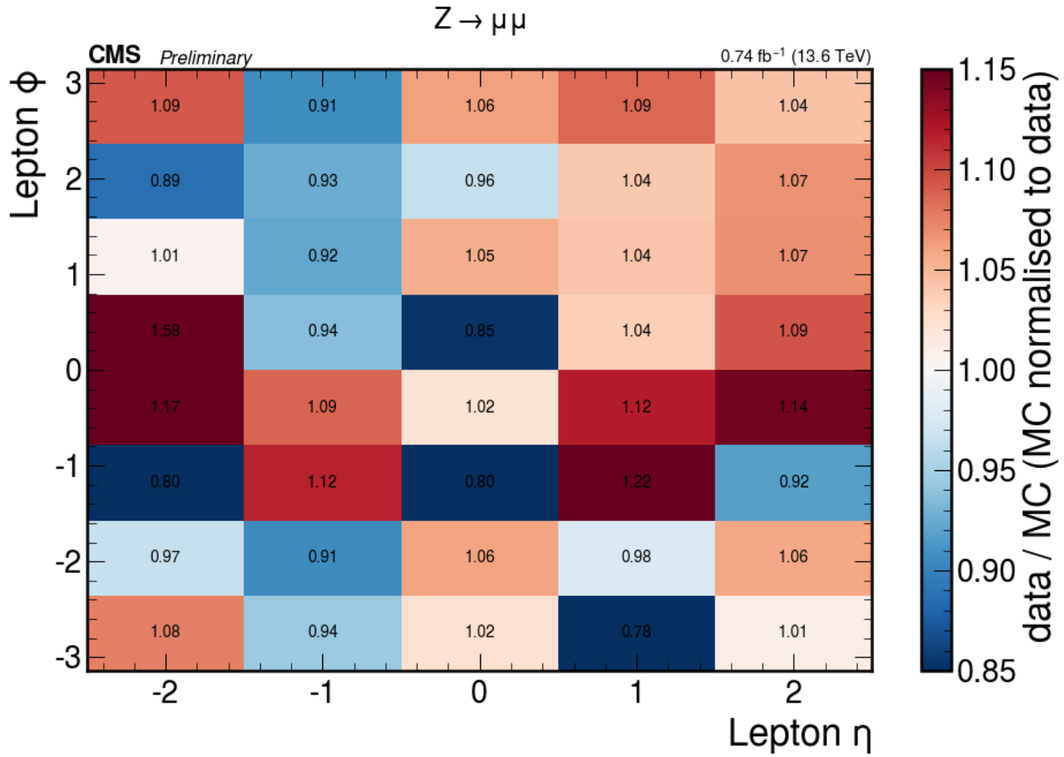




Lepton η–φ map: data / MC

Both Z leptons, MC normalised to data in each map, cells of 1.0 in η × π/4 in φ. Colour scale from 0.85 to 1.15; the number in each cell is data/MC. Few events per cell in a tagged region.





Summary

The three taggers select a similar sample at the Loose working point.

Share of real b jets (DY+b) in all MC, Z→ee / Z→μμ: PNet 21% / 22%, DeepJet 18% / 20%, RobustParT 20% / 22%. Loose is a loose selection: most tagged jets are c or light jets. Medium (~60% b) and Tight (~88% b) are much purer.

Data and MC tag the same fraction of jets within the statistical errors, for all three taggers.

Tag ratio (data/MC), Z→ee / Z→μμ: PNet 0.998 / 0.980; DeepJet 0.976 / 0.979; RobustParT 1.013 / 1.040 (largest deviation from 1: 1.6 standard deviations).

The MC excess at MET 40–120 GeV seen without tagging is also present for all three taggers.

So it is not caused by the tagging (2022 jet/MET issue, see the follow-up slides). Compare the same page number in each section to compare the taggers plot by plot.

The BTV SFs are applied; their uncertainties are not included yet

Applied: the BTV SF (central value) of every tagged jet in MC, for its true flavour (mean light-jet SF 1.18–1.21). Not yet included: the uncertainty on these SFs, typically ±4–10% for light, ±2–8% for c and ±1–3% for b jets at Loose. Light jets are about half of the tagged sample, so this would add a band of a few percent (roughly 3–6%), similar to the statistical errors. Also not yet included: other systematic uncertainties, the loose tau veto, the full dataset (20 files per sample).